Hodge Theory I, II, and III

Pierre Deligne

1970–1974

Note from the translator. This is an English translation of **P. Deligne**'s three papers on Hodge theory:

- "Théorie de Hodge I", *Actes du Congrès intern. math.* **1** (1970) pp. 425–430. `publications.ias.edu/node/359`
- "Théorie de Hodge II", *Pub. Math. de l'IHÉS* **40** (1971) pp. 5–58. `publications.ias.edu/node/361`
- "Théorie de Hodge III", *Pub. Math. de l'IHÉS* **44** (1974) pp. 5–77. `publications.ias.edu/node/373`

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You can view the entire source code of this translation (and contribute or submit corrections) in the GitHub repository `github.com/thosgood/hodge-theory/`. Corrections and comments welcome.

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Hodge Theory I

P. Deligne. "Théorie de Hodge I". *Actes du Congrès intern. math.* **1** (1970) pp. 425–430. publications.ias.edu/node/359

We intend to give a heuristic dictionary between statements in l-adic cohomology and statements in Hodge theory. This dictionary has as its most notable sources [S1960] and the conjectural theory of Grothendieck motives [D1969]. Up until this point, it has mainly served to formulate conjectures in Hodge theory, and it has sometimes even suggested a proof.

1.

Definition 1.1. A *mixed Hodge structure* H consists of

(a) a $\mathbb{Z}$-module $H_{\mathbb{Z}}$ of finite type (the "*integer lattice*");

(b) a finite increasing filtration W of $H_{\mathbb{Q}} = H_{\mathbb{Z}} \otimes \mathbb{Q}$ (the "*weight filtration*");

(c) a finite decreasing filtration F of $H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes \mathbb{C}$ (the "*Hodge filtration*").

This data is subject to the following condition: there exists a (unique) bigradation of $\mathrm{Gr}_W(H_{\mathbb{C}})$ by subspaces $H^{p,q}$ such that

(i) $\mathrm{Gr}_W^n(H_{\mathbb{C}}) = \bigoplus_{p+q=n} H^{p,q}$

(ii) the filtration F induces on $\mathrm{Gr}_W(H_{\mathbb{C}})$ the filtration

$$\mathrm{Gr}_W(F)^p = \bigoplus_{p' \geq p} H^{p',q'}$$

(iii) $\overline{H^{pq}} = H^{qp}$.

A *morphism* $f\colon H \to H'$ is a homomorphism $f_{\mathbb{Z}}\colon H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$ such that $f_{\mathbb{Q}}\colon H_{\mathbb{Q}} \to H'_{\mathbb{Q}}$ and $f_{\mathbb{C}}\colon H_{\mathbb{C}} \to H'_{\mathbb{C}}$ are compatible with the filtrations W and F (respectively).The *Hodge numbers* of H are the integers

$$h^{pq} = \dim H^{pq} = h^{qp}. \tag{1.2}$$

We say that H is pure *of weight* n if $h^{pq} = 0$ for $p+q \neq n$ (i.e. if $\mathrm{Gr}_W^i(H) = 0$ for $i \neq n$). We also say that H is a *Hodge structure of weight* n.

The *Tate Hodge structure* $\mathbb{Z}(1)$ is the Hodge structure of weight -2, of pure degree $(-1,-1)$, for which $\mathbb{Z}(1)_{\mathbb{C}} = \mathbb{C}$ and $\mathbb{Z}(1)_{\mathbb{Z}} = 2\pi i\mathbb{Z} = \mathrm{Ker}(\exp\colon \mathbb{C} \to \mathbb{C}^*) \subset \mathbb{C}$. We set $\mathbb{Z}(n) = \mathbb{Z}(1)^{\otimes n}$.

We can show that mixed Hodge structures form an abelian category. If $f\colon H \to H'$ is a morphism, then $f_{\mathbb{Q}}$ and $f_{\mathbb{C}}$ are strictly compatible with the filtrations W and F (cf. [Hodge Theory II, 2.3.5]).

2.

Let A be a normal integral ring of finite type over $\mathbb{Z}$, with field of fractions K, and $\overline{K}$ an algebraic closure of K.

Let K_{nr} be the largest sub-extension of $\overline{K}$ that is unramified at each prime ideal of A. We know that, or we set,

$$\pi_1(\mathrm{Spec}(A), \overline{K}) = \mathrm{Gal}(K_{\mathrm{nr}}/K).$$

For every closed point x of $\mathrm{Spec}(A)$, defined by some maximal ideal m_x of A, the residue field $k_x = A/m_x$ is finite; the point x defines a conjugation class of "Frobenius substitutions" $\varphi_x \in \pi_1(\mathrm{Spec}(A), \overline{K})$. We set $q_x = \# k_x$ and $F_x = \varphi_x^{-1}$.

Let K be a field of finite type over the prime field of characteristic p, let $\overline{K}$ be an algebraic closure of K, let l be a prime number $\neq p$, and let H be a $\mathbb{Z}_l$-module (or a $\mathbb{Q}_l$-module) of finite type endowed with a continuous action ρ of $\mathrm{Gal}(\overline{K}/K)$. We will still suppose in what follows that there exists an A as above, with l invertible in A, and such that ρ factors through $\pi_1(\mathrm{Spec}(A), \overline{K}) = \mathrm{Gal}(K_{\mathrm{nr}}/K)$. We say that H is ***pure of weight*** n if, for every closed point x of an non-empty open subset of $\mathrm{Spec}(A)$, the eigenvalues α of F_x acting on H are algebraic integers whose complex conjugates are all of absolute value $|\alpha| = q_x^{n/2}$.

Principle 2.1. If the Galois module H "comes from algebraic geometry", then there exists a (unique) increasing filtration W on $H_{\mathbb{Q}_l} = H \otimes_{\mathbb{Z}_l} \mathbb{Q}_l$ (the "***weight filtration***") that is Galois invariant and such that $\mathrm{Gr}_n^W(H)$ is pure of weight n.

We can also further suppose that $\mathrm{Gr}_n^W(H)$ is semi-simple.

When we have a resolution of singularities we can often give a conjectural definition of W, whose validity follows from the Weil conjectures [W1949] (cf. §6).

Let μ be the subgroup of $\overline{K}^*$ given by the roots of unity. The *Tate module* $\mathbb{Z}_l(1)$, defined by

$$\mathbb{Z}_l(1) = \mathrm{Hom}(\mathbb{Q}_l/\mathbb{Z}_l, \mu)$$

is pure of weight -2. We set $\mathbb{Z}_l(n) = \mathbb{Z}_l(1)^{\otimes} n$.

It is trivial that every morphism $f\colon H \to H'$ is strictly compatible with the weight filtration.

Principle 2.1 agrees with the fact that every extension of $\mathbb{G}_m$ ("weight -2") by an abelian variety ("weight $-1 > -2$") is trivial.

3.

Translation. The Galois modules that appear in l-adic cohomology have, as analogues, over $\mathbb{C}$, mixed Hodge structures. We further have the dictionary:

4.

Let X be a complex algebraic variety (i.e. a scheme of finite type over $\mathbb{C}$ that we assume to be separated). Then there exists a subfield K of $\mathbb{C}$, of finite type over $\mathbb{Q}$, such that X can be defined over K (i.e. it comes from an extension of scalars of K to $\mathbb{C}$ applied to a K-scheme X'). Let $\overline{K}$ be the algebraic closure of K in $\mathbb{C}$. The Galois group $\mathrm{Gal}(\overline{K}/K)$ then acts on the l-adic cohomology groups $H^\bullet(X, \mathbb{Z}_l)$; we have

$$H^\bullet(X(\mathbb{C}), \mathbb{Z}) \otimes \mathbb{Z}_l = H^\bullet(X, \mathbb{Z}_l) = H^\bullet(X'_{\overline{K}}, \mathbb{Z}_l).$$

By §3, we should expect for the cohomology groups $H^n(X(\mathbb{C}), \mathbb{Z})$ to carry natural mixed Hodge structures. This is what we can prove (see [Hodge Theory II, 3.2.5] in the case where X is smooth; the proof is algebraic, using classical Hodge theory [W1958]). For X projective and smooth, the Weil conjectures imply that $H^n(X, \mathbb{Z}_l)$ is pure of weight n, while classical Hodge theory endows $H^n(X, \mathbb{Z})$ with a Hodge structure of weight n. For every morphism $f\colon X \to Y$, and for K large enough, $f^\bullet\colon H^\bullet(Y, \mathbb{Z}_l) \to H^\bullet(X, \mathbb{Z}_l)$ Galois-commutes (by structure transport); similarly, $f^\bullet\colon H^\bullet(Y, \mathbb{Z}) \to H^\bullet(X, \mathbb{Z})$ is a morphism of mixed Hodge structures. For X smooth, the cohomology class Z in $H^{2n}(X, \mathbb{Z}_l(n))$ of an algebraic cycle of codimension n defined over K is Galois invariant, i.e. it defines

$$c(Z) \in \mathrm{Hom}_{\mathrm{Gal}}(\mathbb{Z}_l(-n), H^{2n}(X, \mathbb{Z}_l)).$$

Similarly, the cohomology class $c(Z) \in H^{2n}(X(\mathbb{C}), \mathbb{Z})$ is of pure degree (n, n), i.e. it corresponds to

$$c(Z) \in \mathrm{Hom}_{\mathrm{H.M.}}(\mathbb{Z}(-n), H^{2n}(X(\mathbb{C}), \mathbb{Z})).$$

5.

If $f\colon H \to H'$ is a Galois-compatible morphism between $\mathbb{Q}_l$-vector spaces of different weights, then $f = 0$. Similarly, if $f\colon H \to H'$ is a morphism of pure mixed Hodge structures of different weights, then f is torsion. A more useful remark is

Scholium. Let H and H' be Hodge structures of weight n and n' (respectively), with $n > n'$. Let $f\colon H_\mathbb{Q} \to H'_\mathbb{Q}$ be a homomorphism such that $f\colon H_\mathbb{C} \to H'_\mathbb{C}$ respects F. Then $f = 0$.

6.

Let X be a smooth projective variety over $\mathbb{C}$, let $D = \sum_1^n D_i$ be a normal crossing divisor in X, with D_i all smooth divisors, and let j be the inclusion of $U = X \setminus D$ into X. For $Q \subset [1, n]$, we set $D_q = \bigcap_{i \in Q} D_i$.

In l-adic cohomology, we canonically have

$$R^q j_* \mathbb{Z}_l = \bigoplus_{\#Q = q} \mathbb{Z}_l(-q)_{D_Q} \tag{6.1}$$

and the Leray spectral sequence for j is of the form

$$E_2^{pq} = \bigoplus_{\#Q=q} H^p(D_Q, \mathbb{Q}_l) \otimes \mathbb{Z}_l(-q) \Rightarrow H^{p+q}(U, \mathbb{Q}_l). \tag{6.2}$$

By the Weil conjectures [W1949], $H^p(D_Q, \mathbb{Q}_l)$ is pure of weight p, so that E_2^{pq} is pure of weight $p+2q$. As a quotient of a sub-object of E_2^{pq}, E_r^{pq} is also pure of weight $p+2q$. By §5, $d_r = 0$ for $r \geq 3$, since the weights $p+2q$ and $p+2q-r+2$ of E_r^{pq} and $E_r^{p+q,q-r+1}$ (respectively) are different. Thus $E_3^{pq} = E_\infty^{pq}$. Up to renumbering, the weight filtration of $H^\bullet(U, \mathbb{Q}_l)$ is the abutment of Equation 6.2:

$$\operatorname{Gr}_n^W(H^k(U, \mathbb{Q}_l)) = E_3^{2k-n,n-k}. \tag{6.3}$$

7.

In integer cohomology, for the usual topology, the Leray spectral sequence for j is of the form

$$'E_2{}^{pq} = \bigoplus_{\#Q=q} H^p(D_Q, \mathbb{Z}) \Rightarrow H^{p+q}(U, \mathbb{Z}). \tag{7.1}$$

Since each D_Q is a non-singular projective variety, $'E_2{}^{pq}$ is endowed with a Hodge structure of weight p. We set $E_2^{pq} = {}'E_2{}^{pq} \otimes \mathbb{Z}(-q)$ (a Hodge structure of weight $p+2q$). As an abelian group, $E_2^{pq} = {}'E_2{}^{pq}$; it is interesting to consider Equation 7.1 as a spectral sequence with initial page E_2^{pq}. By §3, we should expect for $d_2\colon E_2^{pq} \to E_2^{p+2,q-1}$ to be a morphism of Hodge structures. We prove this by thinking of d_2 as a Gysin morphism. Then E_3^{pq} is endowed with a Hodge structure of weight $p+2q$. By §3, we expect, *modulo torsion*, for the spectral sequence (*[Trans.] The original refers to (6.4) instead of (6.2), but this seems to be a typo.*) Equation 6.2 to degenerate at the E_3 page (i.e. $E_3 = E_\infty$), and for the vanishing of the d_r (for $r \geq 3$) to be an application of §5. This programme was successfully completed in [Hodge Theory II, 3.2]. There, we *define* the weight filtration of $H^\bullet(U, \mathbb{Q})$ as the abutment of Equation 7.1, up to renumbering Equation 6.3.

In fact, to endow the cohomology groups $H^\bullet$ with a mixed Hodge structure, the key point has always been, up until now, to find a spectral sequence E abutting to $H^\bullet$ such that the l-adic analogue of E_2^{pq} be conjecturally pure (of weight $p+2q$); E_2^{pq} should then carry a natural Hodge structure (of weight $p+2q$), and the filtration W is then the abutment of E.

8.

Let $\operatorname{Spec}(V)$ be the spectrum of a Henselian discrete valuation ring (a *Henselian trait*) with field of fractions K, and residue field k that is of finite type over the prime field of characteristic p. Let $\overline{K}$ be an algebraic closure of K, and let H be a vector space of finite dimension over $\mathbb{Q}_l$ (for $l \neq p$) on which $\operatorname{Gal}(\overline{K}/K)$ acts continuously. By Grothendieck, we know ([ST1968, Appendix]) that a subgroup of finite index of the inertia group I acts unipotently. By replacing V with a finite extension, we arrive to the case where the action of all of I is unipotent (the *semi-stable*case); it then factors as the largest pro-l-group I_l that is a quotient of I, and canonically isomorphic to $\mathbb{Z}_l(1)$.

Principle 8.1. In the semi-stable case, if the Galois module H "comes from algebraic geometry", then there exists a (unique) increasing filtration W of H (the "***weight filtration***") such that I acts trivially on $\mathrm{Gr}_n^W(H)$, and such that $\mathrm{Gr}_n^W(H)$, as a Galois module under $\mathrm{Gal}(\overline{k}/k) \simeq \mathrm{Gal}(\overline{K}/K)/I$, is pure of weight n.

We can compare this with Principle 2.1 and with the appendix of Reference **??**.

If we have a resolution of singularities, then we can sometimes give a conjectural definition of W whose validity follows from the Weil conjectures. With the help of the resolution and of Weil, it is sometimes easy to show that, in any case, H splits into pure Galois modules (under $\mathrm{Gal}(\overline{k}/k)$).

Suppose that H is semi-stable. For $T \in I_t$, we define $\log T$ by the *finite* sum $-\sum_{n>0}(\mathrm{Id}-T)^n/n$. The map $(T,x) \mapsto \log T(x)$ can be identified with a homomorphism

$$M\colon \mathbb{Z}_l(1) \otimes H \to H. \tag{8.2}$$

Since $\mathbb{Z}_l(1)$ is of weight -2, we necessarily have (cf. §5)

$$M(\mathbb{Z}_l(1) \otimes W_n(H)) \subset W_{n-2}(H), \tag{8.3}$$

and M induces

$$\mathrm{Gr}(M)\colon \mathbb{Z}_l(1) \otimes \mathrm{Gr}_n^W(H) \to \mathrm{Gr}_{n-2}^W(H). \tag{8.4}$$

If X is a non-singular projective variety over an algebraically closed field k_0, then we define

$$L\colon \mathbb{Z}_l(-1) \otimes H^\bullet(X,\mathbb{Z}_l) \to H^\bullet(X,\mathbb{Z}_l)$$

as being the cup product with the cohomology class of a hyperplane section. We note that there is a formal analogy between L and M; in the same way that M is defined by an action of $\mathbb{Z}_l(1)$, we can think of L as being defined by an action of $\mathbb{Z}_l(-1)$; L increases the degree by 2, and $\mathrm{Gr}\,M$ Equation 8.4 decreases it by 2.

9.

Let D be the unit disc, $D^* = D \setminus \{0\}$, and X

$$\begin{tikzcd} X \arrow[r] \arrow[d, "f"'] & \mathbb{P}^r(\mathbb{C}) \times D \arrow[d, "\mathrm{pr}_2"] \\ D \arrow[r, equal] & D \end{tikzcd}$$

a family of projective varieties parameterised by D, with f proper, and $f|D^*$ smooth.

Keeping the notation of §8, and recalling that, in the analogy between Henselian traits and small neighbourhoods of 0 in the complex line, we have the following dictionary (note that the spectrum of the ring of germs at 0 of holomorphic functions is a Henselian trait):

Dictionary 9.1.

Note that $\widetilde{X}$ is homotopically equivalent to each of the fibres $X_t = f^{-1}(t)$ (for $t \in D^*$): $H^i(X_{\overline{K}}, \mathbb{Z}_l)$ is again analogous to $H^i(X_t, \mathbb{Z})$, and the transformation of the monodromy T corresponds to the action of I.

Here, again, we know that a subgroup of finite index of $\pi_1(D^*)$ acts unipotently on $H^i(\widetilde{X}, \mathbb{Q}) = H^i(X_t, \mathbb{Q})$. We place ourselves in the semi-stable case, where all of $\pi_1(D^*)$ acts unipotently (this reduces to replacing D by a finite covering), and let T be the action of the canonical generator of $\pi_1(D^*)$.

By §3 and §8, we expect for $H^i(\widetilde{X}, \mathbb{Q}) \simeq H^i(X_t, \mathbb{Q})$ to be endowed with an increasing filtration W, for $\mathrm{Gr}_n^W(H^i(\widetilde{X}, \mathbb{Q}))$ to be endowed with a Hodge structure of weight n, for $\log T(W_n) \subset W_{n-2}$, and for $\log T$ to induce a morphism of Hodge structures

$$M_n\colon \mathbb{Z}(-1) \otimes \mathrm{Gr}_n^W(H^i) \to \mathrm{Gr}_{n-2}^W(H^i).$$

We would further like for Equation 8.2, and not just Equation 8.3 and Equation 8.4, to have an analogue.

We have in fact managed to define, for each vector u of the tangent space to D at $\{0\}$, a mixed Hodge structure $\mathscr{H}_u$ on $H^i(\widetilde{X}, \mathbb{Z})$. The filtration W and the Hodge structures on the $\mathrm{Gr}_n^W(H^i)$ are independent of u, and the dependence on u of $\mathscr{H}_u$ can be expressed simply in terms of T.

Analogously to Equation 8.2, we find that, for any u, $\log T$ induces a homomorphism of mixed Hodge structures

$$M\colon \mathbb{Z}(1) \otimes H^i(\widetilde{X}, \mathbb{Z}) \to H^i(\widetilde{X}, \mathbb{Z}).$$

Finally, the analogy in §8 is not misleading (but here, the fact that $f|D^*$ is assumed to be proper and smooth is probably essential). We can prove that

$$(\log T)^k\colon\ \mathrm{Gr}_{n+k}^W(H^n(\widetilde{X}, \mathbb{Q})) \to \mathrm{Gr}_{n-k}^W(H^n(\widetilde{X}, \mathbb{Q}))$$

is an isomorphism for all k (cf. [W1958, IV 6, Corollary to Theorem 5]). This characterises the filtration W. Until now, we have only had an analogue of the positivity theorem of Hodge (cf. [W1958, IV 7, Corollary to Theorem 7]) in very particular cases. We hope that the mixed structures $\mathscr{H}_u$ determine the asymptotic behaviour, for $t \to 0$, of the family of pure structures $H^i(X, \mathbb{Z})$ (for $t \in D^*$).

Hodge Theory II

P. Deligne. "Théorie de Hodge II". *Pub. Math. de l'IHÉS* **40** (1971) pp. 5–58. publications.ias.edu/node/361

0. Introduction

Work presented as a doctoral thesis at l'Université d'Orsay.

By Hodge, the cohomology space $\mathrm{H}^n(X,\mathbb{C})$ of a compact Kähler variety X is endowed with a "Hodge structure" of weight n, i.e. a natural bigrading

$$\mathrm{H}^n(X,\mathbb{C}) = \bigoplus_{p+q=n} \mathrm{H}^{p,q}$$

which satisfies $\overline{\mathrm{H}^{p,q}} = \mathrm{H}^{q,p}$. We will show here that the complex cohomology of a non-singular, not necessarily compact, algebraic variety is endowed with a structure of a slightly more general type, which presents $\mathrm{H}^n(X,\mathbb{C})$ as a "successive extension" of Hodge structures of decreasing weights, contained between $2n$ and n, whose Hodge numbers $h^{p,q} = \dim \mathrm{H}^{p,q}$, are zero for both $p > n$ and $q > n$.

The reader will find an explanation in [Hodge Theory I] of the yoga that underlies this construction.

The proof, which is essentially algebraic, relies on one hand on Hodge theory, and on the other on Hironaka's resolution of singularities, which allows us, via a spectral sequence, to "express" the cohomology of a non-singular quasi-projective algebraic variety in terms of the cohomology of non-singular projective varieties.

§1 contains, apart from reminders on filtrations gathered together for the ease of the reader, two key results:

(a) Theorem 1.2.10, which will only be used via its corollary, Theorem 2.3.5, which gives the fundamental properties of "mixed Hodge structures".

(b) The "two filtrations lemma", Theorem 1.3.16.

§2 recalls Hodge theory and introduces mixed Hodge structures.

The heart of this work is §3.2, which defines the mixed Hodge structure on $\mathrm{H}^n(X,\mathbb{C})$, and establishes some degenerations of spectral sequences.

§4 gives diverse applications, all following from Theorem 4.1.1 and the theory of the (K/k)-trace, for the resulting Hodge structures (Corollary 4.1.2). The principal ones are Theorem 4.2.6 and Corollary 4.4.15.

1. Filtrations

1.1. Filtered objects

Let $\mathscr{A}$ be an abelian category.

We will be considering $\mathbb{Z}$-filtrations, finite, in general, on the objects of $\mathscr{A}$:

Definition 1.1.2. A *decreasing* (resp. *increasing*) *filtration* F of an object A of $\mathscr{A}$ is a family $(F^n(A))_{n\in\mathbb{Z}}$ (resp. $(F_n(A))_{n\in\mathbb{Z}}$) of sub-objects of A satisfying

$$\forall n,m \quad n \leqslant m \implies F^m(A) \subset F^n(A)$$

(resp. $n \leqslant m \implies F_n(A) \subset F_m(A)$).

A *filtered object* is an object endowed with a filtration. When there is no chance of confusion, we often denote by the same letter filtrations on different objects of $\mathscr{A}$.

If F is a decreasing (resp. increasing) filtration on A, then we set $F^{\infty}(A) = 0$ and $F^{-\infty}(A) = A$ (resp. $F_{-\infty}(A) = 0$ and $F_{\infty}(A) = A$).

The *shifted filtrations* of a decreasing filtration W are defined by

$$W[n]^p(A) = W^{n+p}(A)$$

for $n \in \mathbb{Z}$.

If R is a decreasing (resp. increasing) filtration of A, then the $F_n(A) = F^{-n}(A)$ (resp the $F^n(A) = F_{-n}(A)$) form an increasing (resp. decreasing) filtration of A.

This allows us in principal to consider only decreasing filtrations; *unless otherwise explicitly mentioned, when we say "filtration" we always mean "decreasing filtration"*.

A filtration F of A is said to be *finite* if there exist n and m such that $F^n(A) = A$ and $F^m(A) = 0$.

A *morphism* from a filtered object (A, F) to a filtered object (B, F) is a morphism f from A to B that satisfies $f(F^n(A)) \subset F^n(B)$ for all $n \in \mathbb{Z}$.

Filtered objects (resp. finite filtered objects) of $\mathscr{A}$ form an additive category in which inductive limits and finite projective limits exist (and thus kernels, cokernels, images, and coimages of a morphism).

A morphism $f\colon (A, F) \to (B, F)$ is said to be *strict*, or *strictly compatible with the filtrations*, if the canonical arrows from $\mathrm{Coim}(f)$ to $\mathrm{Im}(f)$ is an isomorphism of filtered objects (cf. Proposition 1.1.11).

Let $(-)^{\circ}$ be the contravariant identity functor from $\mathscr{A}$ to the dual category $\mathscr{A}^{\circ}$. If (A, F) is a filtered object of $\mathscr{A}$, then the $(A/F^n(A))^{\circ}$ can be identified with sub-objects of A°. The *dual* filtration on A° is defined by

$$F^n(A^{\circ}) = (A/F^{1-n})^{\circ}.$$

The double dual of (A, F) can be identified with (A, F). This construction identifies the dual of the category of filtered objects of $\mathscr{A}$ with the category of filtered objects of $\mathscr{A}^{\circ}$.

If (A, F) is a filtered object of $\mathscr{A}$, then its *associated graded* is the object of $\mathscr{A}^{\mathbb{Z}}$ defined by

$$\mathrm{Gr}^n(A) = F^n(A)/F^{n+1}(A).$$

The convention §1.1 is justified by the simple formula

$$\mathrm{Gr}^n(A^{\circ}) = \mathrm{Gr}^{-n}(A)^{\circ}$$

which follows from the self-dual diagram

Let (A, F) be a filtered object, and $j \colon X \hookrightarrow A$ a sub-object of A. The *filtration induced by* F (or simply the *induced filtration*) on X is the unique filtration on X such that j is strictly compatible with the filtrations; we have

$$F^n(X) = j^{-1}(F^n(A)) = X \cap F^n(A).$$

Dually, the *quotient filtration* on A/X (the unique filtration such that $p \colon A \to A/X$ is strictly compatible with the filtrations) is given by

$$F^n(A/X) = p(F^n(A)) \cong (X + F^n(A))/X \cong F^n(A)/(X \cap F^n(A)).$$

Lemma 1.1.9. *If X and Y are sub-objects of A, with $X \subset Y$, then on $Y/X \xrightarrow{\sim} \operatorname{Ker}(A/X \to A/Y)$ the quotient filtration of Y agrees with that induced by that of A/X. In the diagram*

$$\begin{tikzcd}
A/Y & & A/X \arrow[ll] & \\
& A \arrow[ul] \arrow[ur] & & Y/X \arrow[ul] \\
X \arrow[ur] \arrow[rr] & & Y \arrow[ul] \arrow[uu] \arrow[ur] &
\end{tikzcd}$$

the arrows are strict.

We call the filtration Lemma 1.1.9 on Y/X the *filtration induced by that of A* (or simply the *induced filtration*). By Lemma 1.1.9, its definition is self-dual.

In particular, if $\Sigma \colon A \xrightarrow{f} B \xrightarrow{G} C$ is a **!!TO-DO: o-suite?!!** sequence, and if B is filtered, then $\mathrm{H}(\Sigma) = \operatorname{Ker}(g)/\operatorname{Im}(f) = \operatorname{Ker}(\operatorname{Coker}(f) \to \operatorname{Coim}(g))$ is endowed with a canonical induced filtration.

The reader can show that:

Proposition 1.1.11.

(i) Let $f \colon (A, F) \to (B, F)$ be a morphism of filtered objects with finite filtrations. For f to be strict, it is necessary and sufficient that the sequence

$$0 \to \operatorname{Gr}(\operatorname{Ker}(f)) \to \operatorname{Gr}(A) \to \operatorname{Gr}(B) \to \operatorname{Gr}(\operatorname{Coker}(f)) \to 0$$

be exact.

(ii) Let $\sigma \colon (A, F) \to (B, F) \to (C, F)$ be a **!!TO-DO: o-suite?!!** *sequence of strict morphisms. We then have*

$$\mathrm{H}(\operatorname{Gr}(\Sigma)) \cong \operatorname{Gr}(\mathrm{H}(\Sigma))$$

canonically. In particular, if Σ is exact in $\mathscr{A}$, then $\operatorname{Gr}(\Sigma)$ is exact in $\mathscr{A}^{\mathbb{Z}}$.

In a category of modules, to say that a morphism $f \colon (A, F) \to (B, F)$ is strict implies that every $b \in B$ of filtration $\geqslant n$ (i.e. $b \in F^n(B)$) that is in the image of A is already in the image of $F^n(A)$:

$$f(F^n(A)) = f(A) \cap F^n(B).$$

If $\otimes\colon \mathscr{A}_1 \times \ldots \times \mathscr{A}_n \to \mathscr{B}$ is a multiadditive right-exact functor, and if A_i is an object of finite filtration of $\mathscr{A}_i$ (for $1 \leqslant i \leqslant n$), then we define a filtration on $\bigotimes_{i=1}^n A_i$ by

$$F^k(\bigotimes_{i=1}^n A_i) = \sum_{\sum k_i = k} \operatorname{Im}(\bigotimes_{i=1}^n F^{k_i}(A_i) \to \bigotimes_{i=1}^n A_i)$$

(a sum of sub-objects).

Dually, if H is left-exact, then we set

$$F^k(H(A_i)) = \bigcap_{\sum k_i = k} \operatorname{Ker}(H(A_i) \to H(A_i/F^{k_i}(A_i))).$$

If H is exact, then the two definitions are equivalent.

We extend these definitions to contravariant functors in certain variables by §1.1. In particular, for the left-exact functor Hom, we set

$$F^k(\operatorname{Hom}(A,B)) = \{f\colon A \to B \mid f(F^n(A)) \subset F^{n+k}(B)\ \forall n\}.$$

We thus have

$$\operatorname{Hom}((A,F),(B,F)) = F^0(\operatorname{Hom}(A,B)).$$

Under the above hypotheses, we have obvious morphisms

$$\bigotimes_{i=1}^n \operatorname{Gr}(A_i) \to \operatorname{Gr}(\bigotimes_{i=1}^n A_i)$$
$$\operatorname{Gr} H(A_i) \to H(\operatorname{Gr}(A_i)).$$

If H is exact, then these are isomorphisms and inverse to one another.

These constructions are compatible with composition of functors, in a sense whose details we leave to the reader.

1.2. Opposite filtrations

Let A be an object of $\mathscr{A}$ endowed with filtrations F and G. By definition, $\operatorname{Gr}_F^n(A)$ is a quotient of a sub-object of A and, as such, is endowed with a filtration induced by G §1.1. Passing to the associated graded defines a bigraded object $(\operatorname{Gr}_G^n \operatorname{Gr}_F^m(A))_{n,m\in\mathbb{Z}}$. By a lemma of Zassenhaus, $\operatorname{Gr}_G^n \operatorname{Gr}_F^m(A)$ and $\operatorname{Gr}_F^m \operatorname{Gr}_G^n(A)$ are canonically isomorphic: if we define the induced filtrations §1.1 as quotient filtrations of the induced filtrations on a sub-object, then we have

$$\begin{gathered}\operatorname{Gr}_G^n \operatorname{Gr}_F^m(A) \cong (F^m(A)\cap G^n(A))/((F^{m+1}(A)\cap G^n(A)) + (F^m(A)\cap G^{n+1}(A)))\\ \Vert\\ \operatorname{Gr}_F^m \operatorname{Gr}_G^n(A) \cong (G^n(A)\cap F^m(A))/((G^{n+1}(A)\cap F^m(A)) + (G^n(A)\cap F^{m+1}(A)))\end{gathered}$$

Let H be a third filtration of A. It induces a filtration on $\operatorname{Gr}_F(A)$, and thus on $\operatorname{Gr}_G\operatorname{Gr}_F(A)$. It also induces a filtration on $\operatorname{Gr}_F\operatorname{Gr}_G(A)$. We note that these filtrations do not in general correspond to one another under the isomorphism §1.2. In the expression $\operatorname{Gr}_H\operatorname{Gr}_G\operatorname{Gr}_F(A)$, G and H thus play a symmetric role, but not F and G.

Definition 1.2.3. Two *finite* filtrations F and $\overline{F}$ on A are said to be n-*opposite* if $\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = 0$ for $p + q \neq n$.

If $A^{p,q}$ is a bigraded object of $\mathscr{A}$ such that

(a) $A^{p,q} = 0$ except for a finite number of pairs (p, q), and

(b) $A^{p,q} = 0$ for $p + q \neq n$

then we define two n-opposite filtrations of $A = \sum_{p,q} A^{p,q}$ by setting

$$F^p(A) = \sum_{p' \geqslant p} A^{p',q'} \tag{1.2.4.1}$$

$$\overline{F}^q(A) = \sum_{q' \geqslant q} A^{p',q'}. \tag{1.2.4.2}$$

We have

$$\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = A^{p,q}. \tag{1.2.4.3}$$

Conversely:

Proposition 1.2.5.

(i) Let F and $\overline{F}$ be finite filtrations on A. For F and $\overline{F}$ to be n-opposite, it is necessary and sufficient that, for all p, q,

$$[p + q = n + 1] \implies [F^p(A) \oplus \overline{F}^q(A) \xrightarrow{\sim} A].$$

(ii) If F and $\overline{F}$ are n-opposite, and if we set

$$\begin{cases} A^{p,q} = 0 & \text{for } p + q \neq n \\ A^{p,q} = F^p(A) \cap \overline{F}^q(A) & \text{for } p + q = n \end{cases}$$

then A is the direct sum of the $A^{p,q}$, and F and $\overline{F}$ come from the bigrading $A^{p,q}$ of A by the procedure of §1.2.

Proof. (i) The condition $\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = 0$ for $p + q > n$ implies that $F^p \cap \overline{F}^q = (F^{p+1} \cap \overline{F}^q) + (F^p \cap \overline{F}^{q+1})$ for $p + q > n$. By hypothesis, $F^p \cap \overline{F}^q$ is zero for large enough $p + q$; by decreasing induction, we thus deduce that the condition $\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = 0$ for $p + q > n$ is equivalent to the condition $F^p(A) \cap \overline{F}^q(A) = 0$ for $p + q > n$. Dually (§1.1, §1.1, §1.1), the condition $\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = 0$ for $p + q < n$ is equivalent to the condition $A = F^p(A) + \overline{F}^q(A)$ for $(1 - p) + (1 - q) > -n$, i.e. for $p + q \leqslant n + 1$, and the claim then follows.

(ii) If F and $\overline{F}$ are n-opposite, then we can prove by decreasing induction on p that

$$\bigoplus_{p' \geqslant p} A^{p',q'} \xrightarrow{\sim} F^p(A). \tag{1.2.5.1}$$

(a) For $F^p(A) = 0$, the claim is evident.

(b) The decomposition $A = F^{p+1}(A) \oplus \overline{F}^{n-p}(A)$ induces on $F^p(A) \supset F^{p+1}(A)$ a decomposition

$$F^p(A) = F^{p+1}(A) \oplus (F^p(A) \cap \overline{F}^{n-p}(A))$$

and we conclude by induction.

For p small enough, we have $F^p(A) = A$. By Equation 1.2.5.1, the $A^{p,q}$ thus form a bigrading of A, and F satisfies Equation 1.2.4.1. The fact that $\overline{F}$ satisfies Equation 1.2.4.2 then follows by symmetry.

□

The constructions §1.2 and Proposition 1.2.5 establish equivalences of categories that are quasi-inverse to one another between objects of $\mathscr{A}$ endowed with two finite n-opposite filtrations and bigraded objects of $\mathscr{A}$ of the type considered in §1.2.

Definition 1.2.7. Three ***finite*** filtrations W, F, and $\overline{F}$ on A are said to be ***opposite*** if

$$\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_W^n(A) = 0$$

for $p + q + n \neq 0$.

This condition is symmetric in F and $\overline{F}$. It implies that F and $\overline{F}$ induce on $W^n(A)/W^{n+1}(A)$ two $(-n)$-opposite filtrations. We set

$$A^{p,q} = \mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_F^{-p-q}(A)$$

whence decompositions §1.2, Proposition 1.2.5

$$W^n(A)/W^{n+1}(A) = \bigoplus_{p+q=-n} A^{p,q} \tag{1.2.7.1}$$

which makes $\mathrm{Gr}_W(A)$ into a bigraded object.

Lemma 1.2.8. *Let W, F, and $\overline{F}$ be three finite opposite filtrations, and σ a sequence $(p_i, q_i)_{i\geqslant 0}$ pairs of integers satisfying*

(a) $p_i \leqslant p_j$ and $q_i \leqslant q_j$ for $i \geqslant j$, and

(b) $p_i + q_i = p_0 + q_0 - i + 1$ for $i > 0$.

Set $p = p_0$, $q = q_0$, $n = -p - q$, and

$$A_\sigma = \left(\sum_{0\leqslant i}(W^{n+i}(A) \cap F^{p_i}(A))\right) \cap \left(\sum_{0\leqslant i}(W^{n+i}(A) \cap \overline{F}^{q_i}(A))\right).$$

Then the projection from $W^n(A)$ to $\mathrm{Gr}_W^n(A)$ induces an isomorphism

$$A_\sigma \xrightarrow{\sim} A^{p,q} \subset \mathrm{Gr}_W^n(A).$$

Proof. We will prove by induction on k the following claim:

The projection from $W^n(A)/W^{n+k}$ to $\mathrm{Gr}^n_W(A)$ induces an isomorphism from

$$\begin{aligned}&\left[\left(\sum_{i<k}(W^{n+i}(A)\cap F^{p_i}(A))+W^{n+k}(A)\right)\right.\\ &\left.\cap\left(\sum_{i<k}(W^{n+i}(A)\cap \overline{F}^{q_i}(A))+W^{n+k}(A)\right)\right]/W^{n+k}(A)\end{aligned} \tag{$*_k$}$$

to $A^{p,q}\subset \mathrm{Gr}^n_W(A)$.

For $k=1$, this is exactly the definition of $A^{p,q}$.

By (i) of Proposition 1.2.5 we have

$$F^{p_k}(\mathrm{Gr}^{n+k}_W(A))\oplus \overline{F}^{q_k}(\mathrm{Gr}^{n+k}_W(A)) \xrightarrow{\sim} \mathrm{Gr}^{n+k}_W(A). \tag{1.2.8.1}$$

Set

$$\begin{aligned}B&=\sum_{i<k}(W^{n+i}(A)\cap F^{p_i}(A))\\ C&=\sum_{i<k}(W^{n+i}(A)\cap \overline{F}^{q_i}(A))\\ B'&=(W^{n+k}(A)\cap F^{p_k}(A))+W^{n+k+1}(A)\\ C'&=(W^{n+k}(A)\cap \overline{F}^{q_k}(A))+W^{n+k+1}(A)\\ D&=W^{n+k}(A)\\ E&=W^{n+k+1}(A).\end{aligned}$$

Equation 1.2.8.1 can then be written as

$$\begin{aligned}B'+C'&=D\\ B'\cap C'&=E.\end{aligned}$$

We also have, since $p_k\leqslant p_i$ (for $i\leqslant k$),

$$B\cap D\subset F^{p_k}(A)\cap W^{n+k}(A)\subset B'$$

and, since $q_k\leqslant q_i$ (for $i\leqslant k$),

$$C\cap D\subset \overline{F}^{q_k}(A)\cap W^{n+k}(A)\subset C'.$$

The claim $(*_{k+1})$ and then follows from $(*_k)$ and the following lemma. ☐

Lemma 1.2.9. *Let B, C, B', C', D, and E be sub-objects of A. Suppose that*

$$\begin{aligned}B'+C'&=D \qquad & B'\cap C'&=E\\ B\cap D&\subset B' \qquad & C\cap D&\subset C'.\end{aligned}$$

Then

$$((B+B')\cap(C+C'))/E\xrightarrow{\sim}((B+D)\cap(C+D))/D.$$

Proof. To prove surjectivity, we write

$$\begin{aligned}((B+B')\cap(C+C'))+D &= (((B+B')\cap(C+C'))+B')+C' \\ &= ((B+B')\cap(C+C'+B'))+C' \\ &= (B+B'+C')\cap(C+C'+B') \\ &= (B+D)\cap(C+D).\end{aligned}$$

To prove injectivity, we write

$$(B+B')\cap(C+C')\cap D = ((B+B')\cap D)\cap((C+C')\cap D).$$

Since $B' \subset D$, we have

$$\begin{aligned}(B+B')\cap D &= (B\cap D)+B' \\ &= B'\end{aligned}$$

and similarly

$$(C+C')\cap D = C'$$

and

$$\begin{aligned}(B+B')\cap(C+C')\cap D &= B'\cap C' \\ &= E.\end{aligned}$$

□

This finishes the proof of Lemma 1.2.8, noting that Lemma 1.2.8 is equivalent to $(*_k)$ for large k.

Theorem 1.2.10. *Let $\mathscr{A}$ be an abelian category, and provisionally denote by $\mathscr{A}'$ the category of objects of $\mathscr{A}$ endowed with three opposite filtrations W, F, and $\overline{F}$. The morphisms in $\mathscr{A}'$ are the morphisms of $\mathscr{A}$ that are compatible with the three filtrations.*

(i) $\mathscr{A}'$ is an abelian category.

(ii) The kernel (resp. cokernel) of an arrow $f\colon A \to B$ in $\mathscr{A}'$ is the kernel (resp. cokernel) of f in $\mathscr{A}$, endowed with the filtrations induced by those of A (resp. the quotients of those of B).

(iii) Every morphism $f\colon A \to B$ in $\mathscr{A}'$ is strictly compatible with the filtrations W, F, and $\overline{F}$; the morphism $\mathrm{Gr}_W(f)$ is compatible with the bigradings of $\mathrm{Gr}_W(A)$ and $\mathrm{Gr}_W(B)$; the morphisms $\mathrm{Gr}_F(f)$ and $\mathrm{Gr}_{\overline{F}}(f)$ are strictly compatible with the filtration induced by W.

(iv) The "forget the filtrations" functors, Gr_W, Gr_F, and $\mathrm{Gr}_{\overline{F}}$, and

$$\begin{aligned}\mathrm{Gr}_W\,\mathrm{Gr}_F &\simeq \mathrm{Gr}_F\,\mathrm{Gr}_W \\ &\simeq \mathrm{Gr}_{\overline{F}}\,\mathrm{Gr}_F\,\mathrm{Gr}_W \\ &\simeq \mathrm{Gr}_{\overline{F}}\,\mathrm{Gr}_W \simeq \mathrm{Gr}_W\,\mathrm{Gr}_{\overline{F}}\end{aligned}$$

from $\mathscr{A}'$ to $\mathscr{A}$ are exact.

Denote by $\sigma_0(p,q)$ and $\sigma_1(p,q)$ the sequences

$$\begin{aligned}\sigma_0(p,q) &= (p,q),(p,q),(p,q-1),(p,q-2),(p,q-3),\ldots\\ \sigma_0(p,q) &= (p,q),(p,q),(p-1,q),(p-2,q),(p-3,q),\ldots\end{aligned}$$

and, with the notation of Lemma 1.2.8, set

$$A_i^{p,q} = A_{\sigma_i(p,q)} \qquad \text{for } i=0,1.$$

If $f\colon A \to B$ is compatible with W, F, and $\overline{F}$, then we have

$$f(A_i^{p,q}) \subset B_i^{p,q} \qquad \text{for } i=0,1. \tag{1.2.10.1}$$

Claim (iii) then follows from the following lemma:

Lemma 1.2.11. *The $A_i^{p,q}$ give a bigrading of A. We have*

$$W^n(A) = \sum_{n+p+q\leqslant 0} A_i^{p,q} \qquad \textit{for } i=0,1 \tag{1.2.11.1}$$

$$F^p(A) = \sum_{p'\geqslant p} A_0^{p',q'} \tag{1.2.11.2}$$

$$\overline{F}^q(A) = \sum_{q'\geqslant q} A_1^{p',q'}. \tag{1.2.11.3}$$

Proof. By symmetry, it suffices to prove the claims concerning $i=0$. Set $A_0 = \bigoplus A_0^{p,q}$ and define filtrations W and F on A_0 by the equations of Lemma 1.2.11. The canonical map i from A_0 to A is compatible with the filtrations W and F. Furthermore, by Lemma 1.2.8, $\mathrm{Gr}_W(i)$ is an isomorphism, and induces isomorphisms of graded objects

$$\sum_{p+q=n} A_0^{p,q} \xrightarrow{\sim} \mathrm{Gr}_W^{-n}(A) = \sum_{p+q=n} A^{p,q}. \tag{1.2.11.4}$$

The morphism i is thus an isomorphism, and the $A_0^{p,q}$ give a bigrading of A.

Equation 1.2.11.1 then says that $\mathrm{Gr}_W(i)$ is an isomorphism. By Equation 1.2.11.4, $\mathrm{Gr}_F\,\mathrm{Gr}_W(i)$ is an isomorphism, and thus so too are $\mathrm{Gr}_W\,\mathrm{Gr}_F(i)$ and $\mathrm{Gr}_F(i)$. Equation Equation 1.2.11.2 then follows. ☐

We now prove Theorem 1.2.10. Let $f\colon A\to B$ in $\mathscr{A}'$ and endow $K=\mathrm{Ker}(f)$ with the filtrations induced by those of A. By Lemma 1.2.11, $\mathrm{Gr}_W(K)\hookrightarrow \mathrm{Gr}_W(A)$; furthermore, the filtration F (resp. $\overline{F}$) on K induces on $\mathrm{Gr}_W(K)$ the inverse image filtration of the filtration F on $\mathrm{Gr}_W(A)$. The sub-object $\mathrm{Gr}_W(K)$ of $\mathrm{Gr}_W(A)$ is then compatible with the bigrading of $\mathrm{Gr}_W(A)$:

$$\mathrm{Gr}_W(K) = \bigoplus_{p,q}(\mathrm{Gr}_W(K)\cap A^{p,q}).$$

We thus deduce that

$$\mathrm{Gr}_F^p\,\mathrm{Gr}_{\overline{F}}^q\,\mathrm{Gr}_W^n(K) \hookrightarrow \mathrm{Gr}_F^p\,\mathrm{Gr}_{\overline{F}}^q\,\mathrm{Gr}_W^n(A);$$

the filtrations of W, F, and $\overline{F}$ on K are thus opposite, and K is a kernel of f in $\mathscr{A}'$. This, combined with the dual result, proves (ii).

If f is an arrow of $\mathscr{A}'$, then the canonical morphism from $\operatorname{Coim}(f)$ to $\operatorname{Im}(f)$ is an isomorphism in $\mathscr{A}$; by (iii), it is also an isomorphism in $\mathscr{A}'$, which is thus abelian.

The "forget the filtrations" functor is exact by (ii). The exactness of the other functors in (iv) follows immediately from (ii), (iii), and (i) or (ii) of Proposition 1.1.11.

Let A be an object of $\mathscr{A}$ endowed with a finite *increasing* filtration $W_\bullet$, and two finite decreasing filtrations F and $\overline{F}$. The construction §1.1 associates to $W_\bullet$ a decreasing filtration $W^\bullet$. We say that the filtrations $W_\bullet$, F, and $\overline{F}$ are *opposite* if the filtrations $W^\bullet$, F, and $\overline{F}$ are, i.e. if, for all n, the filtrations induced by F and $\overline{F}$ on

$$\operatorname{Gr}_n^W(A) = W_n(A)/W_{n-1}(A)$$

are n-opposite.

Theorem 1.2.10 translates trivially to this variation.

1.3. The two filtrations lemma

Let K be a differential complex of objects of $\mathscr{A}$, endowed with a filtration F. The filtration is said to be *biregular* if it induces a finite filtration on each component of K.

We recall the definition of the terms $E_r^{pq}(K,F)$, or simply E_r^{pq}, of the spectral sequence defined by F. We set

$$Z_r^{pq} = \operatorname{Ker}(d\colon F^p(K^{p+q}) \to F^{p+q+1}/F^{p+r}(K^{p+q+1}))$$

and dually we define B_r^{pq} by the formula

$$K^{p+q}/B_r^{pq} = \operatorname{Coker}(d\colon F^{p-r+1}(K^{p+q+1}) \to K^{p+q}/F^{p+1}(K^{p+q})).$$

These formulas still make sense for $r=\infty$.

We note that the use here of the notation B_r^{pq} is different to that of Godement [G1958].

We have, by definition:

$$E_r^{pq} = \operatorname{Im}(Z_r^{pq} \to K^{p+q}/B_r^{pq}) \tag{1.3.1.1}$$

$$= Z_r^{pq}/(B_r^{pq} \cap Z_r^{pq}) \tag{1.3.1.2}$$

$$= \operatorname{Ker}(K^{p+q}/B_r^{pq} \to K^{p+q}/(Z_r^{pq}+B_r^{pq})). \tag{1.3.1.3}$$

We can thus write

$$\begin{aligned} B_r^{p\bullet} \cap Z_r^{p\bullet} &:= (dF^{p-r+1} + F^{p+1}) \cap (d^{-1}F^{p+r} \cap F^p) \\ &= (dF^{p-r+1} \cap F^p) + (F^{p+1} \cap d^{-1}F^{p+r}) \end{aligned} \tag{1.3.1.4}$$

since $dF^{p-r+1} \subset d^{-1}F^{p+r}$ and $F^{p+1} \subset F^p$.

For $r<\infty$, the E_r form a complex graded by the degree $p-r(p+q)$, and E_{r+1} can be expressed as the cohomology of this complex:

$$E_{r+1}^{pq} = \mathrm{H}(E_r^{p-r,q+r-1} \xrightarrow{d_r} E_r^{pq} \xrightarrow{d_r} E_r^{p+r,q-r+1}). \tag{1.3.1.5}$$

For $r = 0$, we have

$$E_0^{\bullet\bullet} = \mathrm{Gr}_F^\bullet(K^\bullet). \tag{1.3.1.6}$$

Proposition 1.3.2. *Let K be a complex endowed with a biregular filtration F. The following conditions are equivalent:*

(i) The spectral sequence defined by F degenerates ($E_1 = E_\infty$).

(ii) The morphisms $d\colon K^i \to K^{i+1}$ are strictly compatible with the filtrations.

Proof. We will prove this in the case where $\mathscr{A}$ is a category of modules. For fixed p and q, the hypothesis that the arrows d_r with domains E_r^{pq} be zero for $r \geqslant 1$ implies that, if $x \in F^p(K^{p+q})$ satisfies $dx \in F^{p+1}(K^{p+q+1})$, then there exists $y \in K^{p+q}$ such that $dy = 0$ and such that x and y have the same image in E_1^{pq}. Modifying y by a boundary, and setting $z = x - y$, we then have

$$\forall x \in F^p(K^{p+q}) \left[dx \in F^{p+1}(K^{p+q+1}) \implies \exists z \text{ s.t. } z \in F^{p+1}(K^{p+q}) \text{ and } dz = dx\right]$$

or, in other words,

$$F^{p+1}(K^{p+q+1}) \cap dF^p(K^{p+q}) = dF^{p+1}(K^{p+q}). \tag{1}$$

If this condition is satisfied for arbitrary p and q, then by induction on r we have

$$F^{p+r} \cap dF^p = dF^{p+r}$$

which, for large $p + r$, can be written as

$$F^p \cap dK = dF^p. \tag{2}$$

Claim (2) trivially implies (1), and is equivalent to (ii), which proves the proposition. □

If (K, F) is a filtered complex, we denote by $\mathrm{Dec}(K)$ the complex K endowed with the *shifted filtration*

$$\mathrm{Dec}(F)^p K^n = Z_1^{p+n,-p}.$$

This filtration is compatible with the differentials:

$$\begin{aligned} dZ_1^{p+n,-p} &\subset F^{p+n+1}(K^{n+1}) \cap \mathrm{Ker}(d) \\ &\subset Z_\infty^{p+n+1,-p} \\ &\subset Z_1^{p+n+1,-p}. \end{aligned}$$

Since

$$\begin{aligned} Z_1^{p+1+n,-p-1} &\subset F^{p+1+n}(K^n) \\ &\subset B_1^{p+n,-p} \\ &\subset Z_1^{p+n,-p} \end{aligned} \tag{1.3.3.1}$$

the evident arrow from $Z_1^{p+n,-p}/Z_1^{p+1+n,-p-1}$ to $Z_1^{p+n,-p}/B_1^{p+n,-p}$ is a morphism

$$u\colon E_0^{p,n-p}(\mathrm{Dec}(K)) \to E_1^{p+n,-p}(K). \tag{1.3.3.2}$$

Proposition 1.3.4.

(i) The morphisms in Equation 1.3.3.2 form a morphism of graded complexes from $E_0(\mathrm{Dec}(K))$ *to* $E_1(K)$.

(ii) This morphism induces an isomorphism on cohomology.

(iii) This morphism induces step-by-step (via Equation 1.3.1.5) isomorphisms of graded complexes $E_r(\mathrm{Dec}(K)) \xrightarrow{\sim} E_{r+1}(K)$ *(for* $r \geqslant 1$*).*

Proof. Let F' be the filtration on K defined by

$$F'^p(K^n) = \mathrm{Dec}(F)^{p-n}(K^n) = Z_1^{p,n-p}.$$

We trivially have isomorphisms

$$E_r^{p,n-p}(\mathrm{Dec}(K)) = E_{r+1}^{p+n,-p}(K,F') \tag{1.3.4.1}$$

that are compatible with the d_r and with Equation 1.3.1.5. The map u comes from Equation 1.3.4.1 and from the identity map

$$(K,F') \to (K,F).$$

This proves (i), and it remains to show that, for $r \geqslant 2$,

$$E_r^{pq}(K,F') \xrightarrow{\sim} E_r^{pq}(K,F).$$

We have

$$\begin{aligned} Z_r^{pq}(K,F') &= Z_r^{pq}(K,F) && \text{for } r \geqslant 1 \\ Z_r^{pq}(K,F') \cap B_r^{pq}(K,F') &= Z_r^{pq}(K,F) \cap B_r^{pq}(K,F) && \text{for } r \geqslant 2 \end{aligned}$$

and we can then apply Equation 1.3.1.2. □

The construction in §1.3 is not self-dual. The dual construction consists of defining

$$\mathrm{Dec}^\bullet(F)^p K^n = B_1^{p+n-1,-p+1}.$$

We then have morphisms

$$E_0^{p,n-p}(\mathrm{Dec}(K)) \to E_1^{p+n,p}(K) \to E_0^{p,n-p}(\mathrm{Dec}^\bullet(K))$$

and, for $r \geqslant 1$, isomorphisms

$$E_r^{p,n-p}(\mathrm{Dec}(K)) \xrightarrow{\sim} E_{r+1}^{p+n,p}(K) \xrightarrow{\sim} E_r^{p,n-p}(\mathrm{Dec}^\bullet(K)).$$

Recall that a morphism of complexes is said to be a *quasi-isomorphism* if it induces an isomorphism on cohomology.

Definition 1.3.6.

(i) A morphism $f\colon (K,F) \to (K',F')$ of filtered complexes with biregular filtrations is a *filtered quasi-isomorphism* if $\mathrm{Gr}_F(f)$ is a quasi-isomorphism, i.e. if the $E_1^{pq}(f)$ are isomorphisms.

(ii) A morphism $f\colon (K,F,W) \to (K,F',W')$ of biregular bifiltered complexes is a *bifiltered quasi-isomorphism* if $\mathrm{Gr}_F\,\mathrm{Gr}_W(f)$ is a quasi-isomorphism.

Let K be a differential complex of objects of $\mathscr{A}$, endowed with two filtrations F and W. Let E_r^{pq} be the spectral sequence defined by W. The filtration F induces on E_r^{pq} various filtrations, which we will compare.

Equation 1.3.1.2 identifies E_r^{pq} with a quotient of a sub-object of K^{p+q}. The E_r^{pq} term is thusly given by endowing with a filtration F_d induced by F, called the *first direct filtration.*

Dually, Equation 1.3.1.3 identifies E_r^{pq} with a sub-object of a quotient of K^{p+q}, whence a new filtration F_{d^*} induced by F, called the *second direct filtration.*

Lemma 1.3.10. *On E_0 and E_1, we have $F_d = F_{d^*}$.*

Proof. For $r = 0, 1$, we have $B_r^{pq} \subset Z_r^{pq}$, and we apply Lemma 1.1.9. □

Equation 1.3.1.5 identifies E_{r+1}^{pq} with a quotient of a sub-object of E_r^{pq}. We define the *recurrent filtration* F_r on the E_r^{pq} by the conditions

(i) On E_0^{pq}, $F_r = F_d = F_{d^*}$.

(ii) On E_{r+1}^{pq}, the recurrent filtration is that induced by the recurrent filtration of E_r^{pq}.

Definitions §1.3 and §1.3 still make sense for $r = \infty$. If the filtration on K is biregular, then the direct filtrations on E_∞^{pq} coincide with those on $E_r^{pq} = E_\infty^{pq}$ for large enough r, and we define the recurrent filtration on E_∞^{pq} as agreeing with that on E_r^{pq} for large enough r.

The filtrations F and W each induce a filtration on $H^\bullet(K)$, and $E_\infty^{\bullet\bullet} = \mathrm{Gr}_W^\bullet(\mathrm{H}^\bullet(K))$. The filtration F on $\mathrm{H}^\bullet(K)$ then induces on E_∞^{pq} a new filtration.

Proposition 1.3.13.

(i) For the first direct filtration, the morphisms d_r are compatible with the filtrations. If E_{r+1}^{pq} is considered as a quotient of a sub-object of E_r^{pq}, then the first direct filtration on E_{r+1}^{pq} is finer than the filtration F' induced by the first direct filtration on E_r^{pq} Y we have $F_d(E_{r+1}^{pq}) \subset F'(E_{r+1}^{pq})$.

(ii) Dually, the morphisms d_r are compatible with the second direct filtration, and the second direct filtration on E_{r+1}^{pq} is less fine than the filtration induced by that of E_r^{pq}.

(iii) $F_d(E_r^{pq}) \subset F_r(E_r^{pq}) \subset F_{d^*}(E_r^{pq})$.

(iv) On E_∞^{pq}, the filtration induced by the filtration F of $\mathrm{H}^\bullet(K)$ §1.3 is finer than the first direct filtration and less fine than the second.

Proof. Claim (i) is evident, (ii) is its dual, and (iii) follows by induction. The first claim of (iv) is easy to verify, and the second is its dual. □

We denote by $\mathrm{Dec}(K)$ (resp. $\mathrm{Dec}^\bullet(K)$) the complex K endowed with the filtrations $\mathrm{Dec}(W)$ and F (resp. $\mathrm{Dec}^\bullet(W)$ and F).

It is clear by Equation 1.3.4.1 that the isomorphism in Proposition 1.3.4 sends the first direct filtration on $E_r(\mathrm{Dec}(K))$ to the second direct filtration on $E_{r+1}(K)$ (for $r \geqslant 1$). The dual isomorphism §1.3 sends the second direct filtration on $E_r(\mathrm{Dec}^\bullet(K))$ to the second direct filtration on $E_{r+1}(K)$.

Lemma 1.3.15. *If the filtration F is biregular, and if, on the $\mathrm{Gr}_W^p(K)$, the morphisms d are strictly compatible with the filtration induced by F, then* *(i) The morphism Equation 1.3.3.2 of graded complexes filtered by F*

$$u\colon \mathrm{Gr}_{\mathrm{Dec}(W)}(K) \to E_1(K, W)$$

is a filtered quasi-isomorphism.

(ii) Dually, the morphism in §1.3

$$u\colon E_1(K, W) \to \mathrm{Gr}_{\mathrm{Dec}^\bullet(W)}(K)$$

is a filtered quasi-isomorphism.

Proof. It suffices, by duality, to prove (i).

By §1.3 and Proposition 1.3.4, the complex $E_1(K, W)$ filtered by F is a quotient of the filtered complex $\mathrm{Gr}_{\mathrm{Dec}(W)}(K)$. Let U be the filtered complex given by the kernel, which is acyclic by (ii) of Proposition 1.3.4. The long exact sequence in cohomology associated to the exact sequence of complexes

$$0 \to \mathrm{Gr}_F(U) \to \mathrm{Gr}_F(\mathrm{Gr}_{\mathrm{Dec}(W)}(K)) \to \mathrm{Gr}_F(E_1(K, W)) \to 0$$

shows that u is a filtered quasi-isomorphism if and only if $\mathrm{Gr}_F(U)$ is an acyclic complex. By Proposition 1.3.2, and since U is acyclic, this reduces to asking that the differentials of U be strictly compatible with the filtration F. From Equation 1.3.3.1 we obtain that U is the sum over p of the complexes

$$(U^p)^n = B_1^{p+n,-p}/Z_1^{p+1+n,-p-1}$$

endowed with the filtration induced by F.

Each differential d of each of the complexes U^p fits into a commutative diagram of filtered objects of the following type, where, for simplicity, we have omitted the total or complementary degree:

$$\begin{tikzcd}
B^p/Z^{p+1} \arrow[rrr, "d"] \arrow[dr] & & & B^{p+1}/Z^{p+1} \arrow[dd] \\
& \mathrm{Coim}(d) \arrow[r, "(5)"] \arrow[r, phantom, "(4)"', yshift=-3ex] & \mathrm{Im}(d) \arrow[ur] \arrow[dr] & \\
W^{p+1}/Z^{p+1} \arrow[uu] \arrow[ur] \arrow[rrr, "(3)"'] \arrow[drrr, phantom, "(2)"] & & & B^{p+1}/W^{p+2} \arrow[d] \\
W^{p+1}/W^{p+2} \arrow[u] \arrow[rrr, "(1)"'] & & & W^{p+1}/W^{p+2}
\end{tikzcd}$$

By hypothesis, the morphism (1) is strict. Since the square (2) is exactly the canonical decomposition of (1), the arrow (3) is a filtered isomorphism. The arrows of the trapezium (4) are isomorphisms; they are thus filtered isomorphisms, since (3) is a filtered isomorphism. The fact that (5) is a filtered isomorphism implies that d is strict. This proves the lemma. □

Theorem 1.3.16. *Let K be a complex endowed with two filtrations, W and F, with the filtration F biregular. Let $r_0 \geqslant 0$ be an integer, and suppose that, for $0 \leqslant r < r_0$, the differentials of the graded complex $E_r(K, W)$ are strictly compatible with the filtration F. Then, for $r \leqslant r_0 + 1$, $F_d = F_r = F_{d^*}$ on E_r^{pq}.*

Proof. We will prove the theorem by induction on r_0. For $r_0 = 0$, the hypothesis is empty, and we apply Lemma 1.3.10 and (iii) of Proposition 1.3.13. For $r_0 \geqslant 1$, by the inductive hypothesis, we have $F_d = F_r = F_{d^*}$ on E_r^{pq} for $r \leqslant r_0$.

By Lemma 1.3.15, the morphism $u\colon E_0(\mathrm{Dec}(K)) \to E_1(K)$ is a filtered quasi-isomorphism. It thus induces a filtered isomorphism from $\mathrm{H}^\bullet(\mathrm{Dec}(K))$ to $\mathrm{H}^\bullet(E_1(K))$:

$$u\colon (E_1(\mathrm{Dec}(K)), F_r) \xrightarrow{\sim} (E_2(K), F_r).$$

Step-by-step, we thus deduce that the canonical isomorphism from $E_s(\mathrm{Dec}(K))$ to E_{s+1} (for $s \geqslant 1$) is a filtered isomorphism for the recurrent filtration.

On $E_1(\mathrm{Dec}(K))$, $F_r = F_d$ (Lemma 1.3.10), and we already know (§1.3) that u' is a filtered isomorphism

$$u'\colon (E_1(\mathrm{Dec}(K)), F_d) \xrightarrow{\sim} (E_2(K), F_d).$$

On $E_2(K)$, we thus have $F_d = F_r$. This, combined with the dual result, proves the theorem for $r_0 = 1$.

Suppose that $r_0 \geqslant 2$. Then the arrows d_1 of $E_1(K)$ are strictly compatible with the filtrations, and thus so too are the arrows d_0 of $E_0(\mathrm{Dec}(K))$ (indeed, u induces an isomorphism of spectral sequences, and we apply the criterion from Proposition 1.3.2).

For $0 < s < r_0 - 1$, the isomorphism $(E_s(\mathrm{Dec}(K)), F_r) \cong (E_{s+1}(K), F_r)$ shows that the d_s are strictly compatible with the recurrent filtrations.

By the induction hypothesis, we thus have $F_d = F_r$ on $E_s(\mathrm{Dec}(K))$ for $s \leqslant s_0$. The isomorphism $(E_s(\mathrm{Dec}(K)), F_d) \cong (E_{s+1}(K), F_d)$ (Proposition 1.3.13) then shows that $F_d = F_r$ on $E_r(K)$ for $r \leqslant r_0 + 1$. This, combined with the dual result, proves the theorem. □

Corollary 1.3.17. *Under the general hypotheses of Theorem 1.3.16, suppose that, for all r, the differentials d_r are strictly compatible with the recurrent filtrations on the E_r. Then, on E_∞, the filtrations F_d, F_r, and F_{d^*} agree, and coincide with the filtration induced by the filtration F of $\mathrm{H}^\bullet(K)$.*

Proof. This follows immediately from Theorem 1.3.16 and (iv) of Proposition 1.3.13. □

1.4. Hypercohomology of filtered complexes

In this section, we recall some standard constructions in hypercohomology. We do not use the language of derived categories, which would be more natural here.

Throughout this entire section, by "complex" we mean "bounded-below complex".

Let T be a left-exact functor from an abelian category $\mathscr{A}$ to an abelian category $\mathscr{B}$. Suppose that every object of $\mathscr{A}$ injects into an injective object; the derived functors $\mathrm{R}^i T\colon \mathscr{A} \to \mathscr{B}$ are then defined. An object A of $\mathscr{A}$ is said to be *acyclic* for T if $\mathrm{R}^i T(A) = 0$ for $i > 0$.

Let (A, F) be a filtered object with finite filtration, and TF the filtration of TA by its sub-objects $TF^p(A)$ (these are sub-objects since T is left exact). If $\mathrm{Gr}_F(A)$ is T-acyclic, then the $F^p(A)$ are T-acyclic as successive extensions of T-acyclic objects. The image under T of the sequence

$$0 \to F^{p+1}(A) \to F^p(A) \to \mathrm{Gr}^p(A) \to 0$$

is thus exact, and

$$\mathrm{Gr}_{FT}\,TA \xrightarrow{\sim} T\,\mathrm{Gr}_F\,A. \tag{1.4.2.1}$$

Let A be an object endowed with finite filtrations F and W such that $\mathrm{Gr}_F\,\mathrm{Gr}_W\,A$ are T-acyclic. The objects $\mathrm{Gr}_F\,A$ and $\mathrm{Gr}_W\,A$ are then T-acyclic, as well as the $F^q(A)\cap W^p(A)$. The sequences

$$0 \to T(F^q\cap W^{p+1}) \to T(F^q\cap W^p) \to T((F^q\cap W^p)/(F^q\cap W^{p+1})) \to 0$$

are thus exact, and $T(F^q(\mathrm{Gr}^p_W(A)))$ is the image in $T(\mathrm{Gr}^p_W(A))$ of $T(F^p\cap W^q)$. The diagram

$$\begin{tikzcd}
T(F^q\cap W^p) \arrow[r] \arrow[d, "\cong"'] & T(F^q\,\mathrm{Gr}^p_W\,A) \arrow[r] & T\,\mathrm{Gr}^p_W\,A \arrow[d, "\cong"] \\
TF^q\cap TW^p \arrow[r, equal] & TF^q\cap TW^p \arrow[r] & \mathrm{Gr}^p_{TW}\,TA
\end{tikzcd}$$

then shows that the isomorphism in Equation 1.4.2.1 relative to W sends the filtration $\mathrm{Gr}_{TW}(TF)$ to the filtration $T(\mathrm{Gr}_W(F))$.

Let K be a complex of objects of $\mathscr{A}$. The ***hypercohomology objects*** $\mathrm{R}^iT(K)$ are calculated as follows:

(a) We choose a quasi-isomorphism $i\colon K\to K$ such that the components of K' are acyclic for T. For example, we can take K' to be the simple complex associated to an injective Cartan–Eilenberg resolution of K.

(b) We set

$$\mathrm{R}^iT(K) = \mathrm{H}^i(T(K')).$$

We can show that $\mathrm{R}^iT(K)$ does not depend on the choice of K', but depends functorially on K, and that a quasi-isomorphism $f\colon K_1\to K_2$ induces *isomorphisms*

$$\mathrm{R}^iT(f)\colon \mathrm{R}^iT(K_1)\to \mathrm{R}^iT(K_2).$$

Let F be a biregular filtration of K. A ***T-acyclic filtered resolution*** of K is a filtered quasi-isomorphism $i\colon K\to K'$ from K to a filtered biregular complex such that the $\mathrm{Gr}^p(K'^n)$ are acyclic for T. If K' is such a resolution, then the K'^n are acyclic for T, and the filtered complex (cf. §1.4) $T(K')$ defines a spectral sequence

$$E_1^{pq} = \mathrm{R}^{p+q}T(\mathrm{Gr}^p(K)) \Rightarrow \mathrm{R}^{p+q}T(K).$$

This is independent of the choice of K'. We call this the ***hypercohomology spectral sequence of the filtered complex*** K. It depends functorially on K, and a filtered quasi-isomorphism induces an isomorphism of spectral sequences.

The differentials d_1 of this spectral sequence are the connection morphisms defined by the short exact sequences

$$0\to \mathrm{Gr}^{p+1}K \to F^pK/F^{p+2}K \to \mathrm{Gr}^p K\to 0.$$

Let K be a complex. We denote by $\tau_{\leqslant p}(K)$ the following subcomplex:

$$\tau_{\leqslant p}(K)^n = \begin{cases} K^n & \text{for } n < p \\ \mathrm{Ker}(d) & \text{for } n = p \\ 0 & \text{for } n > p. \end{cases}$$

The ***filtration***, said to be ***canonical***, of K by the $\tau_{\leqslant p}(K)$ is induced by shifting the trivial filtration G for which $G^0(K) = K$ and $G^1(K) = 0$. We have, for the canonical filtration,

$$E_1^{pq} = 0 \qquad \text{if } p+q \neq -p$$

$$H^{-p} \qquad \text{if } p+q = -p.$$

A quasi-isomorphism $f\colon K \to K'$ is automatically a filtered quasi-isomorphism for the canonical filtrations.

The subcomplexes $\sigma_{\geqslant p}(K)$ of K

$$\sigma_{\geqslant p}(K)^n = \begin{cases} 0 & \text{if } n < p \\ K^n & \text{if } n \geqslant p \end{cases}$$

define a biregular filtration, called the ***stupid filtration*** of K.

The hypercohomology spectral sequences attached to the stupid or canonical filtrations of K are the two ***hypercohomology spectral sequences*** of K.

Example 1.4.8. Let $f\colon X \to Y$ be a continuous map between topological spaces, and let $\mathscr{F}$ be an abelian sheaf on X. Let $\mathscr{F}^\bullet$ be a resolution of $\mathscr{F}$ by f_*-acyclic sheaves. We have $\mathrm{R}^i f_* \mathscr{F} \simeq \mathscr{H}^i(f_*\mathscr{F}^\bullet)$. We take the functor T to be the functor $\Gamma(Y, -)$. The hypercohomology spectral sequence of the complex $f_*\mathscr{F}^\bullet$ endowed with its canonical filtration §1.4

$$E_1^{pq} = \mathrm{H}^{2p+q}(Y, \mathrm{R}^{-p} f_* \mathscr{F}) \Rightarrow \mathrm{H}^{p+q}(X, \mathscr{F})$$

is exactly, up to the renumbering $E_r^{pq} \mapsto E_{r+1}^{2p+q,-p}$, the *Leray spectral sequence* for f and $\mathscr{F}$.

Let (K, W, F) be a biregular bifiltered complex. To this complex, we associate:

(a) A spectral sequence

$$_W E_1^{p,n-p} = \mathrm{H}^n(\mathrm{Gr}_W^p(K)) \Rightarrow \mathrm{H}^n(K)$$

with differentials $_W d_1$ being the connecting morphisms induced by the short exact sequences

$$0 \to \mathrm{Gr}_W^{p+1}(K) \to W^p(K)/W^{p+2}(K) \to \mathrm{Gr}_W^p(K) \to 0;$$

(b) An analogous spectral sequence for the filtration F;

(c) Exact squares

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathrm{Gr}_F^{p+1}\mathrm{Gr}_W^{q+1}K & \longrightarrow & F^p/F^{p+2}(\mathrm{Gr}_W^{q+1}K) & \longrightarrow & \mathrm{Gr}_F^{p}\mathrm{Gr}_W^{q+1}K & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathrm{Gr}_F^{p+1}(W^q/W^{q+2}(K)) & \longrightarrow & F^p/F^{p+2}(W^q/W^{q+2}(K)) & \longrightarrow & \mathrm{Gr}_F^{p}(W^q/W^{q+2}(K)) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathrm{Gr}_F^{p+1}\mathrm{Gr}_W^{q}K & \longrightarrow & F^p/F^{p+2}\mathrm{Gr}_W^{q}K & \longrightarrow & \mathrm{Gr}_F^{p}\mathrm{Gr}_W^{q}K & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

The exterior rows and columns of this square define connection morphisms

$$\begin{aligned}
{}_{F,W}d_1 &\colon \mathrm{H}^n\mathrm{Gr}_F^p\mathrm{Gr}_W^q K \to \mathrm{H}^{n+1}\mathrm{Gr}_F^{p+1}\mathrm{Gr}_W^q K \\
{}_{W,F}d_1 &\colon \mathrm{H}^n\mathrm{Gr}_F^p\mathrm{Gr}_W^q K \to \mathrm{H}^{n+1}\mathrm{Gr}_F^{p}\mathrm{Gr}_W^{q+1} K.
\end{aligned}$$

These morphisms satisfy

$$\begin{aligned}
{}_{FW}d_1 \circ {}_{WF}d_1 + {}_{WF}d_1 \circ {}_{FW}d_1 = 0 \\
{}_{FW}d_1^2 = 0 \\
{}_{WF}d_1^2 = 0
\end{aligned}$$

The morphisms ${}_{FW}d_1$ are the morphisms d_1 of the spectral sequences $E^{(q)}$, with the $E_1^{(q)p,n-p}$ term equal to

$$E_1^{p,q,n-p-q} := \mathrm{H}^n(\mathrm{Gr}_F^p\mathrm{Gr}_W^q K) \Rightarrow \mathrm{H}^n(\mathrm{Gr}_W^q K) = {}_W E_1^{q,n-q} \tag{1.4.9.1}$$

defined by the filtered complex $\mathrm{Gr}_W^q(K)$. This spectral sequence abuts to the filtration induced by F on $\mathrm{H}^\bullet\mathrm{Gr}_W^q K$. Similarly, the ${}_{WF}d_1$ are the d_1 of spectral sequences with the same initial terms

$$E_1^{p,q,n-p-q} := \mathrm{H}^n(\mathrm{Gr}_F^p\mathrm{Gr}_W^q K) \Rightarrow \mathrm{H}^n(\mathrm{Gr}_F^p K). \tag{1.4.9.1}$$

$$\begin{tikzcd}
 & \mathrm{H}^n\mathrm{Gr}_W^q K \arrow[dr, Rightarrow, "{}_W d_1"] & \\
\mathrm{H}^n\mathrm{Gr}_F^p\mathrm{Gr}_W^q K \arrow[ur, Rightarrow, "{}_{F,W}d_1"] \arrow[dr, Rightarrow, "{}_{W,F}d_1"'] & & \mathrm{H}^n K \\
 & \mathrm{H}^n\mathrm{Gr}_F^p K \arrow[ur, Rightarrow, "{}_F d_1"'] &
\end{tikzcd}$$

These constructions are symmetric in F and W via the isomorphism

$$\mathrm{Gr}_F^p\mathrm{Gr}_W^q \sim \mathrm{Gr}_W^q\mathrm{Gr}_F^p.$$

We can also interpret the ${}_{W,F}d_1$ are the initial morphisms of a morphism of spectral sequences Equation 1.4.9.2, abutting to ${}_W d_1$.

Indeed, let C^q be the cone of the morphism

$$W^q(K)/W^{q+2}(K) \to \mathrm{Gr}^q_W(K).$$

In the diagram

$$\Sigma\colon\ \mathrm{Gr}^{q+1}_W(K)[1] \xrightarrow{u} C^q \xleftarrow{i} \mathrm{Gr}^q_W(K)$$

the morphism u is a quasi-isomorphism, and we have

$${}_Wd_1 = \mathrm{H}(u)^{-1} \circ \mathrm{H}(i).$$

In fact, u is even a filtered (for F) quasi-isomorphism, and the above construction defines a morphism from the spectral sequence defined by $(\mathrm{Gr}^q_W(K), F)$ to that defined by $(\mathrm{Gr}^{q+1}_W(K)[1], F)$, and it abuts to ${}_Wd_1$. The initial term of this morphism, induced by $\mathrm{Gr}_F(\Sigma)$ is exactly ${}_{W,F}d_1$.

These constructions pass as they are to hypercohomology. Let K be a complex endowed with two biregular filtrations F and W.

A ***bifiltered** T**-acyclic resolution*** of K is a bifiltered quasi-isomorphism $i\colon K \to K'$ such that the $\mathrm{Gr}^p_F \mathrm{Gr}^n_W(K'^m)$ are T-acyclic. Such a morphism always exists. In the particular case where $\mathscr{A}$ is the category of sheaves of A-modules on a topological space X, and where T is the functor Γ from $\mathscr{A}$ to the category of A-modules, then an example of a bifiltered T-acyclic resolution of K is the simple complex associated to the double complex given by the Godement resolution $\mathscr{C}^\bullet(K)$ of K, filtered by the $\mathscr{C}^\bullet(F^p(K))$ and the $\mathscr{C}^\bullet(W^n(K))$. Since $\mathscr{C}^\bullet$ is exact, we have

$$\mathrm{Gr}_F \mathrm{Gr}_W(\mathscr{C}^\bullet(K)) \simeq \mathscr{C}^\bullet(\mathrm{Gr}_F \mathrm{Gr}_W(K)).$$

We will not have need here of any other case.

If K' is a bifiltered T-acyclic resolution of K, then the complex TK' is filtered by the TF^pK' and the TW^qK' (§1.4). Furthermore, $\mathrm{Gr}^n_W(K')$ is a T-acyclic filtered (for F) resolution of $\mathrm{Gr}^n_F(K)$, and $\mathrm{Gr}^n_F \mathrm{Gr}^m_W(K')$ is a T-acyclic resolution of $\mathrm{Gr}^n_F \mathrm{Gr}^m_W(K)$, and

$$\begin{aligned} T\,\mathrm{Gr}_F K' &\approx \mathrm{Gr}_F TK' \qquad \text{as } W\text{-filtered complexes} \\ T\,\mathrm{Gr}_W K' &\approx \mathrm{Gr}_W TK' \qquad \text{as } F\text{-filtered complexes} \\ T\,\mathrm{Gr}_F \mathrm{Gr}_W K' &\approx \mathrm{Gr}_F \mathrm{Gr}_W TK'. \end{aligned}$$

Lemma 1.4.12. *Under the hypotheses of §1.4:*

(i) The initial terms of the hypercohomology spectral sequences

$${}_WE_1^{q,n-q} = \mathrm{R}^nT(\mathrm{Gr}^q_W K) \Rightarrow \mathrm{R}^nT(K) \tag{1}$$

$${}_FE_1^{p,n-p} = \mathrm{R}^nT(\mathrm{Gr}^p_F K) \Rightarrow \mathrm{R}^nT(K) \tag{2}$$

are abutments of the hypercohomology spectral sequences of the filtered complexes $\mathrm{Gr}^q_W K$ and $\mathrm{Gr}^p_F K$, with E_1 pages given by

$$E_1^{p,q,n-p-q} := \mathrm{R}^nT(\mathrm{Gr}^p_F \mathrm{Gr}^q_W K) \Rightarrow {}_WE_1^{q,n-q} \qquad \textit{for fixed } q \tag{3}$$

$$E_1^{p,q,n-p-q} := \mathrm{R}^nT(\mathrm{Gr}^p_F \mathrm{Gr}^q_W K) \Rightarrow {}_FE_1^{p,n-p} \qquad \textit{for fixed } p \tag{4}$$

(ii) The filtration of ${}_WE_1^{p,n-p}$, abutting to the spectral sequence (3), is the filtration of ${}_WE_1^{p,n-p}(TK')$ induced by the filtration F of TK'.

(iii) For the differentials of the complexes $\mathrm{Gr}_W^n(T(K'))$ *to be strictly compatible with the filtration* F*, it is necessary and sufficient that the hypercohomology spectral sequences* (3) *degenerate on the* E_1 *page.*

(iv) The morphisms d_1 *of the spectral sequence* (4) *are the initial terms of the degree-1 morphisms of the spectral sequences* (3) *that abut to the morphisms* d_1 *of the spectral sequence* (1).

Proof. Claims (i) and (iv) follow from §1.4 and §1.4 applied to TK' via the isomorphisms §1.4. Claim (ii) is then trivial by the definition of the recurrent filtration F (identical to the discrete filtrations by Lemma 1.3.10 and (iii) of Proposition 1.3.13), and claim (iii) follows from Proposition 1.3.2. ☐

2. Hodge structures

2.1. Pure structures

In all the following, we denote by $\mathbb{C}$ an algebraic closure of $\mathbb{R}$, and we do not suppose to have chosen a root i of the equation $x^2+1=0$. The theory will be invariant under complex conjugation (cf. §2.1).

We denote by S the real algebraic group $\mathbb{C}^*$, given by restriction of scalars à la Weil from $\mathbb{C}$ to $\mathbb{R}$ of the group $\mathbb{G}_{\mathrm{m}}$:

$$S = \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{\mathrm{m}}$$
$$S(\mathbb{R}) = \mathbb{C}^*.$$

The group S is a torus, i.e. it is connected and of multiplicative type. It is thus described by the free abelian group of finite type

$$X(S) = \mathrm{Hom}(S_{\mathbb{C}}, \mathbb{G}_{\mathrm{m}}) = \underline{\mathrm{Hom}}(S, \mathbb{G}_{\mathrm{m}})(\mathbb{C})$$

of its complex characters, endowed with the action of $\mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \mathbb{Z}/(2)$.

The group $X(S)$ has generators z and $\overline{z}$, which induce (respectively) the identity and complex conjugation:

$$\mathbb{C}^* = S(\mathbb{R}) \to S(\mathbb{C}) \to \mathbb{G}_{\mathrm{m}}(\mathbb{C}) = \mathbb{C}^*.$$

Complex conjugation exchanges z and $\overline{z}$.

We have a canonical map

$$w\colon \mathbb{G}_{\mathrm{m}} \to S \tag{2.1.3.1}$$

that, on the real points, induces the inclusion of $\mathbb{R}^*$ into $\mathbb{C}^*$. We have

$$zw = \overline{z}w = \mathrm{Id}. \tag{2.1.3.2}$$

We also have a map

$$N\colon S \to \mathbb{G}_{\mathrm{m}} \tag{2.1.3.3}$$

that on the real points can be identified with the norm $N_{\mathbb{C}/\mathbb{R}}\colon \mathbb{C}^* \to \mathbb{R}^*$. We have

$$N = z\overline{z} \tag{2.1.3.4}$$

$$N \circ w = (x \mapsto x^2). \tag{2.1.3.5}$$

Definition 2.1.4. A *real Hodge structure* is a real vector space V of finite dimension endowed with an action of the real algebraic group S.

By the general theory of groups of multiplicative type, giving a real Hodge structure on V is equivalent to giving a bigrading $V^{p,q}$ of $V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$ that satisfies $\overline{V^{pq}} = V^{qp}$. The action of S and the bigrading are mutually determined via the condition:

On V^{pq}, S acts by multiplication by $z^p\overline{z}^q$.

Let $(\mathbb{C}, \times)$ be the multiplicative monoid, and let

$$\overline{S} = \prod_{\mathbb{C}/\mathbb{R}} (\mathbb{C}, \times).$$

If V is a real vector space, then we can show that it is equivalent to either give an action of $\overline{S}$ on V or to give a bigrading V^{pq} on V such that $\overline{V^{pq}} = V^{qp}$ and $V^{pq} = 0$ for $p < 0$ or $q < 0$.

Let V be a real Hodge structure, defined by a representation σ of S and a bigrading V^{pq}. The grading of $V_{\mathbb{C}}$ by the $V^n_{\mathbb{C}} = \sum_{p+q=n} V^{pq}$ is then defined over $\mathbb{R}$. We call this the *weight grading.* On $V^n = V \cap V^n_{\mathbb{C}}$, the representation σw of $\mathbb{G}_{\mathrm{m}}$ is multiplication by x^n.

We say that V is *of weight* n if $V^{pq} = 0$ for $p + q \neq n$, i.e. if σw is multiplication by x^n.

Let V be a real Hodge structure. The *Hodge filtration* on $V_{\mathbb{C}}$ is defined by

$$F^p(V_{\mathbb{C}}) = \sum_{p' \geqslant p} V^{p'q'}.$$

By §1.2, we have:

Proposition 2.1.9. *Let n be an integer. The construction §2.1 establishes an equivalence of categories between:*

(a) the category of real Hodge structures of weight n;

(b) the category of pairs consisting of a real vector space V of finite dimension and of a filtration F on $V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$ that is n-opposite to its complex conjugate $\overline{F}$.

Definition 2.1.10. A *Hodge structure* H, of weight n, consists of

(a) a $\mathbb{Z}$-module $H_{\mathbb{Z}}$ of finite type (the "*integral lattice*");

(b) a real Hodge structure of weight n on $H_{\mathbb{R}} = \mathbb{R} \otimes_{\mathbb{Z}} H_{\mathbb{Z}}$.

A morphism $f\colon H \to H'$ is a homomorphism $f\colon H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$ such that $f_{\mathbb{R}}\colon H_{\mathbb{R}} \to H'_{\mathbb{R}}$ is compatible with the action of S (i.e. such that $f_{\mathbb{C}}$ is compatible with the bigrading, or with the Hodge filtration).

Hodge structures of weight n form an abelian category. If H is of weight n, and H' is of weight n', then we define a Hodge structure $H \otimes H'$ of weight $n + n'$ by the equations:

(a) $(H \otimes H')_{\mathbb{Z}} = H_{\mathbb{Z}} \otimes H'_{\mathbb{Z}}$;

(b) the action of S on $(H \otimes H')_{\mathbb{R}} = H_{\mathbb{R}} \otimes H'_{\mathbb{R}}$ is the tensor product of the actions of S on $H_{\mathbb{R}}$ and on $H'_{\mathbb{R}}$.

The bigrading (resp. Hodge filtration) of $(H \otimes H')_{\mathbb{C}} = H_{\mathbb{C}} \otimes H'_{\mathbb{C}}$ is the tensor product of the bigradings (resp. Hodge filtrations (cf. §1.1)) of $H_{\mathbb{C}}$ and of $H'_{\mathbb{C}}$.

We define in an analogous manner the Hodge structure $\underline{\mathrm{Hom}}(H, H')$ (of weight $n' - n$), the Hodge structures $\wedge^p H$ (of weight pn), and the dual Hodge structure H^*.

The ***internal hom*** $\underline{\mathrm{Hom}}$ above and the homomorphism group, are related by:

Remark 2.1.11.1. $\mathrm{Hom}(H, H')$ is the subgroup of

$$\underline{\mathrm{Hom}}(H, H')_{\mathbb{Z}} = \mathrm{Hom}_{\mathbb{Z}}(H_{\mathbb{Z}}, H'_{\mathbb{Z}})$$

consisting of elements of degree $(0, 0)$.

The actions of S on $H_{\mathbb{R}}$, $H'_{\mathbb{R}}$, and $\underline{\mathrm{Hom}}(H, H')_{\mathbb{R}} = \mathrm{Hom}(H_{\mathbb{R}}, H'_{\mathbb{R}})$ are related by

$$s(f(x)) = s(f)(s(x)).$$

This means that if f is of degree $(0, 0)$, i.e. invariant under S, then it commutes with the action of S.

If A is a Noetherian subring of $\mathbb{R}$, then we define a ***Hodge A-structure of weight*** n to consist of an A-module H_A of finite type along with a real Hodge structure of weight n on $H_{\mathbb{R}} = H_A \otimes_A \mathbb{R}$. This definition is mostly used for $A = \mathbb{Q}$. A Hodge A-structure consists of an A-module H_A of finite type along with a real Hodge structure on $H_{\mathbb{R}} = H_A \otimes_A \mathbb{R}$ such that the weight grading is defined over the field of fractions of A.

Definition 2.1.13. The ***Tate Hodge structure*** $\mathbb{Z}(1)$ is the Hodge structure of weight -2, of rank 1, of pure bidegree $(-1, -1)$, with integral lattice $2\pi i\mathbb{Z} \subset \mathbb{C}$.

The action of S is thus given by multiplication with the inverse of the norm (Equation 2.1.3.3).

For $n \in \mathbb{Z}$, we define $\mathbb{Z}(n)$ as the n-th tensor power of $\mathbb{Z}(1)$, so $\mathbb{Z}(n)$ is the Hodge structure of weight $-2n$, of rank 1, of pure bidegree $(-n, -n)$, with integral lattice $(2\pi i)^n\mathbb{Z} \subset \mathbb{C}$. THe action of S is multiplication by $N(x)^{-n}$.

The choice in $\mathbb{C}$ of a solution i of the equation $x^2 + 1 = 0$ determines, on each complex variety X of pure dimension n, an orientation $\mathrm{or}_i(X)$. Replace i by $-i$ gives

$$\mathrm{or}_{-i}(X) = (-1)^n \mathrm{or}_i(X).$$

The choice of i also defines an element C of order 4 in $S(\mathbb{R})$, given by the image of i under the isomorphism $S(\mathbb{R}) \simeq \mathbb{C}^*$.

Finally, it also defines an isomorphism between $\mathbb{Z}$ and the integral lattice of $\mathbb{Z}(n)$, given by multiplication by $(2\pi i)^n$.

When i, an orientation of X, C, or an identification $\mathbb{Z} \sim \mathbb{Z}(n)_{\mathbb{Z}}$ appear in an equation, it is implicitly understood that they all follow from a single choice of the same i, and that by replacing i with $-i$ we obtain an equivalent equation.

Definition 2.1.15. A ***polarisation*** of a Hodge structure H of weight n is a homomorphism

$$(x, y) \colon H \otimes H \to \mathbb{Z}(-n)$$

such that the real bilinear form $(2\pi i)^n(x, Cy)$ on $H_{\mathbb{R}}$ is symmetric and positive definite.

The *real Tate Hodge structure* is the real Hodge structure $\mathbb{R}(1)$ underlying $\mathbb{Z}(1)$. We similarly define $\mathbb{R}(n)$ as underlying $\mathbb{Z}(1)$. A polarisation of a *real* Hodge structure of weight n is a homomorphism

$$(x,y)\colon H\otimes H\to\mathbb{R}(-n)$$

such that the real bilinear form $(2\pi i)^n(x,Cy)$ on $H_{\mathbb{R}}$ is symmetric and positive definite. A polarisation is entirely defined by the positive definite quadratic form $(2\pi i)^n(x,Cy)$ on $H_{\mathbb{R}}$, imposed with only the condition of being invariant under the compact sub-torus of S given by the kernel of N.

We have

$$(x,y)=(Cx,Cy)=(y,C^2x)=(-1)^n(y,x).$$

The form (x,y) is thus symmetric or alternating, depending on the parity of n.

The reader can generalise these definitions to Hodge A-structures of weight n (§2.1).

2.2. Hodge theory

Let X be a compact Kähler variety (for example, smooth and projective). By the holomorphic Poincaré lemma, the de Rham complex $\Omega_X^\bullet$ is a resolution of the constant sheaf $\mathbb{C}$. We thus have an isomorphism (§1.4)

$$\mathrm{H}^\bullet(X,\mathbb{C})\sim\mathbb{H}^\bullet(X,\Omega_X^\bullet)$$

and the stupid filtration on $\Omega_X^\bullet$ (§1.4) defines the hypercohomology spectral sequence

$$E_1^{pq}=\mathrm{H}^q(X,\Omega_X^q)\Rightarrow\mathrm{H}^{p+q}(X,\mathbb{C}) \tag{2.2.1.1}$$

which abuts to the *Hodge filtration* of $\mathrm{H}^\bullet(X,\mathbb{C})$.

By Hodge theory [H1952; W1958], we have:

(A) The spectral sequence Equation 2.2.1.1 degenerates: $E_1=E_\infty$.

(B) The Hodge filtration of $\mathrm{H}^n(X,\mathbb{C})$ is n-opposite to the complex conjugate filtration.

Let V be a local system of real vector spaces over X, i.e. a sheaf of $\mathbb{R}$-vector spaces that is locally isomorphic to a constant sheaf $\mathbb{R}^n$. Suppose that there exists on V a bilinear form

$$Q\colon V\otimes V\to\mathbb{R}$$

that is locally constant and *defined*. If X is connected, this is the case if V is defined by a representation of a finite quotient of the fundamental group of X.

Statements (A) and (B) above remain true as they are, without needing to change the proofs, for cohomology with coefficients in $V_{\mathbb{C}}=V\otimes_{\mathbb{R}}\mathbb{C}$, since the spectral sequence

$$E_1^{pq}=\mathrm{H}^q(X,\Omega_X^p(V))\Rightarrow\mathrm{H}^{p+q}(X,V_{\mathbb{C}})$$

induced by the de Rham resolution of $V_{\mathbb{C}}$ by $\Omega_X^\bullet(V_{\mathbb{C}})$ degenerates, and abuts to a filtration on $\mathrm{H}^n(X,V_{\mathbb{C}})$ that is n-opposite to the complex conjugate filtration.

The vector space $\mathrm{H}^n(X,V)$ is thus endowed with a canonical real Hodge structure of weight n.

We show in [D1968] that the statements in §2.2 remain true when X is a non-singular complete algebraic, not necessarily Kähler, variety. The proof from *loc. cit.*, based on a reduction to the projective case by Chow's lemma and resolution of singularities, extends to the case of §2.2.

Let $\mathcal{L}$ be an invertible sheaf. Here are two ways of defining the class $c_1(\mathcal{L})$.

The sheaf $\mathcal{L}$ defines an element c in $\mathrm{H}^1(X, \mathcal{O}^*)$. Its image under $\mathrm{d}f/f\colon \mathcal{O}^* \to \Omega^1$ lies in $\mathrm{H}^1(X, \Omega^1)$. More precisely, $\mathrm{d}f/f$ defines a morphism of complexes

$$\mathrm{d}\log\colon \mathcal{O}^*[-1] \to (0 \to \Omega^1_X \to \Omega^2_X \to \ldots) = \sigma_{\geqslant 1}(\Omega^\bullet_X).$$

This complex maps to $\Omega^\bullet_X$, whence

$$\mathrm{d}\log\colon \mathcal{O}^*[-1] \to \Omega^\bullet_X$$

and the image of c under $\mathrm{d}\log$ is in $\mathbb{H}^2(\Omega^\bullet_X)$. This construction still makes sense for an algebraic variety over an arbitrary field k. For $k = \mathbb{C}$, we further have

$$\mathrm{H}^2(X, \mathbb{C}) \xrightarrow{\sim} \mathbb{H}^2(X, \Omega^\bullet_X)$$

whence a class

$$c_1'(\mathcal{L}) \in \mathrm{H}^2(X, \mathbb{C}).$$

The exponential exact sequence

$$0 \to \mathbb{Z}(1) \to \mathcal{O} \to \mathcal{O}^* \to 0$$

defines a homomorphism

$$\partial\colon\ \mathrm{H}^1(X, \mathcal{O}^*) \to \mathrm{H}^2(X, \mathbb{Z}(1))$$

whence a class

$$\partial c = c_1''(\mathcal{L}) \in \mathrm{H}^2(X, \mathbb{Z}(1)).$$

If we have a choice of i, then this class is identified with

$$c_1'''(\mathcal{L}) \in \mathrm{H}^2(X, \mathbb{Z})$$

and for $\mathcal{L} = \mathcal{O}(D)$, this $c_1'''(\mathcal{L})$ is exactly the integer cohomology class defined by D (with the orientations being defined by i).

We will prove:

For α the natural injection of $\mathbb{Z}(1)$ into $\mathbb{C}$, we have

$$-\alpha c_1''(\mathcal{L}) = c_1'(\mathcal{L}).$$

For β the natural injection of $\mathbb{Z}$ into $\mathbb{C}$, we have

$$-\beta c_1'''(\mathcal{L}) = \frac{1}{2\pi i} c_1'(\mathcal{L}).$$

We prove this by considering the following diagram of complexes, in which $\approx$ denotes a quasi-isomorphism, and whose upper triangle anti-commutes up to homotopy:

$$\begin{tikzcd}
\mathbb{C} \arrow[r, "\approx"] \arrow[d, equal] & \Omega^\bullet_X & \sigma_{\geqslant 1}(\Omega^\bullet_X) \arrow[l] \arrow[d, equal] \\
\mathbb{C} & (\mathbb{C} \to \mathcal{O}) \arrow[l] \arrow[r, "\approx", "\mathrm{d}"'] & \sigma_{\geqslant 1}(\Omega^\bullet_X) \\
\mathbb{Z}(1) \arrow[u] & (\mathbb{Z}(1) \to \mathcal{O}) \arrow[l] \arrow[u] \arrow[r, "\approx", "\exp"'] & \mathcal{O}^*[-1] \arrow[u, "\mathrm{d}\log"']
\end{tikzcd}$$

Let X be a non-singular projective variety of pure dimension n. A choice of i defines an orientation of X and an isomorphism $\mathbb{Z}(1) \simeq \mathbb{Z}$. The corresponding trace morphism

$$\mathrm{H}^{2n}(X, \mathbb{Z}(n)) \to \mathbb{Z}$$

that is induced does not depend on the choice of i.

By Hodge, for $i \leqslant n$, the morphism

$$L^{n-i} = - \wedge c_1''(\mathcal{O}(1))^{n-i} \colon \mathrm{H}^i(X, \mathbb{Z}) \to \mathrm{H}^{2n-i}(X, \mathbb{Z}(n-i))$$

is an isomorphism, and, combined with Poincaré duality

$$\mathrm{H}^i(X, \mathbb{Z}) \otimes \mathrm{H}^{2n-i}(X, \mathbb{Z}(n-i)) \to \mathrm{H}^{2n}(X, \mathbb{Z}(n-i)) \to \mathbb{Z}(-i)$$

gives a polarisation on the *primitive part* $\mathrm{Ker}(L^{n-i+1})$ of $\mathrm{H}^i(X, \mathbb{Z})$.

We thus deduce that the rational Hodge structures $\mathrm{H}^i(X, \mathbb{Q})$ are polarisable.

2.3. Mixed structures

Definition 2.3.1. A *mixed Hodge structure* H *consists of*

(a) A $\mathbb{Z}$-module $H_\mathbb{Z}$ of finite type, called the "*integral lattice*";

(b) A finite increasing filtration W_n on $H_\mathbb{Q} = \mathbb{Q} \otimes_\mathbb{Z} H_\mathbb{Z}$, called the *weight filtration*;

(c) A finite filtration F^p on $H_\mathbb{C} = \mathbb{C} \otimes_\mathbb{Z} H_\mathbb{Z}$, called the *Hodge filtration*.

We demand that, on $H_\mathbb{C}$, the filtration $W_\mathbb{C}$ induced by extension of scalars of W, the filtration F, and the complex conjugate $\overline{F}$ form a system $(W_\mathbb{C}, F, \overline{F})$ of three opposite filtrations (Definition 1.2.7 and §1.2).

We also denote by W the filtration of $H_\mathbb{Z}$ given by the inverse image of the filtration W on $H_\mathbb{Q}$. The axiom of mixed Hodge structures implies that, for each n, the filtration F induces on $\mathbb{C} \otimes_\mathbb{Z} \mathrm{Gr}_n^W(H_\mathbb{Z})$ a filtration that is n-opposite to its complex conjugate. By Proposition 2.1.9, $\mathrm{Gr}_n^W(H_\mathbb{Z})$ is endowed with a Hodge structure of weight n, with Hodge filtration induced by F.

Example 2.3.3. If H is a Hodge structure of weight n, then we define a mixed Hodge structure with the same integral lattice and the same Hodge filtration by setting

$$W_i(H_\mathbb{Q}) = \begin{cases} 0 & \text{for } i < n \\ H_\mathbb{Q} & \text{for } i \geqslant n. \end{cases}$$

A *morphism* $f \colon H \to H'$ of Hodge structures is a homomorphism $f \colon H_\mathbb{Z} \to H'_\mathbb{Z}$ that is compatible with the filtrations W and F (and thus compatible with $\overline{F}$). We immediately deduce from Theorem 1.2.10 the following theorem.

Theorem 2.3.5.

(i) The category of mixed Hodge structures is abelian.

(ii) The integral lattice of the kernel (resp. cokernel) of a morphism $f\colon H \to H'$ is the kernel (resp. cokernel) K of $f\colon H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$, with $K \otimes \mathbb{Q}$ and $K \otimes \mathbb{C}$ being endowed with the induced filtrations (resp. quotient filtrations) of the filtrations W and F of $H_{\mathbb{Q}}$ and $H_{\mathbb{C}}$ (resp. of $H'_{\mathbb{Q}}$ and $H'_{\mathbb{C}}$).

(iii) Every morphism $f\colon H \to H'$ is strictly compatible with the filtration W of $H_{\mathbb{Q}}$ and $H'_{\mathbb{Q}}$ and with the filtration F of $H_{\mathbb{C}}$ and $H'_{\mathbb{C}}$. It induces morphisms of Hodge $\mathbb{Q}$-structures

$$\mathrm{Gr}_n^w(f)\colon \mathrm{Gr}_n^W(H_{\mathbb{Q}}) \to \mathrm{Gr}_n^W(H'_{\mathbb{Q}}).$$

It also induces morphisms

$$\mathrm{Gr}_F^p(F)\colon \mathrm{Gr}_F^p(H_{\mathbb{C}}) \to \mathrm{Gr}_F^p(H'_{\mathbb{C}})$$

that are strictly compatible with the filtration induced by $W_{\mathbb{C}}$.

(iv) The functor Gr_n^W is an exact functor from the category of mixed Hodge structures to the category of Hodge $\mathbb{Q}$-structures of weight n.

(v) The functor Gr_F^p is an exact functor.

Let H be a mixed Hodge structure. The $W_n(H_{\mathbb{Z}})$, endowed with the filtrations induced by W and F, then form mixed Hodge substructures $W_n(H)$ of H. The quotient $W_n(H)/W_{n-1}(H)$ can be identified with $\mathrm{Gr}_n^W(H_{\mathbb{Z}})$ endowed with its mixed Hodge structure (§2.3 and Example 2.3.3). We denote this Hodge structure by $\mathrm{Gr}_n^W(H)$.

We set

$$H^{p,q} = \mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_{p+q}^W(H_{\mathbb{C}}) = (\mathrm{Gr}_{p+q}^W(H))^{p,q}.$$

The *Hodge numbers* of H are the integers

$$h^{p,q} = \dim_{\mathbb{C}} H^{p,q}.$$

The Hodge number h^{pq} of H is thus the Hodge number h^{pq} of the Hodge structure $\mathrm{Gr}_{p+q}^W(H)$

We define a *mixed Hodge $\mathbb{Q}$-structure* H to consist of a vector space $H_{\mathbb{Q}}$ of finite dimension over $\mathbb{Q}$, a finite increasing filtration W on $H_{\mathbb{Q}}$, and a finite decreasing filtration F on $H_{\mathbb{C}}$, with the filtrations $W_{\mathbb{C}}$, F, and $\overline{F}$ being opposite. Theorem 2.3.5 generalises trivially to this variation.

3. Hodge theory of non-singular algebraic varieties

3.1. Logarithmic poles and residues

We recall several classical properties of "logarithmic poles" of holomorphic differential forms. The reader will find proofs in [D1970, II, (3.1) to (3.7)], for example.

A divisor Y in a smooth complex analytic variety X is said to be of *normal crossing* if the inclusion of Y into X is locally isomorphic to the inclusion of a union of coordinate hyperplanes in $\mathbb{C}^n$; this does not imply that Y is the union of smooth divisors. Let Y be a normal crossing divisor in X, and j the inclusion of $X^* = X \setminus Y$ into X. We denote by $\Omega_X^1\langle Y\rangle$ the locally free sub-$\mathcal{O}$-module of $j_*\Omega_{X^*}^1$ generated by Ω_X^1 and the $\mathrm{d}z_i/z_i$ for z_i local coordinates of a local irreducible component of Y. The *sheaf $\Omega_X^p\langle Y\rangle$ of differential p-forms on X with logarithmic poles along Y* is by the definition the locally free sub-sheaf $\wedge^p\Omega_X^1\langle Y\rangle$ of $j_*\Omega_{X^*}^p$.

Proposition 3.1.3.

(i) A section α of $j_*\Omega^p_{X^*}$ belongs to $\Omega^p_X\langle Y\rangle$ if and only if α and $\mathrm{d}\alpha$ have at worst simple poles along the divisor Y.

(ii) The $\Omega^p_X\langle Y\rangle$ form the smallest subcomplex of $j_*\Omega^\bullet_{X^*}$ that is stable under exterior product and that contains $\Omega^\bullet_X$ and the logarithmic differential $\mathrm{d}f/f$ of every local section of $j_*\Omega^p_{X^*}$ meromorphic along Y.

We call $\Omega^\bullet_X\langle Y\rangle$ the *logarithmic de Rham complex* of X along Y. By (ii) of Proposition 3.1.3, this complex is contravariant in the pair (X, X^*).

Locally on X, Y is the union of various smooth divisors Y_i, and we denote by Y^n (resp. $\widetilde{Y}^n$) the union (resp. disjoint sum) of the n-fold intersections of the Y_i. The Y^n glue to give a subspace Y^n of X, and the $\widetilde{Y}^n$ glue to give the normalisation variety of Y^n. We have $\widetilde{Y}^0 = Y^0 = X$, and we set $\widetilde{Y} = \widetilde{Y}^1$.

We define the two-element set of *orientations* of an n-element set E as the set of generators of $\wedge^n\mathbb{Z}^E$.

For $n \geqslant 2$, this set is that of the conjugation classes under the alternating group of total orders on E.

If, to each point y of $\widetilde{Y}^n$, we associate the set of n local components of Y that contain the image in X of a neighbourhood of y in $\widetilde{Y}^n$, then we define on $\widetilde{Y}^n$ a local system E_n of n-element sets. The local system of orientations of these sets is a $\mathbb{Z}/(2)$-torsor. This torsor defines, via the inclusion of $\mathbb{Z}/(2)$ into $\mathbb{C}^*$, a complex local system ε^n of rank 1 on $\widetilde{Y}^n$, endowed with an isomorphism $(\varepsilon^n)^{\otimes 2} \simeq \mathbb{C}$. We have

$$\varepsilon^n \simeq \bigwedge^n \mathbb{C}^{E_n}.$$

Locally on $\widetilde{Y}^n$, ε^n is endowed with two opposite isomorphisms $\pm\alpha\colon \varepsilon^n \xrightarrow{\sim} \mathbb{C}$. We set

$$\varepsilon^n_{\mathbb{Z}} = \alpha^{-1}((2\pi i)^{-n}\mathbb{Z}).$$

There is a way of seeing ε^n, endowed with $\varepsilon^n_{\mathbb{Z}}$, as a twisted version of $\mathbb{Z}(-n)$ over $\widetilde{Y}^n$.

We denote by ε^n_X (resp. by $(\varepsilon^n_X)_{\mathbb{Z}}$) the direct image of ε^n (resp. of $\varepsilon^n_{\mathbb{Z}}$) under the mat from $\widetilde{Y}^n$ to X. We have

$$\varepsilon^n_X \simeq \bigwedge^n \varepsilon^1_X \qquad \text{for } n \geqslant 0. \tag{3.1.4.1}$$

If Y is the union of distinct smooth divisors $(Y_i)_{i\in I}$, then the choice of a total order on I trivialises the ε^n.

We denote by $W_n(\Omega^p_X\langle Y\rangle)$ the sub-module of $\Omega^p_X\langle Y\rangle$ consisting of linear combinations of products

$$\alpha \wedge \frac{\mathrm{d}t_{i(1)}}{t_{i(1)}} \ldots \wedge \frac{\mathrm{d}t_{i(m)}}{t_{i(m)}} \qquad \text{for } m \leqslant n$$

with α holomorphic and the $t_{i(j)}$ local coordinates of the distinct local components Y_j of Y. We define the $weight filtration$ of $\Omega^\bullet_X\langle Y\rangle$ to be the *increasing* filtration by the subcomplexes $W_n(\Omega^\bullet_X\langle Y\rangle)$. We have

$$W_n(\Omega^p_X\langle Y\rangle) \wedge W_m(\Omega^p_X\langle Y\rangle) \subset W_{n+m}(\Omega^{p+q}_X\langle Y\rangle). \tag{3.1.5.1}$$

If we denote by i_n the map from $\widetilde{Y}^n$ to X, then we can show that the mapping

$$\alpha \wedge \frac{\mathrm{d}t_{i(1)}}{t_{i(1)}} \ldots \wedge \frac{\mathrm{d}t_{i(m)}}{t_{i(m)}} \longmapsto (\alpha | Y_{i(1)} \cap \ldots \cap Y_{i(n)}) \otimes (\text{orientation } i(1) \ldots i(n))$$

defines an isomorphism of complexes

$$\mathrm{Res}\colon\ \mathrm{Gr}_n^W(\Omega_X^\bullet\langle Y\rangle) \approx (i_n)_*\Omega_{\widetilde{Y}^n}^\bullet(\varepsilon^n)[-n] \tag{3.1.5.2}$$

(the *Poincaré residue*).

The interpretation that follows, in terms of Equation 3.1.5.2, of the Leray spectral sequence for the inclusion of X^* into X, was pointed out to me by N. Katz. It will allow us to prove a point that I had initially considered as evident (the first part of (ii) of Theorem 3.2.5).

Every point of X admits a fundamental system of Stein open neighbourhoods whose intersections on X^* are again Stein. For a coherent analytic sheaf $\mathscr{F}$ on X^*, we thus have $\mathrm{R}^i j_*\mathscr{F} = 0$ for $i > 0$. The de Rham complex $\Omega_{X^*}^\bullet$ is thus a resolution of the constant sheaf $\mathbb{C}$ by sheaves acyclic for the functor j_*. Then

$$\mathrm{H}^\bullet(X^*, \mathbb{C}) \xrightarrow{\sim} \mathbb{H}^\bullet(X^*, \Omega_{X^*}^\bullet) \xleftarrow{\sim} \mathbb{H}^\bullet(X, j_*\Omega_{X^*}^\bullet)$$

and the Leray spectral sequence for the morphism j can be identified with the hypercohomology spectral sequence for $j_*\Omega_{X^*}^\bullet$ corresponding to the filtration τ by the subcomplexes $\tau_{\leqslant -n}(j_*\Omega_{X^*}^\bullet)$ (§1.4).

Proposition 3.1.8. *The morphisms of filtered complexes*

$$(\Omega_X^\bullet\langle Y\rangle, W) \xleftarrow{\alpha} (\Omega_X^\bullet\langle Y\rangle, \tau) \xhookrightarrow{\beta} (j_*\Omega_{X^*}^\bullet, \tau)$$

are filtered quasi-isomorphisms. They define an isomorphism between the Leray spectral sequence for j in complex cohomology and the hypercohomology spectral sequence of the filtered complex $(\Omega_X^\bullet\langle Y\rangle, W)$ on X.

Proof. By §1.4 and §3.1, it suffices to prove the first claim. In [D1970, II, (6.9)] or [AH1955], one can find a proof of the fact that β is a quasi-isomorphism, and thus a filtered quasi-isomorphism. We can also directly calculate the cohomology sheaves of the two sides: those of $\Omega_X^\bullet\langle Y\rangle$ are determined by Equation 3.1.5.2, while those of $j_*\Omega_{X^*}^\bullet$ are the $\mathrm{R}^i j_*\mathbb{C}$, which can be calculated by topological methods.

For $n \geqslant p$, we have

$$W_n(\Omega_X^p\langle Y\rangle) = \Omega_X^p\langle Y\rangle$$

and so α is a morphism from $\Omega_X^\bullet\langle Y\rangle$, endowed with τ, to $\Omega_X^\bullet\langle Y\rangle$, endowed with the decreasing filtration associated to W (§1.1). By Equation 3.1.5.2, we have

$$\mathscr{H}^i(\mathrm{Gr}_n^W(\Omega_X^\bullet\langle Y\rangle)) = \begin{cases} 0 & \text{for } i \neq n \\ \varepsilon_X^n & \text{for } i = n \end{cases} \tag{3.1.8.1}$$

and we deduce from the first line of this equation that α is a filtered quasi-isomorphism. This proves Proposition 3.1.8. ∎

By §3.1, the isomorphism Equation 3.1.8.1 defines an isomorphism

$$\mathrm{R}^n j_* \mathbb{C} \simeq \mathscr{H}^n(j_*\Omega^\bullet_{X^*}) \simeq \mathscr{H}^n(\Omega^\bullet_X\langle Y\rangle) \simeq \varepsilon^n_X. \tag{3.1.8.2}$$

The isomorphisms Equation 3.1.4.1 correspond, via Equation 3.1.8.1, to the cup product.

Proposition 3.1.9. *The canonical morphism from* $\mathrm{R}^n j_*\mathbb{Z}$ *to* $\mathrm{R}^n j_*\mathbb{C}$ *identifies, via Equation 3.1.8.2, the sheaf* $\mathrm{R}^n j_*\mathbb{Z}$ *with* $(\varepsilon^n_X)_\mathbb{Z}$ *(§3.1).*

Proof. The question is local on X. We can thus suppose that X is an open polycylinder D^m, with

$$D = \{z \in \mathbb{C} \mid |z| < 1\}$$

and also that $Y = \bigcup_{k=1}^{\ell} Y_k$ with $Y_k = \mathrm{pr}_k^{-1}(0)$. The fibre at 0 of $\mathrm{R}^n j_*\mathbb{Z}$ is then the integer cohomology of $X^* = (D^*)^\ell \times D^{m-\ell}$, with

$$D^* = \{z \in \mathbb{C} \mid 0 < |z| < 1\}.$$

The space X^* has the homotopy type of a torus; its cohomology is thus torsion free, and the cup product defines isomorphisms

$$\bigwedge^n (\mathrm{R}^1 j_*\mathbb{Z}) \xrightarrow{\sim} (\mathrm{R}^n j_*\mathbb{Z})_0.$$

It thus suffices to prove Proposition 3.1.9 for $n = 1$.

The integer homology $\mathrm{H}_1(X^*)$ is generated by the loops γ_k that go around the various Y_k. We have

$$\oint_{\gamma_k} \frac{\mathrm{d}z_k}{z_k} = \pm 2\pi i$$

and so the integer cohomology is generated by the $\frac{1}{2\pi i}\frac{\mathrm{d}z_k}{z_k}$, and this proves Proposition 3.1.9. ☐

Let $\mathscr{F}$ be a coherent analytic sheaf on X^*, given as the restriction to X^* of a coherent analytic sheaf $\mathscr{F}'$ on X. We define the *meromorphic direct image* $j_*^{\mathrm{m}}\mathscr{F}$ of $\mathscr{F}$ to be the inductive limit

$$j_*^{\mathrm{m}}\mathscr{F} = \varinjlim \mathscr{F}'(nY).$$

Locally on X, Y is the sum of a finite family $(Y_i)_{i\in I}$ of smooth divisors, and we define the *pole-order filtration* P on $j_*^{\mathrm{m}}\mathcal{O}_X^*$ by the equation

$$P^p(j_*^{\mathrm{m}}\mathcal{O}_{X^*}) = \sum_{n\in A_p} \mathcal{O}_X\left(\sum (n_i+1)Y_i\right) \tag{3.1.10.1}$$

where

$$A_p = \left\{(n_i)_{i\in I} \mid \sum_i n_i \leqslant -p \text{ and } n_i \geqslant 0 \text{ for all } i\right\}.$$

This construction globalises by endowing $j_*^{\mathrm{m}}\mathcal{O}_{X^*}$ with an exhaustive filtration such that $P^p = 0$ for $p > 0$.

We define the *pole-order filtration* of the complex $j_*^{\mathrm{m}}\Omega^\bullet_{X^*} = j_*^{\mathrm{m}}\mathcal{O}_{X^*} \otimes \Omega^\bullet_X$ to be the filtration

$$P^p(j_*^{\mathrm{m}}\Omega^k_{X^*}) = P^{p-k}(j_*^{\mathrm{m}}\mathcal{O}_X)\otimes\Omega^k_X. \tag{3.1.10.2}$$

The filtration P induces, on the subcomplex $\Omega^\bullet_X\langle Y\rangle$ of $j_*^{\mathrm{m}}\Omega^\bullet_{X^*}$, the stupid filtration by the $\sigma_{\geqslant p}(\Omega^\bullet_X\langle Y\rangle)$, which we also call the *Hodge filtration* F.

Proposition 3.1.11. *The inclusion morphism*

$$(\Omega^\bullet_X\langle Y\rangle, F) \to (j_*^{\mathrm{m}}\Omega^\bullet_{X_*}, P)$$

is a filtered quasi-isomorphism.

Proof. This statement was suggested to me by [G1969]. A proof can be found in [D1970, II, (3.13)]. □

3.2. Mixed Hodge theory

We recall that, from here on, we say "scheme" to mean a scheme of finite type over $\mathbb{C}$, and "sheaf on S" to mean a sheaf on S^{an}.

Let X be a smooth and separated scheme. By Nagata [N1962], X is a Zariski open of a complete scheme $\overline{X}$. By Hironaka [H1964], we can take $\overline{X}$ to be smooth and such that $Y = \overline{X}\setminus X$ is a normal crossing divisor.

The reader who wishes to avoid the reference to Nagata can suppose X to be quasi-projective.

The smooth completion $\overline{X}$ can then be chosen to be projective and such that Y is the union of smooth divisors. If we limit ourselves to such compactifications, then we only need Hodge theory in its standard form (§2.2).

By §3.1 and Proposition 3.1.8, we have

$$\mathrm{H}^\bullet(X,\mathbb{C}) \simeq \mathbb{H}^\bullet(\overline{X},\Omega^\bullet_{\overline{X}}\langle Y\rangle).$$

We define the *Hodge filtration* F on the complex $\Omega^\bullet_{\overline{X}}\langle Y\rangle$ as the filtration $F^p = \sigma_{\geqslant p}$ given by the stupid truncations (§1.4). On $\Omega^\bullet_{\overline{X}}\langle Y\rangle$, we thus have two filtrations: F and W (§3.1).

We will need to make use of the fact that there exist bifiltered resolutions $i\colon \Omega^\bullet_{\overline{X}}\langle Y\rangle \to K^\bullet$ such that the $\mathrm{Gr}^p_F\,\mathrm{Gr}^W_n(K^j)$ are Γ-acyclic sheaves:

$$\mathrm{H}^i(\overline{X},\mathrm{Gr}^p_F\,\mathrm{Gr}^W_n(K^j)) = 0 \qquad \text{for } i>0.$$

Here are two methods to construct such a resolution:

(a) We can take $K^\bullet$ to be the canonical Godement resolution $\mathscr{C}^\bullet(\Omega^\bullet_{\overline{X}}\langle Y\rangle)$, filtered by the $\mathscr{C}^\bullet(W_n(\Omega_{\overline{X}}\langle Y\rangle))$ and the $\mathscr{C}^\bullet(F^p(\Omega^\bullet_{\overline{X}}\langle Y\rangle))$. This is a bifiltered resolution since $\mathscr{C}^\bullet$ is an exact functor.

(b) We can take $K^\bullet$ to be the d''-resolution of $\Omega^\bullet_{\overline{X}}\langle Y\rangle$. Let $\Omega^{pq}_{\overline{X}}$ be the sheaf of C^∞ forms of degree (p,q); then $K^\bullet$ is the simple complex associated to the double complex of the $\Omega^p_{\overline{X}}\langle Y\rangle\otimes_{\mathcal{O}}\Omega^{0,q}_{\overline{X}}$ (a subcomplex of the $j_*\Omega^{\bullet\bullet}_X$). This complex is filtered by the $F^p(\Omega^\bullet_{\overline{X}}\langle Y\rangle)\otimes\Omega^{0,\bullet}_{\overline{X}}$ and by the $W_n(\Omega^\bullet_{\overline{X}}\langle Y\rangle)\otimes\Omega^{0,\bullet}_{\overline{X}}$; to prove that this is a bifiltered resolution, we use the fact that the sheaf $\mathcal{O}_\infty$ of complex C^∞ functions on $\overline{X}$ is flat over $\mathcal{O}$ (a corollary of the Malgrange C^∞ preparation theorem). The sheaves $\mathrm{Gr}_F\,\mathrm{Gr}_W(K^\bullet)$ are fine, since they are sheaves of modules over the soft sheaf $\mathcal{O}_\infty$.

With the notation of §3.2, the complex cohomology of X appears as the cohomology of the bifiltered complex $\Gamma(\overline{X}, K^\bullet)$. We thus have two spectral sequences abutting to $\mathrm{H}^\bullet(X,\mathbb{C})$. They can be written, with the notation of §3.1, as:

$$ {}_W E_1^{pq} = \mathbb{H}^{p+q}(\overline{X}, \varepsilon_{\overline{X}}^{-p}[p]) = \mathbb{H}^{2p+q}(\widetilde{Y}^p, \varepsilon^{-q}) \Rightarrow \mathrm{H}^n(X,\mathbb{C}) \tag{3.2.4.1}$$

$$ {}_F E_1^{pq} = \mathrm{H}^q(\overline{X}, \Omega^p_{\overline{X}}\langle Y\rangle) \Rightarrow \mathrm{H}^n(X,\mathbb{C}). \tag{3.2.4.2}$$

The first of these, up to the renumbering ${}_W E_1^{pq} \mapsto E_2^{2p+q,-p}$, is exactly the Leray spectral sequence of the inclusion j.

Theorem 3.2.5.

(i) On the pages ${}_W E_r^{pq}$ of the spectral sequence Equation 3.2.4.1, the first direct filtration, the second direct filtration, and the recurrent filtration defined by F all coincide.

(ii) The filtration on $\mathrm{H}^n(X,\mathbb{C})$ that is the abutment of the spectral sequence ${}_W E$ comes from a filtration W of $\mathrm{H}^n(X,\mathbb{Q})$. Neither it, nor the filtration F that is the abutment of the spectral sequence ${}_F E$, depend on the choice of compactification $\overline{X}$ of X or on the choice of $K^\bullet$.

(iii) The filtrations $W[n]$ (Definition 1.1.2) and F define on $\mathrm{H}^n(X,\mathbb{Z})$ a mixed Hodge structure, functorially in X.

By §3.1, the spectral sequence ${}_W E$ is the Leray spectral sequence for j_* (up to renumbering). It is thus induced by tensoring a $\mathbb{Q}$-vectorial spectral sequence with $\mathbb{C}$, and so the first claim of (ii) is true. Complex conjugation acts on the ${}_W E$; it can be calculated via §3.1.

Lemma 3.2.6. *The hypercohomology spectral sequences of the filtered complexes $\mathrm{Gr}_n^W(\Omega^\bullet_{\overline{X}}\langle Y\rangle)$, endowed with the filtration induced by the Hodge filtration, degenerate at the E_1 page.*

Proof. Define the Y^n and the $\widetilde{Y}^n$ as in §3.1, and let $i_n\colon \widetilde{Y}^n \to X$. By Equation 3.1.5.2, we have

$$\mathrm{Gr}_n^W(\Omega^\bullet_{\overline{X}}\langle Y\rangle) \sim (i_n)_*\Omega^\bullet_{\widetilde{Y}^n}(\varepsilon^n)[-n].$$

Furthermore, the Hodge filtration induces the stupid filtration (§1.4), and so the spectral sequence in Lemma 3.2.6 is given by translations of the degrees of the classical spectral sequence

$$E_1^{pq} = \mathrm{H}^q(\widetilde{Y}^n, \Omega^p_{\widetilde{Y}^n}(\varepsilon^n)) \Rightarrow \mathrm{H}^{p+q}(\widetilde{Y}^n, \varepsilon^n).$$

If $\overline{X}$ is projective and Y the union of smooth divisors, then $\widetilde{Y}^n$ is projective, ε^n is a trivial local system, and classical Hodge theory (§2.2) gives the degeneration claimed in Lemma 3.2.6. For the general case, refer to [D1968] (see also §2.2 and §2.2). $\square$

Hodge theory also tells us that the Hodge filtration on $\mathrm{H}^k(\widetilde{Y}^n, \varepsilon^n)$ is k-opposite to its complex conjugate (the complex conjugate being defined in terms of $\varepsilon^n_{\mathbb{Z}}$ (§3.1)). We have, taking into account the translations of the degrees:

Lemma 3.2.7. *The filtration on*

$${}_W E_1^{-n,k+n} = \mathbb{H}^k(\overline{X}, \mathrm{Gr}_n^W(\Omega^\bullet_{\overline{X}}\langle Y\rangle)) \approx \mathrm{H}^{k-n}(\widetilde{Y}^n, \varepsilon^n)$$

which is the abutment of the spectral sequence in Lemma 3.2.6 is $(k+n)$-opposite to its complex conjugate.

Lemma 3.2.8. *The differentials d_1 of the spectral sequence ${}_W E$ are strictly compatible with the filtration F.*

Proof. On the E_1 pages, there is only the filtration induced by F to consider (Lemma 1.3.10 and (iii) of Proposition 1.3.13), and d_1 is compatible with this filtration ((i) of Proposition 1.3.13). This filtration is the abutment of the spectral sequence in Lemma 3.2.6 ((ii) of Example 1.4.8). By Lemma 3.2.7, the arrow

$$d_1\colon \mathbb{H}^k(\overline{X}, \mathrm{Gr}_n^W(\Omega^\bullet_{\overline{X}}\langle Y\rangle)) \to \mathbb{H}^{k+1}(\overline{X}, \mathrm{Gr}_{n-1}^W(\Omega^\bullet_{\overline{X}}\langle Y\rangle))$$

or

$$d_1\colon \mathrm{H}^{k-n}(\widetilde{Y}^n, \varepsilon^n) \to \mathrm{H}^{k-n+1}(\widetilde{Y}^{n-1}, \varepsilon^{n-1}) \tag{3.2.8.1}$$

is compatible with the filtrations that are $(k+n)$-opposite to their complex conjugate. Since d_1 commutes up to complex conjugation, d_1 respects the bigrading (of weight $k+n$) defined by F and $\overline{F}$, which proves Lemma 3.2.8. □

Furthermore, the cohomology of the complex E_1 is again bigraded:

Lemma 3.2.9. *The recurrent filtration F on ${}_WE_2^{pq}$ is q-opposite to its complex conjugate.*

We prove by induction on r that:

Lemma 3.2.10. *For $r \geqslant 0$, the differentials d_r of the spectral sequence ${}_WE$ are strictly compatible with the recurrent filtration F. For $r \geqslant 2$, they are zero.*

Proof. For $r = 0$ (resp. $r = 1$) we apply Lemma 3.2.6 and (iii) of Example 1.4.8 (resp. Lemma 3.2.8). For $r \geqslant 2$, it suffices to prove that $d_r = 0$. By induction, and following Theorem 1.3.16, we can assume that, on the pages ${}_WE^s$ (for $s \geqslant r+1$), we have $F_d = F_r = F_{d^*}$, and that ${}_WE_r = {}_WE_2$. By (i) of Proposition 1.3.13, d_r is thus compatible with the filtration F_r.

On ${}_WE_r^{pq} = {}_WE_2^{pq}$, the filtration F_r is q-opposite to its complex conjugate. The morphism

$$d_r\colon {}_WE_r^{pq} \to {}_WE_r^{p+r,q-r+1}$$

thus satisfies, for $r-1>0$,

$$\begin{aligned} d_r({}_WE_r^{pq}) &= d_r\left(\sum_{a+b=q}\Big(F^a({}_WE_r^{pq}) \cap \overline{F}^b({}_WE_r^{pq})\Big)\right) \\ &\subset \sum_{a+b=q}\Big(F^a({}_WE_r^{p+r,q-r+1}) \cap \overline{F}^b({}_WE_r^{p+r,q-r+1})\Big) \\ &= 0. \end{aligned}$$

This proves Lemma 3.2.10. □

Lemma 3.2.10, using Theorem 1.3.16, proves (i) of Theorem 3.2.5.

By Corollary 1.3.17, the filtration on ${}_WE_\infty^{pq}$ induced by the filtration F of $\mathrm{H}^{p+q}(X,\mathbb{C})$ is q-opposite to its complex conjugate. Since $q = -p + (p+q)$, this proves the first part of (iii) of Theorem 3.2.5

We now prove (ii) and (iii) of Theorem 3.2.5, which will finish the proof.

(A) *Independence from the choice of $K^\bullet$.*

The filtrations F and W of $\mathrm{H}^\bullet(X,\mathbb{C})$ are the abutments of the hypercohomology spectral sequences of $\Omega^\bullet_{\overline{X}}\langle Y\rangle$ for the filtrations F and W. These whole spectral sequences do not depend on the choice of $K^\bullet$.

(B) *Functoriality.*

Let $f\colon X_1 \to X_2$ be a morphism of schemes. Suppose that we are given a morphism of smooth compactifications

$$\begin{tikzcd} X_1 \arrow[r, "f"] \arrow[d, "j_1"'] & X_2 \arrow[d, "j_2"] \\ \overline{X}_1 \arrow[r, "\overline{f}"'] & \overline{X}_2 \end{tikzcd} \tag{3.2.11.1}$$

with the $Y_i = \overline{X}_i \backslash X_i$ normal crossing divisors. The canonical morphism (cf. Proposition 3.1.3) from $\overline{f}^*\Omega^\bullet_{\overline{X}_2}\langle Y_2\rangle$ to $\Omega^\bullet_{\overline{X}_1}\langle Y_1\rangle$ is then a morphism of bifiltered complexes; on hypercohomology, it induces a morphism compatible with F and W, and thus $f^*\colon\ \mathrm{H}^n(X_2,\mathbb{Z}) \to \mathrm{H}^n(X_1,\mathbb{Z})$ is a morphism of mixed Hodge structures, for the structures defined by the compactifications $\overline{X}_i$.

(C) *Independence from the choice of compactification.*

With the notation of (B), if f is an isomorphism, then f^* is a bijective morphism of mixed Hodge structures, and thus an isomorphism (Theorem 2.3.5).

If $\overline{X}_1$ and $\overline{X}_2$ are two smooth compactifications of X, with $Y_i = \overline{X}_i \setminus X$ smooth crossing divisors, then there exists a third smooth compactification $\overline{X}$, with $Y = \overline{X} \setminus X$ a smooth crossing divisor, that fits into a commutative diagram

$$\begin{tikzcd} & X \arrow[dl] \arrow[d, hook] \arrow[dr] & \\ \overline{X}_1 & \overline{X} \arrow[l] \arrow[r] & \overline{X}_2 \end{tikzcd}$$

Namely, we can take $\overline{X}$ to be a resolution of singularities of the closure of the diagonal image of X in $\overline{X}_1 \times \overline{X}_2$. The identity map from $\mathrm{H}^n(X,\mathbb{Z})$ endowed with the mixed Hodge structure defined by $\overline{X}_1$ to $\mathrm{H}^n(X,\mathbb{Z})$ endowed with the mixed Hodge structure defined by $\overline{X}_2$ is thus the composite of two isomorphisms.

To finish the proof of (ii) and (iii), we note that every morphism f fits into a diagram of the form in §3.2: we choose compactifications $\overline{X}'_1$ and $\overline{X}_2$ of X_1 and X_2, then we take $\overline{X}_1$ to be a resolution of singularities of the closure of the image of X_1 in $\overline{X}'_1 \times \overline{X}_2$.

Definition 3.2.12. The *mixed Hodge structure of the cohomology of a smooth separated algebraic variety* is the mixed Hodge structure from (iii) of Theorem 3.2.5.

Corollary 3.2.13. *With the above notation:*

(i) The spectral sequence in Equation 3.2.4.1 degenerates at E_2, i.e. the Leray spectral sequence for the inclusion $j\colon X^* \hookrightarrow X$ degenerates at E_3 (i.e. $E_3 = E_\infty$).

(ii) The spectral sequence in Equation 3.2.4.2

$${}_FE_1^{pq} = \mathrm{H}^q(\overline{X}, \Omega^p_{\overline{X}}\langle Y\rangle) \Rightarrow \mathrm{H}^{p+q}(X,\mathbb{C})$$

degenerates at E_1.

(iii) The spectral sequence defined by the sheaf $\Omega^p_X\langle Y\rangle$*, endowed with the filtration* W

$$\begin{aligned} E_1^{-n,k+n} &= \mathrm{H}^k(\overline{X}, \mathrm{Gr}_n^W(\Omega^p_X\langle Y\rangle)) \\ &\approx \mathrm{H}^k(\widetilde{Y}^n, \Omega^{p-n}_{\widetilde{Y}^n}(\varepsilon^n)) \Rightarrow \mathrm{H}^k(\overline{X}, \Omega^p_X\langle Y\rangle) \end{aligned}$$

degenerates at E_2.

Proof. Claim (i) is proven in Lemma 3.2.10.

Consider the four spectral sequences in §1.4, with respect to the bifiltered complex $\Omega^\bullet_{\overline{X}}\langle Y\rangle$. By (i), we have

$$\sum_n \dim \mathrm{H}^n(X,\mathbb{C}) = \sum_{p,q} \dim {}_W E_2^{p,q}.$$

The ${}_W E_1^{n,k-n}$ pages are, for fixed n, the abutments of a spectral sequence *that degenerates at* E_1 (Lemma 3.2.6) with initial pages $E_1^{p,n,k-p-n}$ (with the notation of Lemma 1.4.12) the initial pages of the spectral sequence in (iii).

Since ${}_W d_1$ is strictly compatible with the filtration F which is the abutment of this spectral sequence (Lemma 3.2.7), and is (Lemma 1.4.12) the abutment of a morphism between these spectral sequences, which starts with the differentials from (iii), we have

$$\mathrm{Gr}^p_F({}_W E_2^{n,k-n}) \approx \mathrm{H}^\bullet(E_1^{p,n-1,k-n-p} \to E_1^{p,n,k-n-p} \to E_1^{p,n+1,k-n-p})$$

and $\mathrm{Gr}^\bullet_F({}_W E_2^{\bullet\bullet})$ is the sum of the pages $E_2^{\bullet\bullet\bullet}$ of the spectral sequences in (iii). We thus have

$$\sum_n \dim \mathrm{H}^n(X,\mathbb{C}) = \sum \dim E_2^{\bullet\bullet\bullet}. \tag{3.2.13.1}$$

We also have

$$\sum_n \dim \mathrm{H}^n(X,\mathbb{C}) \leqslant \dim {}_F E_1^{\bullet\bullet}$$

with equality if and only if the spectral sequence in (ii) degenerates at E_1, and

$$\sum \dim {}_F E_1^{\bullet\bullet} \leqslant \sum \dim E_2^{\bullet\bullet\bullet}$$

with equality if and only if the spectral sequences in (iii) degenerate at E_2. Comparing with Equation 3.2.13.1, we obtain Corollary 3.2.13. □

Corollary 3.2.14. *Let* ω *be a meromorphic differential* p*-form on* $\overline{X}$ *that is holomorphic on* X *and presents at worst logarithmic poles along* Y*. Then the restriction* $\omega|X$ *of* ω *to* X *is closed, and if the cohomology class in* $\mathrm{H}^p(X,\mathbb{C})$ *defined by* ω *is zero then* $\omega = 0$.

Proof. This is the particular case of (ii) of Corollary 3.2.13 where ${}_F E_1^{p0} = {}_F E_\infty^{p0}$. □

Corollary 3.2.15.

(i) If X *is a smooth complete algebraic variety, then the mixed Hodge structure on* $\mathrm{H}^n(X,\mathbb{Z})$ *is the classical Hodge structure of weight* n.

(ii) The Hodge numbers h^{pq} *of the mixed Hodge structure of* $\mathrm{H}^n(X, \mathbb{Z})$ *(with* X *algebraic and smooth) can only be zero for* $p \leqslant n$*,* $q \leqslant n$*, and* $p + q \geqslant n$

Proof. Claim (i) is clear; to prove (ii), note that, if Y is the union of smooth divisors, then the rational Hodge structure $\mathrm{Gr}_k^W(\mathrm{H}^n(X, \mathbb{Q}))$ is the quotient of a sub-object of a Hodge structure of weight $n + k$, namely

$$\mathrm{H}^{n-k}(\widetilde{Y}^n, \mathbb{Q}) \otimes \mathbb{Q}(-k).$$

□

Let X be a smooth separated scheme. We know that X admits smooth compactifications $\overline{X}$ and that the subgroup of $\mathrm{H}^n(X, \mathbb{Z})$ given by the image of $\mathrm{H}^n(\overline{X}, \mathbb{Z})$ is independent of the choice of $\overline{X}$ (cf. [G1968, (9.1) to (9.4)]).

Corollary 3.2.17. *Under the hypotheses of §3.2, the image of* $\mathrm{H}^n(\overline{X}, \mathbb{Q})$ *in* $\mathrm{H}^n(X, \mathbb{Q})$ *is* $W_n(\mathrm{H}^n(X, \mathbb{Q}))$ *(where* W *denotes the weight filtration (Definition 3.2.12)).*

Proof. We can suppose that $\overline{X} \setminus X$ is a normal crossing divisor. The claim then follows from the fact that $W[-n]$ is the abutment of the Leray spectral sequence for the inclusion $j \colon X \hookrightarrow \overline{X}$. □

Corollary 3.2.18. *Let* f *be a morphism from a smooth proper scheme* Y *to a smooth scheme* X*, with* X *admitting a smooth compactification* $j \colon X \hookrightarrow \overline{X}$*. Then the groups* $\mathrm{H}^n(X, \mathbb{Q})$ *and* $\mathrm{H}^n(\overline{X}, \mathbb{Q})$ *have the same image in* $\mathrm{H}^n(Y, \mathbb{Q})$*.*

Proof. Since f^* and $(jf)^*$ are strictly compatible with the weight filtration ((iii) of Theorem 3.2.5), it suffices to prove that $\mathrm{Gr}^W(f^*)$ and $\mathrm{Gr}^W(f^* j^*)$ have the same image in $\mathrm{Gr}^W(\mathrm{H}^n(Y, \mathbb{Q}))$. By Corollary 3.2.17 and Corollary 3.2.15, $\mathrm{Gr}_n^W(j^*)$ is an isomorphism, while $\mathrm{Gr}_m^W(f^*) = 0$ for $m \neq n$ since $\mathrm{Gr}_m^W(\mathrm{H}^n(Y, \mathbb{Q})) = 0$ for $m \neq n$. □

Remark 3.2.19. By Proposition 3.1.11 and §1.4, under the hypotheses of Theorem 3.2.5, the Hodge filtration on $\mathrm{H}^n(X, \mathbb{C})$ is the abutment of the hypercohomology spectral sequence of the complex $j_*^m \Omega^\bullet_{X^*}$ endowed with the filtration given by the pole-order filtration (§3.1): this spectral sequence coincides with Equation 3.2.4.2.

4. Applications and supplements

4.1. The fixed set theorem

Theorem 4.1.1. *Let* S *be a smooth separated scheme, and* $f \colon X \to S$ *a smooth proper morphism.*

(i) In rational cohomology, the Leray spectral sequence

$$E_2^{pq} = \mathrm{H}^p(S, \mathrm{R}^q f_* \mathbb{Q}) \Rightarrow \mathrm{H}^{p+q}(X, \mathbb{Q})$$

degenerates ($E_2 = E_\infty$)*.*

(ii) If $\overline{X}$ is a non-singular compactification of X, then the canonical morphism

$$\mathrm{H}^n(\overline{X}, \mathbb{Q}) \to \mathrm{H}^0(S, \mathrm{R}^n f_* \mathbb{Q})$$

is surjective.

Proof. For f smooth and projective, claim (i) is proven in D1968. The proof in D1968 is rewritten in a more readable manner in G1970; it does not use the smoothness of S.

Let us fix f, and prove that (i) $\implies$ (ii). We can reduce to the case where S is non-empty and connected. If $s \in S$ is a point, then the local system $\mathrm{R}^n f_* \mathbb{Q}$ is entirely described by its fibre $(\mathrm{R}^n f_* \mathbb{Q})_s$ at s and the action of the fundamental group $\pi = \pi_1(S, s)$ on this fibre. We have

$$\mathrm{H}^0(S, \mathrm{R}^n f_* \mathbb{Q}) \xrightarrow{\sim} [(\mathrm{R}^n f_* \mathbb{Q})_s]^\pi.$$

If $X_s = f^{-1}(s)$, then the composite arrow

$$\mathrm{H}^0(S, \mathrm{R}^n f_* \mathbb{Q}) \to (\mathrm{R}^n f_* \mathbb{Q}) \approx \mathrm{H}^n(X_s, \mathbb{Q})$$

is thus injective. Consider the arrows

$$\mathrm{H}^n(\overline{X}, \mathbb{Q}) \xrightarrow{a} \mathrm{H}^n(X, \mathbb{Q}) \xrightarrow{b} \mathrm{H}^0(S, \mathrm{R}^n f_* \mathbb{Q}) \overset{c}{\hookrightarrow} \mathrm{H}^n(X_s, \mathbb{Q}).$$

The arrows cb and cba have the same image, by Corollary 3.2.18. The arrow b is an "edge homomorphism" of the Leray spectral sequence, and is thus surjective by hypothesis. Since c is injective, b and ba have the same image, and ba is surjective.

This proves (ii) for f projective. From this we will deduce the general case. We can again reduce to the case where S is non-empty and connected.

By Chow's lemma and the resolution of singularities, there exists a quasi-projective smooth scheme X' and a projective birational morphism $p\colon X' \to X$.

There thus exists (by Bertini, or Sard) a non-empty Zariski open S_1 of S such that $X'_1 = (fp)^{-1}(S_1)$ is smooth over S_1. Finally, let $X_1 = f^{-1}(S_1)$, let $\overline{X}$ be a smooth compactification of X, and let $\overline{X}'_1$ be a smooth compactification of X'_1 that dominates $\overline{X}$.

$$\begin{tikzcd}
\overline{X} & & \overline{X}'_1 \arrow[ll, "\overline{p}"'] \\
X \arrow[u, hook] & X_1 \arrow[l, hook'] \arrow[d, "f_1"'] & X'_1 \arrow[l, "p"'] \arrow[u, hook] \arrow[d, "f'_1"'] \\
S & S_1 \arrow[l, hook', "i"] \arrow[r, equal] & S_1
\arrow[from=2-1, to=3-1, "f"']
\end{tikzcd}$$

If $s \in S_1$, then the morphism $i_*\colon \pi_1(S_1, s) \to \pi_1(S, s)$ is surjective, since the topological codimension of $S \setminus S_1$ is $\geqslant 2$. We thus have

$$\mathrm{H}^0(S, \mathrm{R}^n f_* \mathbb{Q}) \xrightarrow{\sim} \mathrm{H}^0(S_1, \mathrm{R}^n f_{1*} \mathbb{Q})$$

and it suffices to prove that

$$\mathrm{H}^n(\overline{X}, \mathbb{Q}) \to \mathrm{H}^0(S_1, \mathrm{R}^n f_{1*} \mathbb{Q})$$

is surjective.

The vertical arrows in the diagram

$$\begin{array}{ccccc}
\mathrm{H}^n(\overline{X},\mathbb{Q}) & \longrightarrow & \mathrm{H}^n(X_1,\mathbb{Q}) & \longrightarrow & \mathrm{H}^0(S_1,\mathrm{R}^n f_{1*}\mathbb{Q}) \\
\Big\downarrow{\scriptstyle \overline{p}^*} & & \Big\downarrow{\scriptstyle p^*} & & \Big\downarrow{\scriptstyle p^*} \\
\mathrm{H}^n(\overline{X}_1',\mathbb{Q}) & \longrightarrow & \mathrm{H}^n(X_1',\mathbb{Q}) & \longrightarrow & \mathrm{H}^0(S_1,\mathrm{R}f_{1*}'\mathbb{Q})
\end{array}$$

admit left inverses given by the ***Gysin morphisms*** $\overline{p}_!$, $p_!$, and $p_!$ (respectively), defined by Poincaré duality as the transposes of the arrows analogous to the above but in cohomology with proper support. Furthermore, the diagram

$$\begin{array}{ccc}
\mathrm{H}^n(\overline{X},\mathbb{Q}) & \xrightarrow{u} & \mathrm{H}^0(S_1,\mathrm{R}^n f_{1*}\mathbb{Q}) \\
{\scriptstyle \overline{p}_!}\Big\uparrow & & \Big\uparrow{\scriptstyle p_!} \\
\mathrm{H}^n(\overline{X}',\mathbb{Q}) & \xrightarrow[v]{} & \mathrm{H}^0(S_1,\mathrm{R}^n f_{1*}'\mathbb{Q})
\end{array}$$

commutes. The arrow u is thus a direct factor of the arrow v. Since v is surjective (because f_1' is projective), so too is u.

We now prove (i) in the general case. We can reduce to the case where f is of pure relative dimension n. Let $f\times f$ be the projection of $X\times_S X$ to S.

Let δ be the image of the cohomology class of the diagonal of $X\times_S X$ in $\mathrm{H}^0(S,\mathrm{R}^n(f\times f)_*,\mathbb{Q})$. We have, by Künneth,

$$\mathrm{R}^n(f\times f)_*\mathbb{Q} = \sum_{p+q=n}(\mathrm{R}^p f_*\mathbb{Q}\otimes \mathrm{R}^q f_*\mathbb{Q}).$$

We denote by δ_{pq}' (for $p+q=n$) the components of δ in this decomposition, and by δ_{pq} the classes in $\mathrm{H}^n(X\times_S X)$ given by the images of the δ_{pq}'.

The δ_{pq} define, in the derived category $\mathrm{D}^+(S)$, homomorphisms

$$\delta_p\colon \mathrm{R}f_*\mathbb{Q}\to \mathrm{R}f_*\mathbb{Q}$$

such that $\mathscr{H}^q(\delta_p)$ is zero for $p\neq q$, and is the identity for $p=q$. By [D1968], we thus have

$$\mathrm{R}f_*\mathbb{Q}\approx\sum_p \mathrm{R}^p f_*\mathbb{Q}[-p] \tag{4.1.1.1}$$

in $\mathrm{D}^+(S)$, and the Leray spectral sequence degenerates. □

Corollary 4.1.2. *Let $f\colon X\to S$ be a proper smooth morphism, with S reduced, connected, and separated. Let $(\mathrm{R}^n f_*\mathbb{Q})^0$ be the largest constant local system on $\mathrm{R}^n f_*\mathbb{Q}$, with fibre $\mathrm{H}^0(S,\mathrm{R}^n f_*\mathbb{Q})$. Then, for each $s\in S$, $(\mathrm{R}^n f_*\mathbb{Q})^0_s$ is a Hodge sub-structure of $(\mathrm{R}^n f_*\mathbb{Q})\simeq \mathrm{H}^n(X_s,\mathbb{Q})$, and the induced Hodge structure on $\mathrm{H}^0(S,\mathrm{R}^n f_*\mathbb{Q})$ is independent of s.*

Proof. Let $s\in S$, and let $X_s=f^{-1}(s)$. If S is smooth, and if $\overline{X}$ is a smooth compactification of X, then, by Theorem 4.1.1, the subspace $(\mathrm{R}^n f_*\mathbb{Q})^0_s$ of $(\mathrm{R}^n f_*\mathbb{Q})_s$ is the image of $\mathrm{H}^n(\overline{X},\mathbb{Q})$. Since the restriction map

$$\mathrm{H}^n(\overline{X},\mathbb{Q})\to \mathrm{H}^n(X_s,\mathbb{Q})$$

is a morphism of Hodge structures, its image is a Hodge sub-structure, and the induced Hodge structure on $\mathrm{H}^0(S,\mathrm{R}^n f_*\mathbb{Q})$, given by the quotient of that on $\mathrm{H}^n(\overline{X},\mathbb{Q})$, is independent of s. □

In the general case, Corollary 4.1.2 also implies that, if a is a global section of $\mathrm{R}^n f_*\mathbb{C}$, then its components $a^{p,q}$ of degree (p,q), which are a priori only continuous sections of the complex bundle defined by $\mathrm{R}^n f_*\mathbb{C}$, are in fact locally constant, i.e. are sections of $\mathrm{R}^n f_*\mathbb{C}$. This is the case because $a^{p,q}$ is continuous, and further locally constant on the (dense) open subset of smooth points of S, by the above.

In the case where S is compact, a generalisation of Corollary 4.1.2 is proven by analytic methods in [G1970].

As a corollary to Corollary 4.1.2, we note:

(cf. [G1970]). Let $f\colon X \to S$ be a proper smooth morphism, as in Corollary 4.1.2. If a global section a of $\mathrm{R}^n f_*\mathbb{C}$ is of Hodge degree (p,q) at a point, then a is everywhere of degree (p,q). In particular, if $n = 2$ and if a is, at some point s, the cohomology class of a divisor of X_s, then a is everywhere the cohomology class of a divisor, and, if S is smooth, is even defined by a divisor D on X.

(cf. [G1966]). Let S be a reduced connected scheme of finite type over $\mathbb{C}$, and let $f_1\colon X_1 \to S$ and $f_2\colon X_2 \to S$ be abelian schemes over S. If a morphism $u\colon \mathrm{R}^1 f_{2*}\mathbb{Z} \to \mathrm{R}^1 f_{1*}\mathbb{Z}$ comes from a morphism $\widetilde{u}_s\colon (X_1)_s \to (X_2)_s$ of abelian varieties at some point $s \in S$, then u comes from a (unique) morphism of abelian schemes $\widetilde{u}\colon X_1 \to X_2$.

(cf. [K1971]). Let S be a smooth connected scheme, $s \in S$ a point, $f\colon X \to S$ a proper smooth morphism, and P a direct factor of the local system $\mathrm{R}^i f_*\mathbb{Q}$, which is pointwise a Hodge sub-structure. Then the following conditions are equivalent:

(a) The Hodge structure of P is locally constant.

(b) The representation P_s of $\pi_1(S,s)$ factors through a finite quotient of $\pi_1(S,s)$.

(c) There exists a finite non-empty étale covering $u\colon S' \to S$ such that u^*P is a constant family of Hodge structures.

4.2. The semi-simplicity theorem

Let S be a topological space. A *continuous family of Hodge structures* on S consists of:

(a) A local system $H_{\mathbb{Z}}$ on S of $\mathbb{Z}$-modules of finite type.

(b) For every point $s \in S$, a Hodge structure on the fibre $(H_{\mathbb{Z}})$ that varies continuously in s.

A continuous family H of Hodge structures on S is said to be *of weight* n if the fibres H_s (for $s \in S$) are all of weight n.

We similarly define a *continuous family of Hodge* $\mathbb{Q}$*-structures* as a local system of $\mathbb{Q}$-vector spaces, endowed at each point with a Hodge $\mathbb{Q}$-structure, varying continuously.

A *polarisation* of a continuous family H of Hodge $\mathbb{Q}$-structures of weight n is a morphism of local systems from $H_{\mathbb{Q}} \otimes H_{\mathbb{Q}}$ to the constant local system $\mathbb{Q}(-n)_{\mathbb{Q}}$, which defines at each point $s \in S$ a polarisation of H_s.

Suppose that S is connected, and let $\mathcal{C}$ be a strictly full subcategory of the category of continuous families of Hodge $\mathbb{Q}$-structures on S. We will have need of the following conditions:

Condition 4.2.2.1. $\mathcal{C}$ is stable under direct factors, direct sums, and tensor products; the Tate constant families $\mathbb{Q}(n)$ (for $n \in \mathbb{Z}$) are in $\mathcal{C}$.

Condition 4.2.2.2. Every homogeneous (of some weight n) Hodge structure in $\mathcal{C}$ is polarisable.

Condition 4.2.2.3. For every $H \in \operatorname{Ob} \mathcal{C}$, there exists a local system $H'_{\mathbb{Z}}$ on S of free $\mathbb{Z}$-modules such that $H'_{\mathbb{Z}} \otimes \mathbb{Q} \approx H_{\mathbb{Q}}$.

Condition 4.2.2.4. For every $H \in \operatorname{Ob} \mathcal{C}$, the largest constant local system H^f of H is a constant family of Hodge sub-structures of H.

Lemma 4.2.3. *If $\mathcal{C}$ satisfies Condition 4.2.2.1 and Condition 4.2.2.2, then:*

(i) $\mathcal{C}$ is a semi-simple abelian subcategory of the abelian subcategory of continuous families of Hodge $\mathbb{Q}$-structures on S.

(ii) If $H \in \operatorname{Ob} \mathcal{C}$, then its dual H^* and the $\wedge^p H$ are also in $\mathcal{C}$; if $H_1, H_2 \in \operatorname{Ob} \mathcal{C}$, then $\operatorname{Hom}(H_1, H_2) \in \operatorname{Ob} \mathcal{C}$.

Proof. Let $H \in \operatorname{Ob} \mathcal{C}$, and let H_1 be a sub-object of H, in the category of continuous families of Hodge $\mathbb{Q}$-structures. We will show that H_1 is a direct factor of H in this category. We can suppose H to be homogeneous. If ψ is a polarisation form for H, then the orthogonal of H_1 with respect to ψ is indeed a sub-object of H that is a supplement to H_1. This proves (i)

If $H \in \operatorname{Ob} \mathcal{C}$ is of weight n, then a polarisation of H defines an isomorphism between H^* and $H \otimes \mathbb{Q}(n)$. By Condition 4.2.2.1, we thus have that $H^* \in \operatorname{Ob} \mathcal{C}$. For arbitrary $H \in \operatorname{Ob} \mathcal{C}$, if we decompose H into its homogeneous components, then $H^* = \bigoplus_n (H^n)^*$, whence again $H^* \in \operatorname{Ob} \mathcal{C}$. Finally, $\wedge^p H$ is a direct factor of $\bigotimes^p H$, and $\operatorname{Hom}(H_1, H_2) \simeq H_1^* \otimes H_2$. □

Let $f\colon X \to S$ be a smooth proper morphism of reduced schemes. The sheaf $\mathrm{R}^i f_* \mathbb{Q}$ is then a local system, and, for $s \in S$, $(\mathrm{R}^i f_* \mathbb{Q})_s \simeq \mathrm{H}^i(X_s, \mathbb{Q})$ is endowed with a Hodge $\mathbb{Q}$-structure, which varies continuously in s.

Definition 4.2.4. Let S be a smooth connected scheme. A continuous family H of Hodge $\mathbb{Q}$-structures on S^{an} is said to be *algebraic* if there exists a non-empty Zariski open U of S, an integer k, and a smooth projective morphism $f\colon X \to U$ such that $H|U$ is a direct factor of $\mathrm{R}f_* \mathbb{Q} \otimes \mathbb{Q}(k)$.

Proposition 4.2.5.

(i) The category of algebraic continuous families of Hodge $\mathbb{Q}$-structures on S satisfies the conditions of §4.2.

(ii) If a continuous family H of Hodge structures on S is such that its restriction to a dense Zariski open U of S is algebraic, then H is algebraic.

(iii) If $f\colon X \to S$ is smooth and proper, then $\mathrm{R}f_* \mathbb{Q}$ is algebraic.

Proof. Claim (ii) is evident. We will prove (i). Let $\mathcal{C}_0$ be the set of continuous families of Hodge structures on S that are of the form $\mathrm{R}f_* \mathbb{Q}$ (with f smooth and projective). Then:

(a) By the Künneth formula

$$\mathrm{R}(f \times g)_* \mathbb{Q} \simeq \mathrm{R}f_* \mathbb{Q} \otimes \mathrm{R}g_* \mathbb{Q},$$

$\mathcal{C}_0$ is stable under tensor products.

(b) Since

$$\mathrm{R}(f \sqcup g)_* \mathbb{Q} \simeq \mathrm{R}f_* \mathbb{Q} \oplus \mathrm{R}g_* \mathbb{Q},$$

$\mathcal{C}_0$ is stable under direct sums.

(c) The homogeneous components of each $H \in \operatorname{Ob} \mathcal{C}_0$ are polarisable (cf. §2.2).

(d) The objects $H \in \operatorname{Ob} \mathcal{C}_0$ satisfy Condition 4.2.2.3, since

$$\mathrm{R}f_*\mathbb{Q} \simeq \mathrm{R}f_*\mathbb{Z} \otimes \mathbb{Q}.$$

(e) The objects $H \in \operatorname{Ob} \mathcal{C}_0$ satisfy Condition 4.2.2.4, by Corollary 4.1.2.

Let $\mathcal{C}_1$ be the set of direct factors of objects of $\mathcal{C}_0$. Then $\mathcal{C}_1$ is stable under tensor products, direct sums, and direct factors, and satisfies Condition 4.2.2.2, Condition 4.2.2.3, and Condition 4.2.2.4. Furthermore, $\mathbb{Q}(-1)$ is in $\mathcal{C}_1$, in the form of $\mathrm{R}^2 f_*\mathbb{Q}$ for $f\colon \mathbb{P}^1_S \to S$ the projective bundle. The category $\mathcal{C}_2$ consisting of the $H \otimes \mathbb{Q}(k)$ (for $k \in \mathbb{N}$) thus satisfy all the conditions of §4.2.

To thus deduce (i), it suffices to note that, if H is a continuous family of Hodge structures, and U a dense Zariski open of S, then a sub-object of, a polarisation on, or an integer lattice of $H|U$ uniquely extend to H.

We now prove (iii). If $f\colon X \to S$ is smooth and projective, then there exists a dense Zariski open subset U of S along with a commutative diagram

$$\begin{tikzcd} X' \arrow[rr, "p"] \arrow[dr, "f'"'] & & X|U \arrow[dl, "f|U"] \\ & U & \end{tikzcd}$$

with f' smooth and projective, and p birational and surjective (by applying Chow's lemma and the resolution of singularities to the generic fibre of f). The restriction map

$$p^*\colon (\mathrm{R}f_*\mathbb{Q})|U \to \mathrm{R}f'_*\mathbb{Q}$$

is thus a direct injection of continuous families of Hodge structures, with left inverse given by the Gysin morphism $p_!$, whence the algebraicity of $\mathrm{R}f_*\mathbb{Q}$. □

Theorem 4.2.6. *Let S be a connected topological space that is locally connected and locally simply connected, endowed with a basepoint s. Let $\mathcal{C}$ be a category of continuous Hodge structures on S that satisfy the conditions of §4.2. Then, if $H \in \operatorname{Ob} \mathcal{C}$, the representation of $\pi_1(S,s)$ on the fibre $(H_\mathbb{Q})_s$ is semi-simple.*

Footnote 1. Let S be a smooth connected scheme. A *family of Hodge structures* on S is a continuous family H of Hodge structures on S that satisfies the following conditions:

(a) The Hodge filtration on $(H_\mathbb{C})_s$ varies holomorphically with s, i.e. it corresponds to a filtration F of $H_\mathcal{O} = H_\mathbb{Z} \otimes \mathcal{O}$.

(b) The covariant derivative ∇ satisfies $\nabla F^p \subset \Omega^1_S \otimes F^{p-1}$.

Let $\mathcal{C}$ be the category of such continuous families of Hodge $\mathbb{Q}$-structures on S that underlie a family of Hodge structures and whose homogeneous direct factors are polarisable. It is clear that $\mathcal{C}$ satisfies Condition 4.2.2.1, Condition 4.2.2.2, and Condition 4.2.2.3. We can deduce from results of W. Schmid (August 1970, unpublished) and P.A. Griffiths [G1970] that $\mathcal{C}$ satisfies Condition 4.2.2.4. This theorem, of which Corollary 4.1.2 is a corollary, allows us to apply Theorem 4.2.6 and its corollaries to the objects of $\mathcal{C}$.

Let $H \in \operatorname{Ob} \mathcal{C}$. The local system $H_\mathbb{Q}$ defines a complex local system $H_\mathbb{C} = H_\mathbb{Q} \otimes \mathbb{C}$. For every complex local system V on S, we denote by V^c the complex vector bundle that it defines, identified with the sheaf of its continuous sections.

By definition, the group S (from §2.1) acts on $H_{\mathbb{C}}^{c}$. A sub-bundle of $H_{\mathbb{C}}^{c}$ is said to be *horizontal*, or *locally constant*, if it is defined by a local subsystem of $H_{\mathbb{C}}$.

Lemma 4.2.7. *Let V be a rank-1 local subsystem of $H_{\mathbb{C}}$. Suppose that the n-th tensor power $V^{\otimes n}$ (for some $n \geqslant 1$) of V is a trivial local system, i.e. that $\pi_1(S,s)$ acts on V_s via a finite (necessarily cyclic) group. Then, for all $t \in S$, tV^{c} is again locally constant.*

Proof. In order for tV^{c} to be locally constant, it suffices for $(tV^{c})^{\otimes n} = tV^{c\otimes n} \subset \bigotimes^{n} H_{\mathbb{C}}$ to be locally constant. But $V^{\otimes n}$ is generated by a horizontal global section v, and, by hypothesis, tv is again horizontal Condition 4.2.2.4.

We proceed by induction on $\dim(H_{\mathbb{Q}})_s$. We can suppose H to be homogeneous and non-zero. Let d be the minimal dimension of the non-zero complex local subsystems of $H_{\mathbb{C}}$. The sum W of all the local subsystems of $H_{\mathbb{C}}$ of dimension d (which are automatically simple) is "defined over $\mathbb{Q}$", i.e. is of the form $W_{\mathbb{Q}} \otimes \mathbb{C}$ for $W_{\mathbb{Q}}$ a local subsystem of $H_{\mathbb{Q}}$. By construction, W_s is a semi-simple complex $\pi_1(S,s)$-module, so $(W_{\mathbb{Q}})_s$ is a semi-simple $\pi_1(S,s)$-module over $\mathbb{Q}$.

Let $H_{\mathbb{Z}}$ be a local system of free $\mathbb{Z}$-modules such that $H_{\mathbb{Z}} \otimes \mathbb{Q} \simeq H_{\mathbb{Q}}$, and let $W_{\mathbb{Z}} = H_{\mathbb{Z}} \cap W_{\mathbb{Q}}$. If W is of dimension e, then the local system $(\wedge^{e} W)^{\otimes 2}$ is trivial. Indeed:

$$\bigwedge^{e} W \simeq \bigwedge^{e} W_{\mathbb{Z}} \otimes \mathbb{C}$$

and $\pi_1(S,s)$ can only act on $(\wedge^{e} W_{\mathbb{Z}})$ by ± 1.

Let V be a complex local subsystem of dimension d of $H_{\mathbb{C}}$, and let V' be a complement of V inside W (by the semi-simplicity of W). We have

$$\bigwedge^{e} W \simeq \bigwedge^{d} V \otimes \bigwedge^{e-d} V'.$$

Applying Lemma 4.2.7 to the local subsystem $\wedge^{d} V \otimes \wedge^{e-d} V'$ of $\wedge^{d} H_{\mathbb{C}} \otimes \wedge^{e-d} H_{\mathbb{C}}$, we see that, for $t \in S$,

$$t\left(\bigwedge^{d} V \otimes \bigwedge^{e-d} V'\right)^{c} = \bigwedge^{d} tV^{c} \otimes \bigwedge^{e-d} tV'^{c} \subset \bigwedge^{d} H_{\mathbb{C}} \otimes \bigwedge^{e-d} H_{\mathbb{C}}$$

is locally constant. From this, $\wedge^{d} tV^{c} \subset \wedge^{d} H_{\mathbb{C}}$ and $tV^{c} \subset H_{\mathbb{C}}$ are locally constant.

By the definition of W, we have $tV^{c} \subset W^{c}$, since W^{c} is stable under S, and $W_{\mathbb{Q}}$ is a Hodge sub-structure of H. If ψ is a polarisation of H, then H is thus the direct sum of W and its orthogonal, which are both in $\mathcal{C}$, and we conclude by induction. ☐

Corollary 4.2.8. *Under the hypotheses of Theorem 4.2.6:*

(i) The algebra A of endomorphisms of the local system $H_{\mathbb{Q}}$ is semi-simple. It admits exactly one Hodge $\mathbb{Q}$-structure such that, at each point s, the map $A \otimes H_s \to H_s$ is a morphism of Hodge structures.

(ii) The centre of A is of degree $(0,0)$, and the composition law $\cdot\colon A \otimes A \to A$ is a morphism of Hodge structures.

(iii) Let W be a complex local subsystem of dimension d of $H_{\mathbb{C}}$. Then:

- *(a) For $t \in S$, $tW^{c} \subset H_{\mathbb{C}}$ is locally constant, and defines a local subsystem tW which is isomorphic to W.*

(b) A power $(\wedge^d W)^{\otimes n}$ *(for some* $n \geqslant 1$*) of the local system* $\wedge^d W$ *is a trivial local system.*

Proof. The algebra A is the commutator of $\pi_1(S,s)$ in $(H_{\mathbb{Q}})_s$. Its semi-simplicity thus follows from Theorem 4.2.6 and from Bourbaki, *Algèbre commutative*, Chapter 8, §5.2, Proposition 4. The algebra A is also the fibre of the largest constant local subsystem of $\underline{\mathrm{Hom}}(H_{\mathbb{Q}}, H_{\mathbb{Q}})$. Since $\underline{\mathrm{Hom}}(H,H) \in \mathbb{C}$ (by (ii) of Lemma 4.2.3), the existence of the Hodge structure in (i) follows from the hypotheses (Condition 4.2.2.4). It is clear that the composition law is a morphism of Hodge structures. It thus follows that the centre Z of A is a Hodge substructure: $Z_{\mathbb{C}}$ is bigraded, as the intersection of kernels of bihomogeneous morphisms $[x,-]$, for x bihomogeneous. Finally, Z, and thus $Z_{\mathbb{C}}$, is semi-simple, and, for $(p,q) \neq (0,0)$, Z^{pq} is zero and thus nilpotent.

A complex local subsystem W of $H_{\mathbb{C}}$ is defined by a projector e of $A_{\mathbb{C}}$. For $t \in S$, tW^c is defined by the projector $t \cdot e$, and is thus locally constant. Since the group S is connected, to show that tW is isomorphic to W, it suffices to show that, for t in a small enough neighbourhood of 1, the $\mathbb{C}[\pi_1(S,s)]$ module tW_s is isomorphic to W_s. Let H_i be the isotypic components of the $\mathbb{C}[\pi_1(S,s)]$-module $(H_{\mathbb{C}})_s$. We have

$$W_s = \bigoplus_i (W_s \cap H_i)$$
$$tW_s = \bigoplus_i (tW_s \cap H_i).$$

Furthermore, for t close enough to 1, tW_s is close to W_s in the Grassmannian, and thus

$$\dim(tW_s \cap H_i) \leqslant \dim(W_s \cap H_i). \tag{4.2.8.1}$$

In fact, since $\dim(tW_s) = \dim(W_s)$, we have equality in Equation 4.2.8.1. The various isotypic components of W_s and tW_s thus have, respectively, the same length, and W_s is isomorphic to tW_s.

We now prove (iii.b). It suffices to consider the case where H is homogeneous and W_s is a simple $\mathbb{C}[\pi_1(S,s)]$-module. Let H_1 be the isotypic component of $(H_{\mathbb{C}})_s$ that contains W_s. By (a), H_1 is stable under S. If W_s is of dimension d, and H_1 of length k, then

$$\bigwedge^{kd} H_1 \approx \left(\bigwedge^{d} W_s\right)^{\otimes k}$$

as $\mathbb{C}[\pi_1(S,s)]$-modules. If χ is the character of $\pi_1(S,s)$ defined by $\wedge^d W$, then the character χ_1 defined by $\wedge^{kd} H_1$ is thus χ^k. Let $H_2 = H_1 + \overline{H}_1$. Since H_1 is stable under S, H_2 is defined by a real Hodge substructure of $(H_{\mathbb{R}})_s$. Every polarisation form on H thus induces on H_2 a non-degenerate bilinear form that is $\pi_1(S,s)$-invariant.

If $\dim(H_2) = e$, then the representation $(\wedge^e H_2)^{\otimes 2}$ of $\pi_1(S,s)$ is thus trivial. From this:

(a) For real H_1, χ^{2k} is trivial.

(b) For non-real H_1 (so $H_1 \neq \overline{H}_1$), $\chi^{2k} \cdot \overline{\chi}^{2k}$ is trivial.

In both cases, $|\chi| = 1$.

The representation of $\pi_1(S,s)$ on $(H_{\mathbb{C}})_s$ comes from extension of scalars of a representation defined over $\mathbb{Q}$ (namely $(H_{\mathbb{Q}})_s$). The conjugate representations of W thus appear in $(H_{\mathbb{C}})_s$; there are at most $N = \dim(H)$ of them, and for every automorphism σ of $\mathbb{C}$ we have

$$|\sigma(\chi)| = 1.$$

For $\gamma \in \pi_1(S,s)$, $\chi(\gamma)$ is an algebraic integer with at most N complex conjugates, and these are all of absolute value 1; thus $\chi(\gamma)$ is a k-th root of unity, with $k \leqslant N$. Thus $\chi^{N!} = 1$, which proves (b). □

Corollary 4.2.9. *Let S be a smooth, connected, separated scheme, endowed with a base point $s \in S$. Let $f\colon X \to S$ be a morphism of schemes such that $\mathrm{R}^n f_*\mathbb{Q}$ is a local system on S. Let G be the Zariski closure of the image of $\pi_1(S,s)$ in $\mathrm{Aut}_{\mathbb{C}}((\mathrm{R}^n f_*\mathbb{C})_s)$, and G^0 the identity component of G. Then:*

(a) If f is smooth and proper, then G^0 is semi-simple.

(b) In general, G^0 does not admit any quotient of multiplicative type (i.e. the radical of G^0 is unipotent).

Proof. We first prove (a). By replacing S by a finite étale cover, if necessary, we can reduce to the case where $G = G^0$. By (iii) of Proposition 4.2.5, the local system $\mathrm{R}^n f_*\mathbb{Q}$ underlies an algebraic family H of Hodge structures, and so we can apply Corollary 4.2.8 (by (i) of Proposition 4.2.5). The group G is reductive, since it admits a faithful semi-simple representation (namely $(\mathrm{R}^n f_*\mathbb{C})_s$). If G (which we assume to be connected) were not semi-simple, then it would admit a non-trivial complex representation ρ of rank 1; since H_s is faithful, ρ would then be a direct factor of $((H+H^*)_{\mathbb{C}}^{\otimes m})_s$ for some m. But this would contradict (iii.b) of Corollary 4.2.8, since $(H+H^*)^{\otimes m}$ is algebraic, and no tensor power of ρ is trivial. □

Let π be a group, and consider the following property of a representation ρ of π in a complex vector space V of finite dimension:

Property *. $(*)$ *The identity component G^0 of the Zariski closure G of $\rho(\pi)$ in $\mathrm{Aut}(V)$ does not admit any quotient of multiplicative type.*

Lemma 4.2.10. *Let $0 \to V' \to V \to V'' \to 0$ be an exact sequence of representations of the above type. Then V satisfies Property * if (and only if) V' and V'' satisfy Property *.*

Proof. Let G, G', and G'' be the groups defined by V, V', and V'', respectively. Then V' is stable under G, just as it is under π, whence a morphism

$$u = (u', u'')\colon G \to G' \times G''$$

with unipotent kernel, and such that u' and u'' are surjective. Let N be the identity component of the kernel of u''. Since u' is surjective, $u'(N)$ is distinguished in G', and thus does not admit any quotient of multiplicative type.

If χ is a character of G^0, then χ factors through $u(G^0)$ and vanishes on N; then a power of χ factors through ${G''}^0$, and so χ is trivial. □

Proof. We now prove part (b) of Corollary 4.2.9 by dévissage, and by induction on the relative dimension of f. We can suppose X to be reduced.

Suppose first of all that X is quasi-projective. Then there exists a non-empty open U of S, a dense open X' of $X_U = f^{-1}(U)$ that is smooth over U, and a compactification $\overline{X}'$ of X' that is smooth and proper over U. We can choose $\overline{X}'$ to be such that some open X'' of $\overline{X}'$ contains X' and is proper over X_U.

$$
\begin{tikzcd}
X_U \arrow[ddr, "f_U"'] & & X'' \arrow[ll, "p"'] \arrow[r, hook, "k"] \arrow[ddl, "f''"] & \overline{X}' \arrow[ddll, "\overline{f}'"] \\
& X' \arrow[ul, hook', "i"] \arrow[ur, hook, "j"'] \arrow[d, "f'"] & & \\
& U & &
\end{tikzcd}
$$

We have long exact sequences of sheaves

$$
\begin{gathered}
\ldots \to \mathrm{R}^i f_{U*}(i_!\mathbb{Q}) \to \mathrm{R}^i f_{U*}\mathbb{Q} \to \mathrm{R}^i f_{U*}(\mathbb{Q}/i_!\mathbb{Q}) \to \ldots \\
\ldots \to \mathrm{R}^i f''_{U*}(j_!\mathbb{Q}) \to \mathrm{R}^i f''_{U*}\mathbb{Q} \to \mathrm{R}^i f''_{U*}(\mathbb{Q}/j_!\mathbb{Q}) \to \ldots \\
\ldots \to \mathrm{R}^i f''_!\mathbb{Q} \to \mathrm{R}^i \overline{f}'_*\mathbb{Q} \to \mathrm{R}^i \overline{f}'_*(\mathbb{Q}/k_!\mathbb{Q}) \to \ldots
\end{gathered}
$$

For small enough U, these sheaves are local systems, and the induction hypothesis applies to $\mathrm{R}^\bullet f_{U*}(\mathbb{Q}/i_!\mathbb{Q})$, to $\mathrm{R}^\bullet f''_*(\mathbb{Q}/j_!\mathbb{Q})$, and to $\mathrm{R}^\bullet \overline{f}'_*(\mathbb{Q}/k_!\mathbb{Q})$. By (a) of Corollary 4.2.9, Property * is satisfied by $\mathrm{R}^\bullet \overline{f}'_*\mathbb{Q}$. By Lemma 4.2.10, Property * is then also satisfied by $\mathrm{R}^\bullet f''_!\mathbb{Q}$, and by the dual local system $\mathrm{R}^\bullet \overline{f}'_*\mathbb{Q}$ (Poincaré duality). By Lemma 4.2.10, Property * is satisfied by $\mathrm{R}^\bullet f''_*(j_!\mathbb{Q})$, which is isomorphic to $\mathrm{R}^\bullet f_{U*}(i_!\mathbb{Q})$, since p is proper, as well as by $\mathrm{R}^\bullet f_{U*}\mathbb{Q}$. Since the morphism $\pi_1(U) \to \pi_1(S)$ is surjective, $\mathrm{R}^i f_*\mathbb{Q}$ satisfies (b) of Corollary 4.2.9.

To pass from the case where f is quasi-projective to the general case, it suffices to use the Leray spectral sequence for a cover (U_i) of X by quasi-projective opens:

$$
\mathrm{R}^\bullet f_*\left(\bigcap_i U_i, \mathbb{Q}\right) \Rightarrow \mathrm{R}^\bullet f_*\mathbb{Q}
$$

along with Lemma 4.2.10 □

4.3. Supplement to [D1968]

In the footnote to (5.4) in [D1968], I claim without proof that:

Proposition 4.3.1. *Let X be a smooth proper scheme over $\mathbb{C}$. If a C^∞ form α on X satisfies $\mathrm{d}'\alpha = \mathrm{d}''\alpha = 0$ and is cohomologous to zero, then there exists β such that $\alpha = \mathrm{d}'\mathrm{d}''\beta$.*

Proof. Here is the proof. As in *loc. cit.*, we can reduce to the case where α is bihomogeneous, with bidegree (p,q). Since the spectral sequence of the double complex $\mathrm{H}^0(X, \Omega^{pq})$ degenerates, the exterior differential d is strictly compatible with the filtration by the first degree p (Proposition 1.3.2). If $a \sim 0$, then there thus exists a form β_1 such that $\mathrm{d}\beta_1 = \alpha$ and such that the components $\beta_1^{p'q'}$ of β_1 are zero for $p' < p$.

By complex conjugation, there similarly exists β_2 such that $\mathrm{d}\beta_2 = \alpha$ and such that the components $\beta_2^{p'q'}$ of β_2 are zero for $q' < q$. We have $\mathrm{d}(\beta_1 - \beta_2) = 0$.

Let γ be a form satisfying $\mathrm{d}'\gamma = \mathrm{d}''\gamma = 0$ in the cohomology class of $\beta_1 - \beta_2$ (*loc. cit.*). Let γ_1 (resp. γ_2) be the sum of the components of γ of degree (p', q') for $p' \geqslant p$ (resp. $q' \geqslant q$). If we set $\beta_1' = \beta_1 - \gamma_1$ and $\beta_2' = \beta_2 + \gamma_2$, then we still have that $\mathrm{d}\beta_1' = \mathrm{d}\beta_2' = \alpha$; furthermore, $\beta_1' - \beta_2' = \beta_1 - \beta_2 - \gamma$ is cohomologous to zero: there exists δ such that

$$\mathrm{d}\delta = \beta_1' - \beta_2'.$$

Let δ_1 (resp. δ_2) be the sum of the components of δ of degree (p', q') for $p' < p-1$ (resp. $q' > q-1$); let β be the component of δ of degree $(p-1, q-1)$.

We have that

$$\beta_2' = \mathrm{d}\delta_2 + \mathrm{d}''\beta$$

and

$$\begin{aligned}\alpha = \mathrm{d}\beta_2' &= \mathrm{d}\mathrm{d}\delta_2 + \mathrm{d}\mathrm{d}''\beta \\ &= \mathrm{d}'\mathrm{d}''\beta.\end{aligned}$$

□

4.4. Homomorphisms of abelian schemes

We intend to show how, in some cases, we can remove from §4.1 the assumption of algebraicity at a point.

Let $f\colon X \to S$ be a smooth proper morphism; we denote by $\mathrm{R}_i f_*\mathbb{Z}$ the local system on S^{an} whose homology is the fibres of f. At each point $s \in S$, $(\mathrm{R}_i f_*\mathbb{Z})_s \simeq \mathrm{H}_i(X_s, \mathbb{Z})$ is endowed with a Hodge structure of weight $-i$, dual to the Hodge structure on $\mathrm{H}^i(X_s, \mathbb{Z})$. This structure varies continuously with $s \in S$.

If $f\colon X \to S$ is an abelian scheme, then the exponential defines an exact sequence

$$0 \to \mathrm{R}_1 f_*\mathbb{Z} \to \mathrm{Lie}(X) \to X \to 0$$

of sheaves on S^{an}, and the Hodge filtration is the short exact sequence

$$0 \to \mathrm{Ker}(\alpha) \to \mathrm{R}_1 f_*\mathbb{Z} \otimes \mathcal{O} \xrightarrow{\alpha} \mathrm{Lie}(X) \to 0.$$

It is known that:

Reminder 4.4.3. Let S be a smooth scheme of finite type over $\mathbb{C}$. The construction of §4.4 establishes an equivalence of categories between

(a) the category of abelian schemes over S; and

(b) the category of polarisable continuous families of Hodge structures H of type $(-1,0)+(0,-1)$ on S such that $H_{\mathbb{Z}}$ is torsion free and the Hodge filtration varies holomorphically on S.

Proof. The essential surjectivity of the functor that goes from (a) to (b) follows from Borel (***unpublished***); the full faithfulness follows from the fact that, for two abelian schemes X_1 and X_2 over S, the S-scheme $\underline{\mathrm{Hom}}_S(X_1, X_2)$ is unramified over S, and thus has the same algebraic and holomorphic sections. □

Let Z be a commutative field, H a simple central algebra over Z, and $*$ a Z-linear anti-automorphism of H. After extension of scalars, we have $H \simeq \mathrm{End}(V)$ for some vector space V over Z, and $*$ is then the transposition with respect to a bilinear form Φ on V, unique up to a factor. Suppose that $*$ is involutive; the forms $\Phi(X,Y)$ and $\Phi(Y,X)$ are then proportional, i.e. $\Phi(X,Y) = \lambda\Phi(Y,X)$ for some $\lambda^2 = 1$. *We set* $\varepsilon_* = \lambda$. If $[H : Z] = d^2$, then

$$\mathrm{Tr}(*\colon H \to H) = \varepsilon_* \cdot d.$$

Every Z-linear automorphism C of H is interior: $Cx = cxc^{-1}$. If C is involutive and commutes with $*$, then the elements c^2 and cc^* are central, so that $c^* = \lambda c$ with λ central. Since $c^{**} = c$, we have that $\lambda^2 = 1$. *We set* $\varepsilon_C = \lambda$.

We easily see that the involution $C \circ *$ satisfies

$$\varepsilon_{C\circ *} = \varepsilon_C \cdot \varepsilon_*.$$

Reminder 4.4.5. Let X be a simple abelian variety over a field K, and let H be the field $\mathrm{End}_K(X)\otimes\mathbb{Q}$. Every polarisation of X defines a positive involution $*$ of H, i.e.

$$\mathrm{Tr}(hh^*) > 0 \quad \text{if } h \neq 0.$$

If Z_0 is the $*$-invariant subfield of the centre Z of H, then Z_0 is completely real and we are in one of the following cases:

(a) $Z = Z_0$ and, for every real place ρ of Z, $H \otimes_\rho \mathbb{R}$ is an algebra of matrices over $\mathbb{R}$, and $\varepsilon_* = 1$.

(b) $Z = Z_0$ and, for every real place ρ of Z, $H \otimes_\rho \mathbb{R}$ is an algebra of matrices over $\mathbb{H}$, and $\varepsilon_* = -1$.

(c) Z is a completely imaginary quadratic extension of Z_0.

Let $f_i\colon X_i \to S$ (for $i = 1,2$) be an abelian scheme over a smooth scheme S. The free abelian group

$$\mathrm{Hom}(\mathrm{R}_1 f_{1*}\mathbb{Z}, \mathrm{R}_1 f_{2*}\mathbb{Z}) \simeq \Gamma(S, \underline{\mathrm{Hom}}(\mathrm{R}_1 f_{1*}\mathbb{Z}, \mathrm{R}_1 f_{2*}\mathbb{Z}))$$

is then endowed with a Hodge structure of weight 0 (Corollary 4.2.8). Further, by Remark 2.1.11.1 and Reminder 4.4.3, the group $\mathrm{Hom}_S(X_1, X_2)$ is the degree $(0,0)$ component of this group.

Let S be a non-empty, connected, smooth scheme, with generic point η, and let $f\colon X \to S$ be an abelian scheme over S. The centre Z of $H = \mathrm{End}(\mathrm{R}_1 f_*\mathbb{Z})$ is then of degree $(0,0)$ (Corollary 4.2.8), and thus contained in $\mathrm{End}_S(X) \simeq \mathrm{End}_{k(\eta)}(X_\eta)$.

A polarisation of X defines an involution $*$ of H. The restriction of this to $\mathrm{End}_S(X)$ is positive, so that Z is the product of completely real fields or quadratic imaginary extensions of completely real fields.

Denote by C the actions of $i \in S(\mathbb{R})$ on $H_\mathbb{R} = H \otimes \mathbb{R}$ and on the $(\mathrm{R}_1 f_*\mathbb{R})_s$ for $s \in S$. On $H_\mathbb{R}$, $C^2 = 1$, and C commutes with $*$; on $(\mathrm{R}_1 f_*\mathbb{R})_s$, $C^2 = -1$. If ψ is the alternating form on $\mathrm{R}_1 f_*\mathbb{Z}$ with integer values induced by the polarisation of X, then

$$\psi(hx, Cy) = \psi(x, h^*Cy) = \psi(x, C \cdot C(h^*) \cdot y).$$

We thus deduce that

$$\mathrm{Tr}(h \cdot C(h^*)) > 0 \quad \text{for } h \neq 0. \tag{4.4.7.1}$$

Now suppose that X_η is simple, so that Z is a field. Let $\rho\colon Z \to \mathbb{C}$ be a complex embedding. Then the algebra

$$H_\rho = H \otimes_{Z,\rho} \mathbb{C}$$

is an algebra of matrices and a bigraded direct factor of $H_\mathbb{C} = H \otimes_\mathbb{Q} \mathbb{C}$. Pick an isomorphism

$$H_\rho \simeq \mathrm{End}(V_\rho). \tag{4.4.8.1}$$

From the fact that the infinitesimal generators of S act on H_ρ, by derivations, which are necessarily interior, we deduce that V_ρ admits a bigrading, unique up to translation, such that Equation 4.4.8.1 is a bigraded isomorphism. Let $(\mathrm{R}_1 f_*\mathbb{C})_\rho$ be the bigraded local system

$$(\mathrm{R}_1 f_*\mathbb{C}) \otimes_{Z\otimes\mathbb{C},\rho} \mathbb{C}$$

which is a direct factor of $\mathrm{R}_1 f_*\mathbb{C}$. Denote by T_ρ the local system

$$T_\rho = \mathrm{Hom}_{H_\rho}(V_\rho, (\mathrm{R}_1 f_*\mathbb{C})_\rho).$$

We have the bigraded isomorphism

$$V_\rho \otimes_\mathbb{C} T_\rho \xrightarrow{\sim} (\mathrm{R}_1 f_*\mathbb{C})_\rho. \tag{4.4.8.2}$$

Since $(\mathrm{R}_1 f_*\mathbb{C})_\rho$ is of degree $(-1,0)+(0,-1)$, one of the following conditions is satisfied:

(a) V_ρ is bihomogeneous;

(b) T_ρ is bihomogeneous.

It is clear that condition (a) (resp. (b)) is simultaneously satisfied by ρ and $\overline{\rho}$. If condition (a) is satisfied at each place, then H is of degree $(0,0)$. If condition (b) is satisfied at each place, then the Hodge structure on $\mathrm{R}_1 f_*\mathbb{Q}$ is locally constant.

Now suppose that Z is completely real. For each real embedding φ of Z, we set $\varepsilon_{H,\varphi} = +1$ if $H \otimes_{Z,\varphi} \mathbb{R}$ is an algebra of matrices; we set $\varepsilon_{H,\varphi} = -1$ otherwise. If a complex embedding ρ induces a real embedding φ, then we set $\varepsilon_{H\rho} = \varepsilon_{H\varphi}$.

Let $\rho\colon Z \to \mathbb{C}$ factor through $\varphi\colon Z \to \mathbb{R}$. Complex conjugation on H_ρ is induced by an antilinear automorphism σ_1 of V_ρ, unique up to real scalar, with scalar square. Since σ_1 commutes with σ_1^2, σ_1^2 is real and, by normalising σ_1, we can assume that $\sigma_1^2 = \pm 1$. We then have that

$$\sigma_1^2 = \varepsilon_{H\rho}. \tag{4.4.9.1}$$

The complex conjugation automorphism on

$$(\mathrm{R}_1 f_*\mathbb{C})_\rho = ((\mathrm{R}_1 f_*\mathbb{Q}) \otimes_{Z,\varphi} \mathbb{R}) \otimes_\mathbb{R} \mathbb{C}$$

is "compatible" with Equation 4.4.8.2 and can be written as

$$\sigma = \sigma_1 \otimes \sigma_2.$$

Since $\sigma^2 = 1$, we have, by Equation 4.4.9.1, that

$$\sigma_1^2 = \sigma_2^2 = \varepsilon_{H\rho}. \tag{4.4.9.2}$$

Let Φ be a polarisation form on $\mathrm{R}_1 f_* \mathbb{Z}$. This form is necessarily of the form

$$\Phi = \mathrm{Tr}_{Z/\mathbb{Q}}(\psi)$$

where ψ is an alternating Z-bilinear form on $\mathrm{R}_1 f_* \mathbb{Q}$.

Let ψ_ρ be the form induced on $V_\rho \otimes_{\mathbb{C}} T_\rho$. The alternating form ψ_ρ is invariant under $\pi_1(S, s)$. Since T_ρ is irreducible, we can write ψ_ρ as

$$\psi_\rho = A_\rho \otimes B_\rho \tag{4.4.9.3}$$

with A_ρ on V_ρ, and B_ρ on T_ρ invariant under $\pi_1(S, s)$.

Since T_ρ is irreducible, B_ρ is either symmetric or alternating. Since ψ_ρ is alternating, A_ρ is either symmetric or alternating. Since the T_ρ are conjugate to one another, whether we are in one case or the other does not depend on ρ. We set

$$A_\rho(X, Y) = \varepsilon_V A_\rho(Y, X) \tag{4.4.9.4}$$

$$B_\rho(X, Y) = \varepsilon_T B_\rho(Y, X). \tag{4.4.9.5}$$

We thus have

$$\varepsilon_V \varepsilon_T = -1 \tag{4.4.9.6}$$

and it is clear by §4.4 that if $*$ is the involution of H induced by ψ, then

$$\varepsilon_* = \varepsilon_V. \tag{4.4.9.7}$$

In case (a) (resp. (b)) of §4.4, we normalise the bigradings so that V_ρ (resp. T_ρ) is of bidegree $(0, 0)$. We then have, for non-zero $x \in V_\rho$ and non-zero $y \in T_\rho$, the two cases:

(a) $A_\rho(x, \sigma_1 x) B_\rho(y, C\sigma_2 y) > 0$;

(b) $A_\rho(x, C\sigma_1 x) B_\rho(y, \sigma_2 y) > 0$.

By normalising A_ρ and B_ρ (of which only their tensor product is given), we can suppose, in each respective case, that

(a) $A_\rho(x, \sigma_1 x) > 0$ and $B_\rho(y, C\sigma_2 y) > 0$;

(b) $A_\rho(x, C\sigma_1 x) > 0$ and $B_\rho(y, \sigma_2 y)$.

Furthermore, A_ρ and B_ρ, just like Φ, are invariant under the compact sub-torus $\mathrm{Ker}(N)$ of S, and in particular under C. In case (a), we have that

$$0 < A_\rho(\sigma_1 x, \sigma_1 \sigma_1 x) = \varepsilon_V \cdot \varepsilon_{H,\rho} A_\rho(x, \sigma_1 x)$$

whence $\varepsilon_V \cdot \varepsilon_{H\rho} = 1$. Similarly, in case (b), we see that $\varepsilon_T \cdot \varepsilon_{H\rho} = 1$.

Lemma 4.4.10. *For $\varepsilon_T = \varepsilon_{H\rho}$, we are in case (b); for $\varepsilon_T = -\varepsilon_{H\rho}$, we are in case (a).*

Proposition 4.4.11. *Let S be a non-empty, connected, smooth scheme, with generic point η, and let $f \colon X \to S$ be an abelian scheme over S. Let $H = \mathrm{End}(\mathbb{R}_1 f_* \mathbb{Q})$, and let Z be the centre of H. Suppose that* *(a)* X_η *is simple over* $k(\eta)$.

(b) X *does not become constant on any finite covering of* S.

(c) One of the following conditions is satisfied (cf. Reminder 4.4.5):

> *(1)* Z *is imaginary* quadratic*;*
>
> *(2)* Z *is completely real, and for each real embedding* $\varphi\colon Z \to \mathbb{R}$, $H \otimes_{Z,\varphi} \mathbb{R}$ *is an algebra of matrices over* $\mathbb{R}$*;*
>
> *(3)* Z *is completely real, and for each real embedding* $\varphi\colon Z \to \mathbb{R}$, $H \otimes_{Z,\varphi} \mathbb{R}$ *is an algebra of matrices over* $\mathbb{H}$.

Then

$$\mathrm{End}(X) \xrightarrow{\sim} \mathrm{End}(\mathrm{R}_1 f_* \mathbb{Z}).$$

Proof. We will show that we are either in case (a) or case (b) of §4.4, for every place of Z simultaneously. This is clear (§4.4) under the hypothesis (c.1), and follows from Lemma 4.4.10 under the hypotheses (c.2) or (c.3), since not only ε_T, but also $\varepsilon_{H\rho}$, is then independent of ρ.

Condition (b) of §4.4 cannot be satisfied at every place of Z, since the Hodge structure would then be locally constant, which (by §4.1 and Reminder 4.4.3) contradicts (b). Condition (a) is thus satisfied at each place of Z, and so H is of degree $(0,0)$ (§4.4). Taking Reminder 4.4.3 into account, this finishes the proof. ☐

Proposition 4.4.12. *Let* $f\colon X \to S$ *be an abelian scheme over a smooth scheme* S. *Then the following conditions are equivalent:*

(i) For every abelian scheme $Y \to S$*, we have*

$$\mathrm{Hom}_S(X,Y) \xrightarrow{\sim} \mathrm{Hom}_S(\mathrm{R}_1 f_*\mathbb{Z}, \mathrm{R}_1 g_*\mathbb{Z}).$$

(ii) Condition (i) is satisfied for $X = Y$*, and the centre* Z *of* $\mathrm{End}_S(X) \otimes \mathbb{Q}$ *does not admit any complex place* $\rho\colon Z \to \mathbb{C}$ *such that the direct factor* $\mathrm{R}_1 f_*\mathbb{Q} \otimes_{Z,\rho} \mathbb{C}$ *of* $\mathrm{R}_1 f_*\mathbb{C}$ *is of pure Hodge degree* $(-1,0)$.

Proof. We can reduce to the case where S is non-empty and connected, with generic point η, and where X_η is a simple abelian variety over $k(\eta)$. If (i) or (ii) is satisfied, then $D = \mathrm{End}(X_\eta) \otimes \mathbb{Q}$ is a field, and, for $s \in S$, $(\mathrm{R}_1 f_*\mathbb{Q})_s$ is an irreducible representation of $\pi_1(S,s)$.

To prove that (ii) $\implies$ (i), we can suppose Y_η to be simple, and $(\mathrm{R}_1 g_*\mathbb{Q})_s$ to be isotypic of type $(\mathrm{R}_1 f_*\mathbb{Q})_s$. We then have an isomorphism of continuous families of Hodge $\mathbb{Q}$-structures (with D of degree $(0,0)$)

$$\mathrm{R}_1 g_*\mathbb{Q} \approx \mathrm{R}_1 f_*\mathbb{Q} \otimes_D \mathrm{Hom}(\mathrm{R}_1 f_*\mathbb{Q}, \mathrm{R}_1 g_*\mathbb{Q}).$$

We have that $D \otimes \mathbb{C} \simeq \mathbb{M}_n(Z \otimes \mathbb{C})$. If $e \in D \otimes \mathbb{C}$ is sent by such an isomorphism to the idempotent e_{11}, then we again have an isomorphism of bigraded vector spaces

$$(\mathrm{R}_1 g_*\mathbb{C})_s \approx (\mathrm{R}_1 f_*\mathbb{C})_s \otimes_{Z\otimes\mathbb{C}} e(\mathrm{Hom}(\mathrm{R}_1 f_*\mathbb{C}, \mathrm{R}_1 g_*\mathbb{C})).$$

If condition (ii) is satisfied, and if $e(\mathrm{Hom}(\mathrm{R}_1 f_*\mathbb{C}, \mathrm{R}_1 g_*\mathbb{C}))$ is not of pure degree $(0,0)$, then $(\mathrm{R}_1 g_*\mathbb{C})_s$ cannot be of pure degree $(-1,0)+(0,-1)$, which is a contradiction. We thus deduce that $\operatorname{Hom}(\mathrm{R}_1 f_*\mathbb{Z}, \mathrm{R}_1 g_*\mathbb{Z})$ is of pure degree $(0,0)$, and (i) then follows by Reminder 4.4.3 and Remark 2.1.11.1.

Let Z be a completely imaginary quadratic extension of a completely real number field Z^0. If $Z \otimes \mathbb{R} \approx \mathbb{C} \oplus \ldots \oplus \mathbb{C}$, then we define an action by homotheties of S on $Z \otimes \mathbb{R}$ by sending $s \in S(\mathbb{R}) \simeq \mathbb{C}^*$ to $(s^2 \cdot N(s)^{-1}, 1, \ldots, 1)$. The pair consisting of Z and this action of S on $Z \otimes \mathbb{R}$ is a polarisable Hodge structure Z_ρ of degree $(-1,1)+(0,0)+(1,-1)$ that depends on Z and the chose place $\rho\colon Z \to \mathbb{C}$.

Let X be an abelian scheme over S such that X_η is simple. If the centre Z of $\operatorname{End}(\mathrm{R}_1 f_*\mathbb{Q})$ is not completely real, then it is a completely imaginary quadratic extension of a completely real number field. If the second hypothesis of (ii) is not satisfied, then the direct factor $\mathrm{R}_1 f_*\mathbb{Q} \otimes_Z Z_\rho$ of the continuous family of polarisable Hodge structures $\mathrm{R}_1 f_*\mathbb{Q} \otimes_{\mathbb{Q}} Z_\rho$ is of pure degree $(-1,0)+(0,-1)$.

By Reminder 4.4.3, the choice of an integer lattice in $\mathrm{R}_1 f_*\mathbb{Q} \otimes_Z Z_\rho$ defines an abelian scheme $g\colon Y \to S$, and $\operatorname{Hom}(\mathrm{R}_1 f_*\mathbb{Q}, \mathrm{R}_1 g_*\mathbb{Q})$ is not of pure degree $(0,0)$, which contradicts (i). □

Corollary 4.4.13. *Let S be a smooth connected scheme with generic point η, and let $f\colon X \to S$ be an abelian scheme over S of relative dimension $g \leqslant 3$. Suppose that, on any finite étale covering of S, X does not admit a constant abelian subscheme. Then, for every abelian scheme $g\colon Y \to S$, we have*

$$\operatorname{Hom}(X,Y) \xrightarrow{\sim} \operatorname{Hom}(\mathrm{R}_1 f_*\mathbb{Z}, \mathrm{R}_1 g_*\mathbb{Z})$$
$$\operatorname{Hom}(Y,X) \xrightarrow{\sim} \operatorname{Hom}(\mathrm{R}_1 g_*\mathbb{Z}, \mathrm{R}_1 f_*\mathbb{Z}).$$

Proof. By duality, it suffices to prove the first claim. We can also suppose X_η to be simple. We will show that X satisfies one of the conditions of (c) in Proposition 4.4.11.

Set $H = \mathbb{M}_c(D)$, with D a skew field of rank b^2 over its centre Z, which we know to be either completely real or completely imaginary. If we set $a = [Z : \mathbb{Q}]$, then we have

$$ab^2c \mid 2g \leqslant 6. \tag{4.4.13.1}$$

(i) If Z is completely imaginary, and not imaginary quadratic, then we must have $a = 2g = 4$. For $s \in S$, the centraliser of Z acting on $(\mathrm{R}_1 f_*\mathbb{Q})_s$ is thus commutative, and so the action of $\pi_1(S,s)$ is abelian, and thus factors through a finite group ((iii.b) of Corollary 4.2.8 or Corollary 4.2.9). This is a contradiction (§4.1 and Reminder 4.4.3).

   With the notation of §4.4, we can also note that the T_ρ are then of rank 1, and so condition (b) of §4.4 is satisfied at each place, and the Hodge structure is locally constant.

(ii) If Z is completely real, and if conditions (c.1) and (c.2) are not satisfied, then $a \geqslant 2$ and $b \geqslant 2$, which is a contradiction.

We will show that X satisfies the conditions of Proposition 4.4.12. If Z is imaginary quadratic, and if there exists $\rho\colon Z \to \mathbb{C}$ such that $\mathrm{R}_1 f_*\mathbb{Q} \otimes_{Z,\rho} \mathbb{C}$ is of pure Hodge type $(-1,0)$, then the Hodge structure on $\mathrm{R}_1 f_*\mathbb{Q}$ is locally constant, which is a contradiction. This proves Corollary 4.4.13. □

Let $f\colon X \to S$ be a polarised abelian scheme of dimension g over a smooth connected base S. For $s \in S$, the fundamental group $\pi_1(S,s)$ acts on the fibre $\mathrm{H}^1(X_s,\mathbb{Z})$ of $\mathrm{R}_1 f_*\mathbb{Z}$ over s. The polarisation of X defines an alternating form on $\mathrm{H}^1(X_s,\mathbb{Z})$, with values in $\mathbb{Z}(-1)$, that is non-degenerate over $\mathbb{Q}$, and the action of $\pi_1(S,s)$ respects this form. Consider the hypothesis:

Hypothesis 4.4.14.1. The morphism defined by X from S to one of the irreducible components of the corresponding scheme of modules of polarised abelian varieties is a dominant morphism.

The following result was suggested to me by J.-P. Serre.

Corollary 4.4.15. *Let S be a smooth connected scheme, and $f\colon X \to S$ a polarised abelian scheme over S that satisfies Hypothesis 4.4.14.1. Then, for every abelian scheme $g\colon Y \to S$, we have*

$$\mathrm{Hom}(X, Y) \xrightarrow{\sim} \mathrm{Hom}(\mathrm{R}_1 f_* \mathbb{Z}, \mathrm{R}_1 g_* \mathbb{Z}).$$

Proof. Let $s \in S$. By Proposition 4.4.11 and Proposition 4.4.12, it suffices to show that the representation of $\pi_1(S, s)$ on $(\mathrm{R}_1 f_* \mathbb{Q})_s$ is absolutely irreducible, so that $H = \mathbb{Q}$. This follows from the following lemma. □

Lemma 4.4.16. *If X satisfies Hypothesis 4.4.14.1, then the image of $\pi_1(S, s)$ in the group of symplectic automorphisms of $\mathrm{H}^1(X_s, \mathbb{Z})$ is of finite index.*

Proof. Let n be an integer. Replacing S by a surjective étale covering, defined by a finite-index subgroup of $\pi_1(S, s)$, if necessary, we can assume that the local system ${}_nX = \mathrm{Ker}(n \cdot \mathrm{Id}\colon X \to X)$ is trivial on S. Replacing X by an isogenous abelian scheme, we can then assume that X is polarised in the principal series. Suppose that $n \geqslant 3$.

The abelian scheme X then defines a dominant morphism x from the S to the scheme of "Jacobi" modules M_n of polarised abelian schemes in the principal series endowed with a symplectic isomorphism between ${}_nX$ and $(\mathbb{Z}/n)^{2g}$; furthermore, X is an inverse image under x of the universal abelian scheme on M_n.

We know that $\pi_1(M_n, x(s))$ is sent isomorphically to the subgroup of $\mathrm{Sp}(\mathrm{H}^1(X_s, \mathbb{Z}))$ given by the kernel of the reduction $\bmod n$ map. It thus remains to show that: □

Lemma 4.4.17. *Let $p\colon S \to M$ be a dominant morphism of smooth connected schemes. The subgroup $p(\pi_1(S, s))$ of $\pi_1(M, p(s))$ is then of finite index.*

Proof. Let t be a closed point of the generic fibre of p, and let T' be its closure in S. There exists a dense Zariski open U of M such that $T = p^{-1}(U) \cap T'$ is an étale covering of U.

We lose no generality by assuming that $s \in T$. In the diagram

$$\begin{tikzcd} \pi_1(T, s) \arrow[r] \arrow[d, "a"'] & \pi_1(S, s) \arrow[d, "c"] \\ \pi_1(U, p(s)) \arrow[r, "b"'] & \pi_1(M, p(s)) \end{tikzcd}$$

the image of the arrow a is of finite index, since T is a finite étale covering, and the arrow b is surjective, since $M \setminus U$ is of real codimension $\geqslant 2$. The image of the arrow c is thus of finite index. □

Hodge Theory III

P. Deligne. "Théorie de Hodge III". *Pub. Math. de l'IHÉS* **44** (1974) pp. 5–77. publications.ias.edu/node/361

5. Introduction

This article follows Hodge Theory I and Hodge Theory II; the numbering of the sections here follows that of Volume II (which contains §1 to §4).

In Volume II, we gave an introduction to the yoga that underlies Volume II and the present article Volume III. However, Volume II and Volume III are logically independent of Volume I; they also do not contain all the results that were stated in Volume I.

In Volume II, we introduced the Hodge theory of non-singular (not necessarily complete) algebraic varieties. Here we treat the case of arbitrary singularities. From §9 onwards, we will make essential use of results from §1 to §3.

In the case of a *complete* singular algebraic variety X, the fundamental idea is to use the resolution of singularities to replace X by a simplicial system of smooth projective schemes

$$X_\bullet = \left(\ldots \quad X_2 \;\substack{\longrightarrow\\ \longleftarrow\\ \longrightarrow\\ \longleftarrow\\ \longrightarrow}\; X_1 \;\substack{\longrightarrow\\ \longleftarrow\\ \longrightarrow}\; X_0\right)$$

(or, as we will say, by a smooth projective simplicial scheme). The results of §7 and §8 allow us to find such simplicial schemes $X_\bullet$ that have, in a suitable sense, the same cohomology as X. We can then "express", via a spectral sequence, the cohomology of X in terms of the cohomology of the X_n. Following the general principles of Volume I, we can then, on each X_n, use classical Hodge theory, and obtain an induced mixed Hodge structure on the cohomology of X.

In the case of an arbitrary algebraic variety X, we "replace" X by a smooth simplicial scheme $X_\bullet$ with compactification given by a smooth projective simplicial scheme $\overline{X}_\bullet$. We can further arrange things so that $\overline{X}_n \setminus X_n$ is a normal crossing divisor, given by a union of smooth divisors $D_{n,i}$. We thus "express", via a spectral sequence, the cohomology of X in terms of the cohomology of the p-fold intersections of the $D_{n,i}$ (for $n,p \geqslant 0$), and we obtain an induced mixed Hodge structure on $H^\bullet(X)$.

§7 and §8 constitute a summary (without proofs) of the theory of "cohomological descent". This theory, in the more general framework of a topos, is laid out in [SD].

In §9.1, we restate certain results obtained previously, but in the language of filtered derived categories; §9.2 contains a simpler proof of the two-filtrations lemma than that of §1.3.

The unappealing §10.1 is the heart of this work. The essential result is Theorem 8.1.15.

In its proof, to be able to apply the two-filtrations lemma, we use the fact that every morphism of mixed Hodge structures is *strictly* compatible with the filtrations W and W (Theorem 2.3.5). In §10.2, we define the mixed Hodge structure of $\mathrm{H}^n(X,\mathbb{Z})$ (for X a separated scheme). We show that the Hodge numbers h^{pq} of $\mathrm{H}^n(X,\mathbb{Z})$ can only be non-zero for $(p,q) \in [0,n]\times[0,n]$. For $n > \dim X$, we have a more precise result (Theorem 8.2.4). If X is smooth (resp. complete), they can only be non-zero if, further, $p+q \geqslant n$ (resp. $p+q \leqslant n$). If X is a complete subscheme of Z, with complement U, with Z complete and smooth of dimension n, then we can, by N. Katz, interpret this "duality" between the complete case and the smooth case as coming from the "Alexander duality"

$$\to \ldots \to \mathrm{H}^i(Z,\mathbb{Q}) \to \mathrm{H}^i(X,\mathbb{Q}) \to (\mathrm{H}^{2n-i-1}(U,\mathbb{Q}))^* \to \mathrm{H}^{i+1}(Z,\mathbb{Q}) \to \ldots.$$

From the functorial properties of the theory thus constructed, we obtain the useful corollaries Proposition 8.2.5 to Corollary 8.2.8, which we state in terms independent of any Hodge theory.

In §10.3, we endow the cohomology of any simplicial scheme $X_\bullet$ with a mixed Hodge structure. This generality is not illusory.

(a) We can interpret relative cohomology spaces as being the cohomology of suitable simplicial schemes.

(b) Let B_G be the classifying space of the underlying Lie group of an algebraic group G. We can interpret the cohomology of B_G as being the cohomology of a suitable simplicial scheme.

In §11.1, after having calculated the mixed Hodge structure on the cohomology of B_G (for G linear), we thus obtain that of G. As a corollary, we find that, if a linear group G acts on a non-singular complete variety X, then the map

$$\mathrm{H}^\bullet(X,\mathbb{Q}) \to \mathrm{H}^\bullet(X,\mathbb{Q}) \otimes \mathrm{H}^\bullet(G,\mathbb{Q})$$

factors through $\mathrm{H}^\bullet(X,\mathbb{Q}) \otimes \mathrm{H}^0(G,\mathbb{Q}) \subset \mathrm{H}^\bullet(X,\mathbb{Q}) \otimes \mathrm{H}^\bullet(G,\mathbb{Q})$.

In §12, we interpret in terms of abelian schemes the mixed Hodge structures of pure degree $\{(-1,-1),(-1,0),(0,-1),(0,0)\}$ (see §6), and we consider in detail the H^1 of curves.

The theory developed up until here is an absolute theory (we do not study the functors $\mathrm{R}f_*$), and deals only with constant coefficients. I conjecture that, if $\mathscr{H}$ is a variation of polarisable Hodge structures (in the sense of Griffiths [G1970a]) on a scheme X, then the cohomology of X with coefficients in the local system $\mathscr{H}$ is endowed with a natural mixed Hodge structure. I can only prove this when X is complete.

6. Terminology and notation

Let $u\colon X \to Y$ be a continuous map between topological spaces. We say that u is ***proper*** if it is proper in the sense of Bourbaki (i.e. universally closed) and furthermore separated (i.e. the diagonal of $X \times_Y X$ is closed).

Following Gabriel and Zisman, we say ***simplicial*** where we would have previously said semi-simplicial.

We denote by A a Noetherian subring of $\mathbb{R}$ such that $A \otimes \mathbb{Q}$ is a field. The useful cases are $A = \mathbb{Z}$, $\mathbb{Q}$, or $\mathbb{R}$.

A ***mixed Hodge A-structure*** consists of an A-module H_A of finite type, a finite increasing filtration W on the $A\otimes\mathbb{Q}$-vector space $H_{A\otimes\mathbb{Q}} = H_A \otimes \mathbb{Q}$, and a finite decreasing filtration F on the $\mathbb{C}$-vector space $H_\mathbb{C} = H_A \otimes_A \mathbb{C}$. We demand that the $(\mathrm{Gr}_n^W(H_{A\otimes\mathbb{Q}}), \mathrm{Gr}_n^W(F))$ be Hodge $A\otimes\mathbb{Q}$-structures. For $A = \mathbb{Z}$ (resp. $A = \mathbb{Q}$), we recover Definition 2.3.1 (resp. §2.3); the results of §2.3 carry over as they are.

Let $\mathscr{E}$ be a subset of $\mathbb{Z}\times\mathbb{Z}$. We say that a mixed Hodge structure H is of ***degree*** $\mathscr{E}$ if the Hodge numbers h^{pq} are zero for $(p,q) \notin \mathscr{E}$.

From now on, we denote by $\Omega_X^\bullet(\log D)$ what we previously denoted by $\Omega_X^\bullet\langle D\rangle$ (§3.1).

Unless explicitly stated otherwise, ***scheme*** means "scheme of finite type over $\mathbb{C}$", and a ***sheaf*** on a scheme X is a sheaf on the underlying topological space of X_{an}.

7. Cohomological descent

7.1. Simplicial topological spaces

This section starts with some reminders for which we can refer to the homotopical seminar of Strasbourg, 1963/64.

We will us the following notation, where $n, k \geqslant -1$ are integers.

- $\mathscr{A}^\circ$ = the opposite category of a category $\mathscr{A}$
- $\underline{\mathrm{Hom}}(\mathscr{A}, \mathscr{B})$ = the category of functors from $\mathscr{A}$ to $\mathscr{B}$.
- Δ_n = the finite totally ordered set $[0, n]$.
- $\delta_i \colon \Delta_n \to \Delta_{n+1}$ = the increasing injection such that $i \notin \delta_i(\Delta_n)$ (for $0 \leqslant i \leqslant n+1$).
- $s_i \colon \Delta_{n+1} \to \Delta_n$ = the increasing surjection such that $s_i(i) = s_i(i+1)$ (for $0 \leqslant i \leqslant n$).
- $\varepsilon \colon \Delta_{-1} \to \Delta_n$ = the unique map from Δ_{-1} to Δ_n.
- (Δ^+) = the category whose objects are the Δ_n (for $n \geqslant -1$) and whose morphisms are the increasing maps between the Δ_n.
- (Δ) = the full subcategory of (Δ^+) whose objects are the Δ_n (for $n \geqslant 0$).
- $(\Delta^+)_k$ = the full subcategory of (Δ^+) whose objects are the Δ_n (for $k \geqslant n$).
- $(\Delta)_k$ = the full subcategory of (Δ^+) whose objects are the Δ_n (for $k \geqslant n \geqslant 0$).

For any category $\mathscr{C}$, we define a ***simplicial object*** (resp. ***k-truncated simplicial object***) of $\mathscr{C}$ to be an object of $\underline{\mathrm{Hom}}((\Delta)^0, \mathscr{C})$ (resp. of $\underline{\mathrm{Hom}}((\Delta)_k^0, \mathscr{C})$). Similarly, a ***cosimplicial object*** (resp. ***k-truncated cosimplicial object***) is an object of $\underline{\mathrm{Hom}}((\Delta), \mathscr{C})$ (resp. of $\underline{\mathrm{Hom}}((\Delta)_k, \mathscr{C})$).

For $k \geqslant -1$, the ***k-skeleton*** functor is the restriction functor

$$\mathrm{sq}_k \colon \underline{\mathrm{Hom}}((\Delta)^0, \mathscr{C}) \to \underline{\mathrm{Hom}}((\Delta)_k^0, \mathscr{C})$$

and the ***k-coskeleton*** functor

$$\mathrm{cosq}_k \colon \underline{\mathrm{Hom}}((\Delta)_k^0, \mathscr{C}) \to \underline{\mathrm{Hom}}((\Delta)^0, \mathscr{C}).$$

is the right adjoint to sq_k. Let Y be a simplicial object of $\mathscr{C}$. We also call the functor

$$\mathrm{sq}_k \colon \underline{\mathrm{Hom}}((\Delta)^0, \mathscr{C})/Y \to \underline{\mathrm{Hom}}((\Delta)_k^0, \mathscr{C})/\,\mathrm{sq}_k(Y)$$

the ***skeleton***, and its right adjoint

$$\mathrm{cosq}_k^Y \colon \underline{\mathrm{Hom}}((\Delta)_k^0, \mathscr{C})/\,\mathrm{sq}_k(Y) \to \underline{\mathrm{Hom}}((\Delta)^0, \mathscr{C})/Y$$

is called the ***coskeleton with respect to Y***.

The coskeleton functors exist if finite projective limits exist in $\mathscr{C}$; the relative coskeleton functors exist if fibred products do; we have

$$\mathrm{cosq}_k^Y(X) = \mathrm{cosq}_k(X) \times_{\mathrm{cosq}_k \mathrm{sq}_k(Y)} Y.$$

If $X_\bullet \colon (\Delta)^0 \to \mathscr{C}$ is a simplicial object of $\mathscr{C}$, then we set

- $X_n = X_\bullet(\Delta_n)$
- $\delta_i = X_\bullet(\delta_i \colon \Delta_n \to \Delta_{n+1}) \colon X_{n+1} \to X_n$
- $s_i = X_\bullet(s_i \colon \Delta_{n+1} \to \Delta_n) \colon X_n \to X_{n+1}$.

$$X_\bullet = \Big(\dots \quad X_2 \;\begin{smallmatrix} -\,\delta_0 \to \\ \leftarrow s_0 \,- \\ -\,\delta_1 \to \\ \leftarrow s_1 \,- \\ -\,\delta_2 \to \end{smallmatrix}\; X_1 \;\begin{smallmatrix} -\,\delta_0 \to \\ \leftarrow s_0 \,- \\ -\,\delta_1 \to \end{smallmatrix}\; X_0 \Big)$$

Let $S \in \mathrm{Ob}\,\mathscr{C}$. The ***constant simplicial object*** $S_\bullet$ is the simplicial object for which $S_n = S$ and $\delta_i = s_i = \mathrm{Id}_S$. A simplicial (resp. k-truncated simplicial) object of $\mathscr{C}$ ***augmented over*** S is a morphism $a \colon X_\bullet \to S_\bullet$ (resp. $a \colon X_\bullet \to \mathrm{sq}_k(S_\bullet)$). We identify such a simplicial object (resp. k-truncated simplicial object) augmented over S with the object $X_\bullet^+$ of $\underline{\mathrm{Hom}}((\Delta^+)^0, \mathscr{C})$ (resp. of $\underline{\mathrm{Hom}}((\Delta^+)^0_k, \mathscr{C})$) such that $X_\bullet^+(\Delta_{-1}) = S$. We have that $a_n = X_\bullet^+(\varepsilon \colon \Delta_{-1} \to \Delta_n)$. These objects will be denoted by notation of the form $a \colon X_\bullet \to S$. The relative coskeletons will be mostly used in this setting, and denoted cosq_k^S or simply cosq_k.

The coskeleton of the augmented 0-truncated simplicial object $a_0 \colon X \to S$ is the simplicial object augmented over S whose components are the cartesian powers of X in $\mathscr{C}/S$, i.e.

$$\mathrm{cosq}_0(X \to S) = \big(((X/S)^{\Delta_n})_{n \geqslant 0} \to S\big).$$

Let $u \colon X \to Y$ be a continuous map between topological spaces, and F a sheaf on X and G a sheaf on Y. The set $\mathrm{Hom}_u(G, F)$ of ***u-morphisms*** from G to F is the set $\mathrm{Hom}(u^*G, F) \cong \mathrm{Hom}(G, u_*F)$.

The data of a u-morphism f from G to F consists of the data of, for each pair of opens $(U \subset X, V \subset Y)$ such that $u(U) \subset V$, a map $f_{UV} \colon G(V) \to F(U)$; these maps must satisfy the condition

Condition *. For $U' \subset U \subset X$ and $V' \subset V \subset Y$ such that $u(U) \subset V$ and $u(U') \subset V'$, the diagram

$$\begin{tikzcd} G(V) \arrow[r] \arrow[d] & G(V') \arrow[d] \\ F(U) \arrow[r] & F(U') \end{tikzcd}$$

commutes.

A ***simplicial topological space*** is a simplicial object of the category whose objects are topological spaces and whose morphisms are continuous maps.

A ***sheaf*** $F^\bullet$ on a simplicial topological space $X_\bullet$ consists of

(a) a family of sheaves F^n on the X_n;

(b) for all $f \colon \Delta_n \to \Delta_m$, an $X_\bullet(f)$-morphism $F^\bullet(f)$ from F^n to F^m.

We further require that $F^\bullet(f \circ g) = F^\bullet(f) \circ F^\bullet(g)$.

A morphism u from $F^\bullet$ to $G^\bullet$ is a family of morphisms $u^n \colon F^n \to G^n$ such that, for all $f \in \mathrm{Hom}_{(\Delta)}(\Delta_n, \Delta_m)$, we have that $u^m \circ F^\bullet(f) = G^\bullet(f) \circ u^n$.

For an open U of X_n, let $F^\bullet(U) = F^n(U)$. For $f \colon \Delta_n \to \Delta_m$, $U \subset X_n$, and $V \subset X_m$ such that $X_\bullet(f)(V) \subset (U)$, let $F^\bullet(f, V, U) \colon F^\bullet(U) \to F^\bullet(V)$ be the map induced by $F^\bullet(f)$. We immediately see that the system of sets $F^\bullet(U)$ (indexed by $n \geqslant 0$ and $U \subset X_n$) and of maps $F^\bullet(f, V, U)$ uniquely determines $F^\bullet$ (cf. §7.1). For such a system to come from a sheaf on $X_\bullet$, it is necessary and sufficient that:

(a) $F^\bullet(fg, U, W) = F^\bullet(f, U, V)F^\bullet(g, V, W)$ whenever this makes sense;

(b) for any n, the $F^\bullet(U)$ for $U \subset X_n$ form a sheaf on X_n.

This shows that sheaves on $X_\bullet$ can be interpreted as sheaves on a suitable site, and, in particular, form a topos $(X_\bullet)^\sim$. We freely use, for sheaves on $X_\bullet$, the modern terminology for sheaves on a site. With this in mind, a sheaf $F^\bullet$ of abelian groups (resp. of rings, ...) is a system of sheaves F^n of abelian groups (resp. of rings, ...) on the X_n along with the morphisms $F^\bullet(f)$.

Examples 5.1.9.

(I) Let $X_\bullet$ be a simplicial analytic space. The structure sheaves $\mathcal{O}_{X_n}$ form a sheaf of rings on $X_\bullet$.

(II) Let $X_\bullet$ be a simplicial analytic space augmented over S. The sheaves $\Omega^1_{X_n/S}$ form a sheaf of $\mathcal{O}$-modules on $X_\bullet$. Its i-th exterior power is the sheaf of $\mathcal{O}$-modules $(\Omega^i_{X_n/S})_{n\geqslant 0}$ denoted $\Omega^i_{X_\bullet/S}$.
The de Rham complexes $(\Omega^\bullet_{X_n/S})_{n\geqslant 0}$ form a complex of sheaves on $X_\bullet$.

(III) Let $F^\bullet$ be a sheaf of abelian groups on $X_\bullet$. The canonical flasque Godement resolutions $\mathscr{C}^\bullet(F^n)$ form a complex of sheaves on $X_\bullet$ that is a resolution of $F^\bullet$.

(IV) Let S be a topological space. Sheaves $F^\bullet$ on the constant simplicial space $S_\bullet$ can be identified with cosimplicial sheaves on $S^\bullet$. In particular, an abelian sheaf $F^\bullet$ on $S_\bullet$ defines a chain complex $(F^n, d = \sum_i (-1)^i \delta_i)$. A complex K of abelian sheaves on $S_\bullet$ defines a double complex $(K^{n,m})$ that we again denote by K (here m is the cosimplicial degree) and whose associated simple complex $\mathrm{s}K$ is

$$(\mathrm{s}K)^n = \bigoplus_{p+q=n} K^{pq} \tag{5.1.9.1}$$

with differential given by

$$\mathrm{d}(x^{pq}) = \mathrm{d}_K(x^{pq}) + (-1)^p \sum_i (-1)^i \delta_i x^{pq}. \tag{5.1.9.2}$$

We denote by L the second filtration of $\mathrm{s}K$, i.e.

$$L^r(\mathrm{s}K) = \bigoplus_{q\geqslant r} K^{pq}. \tag{5.1.9.3}$$

Let $u\colon X_\bullet \to Y_\bullet$ be a morphism of simplicial topological spaces, with components $u_n : X_n \to Y_n$. If G (resp. F) is a sheaf on $Y_\bullet$ (resp. on $X_\bullet$), then $(u_n^* G^n)_{n\geqslant 0}$ (resp. $(u_{n*}F^n)_{n\geqslant 0}$) is a sheaf on $X_\bullet$ (resp. on $Y_\bullet$); we denote it by u^*G (resp. by u_*F). The functors u^* and u_* are adjoint to one another; they are the inverse image and direct image morphisms of topos morphism $u\colon (X_\bullet)^\sim \to (Y_\bullet)^\sim$.

Let $a\colon X_\bullet \to S$ be an augmented simplicial topological space. If F is a sheaf on S, then $a^*F = (a_n^*F)_{n\geqslant 0}$ "is" a sheaf on $X_\bullet$. The functor a^* has a right adjoint

$$a_*\colon F^\bullet \mapsto \mathrm{Ker}\left(a_{0*}F^0 \underset{\delta_1^*}{\overset{\delta_0^*}{\rightrightarrows}} a_{1*}F^1\right). \tag{5.1.11.1}$$

The functors a^* and a_* are the inverse image and direct image morphisms of topos morphism $a\colon (X_\bullet)^\sim \to (S)^\sim$.

Let $S_\bullet$ be the constant simplicial space associated to S, and $a_\bullet\colon X_\bullet \to S_\bullet$ the morphism defined by a. For an abelian sheaf $F^\bullet$ on $X_\bullet$, we can identify $a_{\bullet *}F^\bullet$ with a cosimplicial sheaf on S ((IV) of Examples 5.1.9). For a complex K of abelian sheaves on $X_\bullet$, if K^{pq} is the degree p component of the restriction of K to X_q, then the components of the simple complex associated to the double complex defined by $a_{\bullet *}K$ are

$$(\mathrm{s}a_{\bullet *}K)^n = \bigoplus_{p+q=n} a_{q*}K^{pq}. \tag{5.1.12.1}$$

The spectral sequence defined by the filtration L (Equation 5.1.9.3) of $\mathrm{s}a_{\bullet *}K$ can be written as

$$E_1^{pq} = \mathrm{H}^q(a_{p*}(K|X^p)) \Rightarrow \mathrm{H}^{p+q}(\mathrm{s}a_{\bullet *}K). \tag{5.1.12.2}$$

Let $S = P^t$ (the topological space consisting of a single point). We see that, for a sheaf $F^\bullet$ on $X_\bullet$, the $\Gamma(X_n, F^n)$ are the components of a cosimplicial set $\Gamma^\bullet(X_\bullet, F^\bullet)$. The functor Γ (of global sections) is the functor

$$\Gamma\colon F^\bullet \mapsto \operatorname{Ker}\big(\Gamma(X_0, F^0) \rightrightarrows \Gamma(X_1, F^1)\big). \tag{5.1.13.1}$$

If K is a complex of abelian sheaves, we denote by $\Gamma^\bullet(X_\bullet, K)$ the chain complex whose components are the cosimplicial abelian groups $\Gamma^\bullet(X_\bullet, K^n)$, and by $\mathrm{s}\Gamma^\bullet(X_\bullet, K)$ the associated simple complex.

7.2. Cohomology of simplicial topological spaces

To each simplicial topological space $X_\bullet$ is associated a usual topological space $|X_\bullet|$, called its geometric realisation (see below). In the particular case where the X_n are discrete, we recover the usual notion of geometric realisation of a simplicial set.

In [S1968], G. Segal defined the cohomology of $X_\bullet$ with values in an abelian group A as $\mathrm{H}^\bullet(|X_\bullet|, A)$. Under suitable hypotheses, the filtration of $|X_\bullet|$ by successive skeletons gives a spectral sequence

$$E_1^{pq} = \mathrm{H}^q(X_p, A) \Rightarrow \mathrm{H}^{p+q}(X_\bullet, A). \tag{5.2.1.1}$$

One of the interests in using this definition is that it also applies, for example, to the definition of the groups $K(X_\bullet)$. We will adopt another definition, which is better adapted to sheaf-theoretic techniques.Geometric realisation

For an integer $n \geqslant 0$, we denote by $|\Delta_n|$ the simple in $\mathbb{R}^{\Delta_n}$ whose vertices are the set of basis vectors. We identify Δ_n with the set of vertices of $|\Delta_n|$. Every function $f\colon \Delta_n \to \Delta_m$ extends by linearity to $|f|\colon |\Delta_n| \to |\Delta_m|$.

Let $X_\bullet$ be a simplicial topological space. Let

$$Y = \coprod_{n\geqslant 0} (X_n \times |\Delta_n|).$$

Let R be the finest equivalence relation on Y for which, for any $f\colon \Delta_n \to \Delta_m$ in (Δ), any $x_m \in X_m$, and any $a \in |\Delta_n|$, we have

$$(x_m, |f|(a)) \equiv (f(x_m), a) \bmod R.$$

The *geometric realisation* of $X_\bullet$ is by definition

$$|X_\bullet| = Y/R.$$

Definition 5.2.2. Let $X_\bullet$ be a simplicial topological space. The functors $\mathrm{H}^i(X_\bullet, F^\bullet)$ of *cohomology with coefficients in the abelian sheaf $F^\bullet$ on $X_\bullet$* are the derived functors of the functor Γ (Equation 5.1.13.1).

This definition is equivalent to the following, sometimes more convenient.

Let $F^\bullet$ be an abelian sheaf on $X_\bullet$. We can show (by example, using (III) of Examples 5.1.9) that $F^\bullet$ always admits resolutions K on the right such that

$$\mathrm{H}^r(X_q, K^{p,q}) = 0 \quad \text{for } p, q \geqslant 0 \text{ and } r > 0.$$

If K is such a resolution, then we canonically have that

$$\mathrm{H}^n(X_\bullet, F^\bullet) \simeq \mathrm{H}^n(\mathrm{s}\Gamma^\bullet(X_\bullet, K)). \tag{5.2.3.1}$$

It is easy to show directly that the right-hand side is independent (up to unique isomorphism) of the choice of K; for what follows, we are free to *define* $\mathrm{H}^\bullet(X_\bullet, F^\bullet)$ by Equation 5.2.3.1. We can further show that the spectral sequence of the complex $\mathrm{s}\Gamma^\bullet(X_\bullet, K)$, filtered by L (Equation 5.1.9.3 and Equation 5.1.12.2 for $S = P^t$)

$$E_1^{pq} = \mathrm{H}^q(X_p, F^p) \Rightarrow \mathrm{H}^{p+q}(X_\bullet, F^\bullet) \tag{5.2.3.2}$$

is independent (up to unique isomorphism) of the choice of K (cf. Equation 5.2.1.1).

We will need to make precise the above construction when passing to derived categories, and to give a relative variant of it.

Let $a\colon X_\bullet \to S$, let $S_\bullet$ be the constant simplicial space defined by S with augmentation map $\varepsilon\colon S_\bullet \to S$, and let $a_\bullet\colon X_\bullet \to S_\bullet$ be the map induced by a. Then

$$a_* = \varepsilon_* a_{\bullet *}\colon (X_\bullet)^\sim \to (S)^\sim. \tag{5.2.4.1}$$

The derived version of this equation is

$$\mathrm{R}a_* = \mathrm{R}\varepsilon_* \mathrm{R}a_{\bullet *}\colon \mathrm{D}^+(X_\bullet) \to \mathrm{D}^+(S) \to (S)^\sim. \tag{5.2.4.2}$$

where $\mathrm{D}^+(X_\bullet)$ is the bounded-below derived category of the category of abelian sheaves on $X_\bullet$. We will calculate $\mathrm{R}a_{\bullet *}$ and $\mathrm{R}\varepsilon_*$.

Let $u\colon X_\bullet \to Y_\bullet$ be a morphism of simplicial topological spaces. The functor $\mathrm{R}u_*\colon \mathrm{D}^+(X_\bullet) \to \mathrm{D}^+(Y_\bullet)$ can be calculated "component by component": if K is a complex of abelian sheaves on $X_\bullet$, then to calculate $\mathrm{R}u_*K$ we take a resolution $K \xrightarrow{\sim} K'$ such that the components F of K' satisfy $\mathrm{R}^i u_{p_*}(F^p) = 0$ for $i > 0$, $p \geqslant 0$ (cf. (III) of Examples 5.1.9); then $\mathrm{R}u_*K \approx u_*K'$.

The functor $\mathrm{s}\colon\ \mathrm{C}^+(S_\bullet) \to \mathrm{C}^+(S)$ sends acyclic complexes to acyclic complexes. It thus trivially derives to

$$\mathrm{s}\colon \mathrm{D}^+(S_\bullet) \to \mathrm{D}^+(S).$$

If F is an injective sheaf on $S_\bullet$, then the chain complex $\mathrm{s}F$ is a resolution of $\varepsilon_* F$. We thus obtain an isomorphism $\mathrm{R}\varepsilon_* \xrightarrow{\sim} \mathrm{s}$, and Equation 5.2.4.2 becomes

$$\mathrm{R}a_* = \mathrm{sR}a_{\bullet *} \tag{5.2.6.1}$$

Combined with §7.2 and specialised to the case where $S = P^t$, this equation proves Equation 5.2.3.1. In concrete terms, this implies that, to calculate $\mathrm{R}a_* K$, we can proceed in two steps:

(a) We take a resolution $K \xrightarrow{\sim} K'$ such that the components F of K' satisfy $\mathrm{R}^i a_{p_*}(F^p) = 0$ for $i > 0$. The complex $a_{\bullet *} K' \in \mathrm{D}^+(S_\bullet)$ (the derived category of the category of cosimplicial abelian sheaves on S) can be identified with $\mathrm{R}a_{\bullet *} K$.

(b) $\mathrm{R}a_* K$ is the simple complex $\mathrm{s}a_{\bullet *} K'$ associated to the double complex $a_{\bullet *} K'$.

The spectral sequence Equation 5.2.3.2 generalises to a spectral sequence

$$E_1^{pq} = \mathrm{R}^q a_{p_*}(K|X_p) \Rightarrow \mathrm{R}^{p+q} a_* K \tag{5.2.7.1}$$

induced by Equation 5.1.12.2.

7.3. Cohomological descent

Let $a\colon X_\bullet \to S$ be an augmented simplicial topological space. For every sheaf F on S, we have a morphism

$$\varphi\colon F \to a_* a^* F$$

from the adjunction $a^* \dashv a_*$. This morphism derives to a morphism of functors from $\mathrm{D}^+(S)$ to $\mathrm{D}^+(S)$, namely

$$\varphi\colon \mathrm{Id} \to \mathrm{R}a_* a^*. \tag{5.3.1.1}$$

Definition 5.3.2. We say that a is of *cohomological descent* if, for every abelian sheaf F on S, we have

$$F \xrightarrow{\sim} \mathrm{Ker}(a_{0*} a_0{}^* F \to a_{1*} a_1{}^* F)$$

and

$$\mathrm{R}^i a_* a^* F = 0 \quad \text{for } i > 0.$$

This is equivalent to asking that Equation 5.3.1.1 be an isomorphism.

If a is of cohomological descent, then, for $K \in \mathrm{Ob}\,\mathrm{D}^+(S)$, the canonical map

$$\mathrm{R}\Gamma(S, K) \to \mathrm{R}\Gamma(S, \mathrm{R}a_* a^* K) \simeq \mathrm{R}\Gamma(X_\bullet, a^* K) \tag{5.3.3.1}$$

is an isomorphism. In particular, for F an abelian sheaf on S, we have a spectral sequence (Equation 5.2.3.2)

$$E_1^{pq} = \mathrm{H}^q(X_p, a_p^* F) \Rightarrow \mathrm{H}^{p+q}(S, F). \tag{5.3.3.2}$$

For a complex, we again have, in hypercohomology, a spectral sequence

$$E_1^{pq} = \mathbb{H}^q(X_p, a_p^* K) \Rightarrow \mathbb{H}^{p+q}(S, K). \tag{5.3.3.3}$$

In both cases, the $E_1^{\bullet q}$ (for fixed q) form a simplicial group, and

$$d_1 = \sum_i (-1)^i \delta_i \colon E_1^{p,q} \to E_1^{p+1,q}.$$

Definition 5.3.4. A continuous map $a \colon X \to S$ is of *cohomological descent* if the augmentation morphism of $\operatorname{cosq}(X \to S)$, namely

$$\left((X/S)^{\Delta_n}\right)_{n \geqslant 0} \to S,$$

is of cohomological descent. We say that a is of *universal cohomological descent* if, for every $u \colon S' \to S$, the continuous map $a' \colon X \times_S S' \to S'$ is of cohomological descent.

The fundamental results, proven in [SD], are the following.

(I) The continuous maps of universal cohomological descent form a Grothendieck topology on the category of topological spaces, which we call the *universal cohomological descent topology.*

(II) A proper (§6) surjective map is of universal cohomological descent.

(III) A map $a \colon X \to S$ that admits sections locally on S is of universal cohomological descent.

(IV) Let $a \colon X_\bullet \to S$ be a k-truncated augmented simplicial space (with $-1 \leqslant k \leqslant \infty$). For $k \geqslant n \geqslant -1$, let $\varphi_n \colon \operatorname{cosq} X_\bullet \to \operatorname{cosq} \operatorname{sq}_n X_\bullet$ be the evident map. We say that $X_\bullet$ is a *k-truncated hypercover of S, for the universal cohomological descent topology*, if the maps

$$(\varphi_n)_{n+1} \colon X_{n+1} \to (\operatorname{cosq} \operatorname{sq}_n X_\bullet)_{n+1} \tag{5.3.5.1}$$

(for $-1 \leqslant n \leqslant k-1$) are of universal cohomological descent. If $X_\bullet$ is such a hypercover, then the simplicial space $\operatorname{cosq}(X_\bullet)$ augmented over S is of cohomological descent.

(V) Let a be a morphism of simplicial topological spaces augmented over S.

$$\begin{tikzcd} X_\bullet \arrow[r, "a"] \arrow[d, "x"'] & Y_\bullet \arrow[d, "y"] \\ S \arrow[r, equal] & S \end{tikzcd}$$

We say that a is a *hypercover* for the universal cohomological descent topology if the evident maps $X_n \to (\operatorname{cosq}_{n-1}^{Y_\bullet} \operatorname{sq}_{n-1} X_\bullet)_n$ are of universal cohomological descent. If a is such a hypercover then, for every $K \in \operatorname{Ob} \mathrm{D}^+(S)$,

$$a^* \colon \mathrm{R}y_* y^* K \xrightarrow{\sim} \mathrm{R}x_* x^* K.$$

For $k = \infty$, (IV) of §7.3 implies that the $a\colon X_\bullet \to S$ are of cohomological descent if the $(\varphi_n)_{n+1}$ are of universal cohomological descent. For $n = -1, 0$, these maps are

$$(\varphi_n)_{n+1} = \begin{cases} X_0 \xrightarrow{a} S & \text{if } n = -1 \\ X_1 \xrightarrow{(\delta_0,\delta_1)} X_0 \times_S X_0 & \text{if } n = 0. \end{cases}$$

For $n = 1$, $(\operatorname{cosq} \operatorname{sq}_1(X_\bullet))_1$ is the subspace of $X_1 \times_S X_1 \times_S X_1$ consisting of the triples (x, y, z) such that $\delta_0 x = \delta_0 y$, $\delta_1 x = \delta_0 z$, and $\delta_1 y = \delta_1 z$. The map $(\varphi_1)_2$ is $x \mapsto (\delta_0 x, \delta_1 x, \delta_2 x)$.

For $k = 0$, (IV) of §7.3 is Definition 5.3.4.

Example 5.3.7. Let $\mathscr{U} = (U_i)_{i\in I}$ be an open cover, or a finite locally closed cover, of S. Let $X = \coprod_{i\in I} U_i$. Then $a\colon X \to S$ is of cohomological descent. The spectral sequence in Equation 5.3.3.2 for $X_\bullet = \operatorname{cosq}(X \to S)$ is then exactly the Leray spectral sequence of the cover $\mathscr{U}$.

Let $a\colon X_\bullet \to S$ be as in (IV) of §7.3. We say that $X_\bullet$ is a *proper k-truncated hypercover* of S if the arrows in Equation 5.3.5.1 are proper and surjective. For $k = \infty$, we simply say "proper hypercover".

8. Examples of simplicial topological spaces

8.1. Classifying spaces

Let $u\colon X \to S$ be a continuous map. For every sheaf F on S, the sheaf u^*F is endowed with a "descent data" with respect to u, i.e. we have an isomorphism between the two inverse images of u^*F on $X \times_S X$, and this isomorphism satisfies a cocycle condition. If u admits a section locally on S, then this construction defines an equivalence between the category of sheaves on S and that of sheaves on X endowed with a descent data.

Take X to be a (left) principal homogeneous space for the group G on a space S. Then the G-equivariant sheaves on X are exactly the sheaves endowed with a descent data: every equivariant sheaf on X is, in a unique way (as an equivariant sheaf), the inverse image of a sheaf on $S = X/G$.

If a topological group G acts on a space X, then G acts on $G^{\Delta_n} \times X$ by

$$g \cdot (g_0, \ldots, g_n, x) = (g_0 g^{-1}, \ldots, g_n g^{-1}, gx).$$

We denote by $[X/G]_\bullet$ the simplicial space

$$[X/G]_\bullet = \big((G^{\Delta_n} \times X)/G\big)_{n\geqslant 0}. \tag{6.1.2.1}$$

(a) If X is a principal homogeneous space for the group G on $S = X/G$, then the map

$$\begin{aligned} G^{\Delta_n} \times X &\to X^{\Delta_n} \\ (g_0, \ldots, g_n, x) &\mapsto (g_0 x, \ldots, g_n x) \end{aligned}$$

identifies $[X/G]_n$ with the iterated fibre product $(X/S)^{\Delta_n}$:

$$\operatorname{cosq}(X \to S) = ([X/G]_\bullet \to S).$$

In particular, we have (by (III) of §7.3)

$$\mathrm{H}^\bullet([X/G]_\bullet) \simeq \mathrm{H}^\bullet(X/G) \tag{6.1.2.2}$$

(for a principal homogeneous space).

(b) For all n, $G^{\Delta_n} \times X$ is a principal homogeneous space for the group G on $[X/G]_n$. For every equivariant sheaf F on X, $\mathrm{pr}_2^* F$ is an equivariant sheaf on $G^{\Delta_n} \times X$; by §8.1, the latter is the inverse image of F^n on $[X/G]_n$. Every equivariant sheaf F on X thus defines a sheaf on $[X/G]_\bullet$. It is easy to show that we thus obtain an equivalence between the category of equivariant sheaves on X and the category of sheaves $F^\bullet$ on $[X/G]_\bullet$ that satisfy

Property *. $(*)$ *For every $f\colon \Delta_n \to \Delta_m$, the structure morphism $f^*F^n \to F^m$ is an isomorphism.*

(c) Construction (b) above is natural in (G, X, F). We set

$$\mathrm{H}^\bullet(X, G; F) = \mathrm{H}^\bullet([X/G]_\bullet, F^\bullet) \tag{6.1.2.3}$$

(mixed cohomology of X, G with coefficients in F). Under the hypotheses of (a), if F is the inverse image of F^{-1} on $S = X/G$, then

$$\mathrm{H}^\bullet(X, G; F) = \mathrm{H}^\bullet(X/G, F^{-1}) \tag{6.1.2.4}$$

(for a principal homogeneous space). This generalises Equation 6.1.2.2 (which is the case $F = \mathbb{Z}$).

Let P^t be the topological space consisting of a single point. We define the ***simplicial classifying space*** of G, denoted $B_{\bullet G}$, to be the simplicial space

$$B_{\bullet G} = [P^t/G]_\bullet.$$

Let P be a principal homogeneous space for the group G on S. The evident morphism

$$\mathrm{cosq}(P \to S) = [P/G]_\bullet \to [P^t/G]_\bullet = B_{\bullet G}$$

defines a composite morphism

$$[P]\colon \mathrm{H}^\bullet(B_{\bullet G}) \to \mathrm{H}^\bullet([P/G]_\bullet) \xleftarrow[(6.1.2.2)]{\sim} \mathrm{H}^\bullet(S). \tag{6.1.3.1}$$

We will see below that, in good cases, $\mathrm{H}^\bullet(B_{\bullet G}) = \mathrm{H}^\bullet(B_G)$, and that the image of $[P]$ consists of the *characteristic classes* of P.

Let G be a Lie group, B_G a classifying space for G, and $a\colon U_G \to B_G$ the universal principal homogeneous G-space. Let X be a G-space; then $X \times U_G$ is a principal homogeneous G-space over $X \times U_G/G$ such that, for every equivariant sheaf F on X, $\mathrm{pr}_1^* F$ is the inverse image of a sheaf F^G on $X \times U_G/G$. Since U_G is contractible, and by Equation 6.1.2.2, we have

$$\mathrm{H}^\bullet(X, G; F) \xrightarrow{\sim} \mathrm{H}^\bullet(X \times U_G, G; \mathrm{pr}_1^* F) \xleftarrow{\sim} \mathrm{H}^\bullet(X \times U_G/G, F^G). \tag{6.1.4.1}$$

In particular, for $X = P^t$,

$$\mathrm{H}^\bullet(B_{\bullet G}) = \mathrm{H}^\bullet(B_G). \tag{6.1.4.2}$$

We can see that the isomorphism in Equation 6.1.4.2 is a particular case of Equation 6.1.3.1 where $S = B_G$ and $P = U_G$.

The spectral sequence in Equation 5.2.3.2 for $B_{\bullet G}$

$$E_1^{pq} = \mathrm{H}^q(G^{\Delta_p}/G) \Rightarrow \mathrm{H}^{p+q}(B_{\bullet G}) = \mathrm{H}^{p+q}(B_G)$$

is essentially the ***Eilenberg–Moore spectral sequence***. We briefly recall how it allows us to relate the rational cohomologies of G and of B_G, for connected G.

(a) The algebra $\mathrm{H}^\bullet(G, \mathbb{Q})$ is a connected graded Hopf algebra of finite dimension over $\mathbb{Q}$. If $P^\bullet(G)$ is the graded module of its primitive elements, then we have

$$H^\bullet(G, \mathbb{Q}) = \bigwedge P^\bullet(G)$$

and the generators of $P^\bullet(G)$ are of odd degree.

(b) The simplicial algebra $(E_1^{p\bullet})_{p\geqslant 0}$ is

$$E_1^{p\bullet} = \bigwedge \left(P^\bullet(G)^{\Delta_p}/P^\bullet(G)\right)$$

which is the exterior algebra of the suspension of the constant cosimplicial module $P^\bullet(G)$; we thus have (by Quillen [Q1968]) that $E_2^{p\bullet} = \mathrm{Sym}^p(P^\bullet(G))$. The pages E_2^{pq} are only zero for $p+q$ even; we thus have that $E_2^{pq} = E_\infty^{pq}$, and, for a suitable filtration, we canonically have that

$$\mathrm{Gr}\,\mathrm{H}^\bullet(B_G, \mathbb{Q}) \simeq \mathrm{Sym}^\bullet(P^\bullet(G)[-1])$$

and non-canonically that

$$\mathrm{H}^\bullet(B_G, \mathbb{Q}) \simeq \mathrm{Sym}^\bullet(P^\bullet(G)[-1]).$$

Let G be a (complex) linear algebraic group.

(a) If T is a maximal torus of G, with Weyl group W, then

$$\mathrm{H}^\bullet(B_G, \mathbb{Q}) \xrightarrow{\sim} \mathrm{H}^\bullet(B_T, \mathbb{Q})^W. \tag{6.1.6.1}$$

(b) If T is a torus with character group $X(T)$, then

$$\mathrm{H}^\bullet(T, \mathbb{Z}) \simeq \bigwedge^\bullet X(T) \tag{6.1.6.2}$$

(an isomorphism of graded Hopf algebras).

We will only use (a) in the following weaker form:

(a') *(The splitting principle).* The map $\mathrm{H}^\bullet(B_G, \mathbb{Q}) \to \mathrm{H}^\bullet(B_T, \mathbb{Q})$ is injective.

For completion, we recall a proof of (a').

If B is a Borel subgroup of G, then the bundle U_G/B on B_G is a fibre in flag spaces. By [2.1 and 2.6.3, D1968], which is better explained in [Proposition 3.1, G1970], or by [B1956], the Leray spectral sequence of $U_G/B \to B_G$ degenerates to rational cohomology. We thus have that $\mathrm{H}^\bullet(B_G, \mathbb{Q}) \hookrightarrow \mathrm{H}^\bullet(U_G/B, \mathbb{Q})$. We conclude by noting that $U_G/B \sim B_B \sim B_T$.

8.2. Construction of hypercovers

Let $X_\bullet$ be a simplicial set. Denote by $D(\Delta_n, \Delta_M)$ the set of increasing surjective maps from Δ_n to Δ_m (degeneracy operators), and set

$$N(X_n) = X_n \setminus \bigcup_{s \in D(\Delta_n, \Delta_{n-1})} s(X_{n-1}). \tag{6.2.1.1}$$

Recall that, for all n, the map

$$\coprod_{\substack{m \leqslant n, \\ s \in D(\Delta_n, \Delta_m)}} s \colon \coprod N(X_m) \to X_n \tag{6.2.1.2}$$

is bijective.

Definition 6.2.2. We say that a simplicial topological space is s-*split* if the maps in Equation 6.2.1.2 are homeomorphisms.

Let $X_\bullet$ be a k-truncated simplicial set. For $n \leqslant k$, we again define $N(X_n)$ by Equation 6.2.1.1, and then Equation 6.2.1.2 is a bijection. We say that a k-truncated simplicial topological space is s-*split* if Equation 6.2.1.2 is a homeomorphism for $n \leqslant k$.

For an s-split $(n+1)$-truncated topological simplicial space X augmented over S, let $\alpha(X)$ be the triple consisting of $\mathrm{sq}_n(X)$, NX_{n+1}, and the evident map from NX_{n+1} to $(\operatorname{cosq}\mathrm{sq}_n X)_{n+1}$.

This triple $\alpha(X) = (Y, N, \beta)$ satisfies the following:

Property *. *$(*)$ Y is an s-split n-truncated simplicial topological space augmented over S, and β is a continuous map from N to $(\operatorname{cosq} Y)_{n+1}$.*

Proposition 6.2.4. *Let (Y, N, β) satisfy Property * above.*

(i) Up to unique isomorphism, there exists exactly one s-split $(n+1)$-truncated topological space X augmented over S such that $\alpha(X) \simeq (Y, N, \beta)$.

(ii) It is equivalent to give either $f\colon X \to Z$ or:

(a) a morphism $f'\colon Y \to \mathrm{sq}_n(Z)$; and

(b) a morphism $f''\colon N \to Z_{n+1}$, such that the diagram

$$\begin{array}{ccc} N & \xrightarrow{\beta} & (\operatorname{cosq} Y)_{n+1} \\ {\scriptstyle f''}\downarrow & & \downarrow{\scriptstyle f'} \\ Z_{n+1} & \longrightarrow & (\operatorname{cosq}\mathrm{sq}_n Z)_{n+1} \end{array}$$

commutes.

Proof. [5.1.3, SD]. ☐

This proposition also applies to simplicial objects in other categories $\mathscr{C}$ apart from that of topological spaces; it suffices that $\mathscr{C}$ satisfy the following:

Finite projective limits exist in $\mathbb{C}$. Finite projective sums exist in $\mathbb{C}$, and they are disjoint and universal.

Proposition 6.2.4 allows us to construct, by induction, proper hypercovers of S.

(0) We take $f_0\colon X_0 \to S$, proper and surjective. Then $\{X_0\}$ is a 0-truncated proper hypercover of S (§7.3), and it is s-split.

(1) We take $f_1 \colon N_1 \to \operatorname{cosq}(\{X_0\})_1$, i.e. $f_1 \colon N_1 \to X_0 \times_S X_0$. Applying Proposition 6.2.4, we associate to f_1 the s-split 1-truncated augmented simplicial topological space

$$ {}_1X_\bullet = \left(N_1 \coprod X_0 \overset{\rightarrow}{\underset{\rightarrow}{\leftarrow}} X_0 \to S \right). $$

Suppose f_1 to be chosen such that

$$ f_1' \colon N_1 \coprod X_0 \to X_0 \times_S X_0 $$

is proper and surjective (for example, if f_1 is proper and surjective). Then ${}_1X_\bullet$ is an s-split 1-truncated proper hypercover of S.

(k+1) Assume that we have already constructed an s-split k-truncated proper hypercover ${}_kX_\bullet \to S$. We take $f_{k+1} \colon N_{k+1} \to (\operatorname{cosq}({}_kX_\bullet))_{k+1}$ and, applying Proposition 6.2.4, we associate to f_{k+1} an s-split $(k+1)$-truncated augmented semi-simplicial space ${}_{k+1}X_\bullet$. Suppose that f_{k+1} is such that

$$ f_{k+1}' \colon {}_{k+1}X_{k+1} \to \operatorname{cosq}({}_kX_\bullet)_{k+1} $$

is proper and surjective (for example, if f_{k+1} is proper and surjective). Then ${}_{k+1}X_\bullet$ is an s-split $(k+1)$-truncated proper hypercover of S.

(∞) The ${}_kX_\bullet$ thus constructed are the successive skeletons of an s-split proper hypercover of S.

We say that a simplicial scheme $X_\bullet$ over $\mathbb{C}$ is *smooth* if the X_n are smooth; it is said to be *proper* if the X_n are compact. A *normal crossing divisor* $D_\bullet$ of $X_\bullet$, assumed to be smooth, is a family $D_n \subset X_n$ of normal crossing divisors (§3.1) such that the $U_n = X_n \setminus D_n$ form a simplicial subscheme $U_\bullet$ of $X_\bullet$. This definition is justified by the following lemma.

Lemma 6.2.7. *If $D_\bullet$ is a normal crossing divisor of $X_\bullet$, then the logarithmic de Rham complexes $(\Omega^\bullet_{X_n}(\log D_n))_{n\geqslant 0}$, endowed with the weight filtration (§3.1), form a filtered complex on $X_\bullet$.*

Proof. This follows from (ii) of Proposition 3.1.3. □

We denote the complex $(\Omega^\bullet_{X_n}(\log D_n))_{n\geqslant 0}$ by $\Omega^\bullet_{X_\bullet}(\log D_\bullet)$.

Using §8.2, we can show that, for every separated scheme S over $\mathbb{C}$, there exists:

(α) a simplicial scheme $X_\bullet$ over $\mathbb{C}$, smooth and proper, that we can take to be s-split;

(β) a normal crossing divisor $D_\bullet$ of $X_\bullet$; we set $U_\bullet = X_\bullet \setminus D_\bullet$;

(γ) an augmentation $a \colon U_\bullet \to S$ that realises $U_\bullet^{\mathrm{an}}$ as a proper hypercover of S^{an}.

Furthermore, any two such systems are covered by a third, and a morphism $u \colon S \to T$ can be covered by a morphism

$$ \begin{array}{ccc} X_\bullet & \longrightarrow & Y_\bullet \\ \uparrow & & \uparrow \\ U_\bullet & \longrightarrow & V_\bullet \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ S & \xrightarrow[u]{} & T \end{array} $$

of such systems (see [SD]).

8.3. Relative cohomology

The *mapping cone* construction for morphisms of simplicial sets works in the same way for simplicial objects in any category $\mathscr{C}$ that has a final object e and finite sums. For $u\colon Y_\bullet \to X_\bullet$, the cone $C(u)$ satisfies

$$C(u)_n = X_n \coprod \coprod_{i<n} Y_i \coprod e.$$

We take $\mathscr{C}$ to be:

(a) the category of topological spaces and continuous maps, with final object $e = P^t$;

(b) the category of pairs (X, F) where X is a topological space and F is an abelian sheaf on X, with an arrow $(u, f)\colon (Y, F) \to (X, G)$ consisting of a continuous map $u\colon Y \to X$ and a u-morphism (§7.1) $f\colon G \to F$, with final object $e = (P^t, 0)$.

Let $u\colon Y_\bullet \to X_\bullet$ be a morphism of topological simplicial spaces, with cone $C(u)$. Let F be an abelian sheaf on $X_\bullet$ and G an abelian sheaf on $Y_\bullet$, and let $f\colon G \to F$ be a u-morphism. The cone $C(f)$ of f is an abelian sheaf on $C(u)$, and we set

$$\mathrm{H}^n(C(u), C(f)) = \mathrm{H}^n(X_\bullet \bmod Y_\bullet, F \bmod G). \tag{6.3.2.1}$$

These are the *relative cohomology groups.* We can easily show that they fit into a long exact sequence

$$\ldots \to \mathrm{H}^i(X_\bullet \bmod Y_\bullet, F \bmod G) \to \mathrm{H}^i(X_\bullet, F) \to \mathrm{H}^i(Y_\bullet, G) \to \ldots. \tag{6.3.2.2}$$

More generally, let L and K be bounded-below complexes of abelian sheaves on $Y_\bullet$ and $X_\bullet$ (respectively), and let $f\colon K \to L$ be a u-morphism. We thus obtain a complex $C(f)$ on $C(u)$. We again define the hypercohomology

$$\mathbb{H}^n(C(u), C(f)) = \mathbb{H}^n(X_\bullet \bmod Y_\bullet, K \bmod L).$$

These groups appear in an exact sequence analogous to that of Equation 6.3.2.2, coming, in the suitable derived category, from a distinguished triangle.

The construction presented above is not the only one possible. It has the inconvenience that, even if we start with true topological spaces X and Y (i.e. constant simplicial spaces), we are led to consider non-constant simplicial spaces.

8.4. Multisimplicial spaces

Let $r \geqslant 0$ be an integer. An *r-simplicial object* $Z_\bullet$ in a category $\mathscr{C}$ is a contravariant functor from the r-fold product $(\Delta)^r$ to $\mathscr{C}$. The *diagonal* simplicial object $\delta Z_\bullet$ is the composite functor $(\Delta) \to (\Delta)^r \to \mathscr{C}$.

As in §7.1, we define the topos of *sheaves* on an r-simplicial topological space. Let $\Gamma^\bullet$ be the functor

$$F \mapsto (\Gamma(X_{n_1\ldots n_r}, F^{n_1\ldots n_r}))$$

from sheaves on $X_\bullet$ to r-cosimplicial sets. For small r, we often prefer to write $\Gamma^{\bullet\cdots\bullet}$ (with r-many copies of $\bullet$). We have a co-augmentation $\Gamma(X_\bullet, F^\bullet) \to \Gamma^\bullet(X_\bullet, F^\bullet)$. The functors of *cohomology with* *values in an abelian sheaf* F are the derived functors of the "global sections" functor Γ. They can be calculated by a procedure parallel to that of §7.2, i.e.

$$\mathrm{R}\Gamma = \mathrm{sR}\Gamma^{\bullet}\colon \mathrm{D}^+(X_\bullet) \to \mathrm{D}^+((\mathsf{Ab})). \tag{6.4.2.1}$$

A sheaf F on an r-simplicial topological space $X_\bullet$ induces a sheaf δF on the diagonal simplicial space $\delta X_\bullet$. It follows from the Cartier–Eilenberg–Zilber theorem that

$$\mathrm{R}\Gamma(X_\bullet, K) \xrightarrow{\sim} \mathrm{R}\Gamma(\delta X_\bullet, \delta K). \tag{6.4.2.2}$$

We restrict to the case $r = 2$. A ***bisimplicial object 1-augmented over a simplicial object*** $S_\bullet$ is a contravariant functor from $(\Delta^+) \times (\Delta)$ to $\mathscr{C}$ such that $S_\bullet$ is the composite functor

$$(\Delta) \xrightarrow{(\Delta_{-1},\bullet)} (\Delta^+) \times (\Delta) \to \mathscr{C}.$$

To denote a bisimplicial object 1-augmented over $S_\bullet$, with underlying bisimplicial object $X_{\bullet\bullet}$, we will use notation of the form

$$a\colon X_{\bullet\bullet} \to S_\bullet.$$

For $n \geqslant 0$, $a\colon (X_{\bullet n}) \to S_n$ is a simplicial object augmented over S_n. If F is a sheaf on $X_{\bullet\bullet}$, then the $(a_*(F|X_{\bullet n})$ on $S_n)_{n\geqslant 0}$ form a sheaf on $S_\bullet$; we thus define a topos morphism

$$a\colon X_{\bullet\bullet}^\sim \to S_\bullet^\sim.$$

We explain in [SD] that $\mathrm{R}a_*$ can be calculated "component by component", i.e.

$$\mathrm{R}a_*K|S_n = \mathrm{R}a_*(K|X_{\bullet n}).$$

It thus follows that if, for each n, the morphism $a\colon (X_{\bullet n}) \to S_n$ is of cohomological descent, then $a\colon X_{\bullet\bullet} \to S_\bullet$ is of cohomological descent: for every complex $K \in \mathrm{D}^+(S_\bullet)$ of abelian sheaves, we have

$$K \xrightarrow{\approx} \mathrm{R}a_*a^*K.$$

In [SD], we show that, for every separated simplicial scheme $S_\bullet$, there exists a bisimplicial scheme $X_{\bullet\bullet}$ that is 1-augmented over $S_\bullet$, along with $i\colon X_{\bullet\bullet} \hookrightarrow \overline{X}_{\bullet\bullet}$ such that:

(a) The $\overline{X}_{nm}$ are smooth and projective; X_{nm} is the complement of a normal crossing divisor D_{nm} in $\overline{X}_{nm}$, and we can suppose the D_{nm} to be a union of smooth divisors.

(b) For $n \geqslant 0$, $X_{\bullet n}$ is a proper hypercover of S_n, and we can take it to be s-split.

The construction proceeds as in §8.2, but the induction is more complicated. The claims of uniqueness (§8.2) remain true, mutatis mutandis.

9. Supplements to §1

9.1. Filtered derived category

This section completes §1.4.
Let $\mathscr{A}$ be an abelian category. We set:

- $\mathrm{F}\mathscr{A}$ (resp. $\mathrm{F}_2\mathscr{A}$) = the category of filtered (resp. bifiltered) objects with finite filtration(s) of $\mathscr{A}$.
- $\mathrm{K}^+\mathrm{F}\mathscr{A}$ (resp. $\mathrm{K}^+\mathrm{F}_2\mathscr{A}$) = the category of filtered (resp. bifiltered) bounded-below complexes of objects of $\mathscr{A}$, up to homotopy that respects the filtration(s).
- $\mathrm{D}^+\mathrm{F}\mathscr{A}$ (resp. $\mathrm{D}^+\mathrm{F}_2\mathscr{A}$) = the triangulated category induced from $\mathrm{K}^+\mathrm{F}\mathscr{A}$ (resp. from $\mathrm{K}^+\mathrm{F}_2\mathscr{A}$) by inverting the filtered (resp. bifiltered) quasi-isomorphisms (Definition 1.3.6); this is the *derived filtered category* (resp. *derived bifiltered category*).

A filtered quasi-isomorphism $u\colon (K,F)\to(K',F')$ induces an isomorphism of spectral sequences $u\colon E_\bullet^{\bullet\bullet}(K,F)\to E_\bullet^{\bullet\bullet}(K',F')$. An object K of $\mathrm{D}^+\mathrm{F}\mathscr{A}$ thus defines a spectral sequence $E_\bullet^{\bullet\bullet}(K)$.
Similarly, an object L of $\mathrm{D}^+\mathrm{F}_2\mathscr{A}$ defines an accumulation of spectral sequences of the type considered in §1.4.

Let T be a left exact functor from $\mathscr{A}$ to an abelian category $\mathscr{B}$. Suppose that every object of $\mathscr{A}$ injects into an injective object. The functor T can then be "derived" to give the functors

$$\mathrm{R}\colon \mathrm{D}^+(\mathscr{A})\to\mathrm{D}^+(\mathscr{B}) \tag{7.1.3.1}$$

$$\mathrm{R}\colon \mathrm{D}^+\mathrm{F}(\mathscr{A})\to\mathrm{D}^+\mathrm{F}(\mathscr{B}) \tag{7.1.3.2}$$

$$\mathrm{R}\colon \mathrm{D}^+\mathrm{F}_2(\mathscr{A})\to\mathrm{D}^+\mathrm{F}_2(\mathscr{B}). \tag{7.1.3.3}$$

They can be calculated as follows: if K' is a T-acyclic resolution (resp. a filtered resolution, resp. a bifiltered resolution) of K (§1.4 and §1.4), then $\mathrm{R}T(K)=T(K')$.

The hypercohomology spectral sequence (for T) of $K\in\operatorname{Ob}\mathrm{D}^+\mathrm{F}(\mathscr{A})$ is the spectral sequence of $\mathrm{R}TK\in\operatorname{Ob}\mathrm{D}^+\mathrm{F}(\mathscr{B})$ (cf. §1.4).

We will need more precise results for the functors $\mathrm{R}a_*$, where $a\colon X_\bullet\to S$ is an augmentation of a simplicial topological space. The case $S=P^t$, where $\mathrm{R}a_*=\mathrm{R}\Gamma$ will suffice.

We reuse the notation of §7.1, and recall Equation 5.2.6.1:

$$\mathrm{R}a_*=\mathrm{sR}a_{\bullet *}.$$

For every complex $K\in\operatorname{Ob}\mathrm{C}^+(S_\bullet)$, the simple complex $\mathrm{s}K$ is endowed with a natural filtration L (Equation 5.1.9.3). A quasi-isomorphism $u\colon K'\xrightarrow{\sim}K''$ induces a filtered quasi-isomorphism $u\colon(\mathrm{s}K',L)\xrightarrow{\sim}(\mathrm{s}K'',L)$. Then s factors as

$$\mathrm{s}\colon \mathrm{D}^+(S_\bullet)\to\mathrm{D}^+\mathrm{F}(S)$$

and $\mathrm{R}a_*$ factors as

$$\mathrm{R}a_*\colon \mathrm{D}^+(X_\bullet)\to\mathrm{D}^+\mathrm{F}(S).$$

The spectral sequence of the filtered complex $(\mathrm{R}a_*K, L) \in \mathrm{D}^+\mathrm{F}(S)$ is exactly Equation 5.2.7.1.

If K is filtered (resp. bifiltered), then $\mathrm{R}a_{\bullet *}K$ is filtered (resp. bifiltered): we have

$$\mathrm{R}a_{\bullet *}\colon \mathrm{D}^+\mathrm{F}(X_\bullet) \to \mathrm{D}^+\mathrm{F}(S_\bullet) \tag{7.1.5.1}$$

$$\mathrm{R}a_{\bullet *}\colon \mathrm{D}^+\mathrm{F}_2(X_\bullet) \to \mathrm{D}^+\mathrm{F}_2(S_\bullet). \tag{7.1.5.2}$$

Let K be a complex of cosimplicial sheaves on S, endowed with an increasing filtration W. We define the *diagonal filtration* $\delta(W, L)$ of W and L to be the increasing filtration of $\mathrm{s}K$ given by

$$\begin{aligned} \delta(W, L)_n(\mathrm{s}K) &= \bigoplus_{p,q} W_{n+p}(K^{p,q}) \\ &= \sum_p \mathrm{s}(W_{n+p}(K)) \cap L^p(\mathrm{s}K). \end{aligned} \tag{7.1.6.1}$$

We have

$$\mathrm{Gr}_n^{\delta(W,L)}(\mathrm{s}K) \simeq \bigoplus_p \mathrm{Gr}^W_{n+p}(K^{\bullet p})[-p]. \tag{7.1.6.2}$$

The functor $(K, W) \mapsto (\mathrm{s}K, \delta(W, L))$ sends filtered quasi-isomorphisms to filtered quasi-isomorphisms, and defines

$$\begin{aligned} (s, \delta)\colon \mathrm{D}^+\mathrm{F}(S_\bullet) &\to \mathrm{D}^+\mathrm{F}(S) \\ (K, W) &\mapsto (\mathrm{s}K, \delta(W, L)) \end{aligned} \tag{7.1.6.3}$$

whence, by composition with Equation 7.1.5.1,

$$\begin{aligned} (\mathrm{R}\Gamma, \delta(-, L))\colon \mathrm{D}^+\mathrm{F}(X_\bullet) &\to \mathrm{D}^+\mathrm{F}(S) \\ (K, W) &\mapsto (\mathrm{R}\Gamma K, \delta(W, L)). \end{aligned} \tag{7.1.6.4}$$

From Equation 7.1.6.2, we see that

$$\mathrm{Gr}_n^{\delta(W,L)} = \bigoplus_p \mathrm{R}a_{p*}(\mathrm{Gr}^W_{n+p} K)[-p]. \tag{7.1.6.5}$$

If (K, W, F) is a bifiltered complex of cosimplicial sheaves, then $\mathrm{s}K$ is endowed with the three filtrations W, F, and L, and defines different bifiltered complexes.

For example, for increasing W, the functor $(K, W, F) \mapsto (K, \delta(W, L), F)$ sends bifiltered quasi-isomorphisms to bifiltered quasi-isomorphisms, and thus defines

$$\begin{aligned} \mathrm{D}^+\mathrm{F}_2(S_\bullet) &\to \mathrm{D}^+\mathrm{F}_2(S) \\ (K, W, F) &\mapsto (K, \delta(W, L), F). \end{aligned} \tag{7.1.7.1}$$

By composition with Equation 7.1.5.2, we thus obtain

$$\begin{aligned} \mathrm{D}^+\mathrm{F}_2(X_\bullet) &\to \mathrm{D}^+\mathrm{F}_2(S) \\ (K, W, F) &\mapsto (\mathrm{R}\Gamma K, \delta(W, L), F). \end{aligned} \tag{7.1.7.2}$$

and we have that

$$\mathrm{Gr}_n^{\delta(W,L)}(\mathrm{R}\Gamma K, F) = \bigoplus_p \mathrm{R}a_{p*}(\mathrm{Gr}^W_{n+p} K, F)[-p] \tag{7.1.7.3}$$

in $\mathrm{D}^+\mathrm{F}(S)$.

9.2. Supplements to the two filtrations lemma

In this section, we give a new proof of the two filtrations lemma (Theorem 1.3.16) and some supplements.

Let (K, F, W) be a bounded-below bifiltered complex with objects in an abelian category $\mathscr{A}$, with F assumed to be biregular.

We say that (K, F, W) is F-*splitable* if the filtered complex (K, W) can be written as a sum of filtered complexes

$$(K, W) = \bigoplus_{n\in\mathbb{Z}}(K_n, W_n)$$

with

$$F^n K = \bigoplus_{n'\geqslant n} K_{n'}.$$

Let $r_0 \geqslant 0$ be an integer or $+\infty$. The following condition was considered in Theorem 1.3.16 and Corollary 1.3.17:

Condition 7.2.2. For every non-negative integer $r < r_0$, the differentials d_r of the graded complex $E_r(K, W)$ are strictly compatible with the recurrent filtration defined by F.

It is clear that, if (K, F, W) is F-splitable, then Condition 7.2.2 is satisfied for $r_0 = \infty$. Conversely, it seems that if Condition 7.2.2 is satisfied for $r_0 = \infty$, then everything is as if the functor Gr_F were exact. For example, we will show that the Gr_F of the spectral sequence $E(K, W)$ can then be identified with the spectral sequence $E(\mathrm{Gr}_F K, W)$, and that the spectral sequence $E(K, F)$ degenerates ($E_1 = E_\infty$).

We immediately deduce from the definition (§1.3) that the first direct filtration of $E_r(K, W)$ is the filtration F_d of $E_r(K, W)$ by the images

$$F_d^p(E_r(K, W)) = \mathrm{Im}\left(E_r(F^pK, W) \to E_r(K, W)\right).$$

Dually, the second direct filtration (§1.3) of $E_r(K, W)$ is the filtration F_{d^*} of $E_r(K, W)$ by the kernels

$$F_{d^*}^p(E_r(K, W)) = \mathrm{Ker}\left(E_r(K, W) \to E_r(K/F^pK, W)\right).$$

The recurrent filtration F_{rec} of $E_r(K, W)$ is intermediary between these two filtrations ((iii) of Proposition 1.3.13).

Proposition 7.2.5. *Suppose that (K, F, W) satisfies Condition 7.2.2 for some $r_0 \geqslant 0$. Then*

(i) $(F^aK/F^bK, F, W)$ also satisfies Condition 7.2.2 for r_0.

(ii) For $r \leqslant r_0$, the sequence

$$0 \to E_r(F^pK, W) \to E_r(K, W) \to E_r(K/F^pK, W) \to 0$$

is exact; for $r = r_0 + 1$, the sequence

$$E_r(F^pK, W) \to E_r(K, W) \to E_r(K/F^pK, W)$$

is exact. In particular, for $r \leqslant r_0+1$, the two direct filtrations and the recurrent filtration of $E_r(K, W)$ agree.

Proof. Fix an integer p. We will prove the claim

$(*)_r$ *If* $r \leqslant r_0$, *then* $E_r(F^pK, W)$ *injects into* $E_r(K, W)$; *if* $r \leqslant r_0+1$, *then its image is* $F_{\text{rec}}(E_r(K, W))$.

by induction on r.

The claim $(*)_0$ is always true. Assume $(*)_r$, and let us prove $(*)_{r+1}$. We can suppose that $0 \leqslant r \leqslant r_0$. We have a diagram

$$\begin{tikzcd}
E_r(K,W) \arrow[r, "d_r"] & E_r(K,W) \arrow[r, "d_r"] & E_r(K,W) \\
E_r(F^pK,W) \arrow[r] \arrow[u, hook] & E_r(F^pK,W) \arrow[r] \arrow[u, hook] & E_r(F^pK,W) \arrow[u, hook, "i"']
\end{tikzcd}$$

and the image of the vertical inclusions is $F^p_{\text{rec}}(E_r(K, W))$. Since i is injective,

$$\begin{aligned}
F^p_{\text{rec}}(E_{r+1}(K,W)) &= \operatorname{Im}\Big(\operatorname{Ker}\big(F^p_{\text{rec}}E_r(K,W) \xrightarrow{d_r} E_r(K,W)\big) \to E_{r+1}(K,W)\Big) \\
&= \operatorname{Im}\Big(\operatorname{Ker}\big(E_r(F^pK,W) \xrightarrow{d_r} E_r(F^pK,W)\big) \to E_{r+1}(K,W)\Big) \\
&= \operatorname{Im}\big(E_{r+1}(F^pK,W) \to E_{r+1}(K,W)\big).
\end{aligned}$$

If $r < r_0$, then d_r is strictly compatible with F_{rec}, whence

$$d_rE_r(K,W) \cap E_r(F^pK,W) = d_rE_r(F^pK,W)$$

and $E_{r+1}(F^pK, W)$ thus injects into $E_{r+1}(K, W)$. This proves $(*)_{r+1}$.

The statements $(*)_r$, combined with the dual statements, implies (ii).

It follows from (ii) that, for $r < r_0$ and $a \leqslant b$, $E_r(F^bK, W)$ injects into $E_r(F^aK, W)$, and that the differential d_r of $E_r(F^aK, W)$ is strictly compatible with the first direct filtration of $E_r(F^aK, W)$. We thus deduce, by induction on $r < r_0$, that the first direct filtration of $E_r(F^aK, W)$ coincides with the recurrent filtration, and (i) then follows. ☐

If Condition 7.2.2 is satisfied for r_0, and $r \leqslant r_0 + 1$, then we denote by F the filtration $F_d = F_{d^*} = F_{\text{rec}}$ of $E_r(W, K)$. The exact sequences from (ii) of Proposition 7.2.5

$$\begin{tikzcd}
0 & E_r(K/F^pK,W) \arrow[l] & & E_r(K/F^{p+1}K,W) \arrow[ll] & & 0 \\
& & E_r(K,W) \arrow[ul] \arrow[ur] & & E_r(\mathrm{Gr}^p_F K,W) \arrow[ul] \arrow[ur] & \\
0 \arrow[r] & E_r(F^{p+1}K,W) \arrow[ur] \arrow[rr] & & E_r(F^pK,W) \arrow[ul] \arrow[ur] & & 0 \arrow[ul]
\end{tikzcd}$$

define, for $r \leqslant r_0$, an (autodual) isomorphism that is compatible with the differentials d_r, namely

$$\mathrm{Gr}^p_F(E_r(K,W)) \simeq E_r(\mathrm{Gr}^p_F K,W) \qquad \text{for } r \leqslant r_0. \tag{7.2.6.1}$$

If Condition 7.2.2 is satisfied for $r_0 = \infty$, then the sequences

$$0 \to E_r(F^pK,W) \to E_r(K,W) \to E_r(K/F^pK,W) \to 0$$

are exact for all r. If the filtration W is biregular, then the sequence

$$0 \to E_\infty(F^pK,W) \to E_\infty(K,W) \to E_\infty(K/F^pK,W) \to 0$$

is thus exact. We can rewrite this sequence as

$$0 \to \mathrm{Gr}_W(\mathrm{H}(F^pK)) \to \mathrm{Gr}_W(\mathrm{H}(K)) \to \mathrm{Gr}_W(\mathrm{H}(K/F^pK)) \to 0.$$

Lemma 7.2.7.1. *Let L be a complex endowed with a biregular filtration G. For $\mathrm{Gr}_G(L)$ to be acyclic, it is necessary and sufficient that L be acyclic and that the differentials of L be strictly compatible with the filtration G.*

Proof. This is a particular case of Proposition 1.3.2. □

Applying Lemma 7.2.7.1, taking L to be the complex

$$0 \to \mathrm{H}(F^pK) \to \mathrm{H}(K) \to \mathrm{H}(K/F^pK) \to 0,$$

we see that:

(a) This sequence is exact, i.e. the differentials of K are strictly compatible with the filtration F;

(b) the filtration of $\mathrm{H}(F^pK)$ induced by the filtration W of F^pK coincides with the filtration induced by the filtration W of $\mathrm{H}(K)$;

(c) an analogous statement holds for K/F^pK.

In conclusion:

Proposition 7.2.8. *If (K, F, W) satisfies Condition 7.2.2 for $r_0 = \infty$, then the spectral sequence $E(K, F)$ degenerates ($E_1 = E_\infty$). Furthermore, we have an isomorphism of spectral sequences*

$$\mathrm{Gr}_F^p(E_\bullet(K, W)) \simeq E_\bullet(\mathrm{Gr}_F^p K, W).$$

10. Hodge theory of algebraic spaces

10.1. Hodge complexes

Let A be as in §6. Then a *Hodge A-complex* K of weight n consists of

(α) a complex $K_A \in \mathrm{Ob}\,\mathrm{D}^+(A)$ such that $\mathrm{H}^k(K_A)$ is an A-module of finite type for all k;

(β) a filtration F on $K_A \otimes \mathbb{C}$, i.e. a filtered complex $(K_\mathbb{C}, F) \in \mathrm{Ob}\,\mathrm{D}^+F(\mathbb{C})$, and an isomorphism $\alpha\colon K_\mathbb{C} \simeq K_A \otimes \mathbb{C}$ in $\mathrm{D}^+(\mathbb{C})$.

The following axioms have to be satisfied:

Axiom CH1. The differential d of $K_\mathbb{C}$ is strictly compatible with the filtration F; in other words, the spectral sequence defined by $(K_\mathbb{C}, F)$ degenerates at E_1 (i.e. $E_1 = E_\infty$) Proposition 1.3.2.

Axiom CH2. For all k, the filtration F on $\mathrm{H}^k(K_\mathbb{C}) \simeq \mathrm{H}^k(K_A) \otimes \mathbb{C}$ defines a Hodge A-structure of weight $n + k$ on $\mathrm{H}^k(K_A)$, i.e. the filtration F is $(n + k)$-opposite to its complex conjugate (which makes sense, since $A \subset \mathbb{R}$).

Let X be a topological space. Then a *cohomological Hodge A-complex K of weight n* on X consists of:

(α) a complex $K_A \in \operatorname{Ob} \mathrm{D}^+(X, A)$,

(β) a filtered complex $(K_{\mathbb{C}}, F) \in \operatorname{Ob} \mathrm{D}^+ F(X, \mathbb{C})$,

(γ) an isomorphism $\alpha\colon K_{\mathbb{C}} \simeq K_A \otimes \mathbb{C}$ in $\mathrm{D}^+(X, \mathbb{C})$.

The following axiom has to be satisfied:

Axiom CHC. The triple $(\mathrm{R}\Gamma(K_A), \mathrm{R}\Gamma((K_{\mathbb{C}}, F)), \mathrm{R}\Gamma(\alpha))$ is a Hodge complex of weight n.

When $A = \mathbb{Z}$, we simply speak of Hodge complexes and cohomological Hodge complexes. The following statement reformulates a part of classical Hodge theory (cf. §2.2).

Scholium 8.1.3. Let X be a complex Kähler variety. Let $K_{\mathbb{Z}}$ be the complex $\mathbb{Z}[0]$ consisting of the constant sheaf $\mathbb{Z}$ in degree 0, and let $K_{\mathbb{C}}$ be the holomorphic de Rham complex $\Omega_X^\bullet$, with F its stupid filtration. We have $\alpha\colon K_{\mathbb{Z}} \otimes \mathbb{C} \simeq \mathbb{C}[0] \xrightarrow{\approx} \Omega_X^\bullet$ (the Poincaré lemma), and $(K_{\mathbb{Z}}, (K_{\mathbb{C}}, F), \alpha)$ is a cohomological Hodge complex of weight 0.

Remark 8.1.4. If K is a Hodge complex (resp. cohomological Hodge complex) of weight n, then $K[m]$ is a Hodge complex (resp. cohomological Hodge complex) of weight $n+m$.

A *mixed Hodge A-complex* K consists of:

(α) a complex $K_A \in \operatorname{Ob} \mathrm{D}^+(A)$ such that $\mathrm{H}^k(K_A)$ is an A-module of finite type for all k;

(β) a filtered complex $(K_{A\otimes\mathbb{Q}}, W) \in \operatorname{Ob} \mathrm{D}^+ F(A \otimes \mathbb{Q})$ (for an *increasing* filtration W) and an isomorphism $K_{A\otimes\mathbb{Q}} \simeq K_A \otimes \mathbb{Q}$ in $\mathrm{D}^+(A \otimes \mathbb{Q})$;

(γ) a bifiltered complex $(K_{\mathbb{C}}, W, F) \in \operatorname{Ob} \mathrm{D}^+ F_w(\mathbb{C})$ (for an *increasing* filtration W and a *decreasing* filtration F) and an isomorphism $\alpha\colon (K_{\mathbb{C}}, W) \simeq (K_{A\otimes\mathbb{Q}}, W) \otimes \mathbb{C}$ in $\mathrm{D}^+ F(\mathbb{C})$.

The following axiom must be satisfied:

Axiom CHM. For all n, the system consisting of the complex $\operatorname{Gr}_n^W(K_{A\otimes\mathbb{Q}}) \in \operatorname{Ob} \mathrm{D}^+(A \otimes \mathbb{Q})$, the complex $\operatorname{Gr}_n^W(K_{\mathbb{C}}) \in \operatorname{Ob} \mathrm{D}^+ F(\mathbb{C})$ filtered by F, and the isomorphism

$$\operatorname{Gr}_n^W(\alpha)\colon\ \operatorname{Gr}_n^W(K_{A\otimes\mathbb{Q}}) \otimes \mathbb{C} \simeq \operatorname{Gr}_n^W(K_{\mathbb{C}}),$$

is a Hodge $(A \otimes \mathbb{Q})$-complex of weight n.

A *cohomological mixed Hodge A-complex* K on a topological space X consists of:

(α) a complex $K_A \in \operatorname{Ob} \mathrm{D}^+(X, A)$ such that $\mathbb{H}^k(X, K_A)$ is an A-module of finite type for all k, and such that $\mathbb{H}(X, K_A) \otimes \mathbb{Q} \xrightarrow{\sim} \mathbb{H}(X, K_A \otimes \mathbb{Q})$;

(β) a filtered complex $(K_{A\otimes\mathbb{Q}}, W) \in \operatorname{Ob} \mathrm{D}^+ F(X, A \otimes \mathbb{Q})$ (for an *increasing* filtration W) and an isomorphism $K_{A\otimes\mathbb{Q}} \simeq K_A \otimes \mathbb{Q}$ in $\mathrm{D}^+(X, A \otimes \mathbb{Q})$;

(γ) a bifiltered complex $(K_{\mathbb{C}}, W, F) \in \operatorname{Ob} \mathrm{D}^+ F_w(X, \mathbb{C})$ (for an *increasing* filtration W and a *decreasing* filtration F) and an isomorphism $\alpha\colon (K_{\mathbb{C}}, W) \simeq (K_{A\otimes\mathbb{Q}}, W) \otimes \mathbb{C}$ in $\mathrm{D}^+ F(X, \mathbb{C})$.

The following axiom must be satisfied:

Axiom CHMC. For all n, the system consisting of the complex $\mathrm{Gr}_n^W(K_{A\otimes\mathbb{Q}}) \in \mathrm{Ob}\,\mathrm{D}^+(X, A\otimes\mathbb{Q})$, the complex $\mathrm{Gr}_n^W(K_{\mathbb{C}}) \in \mathrm{Ob}\,\mathrm{D}^+F(X,\mathbb{C})$ filtered by F, and the isomorphism

$$\mathrm{Gr}_n^W(\alpha)\colon\ \mathrm{Gr}_n^W(K_{A\otimes\mathbb{Q}})\otimes\mathbb{C}\simeq\mathrm{Gr}_n^W(K_{\mathbb{C}}),$$

is a cohomological Hodge $(A\otimes\mathbb{Q})$-complex of weight n.

We can trivially show:

Proposition 8.1.7. *If $K=(K_A,(K_{A\otimes\mathbb{Q}},W),(K_{\mathbb{C}},W,F))$ is a cohomological mixed Hodge A-complex, then $\mathrm{R}\Gamma K=(\mathrm{R}\Gamma K_A,\mathrm{R}\Gamma(K_{A\otimes\mathbb{Q}},W),\mathrm{R}\Gamma(K_{\mathbb{C}},W,F))$ is a mixed Hodge A-complex.*

Scholium 8.1.8. Let U be the complement in a smooth proper scheme X, of a normal crossing divisor D, with $j\colon U\hookrightarrow X$ the inclusion morphism. Let W be the canonical filtration on $\mathrm{R}j_*\mathbb{Q}=\mathrm{R}j_*\mathbb{Z}\otimes\mathbb{Q}$: $W_n(\mathrm{R}j_*\mathbb{Q})=\tau_{\leqslant n}(\mathrm{R}j_*\mathbb{Q})$. We have (Proposition 3.1.8)

$$(\mathrm{R}j_*\mathbb{Q})\otimes\mathbb{C}\xrightarrow{\approx}\mathrm{R}j_*\mathbb{C}\xrightarrow{\approx}j_*\Omega_U^\bullet\xleftarrow{\approx}\Omega_X^\bullet(\log D).$$

Let W be the weight filtration of $\Omega_+X^\bullet(\log D)$, and F its Hodge filtration (by the $\sigma_{\geqslant p}$). Then Proposition 3.1.8 gives an isomorphism $(\mathrm{R}j_*\mathbb{Q},W)\otimes\mathbb{C}\simeq(\Omega_X^\bullet(\log D),W)$ in $\mathrm{D}^+F(X,\mathbb{C})$, and, by §3.1 and Proposition 3.1.9,

$$(\mathrm{R}j_*\mathbb{Z},(\mathrm{R}j_*\mathbb{Q},W),(\Omega_X^\bullet(\log D),W,F))$$

is a cohomological mixed Hodge complex on X.

The following result is an "abstract" version of §3.2.

Scholium 8.1.9. Let K be a mixed Hodge A-complex.

(i) On the pages ${}_WE_r^{pq}$ of the spectral sequence of $(K_{\mathbb{C}},W)$, the recurrent filtration and the two direct filtrations defined by F all coincide.

(ii) The filtration $W[n]$ (Definition 1.1.2) on $\mathrm{H}^n(K_{A\otimes\mathbb{Q}})\simeq\mathrm{H}^n(K_A)\otimes\mathbb{Q}$ and the filtration F on $\mathrm{H}^n(K_{\mathbb{C}})\simeq\mathrm{H}^n(K_A)\otimes_A\mathbb{C}$ define a mixed Hodge A-structure on $\mathrm{H}^n(K_A)$ (Definition 2.3.1 along with §6).

(iii) The morphisms ${}_Wd_1\colon {}_WE_1^{pq}\to{}_WE_1^{p+1,q}$ are compatible with the Hodge bigrading; in particular, they are strictly compatible with the Hodge filtration.

(iv) The spectral sequence of $(K_{A\otimes\mathbb{Q}},W)$ degenerates at E_2 (i.e. ${}_WE_2={}_WE_\infty$).

(v) The spectral sequence of $(K_{\mathbb{C}},F)$ degenerates at E_1 (i.e. ${}_FE_2={}_FE_\infty$).

(vi) The spectral sequence of the complex $\mathrm{Gr}_F^p(K_{\mathbb{C}})$, endowed with the filtration induced by W, degenerates at E_2.

Proof. The proof of (i), (ii), (iii), (iv) is parallel to that of Theorem 3.2.5. The analogy of Lemma 3.2.6 and Lemma 3.2.7 has been taken as an axiom (Axiom CH1 and Axiom CH2). The proofs of Lemma 3.2.8, Lemma 3.2.9, and Lemma 3.2.10 can be carried over as they are, and, as in Lemma 3.2.10, we thus deduce (i), (ii), (iii), and (iv). Claims (v) and (vi) then follow from Proposition 7.2.8. □

We abbreviate "differential graded" to DG, and "bounded-below differential graded" to DG+. A DG complex of A-modules can be seen as a double complex of A-modules; the *first* degree is that of the complex, and the *second* degree is that defined by the grading of DG modules.

We denote by $\mathrm{D}^+(A-\mathsf{mod}_{\mathsf{DG}^+})$ the derived category of the category of bounded-below complexes of DG A-modules, with degrees uniformly bounded below.

A *DG mixed Hodge A-complex* K consists of:

(a) a complex $K_A \in \operatorname{Ob}\mathrm{D}^+(A-\mathsf{mod}_{\mathsf{DG}^+})$;

(b) a filtered complex $(K_{A\otimes\mathbb{Q}}, W) \in \operatorname{Ob}\mathrm{D}^+F((A\otimes\mathbb{Q})-\mathsf{mod}_{\mathsf{DG}^+})$, and is isomorphism $K_A\otimes\mathbb{Q}\simeq K_{A\otimes\mathbb{Q}}$ in D^+;

(c) a bifiltered complex $(K_\mathbb{C}, F, W) \in \operatorname{Ob}\mathrm{D}^+F_2(\mathbb{C}-\mathsf{mod}_{\mathsf{DG}^+})$, and an isomorphism $K_{A\otimes\mathbb{Q}}\otimes\mathbb{C}\simeq K_\mathbb{C}$ in D^+F.

We require that, for each n, the component $(K^{\bullet n}, F, W)$ of K whose second degree is n be a mixed Hodge A-complex.

We denote by L the filtration by the second degree of $\mathrm{s}K$. We denote by $\operatorname{Dec}_1 W$ the filtration of $K_{A\otimes\mathbb{Q}}$ induced by shifting (§1.3) W. We will not confuse the filtration that this induces on $\mathrm{s}K_{A\otimes\mathbb{Q}}$ (also denoted by $\operatorname{Dec}_1 W$) with the shifted filtration of the filtration induced on $\mathrm{s}K_{A\otimes\mathbb{Q}}$ by W.

Variant 8.1.10.1. We similarly define *cosimplicial* (resp. *r-cosimplicial*) *mixed Hodge A-complexes* by replacing "DG" with "cosimplicial" (resp. r-cosimplicial) everywhere in the definition. The functor from cosimplicial A-modules to DG A-modules sends cosimplicial mixed Hodge A-complexes to DG mixed Hodge A-complexes (and similarly for r-cosimplicial).

Let $X_\bullet$ be a simplicial topological space. A *cohomological mixed Hodge A-complex* K on $X_\bullet$ consists of:

(α) a complex $K_A \in \operatorname{Ob}\mathrm{D}^+(X_\bullet, A)$;

(β) a filtered complex $(K_{A\otimes\mathbb{Q}}, W) \in \operatorname{Ob}\mathrm{D}^+F(X_\bullet, A\otimes\mathbb{Q})$, and an isomorphism $K_{A\otimes\mathbb{Q}}\simeq K_A\otimes\mathbb{Q}$ in $\mathrm{D}^+(X_\bullet, A\otimes\mathbb{Q})$;

(γ) a bifiltered complex $(K_\mathbb{C}, W, F) \in \operatorname{Ob}\mathrm{D}^+F_2(X_\bullet, \mathbb{C})$, and an isomorphism $(K_\mathbb{C}, W)\simeq(K_{A\otimes\mathbb{Q}}, W)\otimes\mathbb{C}$ in $\mathrm{D}^+F(X_\bullet, \mathbb{C})$.

The following axiom has to be satisfied:

Axiom CHM•. The restriction of K to each of the X_n is a cohomological mixed Hodge A-complex.

For $A=\mathbb{Z}$, we simply speak of a cohomological mixed Hodge complex.

Example 8.1.12. Let $X_\bullet$ be a smooth proper scheme (over $\mathbb{C}$), and let $Y_\bullet$ be a normal crossing divisor in $X_\bullet$. We set $U_\bullet = X_\bullet \setminus Y_\bullet$, and we denote by j the inclusion of $U_\bullet$ into $X_\bullet$. The constructions in Scholium 8.1.8 can be performed "uniformly in X_n", as §7.2 and Lemma 6.2.7 show.

They thus give a cohomological mixed Hodge complex

$$\big(\mathrm{R}j_*\mathbb{Z}, (\mathrm{R}j_*\mathbb{Q}, \tau_\leqslant), (\Omega^\bullet_{X_\bullet}(\log Y_\bullet), W, F)\big)$$

on $X_\bullet$.

Let K be a cohomological mixed Hodge A-complex on $X_\bullet$. Write $A-\mathsf{mod}_{\mathsf{cosimp}}$ to mean the category of cosimplicial A-modules. If we apply the functor $\mathrm{R}\Gamma^\bullet$ to K, then we obtain

(a) a cosimplicial complex $\mathrm{R}\Gamma^\bullet K_A \in \mathrm{Ob}\,\mathrm{D}^+(A{-}\mathsf{mod}_{\mathsf{cosimp}})$;

(b) a cosimplicial filtered complex $\mathrm{R}\Gamma^\bullet(K_{A\otimes\mathbb{Q}}, W) \in \mathrm{Ob}\,\mathrm{D}^+F(A\otimes\mathbb{Q}{-}\mathsf{mod}_{\mathsf{cosimp}})$;

(c) a cosimplicial bifiltered complex $\mathrm{R}\Gamma^\bullet(K_{\mathbb{C}}, W, F) \in \mathrm{Ob}\,\mathrm{D}^+F_2(\mathbb{C}{-}\mathsf{mod}_{\mathsf{cosimp}})$;

(d) isomorphisms as in (b) and (c) of §10.1 between these objects.

It is clear that the $\mathrm{R}\Gamma^\bullet(K, W, F)$ described above is a cosimplicial mixed Hodge A-module. It defines a DG mixed Hodge A-complex (Variant 8.1.10.1).

Let K be a DG mixed Hodge A-complex, and consider the spectral sequence abutting to the cohomology of $\mathrm{s}K$, which is defined by the filtration L. By hypothesis, the E_1 pages of this spectral sequence are endowed with mixed Hodge A-structures. The content of Theorem 8.1.15 is that every spectral sequence, including its abutment, is endowed with a mixed Hodge A-structure. In other words, c' is a spectral sequence in the abelian category of mixed Hodge A-structures. The structure on the abutment corresponds to a natural mixed Hodge A-complex "structure" on $\mathrm{s}K$.

Theorem 8.1.15. *Let K be a DG mixed Hodge A-complex.*

(i) With the notation of §10.1, Variant 8.1.10.1, and §9.1, $(\mathrm{s}K, \delta(W, L), F)$ is a mixed Hodge A-complex.

(ii) We have

$$ {}_{\delta(W,L)}E_1^{ab}(\mathrm{s}K\otimes\mathbb{Q}) = \sum_{\substack{\alpha=\gamma-\beta \\ b=\alpha+\beta}} \mathrm{H}^\alpha(\mathrm{Gr}^W_\beta\, K^{\bullet\gamma}) \tag{8.1.15.1} $$

where the complex $({}_{\delta(W,L)}E_1^{\bullet b}, d_1)$ is the simple complex associated to the double complex of Hodge $A\otimes\mathbb{Q}$-structures of weight b given by

$$ \begin{array}{ccccc} \mathrm{H}^{b-(\beta+1)}(\mathrm{Gr}^W_{\beta+1} K^{\bullet,\gamma+1}) & \xrightarrow{\partial} & \mathrm{H}^{b-\beta}(\mathrm{Gr}^W_{\beta} K^{\bullet,\gamma+1}) & \xrightarrow{\partial} & \mathrm{H}^{b-(\beta-1)}(\mathrm{Gr}^W_{\beta-1} K^{\bullet,\gamma+1}) \\ & & {\scriptstyle d''}\uparrow & & \uparrow{\scriptstyle d''} \\ & & \mathrm{H}^{b-\beta}(\mathrm{Gr}^W_{\beta} K^{\bullet,\gamma}) & \xrightarrow[\partial]{} & \mathrm{H}^{b-(\beta-1)}(\mathrm{Gr}^W_{\beta-1} K^{\bullet,\gamma}) \end{array} \tag{8.1.15.2} $$

(iii) On the ${}_LE_r$ pages of the spectral sequence defined by $(\mathrm{s}K_{A\otimes\mathbb{Q}}, L)$, the recurrent filtration and the two direct filtrations defined by $\mathrm{Dec}_1(W)$ agree; similarly, on the ${}_LE_r(\mathrm{s}K_{\mathbb{C}})$, the recurrent filtration and the two direct filtrations induced by F agree. We denoted these filtrations again by W and F (respectively).

(iv) For $r \geqslant 1$, $({}_LE_r, W, F)$ is a mixed Hodge A-structure (§6). The differentials d_r are morphisms of mixed Hodge A-structures.

(v) The filtration L on $\mathrm{H}^\bullet(\mathrm{s}K)$ is a filtration in the category of mixed Hodge structures, and

$$ \mathrm{Gr}^p_L\left(\mathrm{H}^{p+q}(\mathrm{s}K), \delta(W, L)[p+q], F\right) = ({}_LE_\infty^{pq}, W, F). \tag{8.1.15.3} $$

Remark 8.1.16. In Theorem 8.1.15, the filtration $\mathrm{Dec}_1(W)$ (§10.1) plays a more important role than W. We have

$$\mathrm{Dec}(\delta(W,L)) = \mathrm{Dec}_1(W) \qquad \text{on } \mathrm{s}K \tag{8.1.16.1}$$

and Equation 8.1.15.3 can be rewritten as

$$\mathrm{Gr}_L(\mathrm{H}(K), \mathrm{Dec}_1(W,F)) = ({}_LE_\infty, W, F). \tag{8.1.16.2}$$

In the same spirit, we note that, with the notation of Scholium 8.1.8,

$$\mathrm{R}\Gamma\big(\mathrm{R}j_*\mathbb{Z}, (\mathrm{R}j_*\mathbb{Q}, \mathrm{Dec}\, W), (\Omega_X^\bullet(\log D), \mathrm{Dec}\, W, F)\big)$$

depends only on U, and not on the choice of compactification X.

We now prove Theorem 8.1.15. Claims (i) and (ii) follow from the fact that

$$\begin{aligned}\mathrm{Gr}_n^{\delta(W,L)}(\mathrm{s}K \otimes \mathbb{Q}) &= \sum_{p-q=n} \mathrm{Gr}_p^W \mathrm{Gr}_L^q(\mathrm{s}K \otimes \mathbb{Q}) \\ &= \sum_{p-q=n} (\mathrm{Gr}_p^W K^{\bullet q})[-q]\end{aligned}$$

(this isomorphism is compatible with F, and we apply Remark 8.1.4). The proof of Equation 8.1.15.2 is left to the reader.

Consider the claims:

($\mathrm{a_r}$) The morphisms d_r of ${}_LE_r(\mathrm{s}K_{A\otimes\mathbb{Q}})$ are strictly compatible with the recurrent filtration defined by $\mathrm{Dec}_1(W)$.

($\mathrm{b_r}$) The morphisms d_r of ${}_LE_r(\mathrm{s}K_{\mathbb{C}})$ are strictly compatible with the recurrent filtration defined by F.

($\mathrm{c_r}$) ${}_LE_r$, endowed with the filtrations considered in (a) and (b) is a mixed Hodge A-structure.

($\mathrm{d_r}$) The d_r are morphisms of mixed Hodge A-structures.

We will prove (a_r) and (b_r) for $r \geqslant 0$, and (c_r) and (d_r) for $r \geqslant 1$, by simultaneous induction.

(A) *Proof of (a_0).* We have

$$\big({}_LE_0^{p\bullet}(\mathrm{s}K_{A\otimes\mathbb{Q}}), \mathrm{Dec}_1 W\big) = (K_{A\otimes\mathbb{Q}}^{\bullet p}, \mathrm{Dec}\, W).$$

By (iv) of Scholium 8.1.9, the spectral sequence of $(K_{A\otimes\mathbb{Q}}^{\bullet p}, W)$ degenerates at E_2. By Proposition 1.3.4, the spectral sequence of $(K_{A\otimes\mathbb{Q}}^{\bullet p}, \mathrm{Dec}\, W)$ degenerates at E_1, and this proves (a_0) (Proposition 1.3.2).

(B) *Proof of (b_0).* We have

$$\big({}_LE_0^{p\bullet}(\mathrm{s}K_{\mathbb{C}}), F\big) = (K_{\mathbb{C}}^{\bullet p}, F).$$

We apply (v) of Scholium 8.1.9 and Proposition 1.3.2.

(C) *Proof of (c$_1$).* We have to show that $\mathrm{H}^n(K_A^{\bullet p})$, endowed with the filtrations induced by $\operatorname{Dec} W$ on $\mathrm{H}^n(K_{A\otimes\mathbb{Q}}^{\bullet p}) = \mathrm{H}^n(K_A^{\bullet p})\otimes\mathbb{Q}$ and by F on $\mathrm{H}^n(K_{\mathbb{C}}^{\bullet p}) = \mathrm{H}^n(K_A^{\bullet p})\otimes_A\mathbb{C}$, is a mixed Hodge A-structure. But $\operatorname{Dec} W$ induces on $\mathrm{H}^n(K_{A\otimes\mathbb{Q}}^{\bullet p})$ the same filtration as $W[n]$, and so we conclude by (ii) of Scholium 8.1.9.

(D) *Proof of (c$_r$)+(d$_r$) $\Longrightarrow$ (a$_r$)+(b$_r$)+(c$_{r+1}$).* This is the key point of the proof, and is a simple application of Theorem 2.3.5 (along with §6).

(E) *Proof of (a$_r$)$_{r\leqslant r_0-2}$+(b$_r$)$_{r\leqslant r_0-2}$+(c$_{r_0}$) $\Longrightarrow$ (d$_{r_0}$).* By Theorem 1.3.16, the recurrent filtration and the direct filtrations induced by $\operatorname{Dec}_1 W$ (resp. F) on ${}_LE_r$ agree for $r\leqslant r_0$, and we apply (i) of Proposition 1.3.13.

This finishes the proof by induction of (a), (b), (c), and (d).

Claim (iii) was proven in part (E) of the proof by induction, and the claims of (iv) are (c$_r$) and (d$_r$). We will prove Equation 8.1.15.3 in the form of Equation 8.1.16.2.

The fact that

$$\operatorname{Gr}_L(\mathrm{H}(\mathrm{s}K_{A\otimes\mathbb{Q}}), \operatorname{Dec}_1 W) = ({}_LE_\infty(\mathrm{s}K_{A\otimes\mathbb{Q}}), W)$$

follows from (a$_r$) and Corollary 1.3.17. The fact that

$$\operatorname{Gr}_L(\mathrm{H}(\mathrm{s}K_{\mathbb{C}}), F) = ({}_LE_\infty(\mathrm{s}K_{\mathbb{C}}), F)$$

follows from (b$_r$) and Corollary 1.3.17.

We conclude by the following lemma, whose proof is left to the reader:

Lemma 8.1.18. *Let $H = (H_A, W, F)$ be a mixed Hodge A-structure. For a filtration L of H_A to come from a filtration of H, in the abelian category of mixed Hodge A-structures, it is necessary and sufficient that, for all n, $(\operatorname{Gr}_L^n(H_A), \operatorname{Gr}_L^n(W), \operatorname{Gr}_L^n(F))$ be a mixed Hodge A-structure.*

We now apply §10.1 and Theorem 8.1.15 to Example 8.1.12.

We obtain a mixed Hodge complex consisting of (a), (b), and (c) below.

(a) The complex $\mathrm{R}\Gamma\mathrm{R}j_*\mathbb{Z}\simeq\mathrm{R}\Gamma(U_\bullet,\mathbb{Z})$, whose cohomology groups are the $\mathrm{H}^n(U_\bullet,\mathbb{Z})$.

(b) The filtration $\delta(W,L)$ of $\mathrm{R}\Gamma\mathrm{R}j_*\mathbb{Q}\simeq\mathrm{R}\Gamma(U_\bullet,\mathbb{Q})$. With the notation of §3.1, we have

$$\begin{aligned}\operatorname{Gr}_n^{\delta(W,L)}\mathrm{R}\Gamma(U_\bullet,\mathbb{Q}) &\underset{(7.1.6.5)}{\simeq} \bigoplus_m \operatorname{Gr}_{n+m}^W \mathrm{R}\Gamma(U_m,\mathbb{Q})[-m]\\ &\simeq \bigoplus_m \mathrm{R}\Gamma(\widetilde{Y}_m^{n+m}, \varepsilon_{\mathbb{Q}}^{n+m}[-n-m])[-m]\\ &= \bigoplus_m \mathrm{R}\Gamma(\widetilde{Y}_m^{n+m}, \varepsilon_{\mathbb{Q}}^{n+m})[-n-2m].\end{aligned}$$

The E_1 page of the corresponding spectral sequence is the sum of the groups of form $\mathrm{H}^p(\widetilde{Y}_q^r,\varepsilon_{\mathbb{Q}}^r)$; this contributes to the $\mathrm{H}^{p+q+r}(U_\bullet)$; it corresponds to $\operatorname{Gr}_a^{\delta(W,F)}$ for $p+2r = (p+q+r)+a$:

$$ {}_{\delta(W,L)}E_1^{-a,b} = \bigoplus_{\substack{p+2r=b\\ q-r=-a}} \mathrm{H}^p(\widetilde{Y}_q^r,\varepsilon_{\mathbb{Q}}^r) \Rightarrow \mathrm{H}^{-a+b}(U_\bullet,\mathbb{Q}). \tag{8.1.19.1}$$

(c) The bifiltration $(\delta(W,L),F)$ of $\mathrm{R}\Gamma\mathrm{R}j_*\mathbb{C}\simeq\mathrm{s}\mathrm{R}\Gamma^\bullet\Omega_X^\bullet(\log D)$.

The filtration F endows ${}_{\delta(W,L)}E_1^{-a,b}$ with a Hodge structure of weight b, and the differentials d_1 of ${}_{\delta(W,L)}E$ are the morphisms of the Hodge $\mathbb{Q}$-structures. The complex ${}_{\delta(W,L)}E_q^{\bullet b}$ is the simple complex associated the double complex of Hodge structures of weight b given by:

$$\begin{tikzcd}
\mathrm{H}^{b-2(r+2)}(\widetilde{Y}_{q+1}^{r+1},\varepsilon_{\mathbb{Q}}^{r+1})[-r-1] \arrow[r, "\text{Gysin}"] & \mathrm{H}^{b-2r}(\widetilde{Y}_{q+1}^{r},\varepsilon_{\mathbb{Q}}^{r})[-r] \arrow[r, "\text{Gysin}"] & \mathrm{H}^{b-2(r-2)}(\widetilde{Y}_{q+1}^{r-1},\varepsilon_{\mathbb{Q}}^{r-1})[-r+1] \\
& \mathrm{H}^{b-2r}(\widetilde{Y}_{q}^{r},\varepsilon_{\mathbb{Q}}^{r})[-r] \arrow[u, "\sum_i(-1)^i\delta_i"] \arrow[r, "\text{Gysin}"'] & \mathrm{H}^{b-2(r-2)}(\widetilde{Y}_{q}^{r-1},\varepsilon_{\mathbb{Q}}^{r-1})[-r+1] \arrow[u, "\sum_i(-1)^i\delta_i"']
\end{tikzcd}$$

The indices in the terms in E_1^{ab} satisfy $-a = q - r$.

In particular, the $\mathrm{H}^n(U_\bullet,\mathbb{Z})$ are endowed with mixed Hodge structures (Scholium 8.1.9).

Proposition 8.1.20. *Under the above hypotheses:*

(i) The mixed Hodge structure of $\mathrm{H}^n(U_\bullet,\mathbb{Z})$ *is functorial in the pair* $(U_\bullet,X_\bullet)$.

(ii) In rational cohomology, the spectral sequence Equation 8.1.19.1 degenerates at E_2 *(i.e.* $E_2 = E_\infty$*) and abuts to the weight filtration of* $\mathrm{H}^n(U_\bullet,\mathbb{Q})$. *The Hodge structure of* E_2 *induced by that of* E_1 *is also that of* $\mathrm{Gr}_W \mathrm{H}^n(U_\bullet,\mathbb{Q})$.

(iii) The Hodge numbers of $\mathrm{H}^n(U_\bullet,\mathbb{Z})$ *satisfy*

$$h^{pq} \neq 0 \implies 0 \leqslant p,q \leqslant n.$$

(iv) For $Y_\bullet = \varnothing$, *the Hodge numbers of* $\mathrm{H}^n(X_\bullet,\mathbb{Z})$ *satisfy*

$$h^{pq} \neq 0 \implies [0 \leqslant p,q \leqslant n \quad \textit{and} \quad p+q \leqslant n].$$

Proof. Both (i) and (ii) follow from the general theory (Scholium 8.1.9). To prove (iii), we contemplate §10.1 and note that the Hodge structures $\mathrm{H}^p(\widetilde{Y}_q^r,\varepsilon_{\mathbb{Q}}^r)[-r]$ (for $p+q+r=n$) satisfy (iii). To prove (iv), we note that the Hodge structures $\mathrm{H}^p(X_q)$ (for $p+q=n$) satisfy (iv). ☐

We define a *cohomological mixed Hodge A-complex on an r-simplicial topological space* by transposing §10.1. If K is such a complex, then we see, as in §10.1, that $\mathrm{R}\Gamma^\bullet K$ is an r-cosimplicial mixed Hodge A-complex (Variant 8.1.10.1).

Let $X_\bullet$ be a smooth and proper r-simplicial scheme. A normal crossing divisor $D_\bullet$ on $X_\bullet$ is a system of normal crossing divisors on the X_n such that $X_\bullet \setminus D_\bullet = U_\bullet$ is an r-simplicial subscheme of $X_\bullet$. Let $j\colon U_\bullet \hookrightarrow X_\bullet$. We define, as in Example 8.1.12, a cohomological mixed Hodge complex $\mathrm{R}j_*\mathbb{Z}$ on $X_\bullet$.

We can, as in Proposition 8.1.20, thus deduce a mixed Hodge structure on $\mathrm{H}^\bullet(U_\bullet,\mathbb{Z})$, but, since

$$\mathrm{H}^\bullet(U_\bullet,\mathbb{Z}) = \mathrm{H}^\bullet(\delta U_\bullet,\mathbb{Z}),$$

this does not teach us anything new (the above equality would be an identity between mixed Hodge structures).

Set $r=2$. For every bounded-below bicosimplicial complex $K = (K^{pn_1n_2})$ (where p is the differential degree, and n_i is the i-th simplicial degree), we denote by s_1K the cosimplicial complex of simplicial degree n_2 given by

$$(\mathrm{s}_1K)^{\bullet n_2} = \mathrm{s}K^{\bullet\bullet n_2}.$$

We denote by L_1 the filtration on s_1K that, on each $\mathrm{s}K^{\bullet\bullet n_2}$, induces the filtration by the simplicial degree n_1.

Let K be a bicosimplicial mixed Hodge A-complex. It follows from Theorem 8.1.15 that $(\mathrm{s}_1K, \delta(W, L_1), F)$ is a cosimplicial mixed Hodge complex. Let L_2 be the filtration of $\mathrm{s}K = \mathrm{ss}_1K$ by the simplicial degree n_2. By §10.1 and Theorem 8.1.15, the spectral sequence defined by L_2 comes from a spectral sequence in the abelian category of mixed Hodge A-structures

$$ {}_{L_2}E_1^{pq} = \mathrm{H}^q(\mathrm{s}K^{\bullet\bullet p}) \Rightarrow \mathrm{H}^{p+q}(\mathrm{s}K). \tag{8.1.22.1} $$

In this spectral sequence, the mixed Hodge A-structure on E_1^{pq} is the mixed Hodge A-structure ((ii) of Scholium 8.1.9) of the cohomology of the mixed Hodge A-complex $(\mathrm{s}K^{\bullet\bullet p}, \delta(W, L_1), F)$. The mixed Hodge A-structure of the abutment is the mixed Hodge A-structure of the cohomology of the mixed Hodge A-complex $(\mathrm{s}K, \delta(\delta(W, L_1), L_2), F)$. The filtration $\delta(\delta(W, L_1), L_2) =: \delta(W, L_1, L_2)$ is given by

$$ \delta(W, L_1, L_2)_k(K^{pn_1n_2}) = W_{k+n_1+n_2}K^{pn_1n_2}. $$

In particular, a cohomological mixed Hodge A-complex on a bisimplicial topological space $X_{\bullet\bullet}$ defines a spectral sequence of mixed Hodge A-structures

$$ E_1^{pq} = \mathrm{H}^q(X_{\bullet p}, K) \Rightarrow \mathrm{H}^{p+q}(X_{\bullet\bullet}, K). \tag{8.1.23.1} $$

In the particular case considered in §10.1, for $r = 2$, we have $A = \mathbb{Z}$, and Equation 8.1.23.1 can be written as

$$ E_1^{pq} = \mathrm{H}^q(U_{\bullet p}, \mathbb{Z}) \Rightarrow \mathrm{H}^{p+q}(U_{\bullet\bullet}, \mathbb{Z}). \tag{8.1.23.2} $$

If K' and K'' are mixed Hodge A-complexes, then then their product $K' \otimes K''$, defined by

$$ \begin{aligned} (K' \otimes K'')_A &= K'_A \overset{\mathrm{L}}{\otimes} K''_A \\ ((K' \otimes K'')_{A\otimes\mathbb{Q}}, W) &= (K'_{A\otimes\mathbb{Q}}, W') \otimes (K''_{A\otimes\mathbb{Q}}, W'') \\ ((K' \otimes K'')_{\mathbb{C}}, W, F) &= (K'_{\mathbb{C}}, W', F') \otimes (K''_{\mathbb{C}}, W'', F'') \end{aligned} $$

is again a mixed Hodge A-complex.

Let U' (resp. U'') be the complement of a normal crossing divisor Y' (resp. Y'') in a smooth and proper scheme X' (resp. X''). Then $U = U' \times U''$ is the complement of a normal crossing divisor Y in $X = X' \times X''$. Let j (resp. j', j'') be the inclusion of U (resp. U', U'') into X (resp. X', X'').

We have a quasi-isomorphism

$$ \mathrm{R}j'_*\mathbb{Z} \overset{\mathrm{L}}{\boxtimes} \mathrm{R}j''_*\mathbb{Z} := \mathrm{pr}_1^* \mathrm{R}j'_*\mathbb{Z} \overset{\mathrm{L}}{\otimes} \mathrm{pr}_2^* \mathrm{R}j''_*\mathbb{Z} \xrightarrow{\sim} \mathrm{R}j_*\mathbb{Z}, $$

a filtered quasi-isomorphism

$$ (\mathrm{R}j'_*\mathbb{Q}, \tau_\leqslant) \overset{\mathrm{L}}{\boxtimes} (\mathrm{R}j''_*\mathbb{Q}, \tau_\leqslant) \xrightarrow{\sim} (\mathrm{R}j_*\mathbb{Q}, \tau_\leqslant), $$

and a bifiltered morphism

$$ (\Omega^\bullet_{X'}(\log Y'), W, F) \boxtimes (\Omega_{X''}(\log Y''), W, F) \to (\Omega^\bullet_X(\log Y), W, F). $$

Applying the functor $\mathrm{R}\Gamma$ gives us, by the Künneth formula, and by its analogue for cohomology of coherent analytic sheaves, an isomorphism of mixed Hodge complexes

$$\mathrm{R}\Gamma(U', \mathbb{Z}) \overset{\mathrm{L}}{\otimes} \mathrm{R}\Gamma(U'', \mathbb{Z}) \xrightarrow{\sim} \mathrm{R}\Gamma(U, \mathbb{Z})$$

whence an isomorphism of mixed Hodge $\mathbb{Q}$-structures

$$\mathrm{H}^\bullet(U', \mathbb{Q}) \otimes \mathrm{H}^\bullet(U'', \mathbb{Q}) \xrightarrow{\sim} \mathrm{H}^\bullet(U, \mathbb{Q}).$$

Standard arguments, based on the Cartier–Eilenberg–Zilber theorem, give us the following simplicial variant of §10.1.

Let $U'_\bullet$ (resp. $U''_\bullet$) be the complement of a normal crossing divisor in a smooth and proper simplicial scheme $X'_\bullet$ (resp. $X''_\bullet$). Let $U_\bullet = U'_\bullet \times U''_\bullet$ and $X_\bullet = X'_\bullet \times X''_\bullet$.

We have an isomorphism of mixed Hodge complexes

$$\mathrm{R}\Gamma(U'_\bullet, \mathbb{Z}) \overset{\mathrm{L}}{\otimes} \mathrm{R}\Gamma(U''_\bullet, \mathbb{Z}) \xrightarrow{\sim} \mathrm{R}\Gamma(U_\bullet, \mathbb{Z})$$

and, in particular, an isomorphism of mixed Hodge $\mathbb{Q}$-structures

$$\mathrm{H}^\bullet(U'_\bullet, \mathbb{Q}) \otimes \mathrm{H}^\bullet(U''_\bullet, \mathbb{Q}) \xrightarrow{\sim} \mathrm{H}^\bullet(U_\bullet, \mathbb{Q}).$$

Let $U'_{\bullet\bullet}$ (resp. $U''_{\bullet\bullet}$) be the complement of a normal crossing divisor in a smooth and proper bisimplicial scheme $X'_{\bullet\bullet}$ (resp. $X''_{\bullet\bullet}$). Let $U_{\bullet\bullet} = U'_{\bullet\bullet} \times U''_{\bullet\bullet}$ and $X_{\bullet\bullet} = X'_{\bullet\bullet} \times X''_{\bullet\bullet}$.

Standard arguments show that, in rational cohomology, the spectral sequence Equation 8.1.23.2 for $U_{\bullet\bullet}$ is the tensor product of the spectral sequences Equation 8.1.23.2 for $U'_{\bullet\bullet}$ and $U''_{\bullet\bullet}$.

10.2. Separated algebraic spaces

Let X be a scheme (or an algebraic space) of finite type over $\mathbb{C}$, and assume it to be separated. Then there exists a diagram

$$X \xleftarrow{\alpha} Y_\bullet \hookrightarrow \overline{Y}_\bullet \tag{8.2.1.1}$$

in which $Y_\bullet$ is the complement of a normal crossing divisor in a smooth and proper simplicial scheme $\overline{Y}_\bullet$, and is a proper hypercover of X. We have ((II) of §7.3)

$$\mathrm{H}^\bullet(X) \xrightarrow{\sim} \mathrm{H}^\bullet(Y_\bullet). \tag{8.2.1.2}$$

In §10.1, we endow $\mathrm{H}^\bullet(Y_\bullet)$ with a mixed Hodge structure. Let $\mathscr{H}(\alpha)$ be the mixed Hodge structure on $\mathrm{H}^\bullet(X)$ induced by Equation 8.2.1.2.

Proposition 8.2.2. *With the above notation, the mixed Hodge structure $\mathscr{H}(\alpha)$ on $\mathrm{H}^\bullet(X)$ is independent of the choice of α. We call it the* mixed Hodge structure of $\mathrm{H}^\bullet(X, \mathbb{Z})$. *For every morphism $f\colon X_1 \to X_2$ of schemes, $f^*\colon \mathrm{H}^\bullet(X_2) \to \mathrm{H}^\bullet(X_1)$ is a morphism of mixed Hodge structures.*

Proof. The proof uses §8.2, and is parallel to that of (C) of §3.2 (cf. also §10.3). □

For smooth X, we can show, by taking $\overline{Y}_\bullet$ to be a constant simplicial scheme, that the mixed Hodge structure in Proposition 8.2.2 coincides with that defined in Definition 3.2.12.

Theorem 8.2.4. *The pairs (p,q) such that the Hodge number h^{pq} of $\mathrm{H}^n(X)$ are non-zero satisfy the following conditions:*

(i) $(p,q) \in [0,n] \times [0,n]$.

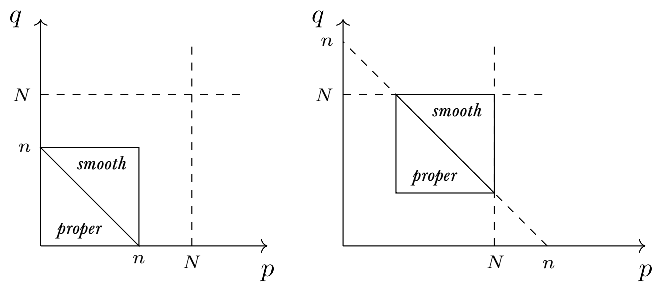
(ii) If $\dim X = N$*, and* $n \geqslant N$*, then*

$$(p,q) \in [n-N,N] \times [n-N,N].$$

(iii) If X is proper, then $p+q \leqslant n$.

(iv) If X is smooth, then $p+q \geqslant n$. The same conclusion holds if X is an equidimensional "rational homology manifold" of dimension N, i.e. if, for all $x \in X$,

$$\mathrm{H}^i_{(x)}(X,\mathbb{Q}) = \begin{cases} 0 & \text{if } i \neq 2N \\ \mathbb{Q} & \text{if } i = 2N. \end{cases}$$



Proof. Claim (ii) will be proven in §10.3. Claim (i) follows from (iii) of Proposition 8.1.20. For proper X, we can find a diagram as in Equation 8.2.1.1 in which $Y_\bullet = \overline{Y}_\bullet$, and (iii) then follows from (iv) of Proposition 8.1.20. For smooth X, (iv) follows from §10.2 and Corollary 3.2.15. The general case then follows: if $p\colon \widetilde{X} \to X$ is a resolution of singularities of X, then $p^*\colon \mathrm{H}^\bullet(X,\mathbb{Q}) \to \mathrm{H}^\bullet(\widetilde{X},\mathbb{Q})$ is injective, since the Poincaré dual transpose $p_!$ of $p^*\colon \mathrm{H}^\bullet_\mathrm{c}(X,\mathbb{Q}) \to \mathrm{H}^\bullet_\mathrm{c}(\widetilde{X},\mathbb{Q})$ is a retraction for it. □

Proposition 8.2.5. *Suppose that X is proper. If $u\colon Y \to X$ is a surjective proper morphism, with Y smooth, then the weight-n quotient of $\mathrm{H}^n(X,\mathbb{Q})$ is the image of $\mathrm{H}^n(X,\mathbb{Q})$ in $\mathrm{H}^n(Y,\mathbb{Q})$. The sequence*

$$\mathrm{H}^n(X,\mathbb{Q}) \to \mathrm{H}^n(Y,\mathbb{Q}) \xrightarrow{\mathrm{pr}_1^* - \mathrm{pr}_2^*} \mathrm{H}^n(Y \times_X Y,\mathbb{Q}) \tag{8.2.5.1}$$

is exact.

Proof. Let $Y_\bullet$ be as in Equation 8.2.1.1, with $Y_0 = Y$ and $Y_\bullet = \overline{Y}_\bullet$. By (ii) of Proposition 8.1.20, we have an exact sequence

$$0 \to \mathrm{H}^n(X,\mathbb{Q})/W^{n-1}\,\mathrm{H}^n(X,\mathbb{Q}) \to \mathrm{H}^n(Y_0,\mathbb{Q}) \underset{\delta_1}{\overset{\delta_0}{\rightrightarrows}} \mathrm{H}^n(Y_1,\mathbb{Q}) \tag{8.2.5.2}$$

which implies Proposition 8.2.5 (conversely, this exact sequence can easily be deduced from Proposition 8.2.5 and Proposition 8.2.7 below). □

Proposition 8.2.6. *Let*

$$Y \xrightarrow{f} X \xhookrightarrow{j} \overline{X}$$

be morphisms of schemes. Suppose that Y is proper and X is smooth, and that $\overline{X}$ is a smooth compactification of X. Then $\mathrm{H}^\bullet(\widetilde{X}, \mathbb{Q})$ and $\mathrm{H}^\bullet(X, \mathbb{Q})$ have the same image in $\mathrm{H}^\bullet(Y, \mathbb{Q})$.

Proof. The proof is parallel to that of Corollary 3.2.18. It suffices to prove, for any i and any n, that $\mathrm{Gr}_i^W(f^*)$ and $\mathrm{Gr}_i^W((jf)^*)$ have the same image in $\mathrm{Gr}_i^W \mathrm{H}^n(Y, \mathbb{Q})$.

(a) For $i > n$, these images are both zero since $\mathrm{Gr}_i^W \mathrm{H}^n(Y, \mathbb{Q}) = 0$ ((iii) of Theorem 8.2.4).

(b) For $i < n$, these images are both zero since $\mathrm{Gr}_i^W \mathrm{H}^n(X, \mathbb{Q}) = 0$ ((iv) of Theorem 8.2.4).

(c) For $i = n$, these images are equal since $\mathrm{Gr}_n^W(j^*)$ is surjective (Corollary 3.2.17).

□

Proposition 8.2.7. *Let*

$$\widetilde{X} \xrightarrow{\pi} X \xrightarrow{f} Y$$

be morphisms of schemes. Suppose that Y is smooth, X is proper, $\widetilde{X}$ is smooth and proper, and that π is surjective. Then the kernels of f^* and $(f\pi)^*$ in $\mathrm{H}^\bullet(Y, \mathbb{Q})$ are equal.

Proof. We will prove that $\mathrm{Ker}(\mathrm{Gr}_i^W(f^*)) = \mathrm{Ker}(\mathrm{Gr}_i^W((f\pi)^*))$ in $\mathrm{Gr}_i^W \mathrm{H}^n(Y, \mathbb{Q})$.

(a) For $i < n$, these kernels are both zero since $\mathrm{Gr}_i^W \mathrm{H}^n(Y, \mathbb{Q}) = 0$ ((iv) of Theorem 8.2.4).

(b) For $i > n$, these kernels are both all of $\mathrm{Gr}_i^W \mathrm{H}^n(Y, \mathbb{Q})$ since $\mathrm{Gr}_i^W \mathrm{H}^n(X, \mathbb{Q}) = 0$ ((iii) of Theorem 8.2.4).

(a) For $i = n$, these kernels are equal since $\mathrm{Gr}_n^W(\pi^*)$ is injective (Equation 8.2.5.2).

□

Corollary 8.2.8. *Let*

$$\widetilde{X} \xrightarrow{\pi} X \xhookrightarrow{f} Y$$

be morphisms of schemes. Suppose that Y is smooth and proper, that X is a closed subscheme of Y, and that $\widetilde{X}$ is a resolution of singularities of X. Let $q = f\pi$ and $U = Y \setminus X$. Then the sequence

$$\mathrm{H}^\bullet(\widetilde{X}, \mathbb{Q}) \xrightarrow{q_!} \mathrm{H}^\bullet(Y, \mathbb{Q}) \to \mathrm{H}^\bullet(U, \mathbb{Q}) \tag{8.2.8.1}$$

is exact.

Proof. By Proposition 8.2.7, we have that

$$\operatorname{Ker}(f^*\colon \mathrm{H}^\bullet(Y,\mathbb{Q}) \to \mathrm{H}^\bullet(X,\mathbb{Q})) = \operatorname{Ker}(q^*\colon \mathrm{H}^\bullet(Y,\mathbb{Q}) \to \mathrm{H}^\bullet(\widetilde{X},\mathbb{Q})).$$

The long exact sequence in cohomology of the pair Y, X thus gives an exact sequence

$$\mathrm{H}^\bullet_c(U,\mathbb{Q}) \to \mathrm{H}^\bullet(Y,\mathbb{Q}) \to \mathrm{H}^\bullet(\widetilde{X},\mathbb{Q}) \tag{8.2.8.2}$$

and Equation 8.2.8.1 is the transpose sequence of Equation 8.2.8.2 under Poincaré duality. □

Remark 8.2.9. Corollary 8.2.8 is used under more general hypotheses in [G1968, (9.7), p. 161]. As Lieberman remarked, the justification given loc. cit. is insufficient.

From §10.1 we deduce the following:

Proposition 8.2.10. *The Künneth isomorphisms*

$$\mathrm{H}^\bullet(X\times Y,\mathbb{Q}) \simeq \mathrm{H}^\bullet(X,\mathbb{Q}) \otimes \mathrm{H}^\bullet(Y,\mathbb{Q})$$

are isomorphisms of mixed Hodge structures.

Corollary 8.2.11. *The cup product*

$$\mathrm{H}^\bullet(X) \otimes \mathrm{H}^\bullet(X) \to \mathrm{H}^\bullet(X)$$

is a morphism of mixed Hodge structures.

Proof. This follows from Proposition 8.2.10 and from Proposition 8.2.2 applied to the diagonal morphism $\Delta\colon X \to X\times X$. □

10.3. Hodge theory of simplicial schemes

For a separated simplicial scheme (or separated simplicial algebraic space) $X^\bullet$, we will have to consider diagrams
of simplicial schemes satisfying:

(a) The morphism j is an open immersion with dense image; $\overline{Z}_\bullet$ is proper.

(b) The evident morphisms $Z_n \to (\operatorname{cosq}^{X_\bullet}_{n-1} \operatorname{sq}_{n-1} Z_\bullet)_n$ are proper and surjective.

For reduced $\overline{Z}_\bullet$, we can consider j as a simplicial object in the category $\mathscr{C}$ of pairs $(Y,\overline{Y})$ consisting of a proper reduced scheme $\overline{Y}$ and a dense (Zariski) open Y of $\overline{Y}$. The category $\mathscr{C}$ satisfies §8.2, which allows us to apply the analogue of Proposition 6.2.4.

Proposition 8.3.2.

(i) For every separated simplicial scheme $X_\bullet$, there exists a diagram of the form Equation 8.3.1.1 that satisfies (a) and (b), and such that $\overline{Z}_\bullet$ is smooth and proper, and $Z_\bullet$ is the complement of a normal crossing divisor.

(ii) Let $F\colon X_{\bullet 1} \to X_{\bullet 2}$ be a morphism of separated simplicial schemes. Let $(X_{\bullet i}, Z_{\bullet i}, \overline{Z}_{\bullet i})_{i=1,2}$ be diagrams of the form Equation 8.3.1.1 that satisfy (a) and (b). Then there exists $(X_{\bullet 1}, Z_{\bullet}, \overline{Z}_{\bullet})$ as in (i), and a commutative diagram

$$\begin{tikzcd}
Z_\bullet \arrow[r, hook] \arrow[d] \arrow[drr] & \overline{Z}_\bullet \arrow[d] \arrow[drr] & & \\
Z_{\bullet 1} \arrow[r, hook, "j_1"'] \arrow[d, "p_1"'] & \overline{Z}_{\bullet 1} & Z_{\bullet 2} \arrow[r, hook, "j_2"'] \arrow[d, "p_2"'] & \overline{Z}_{\bullet 2} \\
X_{\bullet 1} \arrow[rr, "f"] & & X_{\bullet 2} &
\end{tikzcd}$$

Proof. The proof, which is a more complicated analogue of §8.2, is left to the reader. □

Consider a diagram of the form Equation 8.3.1.1 as in (ii) of Proposition 8.3.2. By (V) of §7.3, the map $\mathrm{H}^\bullet(X_\bullet, \mathbb{Z}) \to \mathrm{H}^\bullet(Z_\bullet, \mathbb{Z})$ is an isomorphism. In §10.1 and Proposition 8.1.20, we endowed $\mathrm{H}^\bullet(Z_\bullet, \mathbb{Z})$ with a mixed Hodge structure, and obtained a mixed Hodge structure on $\mathrm{H}^\bullet(X_\bullet, \mathbb{Z})$.

This is independent of the choice of diagram of the form Equation 8.3.1.1: by (ii) of Proposition 8.3.2 for $f = \mathrm{Id}$. any two systems $(Z_{\bullet i}, \overline{Z}_{\bullet i}, p_i)_{i=1,2}$ are always covered by a third $(Z_\bullet, \overline{Z}_\bullet, p)$, whence a commutative diagram of isomorphisms

$$\begin{tikzcd}
& \mathrm{H}^\bullet(X_\bullet, \mathbb{Z}) \arrow[dl] \arrow[d] \arrow[dr] & \\
\mathrm{H}^\bullet(Z_{\bullet 1}, \mathbb{Z}) \arrow[r] & \mathrm{H}^\bullet(Z_\bullet, \mathbb{Z}) & \mathrm{H}^\bullet(Z_{\bullet 2}, \mathbb{Z}) \arrow[l]
\end{tikzcd}$$

Definition 8.3.4. The *mixed Hodge structure* of $\mathrm{H}^\bullet(X_\bullet, \mathbb{Z})$ is the one defined above.

It follows from (ii) of Proposition 8.3.2 that this mixed Hodge structure is functorial in $X_\bullet$.

Proposition 8.3.5. *Let $X_\bullet$ be a separated simplicial scheme. Then the spectral sequence*

$$E_1^{pq} = \mathrm{H}^q(X_p, \mathbb{Z}) \Rightarrow \mathrm{H}^{p+q}(X_\bullet, \mathbb{Z})$$

is a spectral sequence of mixed Hodge structures.

For a diagram as in Equation 8.3.1.1, the maps $p_n\colon Z_n \to X_n$ are proper and surjective. We can thus apply Proposition 6.2.4 to the category of simplicial objects in the category $\mathscr{C}$ (§10.3) to obtain, as in §8.2, the following lemma:

Lemma 8.3.6. *Let $X_\bullet$ be a separated simplicial scheme. Then there exists a bisimplicial scheme $Z_{\bullet\bullet}$ augmented over $X_\bullet$*

$$\varepsilon_{n,m}\colon Z_{n,m} \to X_n$$

and $i\colon Z_{\bullet\bullet} \hookrightarrow \overline{Z}_{\bullet\bullet}$ such that:

(a) $\overline{Z}_{\bullet\bullet}$ is a smooth and proper bisimplicial scheme, and $Z_{\bullet\bullet}$ is the complement of a normal crossing divisor in $\overline{Z}_{\bullet\bullet}$;

(b) for all n, the augmented simplicial scheme $Z_{n\bullet} \to X_n$ is a proper hypercover of X_n.

In §10.1, we endowed $\mathrm{H}^\bullet(Z_{\bullet\bullet},\mathbb{Z})$ with a mixed Hodge structure. If $\delta\mathbb{Z}_{\bullet\bullet}$ is the diagonal simplicial scheme of $Z_{\bullet\bullet}$, then we have a commutative diagram of morphisms

$$\begin{tikzcd} \mathrm{H}^\bullet(X_\bullet,\mathbb{Z}) \arrow[r, "\gamma"] \arrow[dr, "\alpha"'] & \mathrm{H}^\bullet(Z_{\bullet\bullet},\mathbb{Z}) \arrow[d, equal, "\beta"] \\ & \mathrm{H}^\bullet(\delta Z_{\bullet\bullet},\mathbb{Z}) \end{tikzcd}$$

in which β and γ are isomorphisms, and α and β are morphisms of mixed Hodge structures; γ is thus an isomorphism of mixed Hodge structures.

We have a morphism of spectral sequences

$$\begin{tikzcd} \mathrm{H}^q(X_p,\mathbb{Z}) \arrow[r, Rightarrow] \arrow[d] & \mathrm{H}^{p+q}(X_\bullet,\mathbb{Z}) \arrow[d] & (5.2.3) \\ \mathrm{H}^q(Z_{p\bullet},\mathbb{Z}) \arrow[r, Rightarrow] & \mathrm{H}^{p+q}(Z_{\bullet\bullet},\mathbb{Z}) & (8.1.23.1) \end{tikzcd}$$

and this morphism is an isomorphism that is compatible with the mixed Hodge structures, so that Proposition 8.3.5 follows from §10.1.

Example 8.3.7. Let $X_\bullet$ be a simplicial scheme, not necessarily separated. Then there exists a separated simplicial scheme $Y_\bullet$ and $u\colon Y_\bullet \to X_\bullet$ such that, for $n \geq 0$, the map $u'_n\colon Y_n \to (\operatorname{cosq}^{X_\bullet}_{n-1}\operatorname{sq}_{n-1}Y_\bullet)_n$ is étale and surjective (for example, the sum of opens of an open cover).

The map $u^\bullet\colon\ \mathrm{H}^\bullet(X_\bullet,\mathbb{Z}) \to \mathrm{H}^\bullet(Y_\bullet,\mathbb{Z})$ is an isomorphism ((V) and (III) of §7.3). By definition, the *mixed Hodge structure* of $\mathrm{H}^\bullet(X_\bullet,\mathbb{Z})$ is induced from that of $\mathrm{H}^\bullet(Y_\bullet,\mathbb{Z})$ *via* $u^\bullet$. It does not depend on the choice of $Y_\bullet$.

For X a non-separated scheme, the *mixed Hodge structure* of $\mathrm{H}^\bullet(X,\mathbb{Z})$ is that of $\mathrm{H}^\bullet(X_\bullet,\mathbb{Z})$, where $X_\bullet$ is the constant simplicial scheme defined by X. It is plausible, but not proven, that (iv) of Theorem 8.2.4 remains true for X smooth and non-separated.

Example 8.3.8. Let $u\colon X_\bullet \to Y_\bullet$ be a morphism of simplicial schemes. The relative cohomology groups $\mathrm{H}^n(Y_\bullet \operatorname{mod} X_\bullet,\mathbb{Z})$ are, by definition (§8.3), the cohomology groups of the simplicial scheme $C(u)$. They are thus endowed with a mixed Hodge structure.

The long exact sequence of relative cohomology

$$\mathrm{H}^n(Y_\bullet,\mathbb{Z}) \to \mathrm{H}^n(X_\bullet,\mathbb{Z}) \xrightarrow{j} \mathrm{H}^{n+1}(Y_\bullet \operatorname{mod} X_\bullet,\mathbb{Z}) \to \mathrm{H}^{n+1}(Y_\bullet,\mathbb{Z})$$

is a long exact sequence of mixed Hodge structures.

Proof. We can reduce to the case where $X_\bullet$ and $Y_\bullet$ are smooth, and where, furthermore, there exists a commutative diagram

$$\begin{tikzcd} X_\bullet \arrow[r, "u"] \arrow[d, hook] & Y_\bullet \arrow[d, hook] \\ \overline{X}_\bullet \arrow[r, "\overline{u}"'] & \overline{Y}_\bullet \end{tikzcd}$$

with $\overline{X}_\bullet$ and $\overline{Y}_\bullet$ both smooth and proper, and $X_\bullet$ and $Y_\bullet$ both complements of normal crossing divisors $D_\bullet$ and $E_\bullet$ in $\overline{X}_\bullet$ and $\overline{Y}_\bullet$ (respectively).

There thus exist bifiltered resolutions (graded with acyclic components) $a\colon \Omega^\bullet_{\overline{X}_\bullet}(\log D_\bullet) \to K$ and $b\colon \Omega^\bullet_{\overline{Y}_\bullet}(\log E_\bullet) \to L$ that fit into a commutative diagram of bifiltered complexes

$$\begin{tikzcd}
\Omega^\bullet_{\overline{X}_\bullet}(\log D_\bullet) \arrow[d, "a"'] & \Omega^\bullet_{\overline{Y}_\bullet}(\log E_\bullet) \arrow[l, "\overline{u}^*"'] \arrow[d, "b"] \\
K & L \arrow[l, "v"]
\end{tikzcd}$$

(where $\overline{u}^*$ and v are $\overline{u}$-morphisms). The long exact sequence in §10.3 is then identified with the exact sequence given by the distinguished triangle

$$\begin{tikzcd}
& C(\Gamma v) \arrow[dl] & \\
\Gamma L \arrow[rr] & & \Gamma K \arrow[ul]
\end{tikzcd}$$

The complex $C(v)$ over $C(\overline{u})$ is a bifiltered resolution of $C(u^*\colon \Omega^\bullet_{\overline{Y}_\bullet}(\log E_\bullet) \to \Omega^\bullet_{\overline{X}_\bullet}(\log D_\bullet))$. From this it results that the Hodge and weight filtrations of $\mathrm{H}^\bullet(Y_\bullet \bmod X_\bullet, \mathbb{C})$ are induced by the filtrations of the cone $C(\Gamma v)$, and §10.3 then follows formally. $\square$

Proof of (ii) of (8.2.4)

Proof. The proof proceeds by induction on $N = \dim(X)$. We can assume X to be reduced. Let U be a smooth dense Zariski open of X, and let $Y = X \setminus U$. By the resolution of singularities, there exists a proper birational morphism $p\colon X' \to X$, with smooth source X', that induces an isomorphism from $p^{-1}(U)$ to U. Let $Y' = p^{-1}(Y)$. Consider, for $n \geqslant N$, the diagram of mixed Hodge structures

$$\begin{tikzcd}
& \mathrm{H}^n(X \bmod Y, \mathbb{Z}) \arrow[r] \arrow[d, "p^*"'] & \mathrm{H}^n(X, \mathbb{Z}) \arrow[r] \arrow[d] & \mathrm{H}^n(Y, \mathbb{Z}) \arrow[d] \\
\mathrm{H}^{n-1}(Y', \mathbb{Z}) \arrow[r] & \mathrm{H}^n(X' \bmod Y', \mathbb{Z}) \arrow[r] & \mathrm{H}^n(X', \mathbb{Z}) \arrow[r] & \mathrm{H}^n(Y', \mathbb{Z})
\end{tikzcd}$$

Let Φ be the family of supports of U consisting of those $A \subset U$ such that, in X (or, equivalently, in X'), $\overline{A} \subset U$. We have that

$$\mathrm{H}^n(X \bmod Y, \mathbb{Z}) = \mathrm{H}^n_\Phi(U, \mathbb{Z}) = \mathrm{H}^n(X' \bmod Y', \mathbb{Z})$$

and so the arrow denoted p^* is an isomorphism.

For h^{pq} a non-zero Hodge number of one of the mixed Hodge structures in play, we have:

(a) for $\mathrm{H}^n(X', \mathbb{Z})$, $(p,q) \in [n-N, N] \times [n-N, N]$

(b) for $\mathrm{H}^{n-1}(Y', \mathbb{Z})$, $(p,q) \in [n-N, N-1] \times [n-N, N-1]$

(c) for $\mathrm{H}^n(X' \bmod Y', \mathbb{Z}) = \mathrm{H}^n(X \bmod Y, \mathbb{Z})$, $(p,q) \in [n-N, N] \times [n-N, N]$

(d) for $\mathrm{H}^n(Y, \mathbb{Z})$, $(p,q) \in [n-N+1, N-1] \times [n-N+1, N-1]$

(e) for $\mathrm{H}^n(X, \mathbb{Z})$, $(p,q) \in [n-N, N] \times [n-N, N]$

(a) follows from the fact that X' is smooth, and from the fact that the claim is true for the ${}_WE_1$ pages of the spectral sequence Equation 3.2.4.1. (b) and (d) follow from the inductive hypothesis: we have that $\dim Y, \dim Y' \leqslant N-1$. (c) follows from (a) and (b), and (e) follows from (c) and (d). This finishes the proof. $\square$

11. Examples and applications

11.1. Cohomology of groups and of classifying spaces

Let G be a complex connected linear algebraic group. We intend to calculated the mixed Hodge structure of $\mathrm{H}^\bullet(G)$. We will start by calculating that of $\mathrm{H}^\bullet(\mathrm{B}_{\bullet G})$, by reduction to the case of tori; we will then calculate that of $\mathrm{H}^\bullet(G)$ via the Eilenberg–Moore spectral sequence §8.1. As a bonus, we obtain the strange Theorem Theorem 9.1.7.

Theorem 9.1.1. *Let G be a linear algebraic group. We have that $\mathrm{H}^{2n+1}(\mathrm{B}_{\bullet G}, \mathbb{Q}) = 0$, and that $\mathrm{H}^{2n}(\mathrm{B}_{\bullet G}, \mathbb{Q}) \otimes \mathbb{C}$ is pure of type (n, n) (§6)*

Proof. Let G^0 be the identity component of G, and let $T \subset G^0$ be a maximal torus. Since $\mathrm{H}^\bullet(\mathrm{B}_{\bullet G}, \mathbb{Q}) = \mathrm{H}^\bullet(\mathrm{B}_{\bullet G^0}, \mathbb{Q})^{G/G^0}$, we have, by the *splitting principle* (see §8.1)

$$\mathrm{H}^\bullet(\mathrm{B}_{\bullet G}, \mathbb{Q}) \hookrightarrow \mathrm{H}^\bullet(\mathrm{B}_{\bullet T}, \mathbb{Q}).$$

This inclusion is a morphism of mixed Hodge structures: it suffices to consider the case of a torus, or even, by the Künneth formula, the case where $G = \mathbb{G}_{\mathrm{m}}$. Here are two proofs of Theorem 9.1.1 in this case.

1st proof. — The cohomology of $\mathbb{G}_{\mathrm{m}}$ (i.e. of the projective line minus two points) is $\mathrm{H}^0 = \mathrm{H}^1 = \mathbb{Z}$. The only Hodge numbers h^{pq} of $\mathrm{H}^1(\mathbb{G}_{\mathrm{m}})$ that can be non-zero are $h^{01} = h^{10}$ and h^{11} ((i) and (iv) of Theorem 8.2.4). Since $h^{01} + h^{10} + h^{11} = 2h^{01} + h^{11} = 1$, we have that $h^{01} = 0$, and so $\mathrm{H}^1(\mathbb{G}_{\mathrm{m}})$ is pure of type $(1, 1)$ (for another proof, see Construction 10.3.8). The $E_2^{p,q}$ page of the spectral sequence §8.1, which is zero for odd $p + q$, is thus pure of type $((p + q)/2, (p + q)/2)$. Its abutment has the same homogeneity.

2nd proof. — We know that $\mathbb{P}^r(\mathbb{C})$, endowed with $\mathcal{O}(1)$, is an "approximation" of $\mathrm{B}_{\mathbb{G}_{\mathrm{m}}}$: the homomorphism $[\mathcal{O}(1)]\colon\ \mathrm{H}^i(\mathrm{B}_{\bullet\mathbb{G}_{\mathrm{m}}}) \to \mathrm{H}^i(\mathbb{P}^r(\mathbb{C}))$ is bijective for $i \leqslant 2r$. The cohomology group $\mathrm{H}^{2i}(\mathbb{P}^r(\mathbb{C}))$ is generated by the cohomology class of an algebraic cycle (that is, a linear subspace), and is thus of type (i, i). The following proposition allows us to deduce that $\mathrm{H}^{2i}(\mathrm{B}_{\mathbb{G}_{\mathrm{m}}})$ is of type (i, i). ☐

Proposition 9.1.2. *Let G be an algebraic group (over $\mathbb{C}$), and P a (algebraic) principal homogeneous space of group G over a scheme X. Then the morphism*

$$[P]\colon\ \mathrm{H}^\bullet(\mathrm{B}_G) = \mathrm{H}^\bullet(\mathrm{B}_{\bullet G}) \to \mathrm{H}^\bullet(X)$$

is a morphism of mixed Hodge structures.

Proof. This follows from the definition (Equation 6.1.3.1) of $[P]$ and from functoriality (Definition 8.3.4). ☐

Corollary 9.1.3. *The characteristic classes, in $\mathrm{H}^{2n}(S)$, of a (algebraic) principal homogeneous space P over S, with structure group a linear group G, are pure of type (n, n): they come from morphisms $\mathbb{Q}(-n) \to \mathrm{H}^{2n}(S, \mathbb{Q})$.*

In the particular case where S is smooth and quasi-projective, we know that the characteristic classes of P are even algebraic, i.e. in the image of the Chow ring $(\otimes\mathbb{Q})$.

Let G be a linear algebraic group over a finite field $\mathbb{F}_q$, and let $\overline{G}/\overline{\mathbb{F}}_q$ be the group induced by extension of scalars to an algebraic closure of $\mathbb{F}_q$. A proof parallel to that of Theorem 9.1.1 shows that, in ℓ-adic cohomology, the proper values of the (geometric) Frobenius acting on $\mathrm{H}^{2n}(\mathrm{B}_{\bullet\overline{G}}, \mathbb{Q}_\ell)$ are of the form $q^n\zeta$ (with ζ a root of unity). This agrees with the philosophy of §3. Similarly, the statement of Theorem 9.1.5 below is inspired by the formulas that give the number of points of reductive groups over finite fields.

Theorem 9.1.5. *Let G be a connected linear algebraic group. Let $P^\bullet \subset \mathrm{H}^\bullet(G, \mathbb{Q})$ be the primitive part of the Hopf algebra $\mathrm{H}^\bullet(G, \mathbb{Q})$. It is a mixed Hodge substructure. We have that $P^{2i}(G) = 0$, and $P^{2i-1}(G)$ is pure of type (i, i). Finally, the isomorphism*

$$\bigwedge P^\bullet(G) \to \mathrm{H}^\bullet(G, \mathbb{Q}) \tag{9.1.5.1}$$

is an isomorphism of mixed Hodge structures.

Proof. In the spectral sequence §8.1, $E^{0,q} = 0$ for $q > 0$. For $n > 0$, the *edge homomorphism*

$$\mathrm{H}^{2n}(\mathrm{B}_G, \mathbb{Q}) \to E_1^{1,2n} = \mathrm{H}^{2n-1}(G, \mathbb{Q})$$

is a morphism of mixed Hodge structures (Proposition 8.3.5); its image $P^{2n-1}(G)$ is thus a mixed Hodge substructure, and is pure of type (n, n) by Theorem 9.1.1. The isomorphism Equation 9.1.5.1 is given by the cup product, and we conclude by Corollary 8.2.11. □

Corollary 9.1.6. *Let G be a linear algebraic group. For $n > 0$, we have that*

$$W_n(\mathrm{H}^n(G, \mathbb{Q})) = 0.$$

Proof. We can reduce to the case where G is connected, and then apply Theorem 9.1.5. □

Theorem 9.1.7. *Let X be a smooth and proper algebraic space on which a connected linear algebraic group G acts via $\rho\colon G \times X \to X$. Then the diagram*

$$\begin{tikzcd}
\mathrm{H}^\bullet(X, \mathbb{Q}) \arrow[r, "\rho^*"] \arrow[d, equal] & \mathrm{H}^\bullet(X, \mathbb{Q}) \otimes \mathrm{H}^\bullet(G, \mathbb{Q}) \\
\mathrm{H}^\bullet(X, \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{Q} \arrow[r, equal] & \mathrm{H}^\bullet(X, \mathbb{Q}) \otimes \mathrm{H}^0(G, \mathbb{Q}) \arrow[u]
\end{tikzcd}$$

commutes.

Proof. Since the composite map

$$\mathrm{H}^\bullet(X, \mathbb{Q}) \xrightarrow{p^*} \mathrm{H}^\bullet(X, \mathbb{Q}) \otimes \mathrm{H}^\bullet(G, \mathbb{Q}) \to \mathrm{H}^\bullet(X, \mathbb{Q}) \otimes \mathrm{H}^0(G, \mathbb{Q}) = \mathrm{H}^\bullet(X, \mathbb{Q})$$

is the identity (the composite $X = \{e\} \times X \to G \times X \to X$ is the identity), it suffices to prove that, for $i > 0$, the i-th Künneth component of ρ^*

$$\rho_i^*\colon\ \mathrm{H}^n(X, \mathbb{Q}) \to \mathrm{H}^{n-i}(X, \mathbb{Q}) \otimes \mathrm{H}^i(G, \mathbb{Q})$$

is zero. Since $\mathrm{H}^n(X, \mathbb{Q})$ is pure of weight n, and $\mathrm{H}^{n-i}(X, \mathbb{Q})$ is pure of weight $n - i$, we have that

$$\mathrm{Im}(\mathrm{H}^n(X, \mathbb{Q})) \subset W_n(\mathrm{H}^{n-i}(X, \mathbb{Q}) \otimes \mathrm{H}^i(G, \mathbb{Q})) = \mathrm{H}^{n-i}(X, \mathbb{Q}) \otimes W_i\,\mathrm{H}_i(G, \mathbb{Q}).$$

By Corollary 9.1.6, this image is thus null. □

Variant 9.1.8. Let X be a separated algebraic space on which a connected linear group G acts.

(i) If X is smooth, and $\overline{X}$ is a compactification of X, then the diagram

$$\begin{tikzcd} \mathrm{H}^\bullet(\overline{X},\mathbb{Q}) \arrow[r] \arrow[d] & \mathrm{H}^\bullet(X,\mathbb{Q}) \arrow[r, "\rho^*"] & \mathrm{H}^\bullet(X,\mathbb{Q})\otimes\mathrm{H}^\bullet(G,\mathbb{Q}) \\ \mathrm{H}^\bullet(X,\mathbb{Q}) \arrow[rr, equal] & & \mathrm{H}^\bullet(X,\mathbb{Q})\otimes\mathrm{H}^0(G,\mathbb{Q}) \arrow[u, hook] \end{tikzcd}$$

commutes.

(ii) If X is proper, and $\widetilde{X}$ is a resolution of singularities of X, then the diagram

$$\begin{tikzcd} \mathrm{H}^\bullet(X,\mathbb{Q}) \arrow[r, "\rho^*"] \arrow[d] & \mathrm{H}^\bullet(X,\mathbb{Q})\otimes\mathrm{H}^\bullet(G,\mathbb{Q}) \arrow[r] & \mathrm{H}^\bullet(\widetilde{X},\mathbb{Q})\otimes\mathrm{H}^\bullet(G,\mathbb{Q}) \\ \mathrm{H}^\bullet(\widetilde{X},\mathbb{Q}) \arrow[rr, equal] & & \mathrm{H}^\bullet(\widetilde{X},\mathbb{Q})\otimes\mathrm{H}^0(G,\mathbb{Q}) \arrow[u, hook] \end{tikzcd}$$

commutes.

Proof. It suffices to prove that, for $i>0$, the Künneth components

$$\mathrm{H}^n(\overline{X},\mathbb{Q})\to\mathrm{H}^n(X,\mathbb{Q})\to\mathrm{H}^{n-i}(X,\mathbb{Q})\otimes\mathrm{H}^i(G,\mathbb{Q})$$
$$(\text{resp.}\quad \mathrm{H}^n(X,\mathbb{Q})\to\mathrm{H}^{n-i}(X,\mathbb{Q})\otimes\mathrm{H}^i(G,\mathbb{Q})\to\mathrm{H}^{n-i}(\widetilde{X},\mathbb{Q})\otimes\mathrm{H}^i(G,\mathbb{Q}))$$

are zero. In case (i) (resp. case (ii)), we have (Theorem 8.2.4)

$$W_n\,\mathrm{H}^n(\overline{X},\mathbb{Q})=\mathrm{H}^n(\overline{X},\mathbb{Q})$$
$$(\text{resp.}\quad W_n\,\mathrm{H}^n(X,\mathbb{Q})=\mathrm{H}^n(X,\mathbb{Q}))$$

and so, by Theorem 8.2.4 and Corollary 9.1.6,

$$W_n(\mathrm{H}^{n-i}(X,\mathbb{Q})\otimes\mathrm{H}^i(G,\mathbb{Q}))=0$$
$$(\text{resp.}\quad W_n(\mathrm{H}^{n-i}(\widetilde{X},\mathbb{Q})\otimes\mathrm{H}^i(G,\mathbb{Q}))=0)$$

whence the claim. □

11.2. Hodge theory of smooth hypersurfaces, following Griffiths

This section depends only on §1 to §3. We will explain Theorem (8.6) of [G1969].

Let X be a smooth and proper algebraic variety, and Y a smooth divisor in X. Let j be the inclusion of $U=X\setminus Y$ into X. We have that $\mathrm{R}^i j_*\mathbb{Z}=0$ for $i\neq 0,1$, and the Leray spectral sequence for j can be identified with the long exact sequence of cohomology

$$\ldots\to\mathrm{H}^n_Y(X)\to\mathrm{H}^n(X)\to\mathrm{H}^n(U)\xrightarrow{\partial}\ldots \tag{9.2.1.1}$$

with

$$\mathrm{H}^n_Y(X)=\mathrm{H}^{n-2}(Y,\mathscr{H}^2_Y).$$

For a reason that was made clear in Proposition 3.1.9, we identify $\mathscr{H}^2_Y$ with the constant sheaf $\mathbb{Z}(-1)_{\mathbb{Z}}$. Since the Leray spectral sequence for j converges, by definition, to the mixed Hodge structure of $\mathrm{H}^n(U)$, we find that the exact sequence

$$\ldots \to \mathrm{H}^{n-2}(Y,\mathbb{Z})(-1) \to \mathrm{H}^n(X,\mathbb{Z}) \to \mathrm{H}^n(U,\mathbb{Z}) \xrightarrow{\partial} \ldots \tag{9.2.1.2}$$

is an exact sequence of mixed Hodge structures.

The Leray spectral sequence for j is also the spectral sequence of the filtered complex $(\Omega^\bullet_X(\log Y), W)$; the exact sequence above can thus also be identified with that induced by the short exact sequence

$$0 \to \Omega^\bullet_X \to \Omega^\bullet_X(\log Y) \xrightarrow{\mathrm{Res}} \Omega^\bullet_Y \to 0. \tag{9.2.1.3}$$

For this reason, the homomorphism ∂ in Equation 9.2.1.2 is also called the *residue operator*.

The residue operator

$$\mathrm{Res}\colon\ \mathrm{H}^{n+1}(U,\mathbb{Z}) \to \mathrm{H}^n(Y,\mathbb{Z})(-1) \tag{9.2.2.1}$$

is strictly compatible with the Hodge filtration. The Hodge filtration of $\mathrm{H}^n(U,\mathbb{C})$ thus determines that of $\mathrm{Ker}(\mathrm{H}^n(Y,\mathbb{C})) \to \mathrm{H}^{n+2}(X,\mathbb{C})$.

Suppose that Y is ample. If X is pure of dimension $N+1$, then $\mathrm{H}^i(X,\mathbb{Z}) \xrightarrow{\sim} \mathrm{H}^i(Y,\mathbb{Z})$ for $i < N$ (Lefschetz). By Poincaré duality, for $i \neq N$, $\mathrm{H}^i(Y,\mathbb{C})$ can thus be calculated from $\mathrm{H}^\bullet(X,\mathbb{C})$. Set

$$\begin{aligned}\mathrm{H}^N_{\mathrm{ev}}(Y,\mathbb{C}) &= \mathrm{Ker}(\mathrm{H}^N(Y,\mathbb{C}) \to \mathrm{H}^{N+2}(X,\mathbb{C}))\\ &= \text{orthogonal complement of } \mathrm{H}^N(X,\mathbb{C}) \hookrightarrow \mathrm{H}^N(Y,\mathbb{C})\end{aligned}$$

(the evanescent part of the cohomology). We have that

$$\mathrm{H}^N(Y,\mathbb{C}) = \mathrm{H}^N_{\mathrm{ev}}(Y,\mathbb{C}) \oplus \mathrm{H}^N(X,\mathbb{C})$$

and we thus obtain a lot of information about the Hodge structure of $\mathrm{H}^\bullet(Y,\mathbb{C})$ when we know about that of $\mathrm{H}^\bullet(X,\mathbb{C})$ and the Hodge filtration of $\mathrm{H}^{N+1}(U,\mathbb{C})$.

By definition, the Hodge filtration of $\mathrm{H}^\bullet(U,\mathbb{C})$ is the abutment of the spectral sequence that *degenerates* at E_1

$$E_1^{pq} = \mathrm{H}^q(X,\Omega^p_X(\log Y)) \Rightarrow \mathrm{H}^{p+q}(U,\mathbb{C}).$$

By Proposition 3.1.11, this spectral sequence agrees with the spectral sequence of the complex $j_*\Omega^\bullet_U$ endowed with the filtration given by the order of the pole P.

In particular, we have that

$$F^p(\mathrm{H}^\bullet(U,\mathbb{C})) = \mathbb{H}^\bullet(X, P^p(j_*\Omega^\bullet_U)).$$

Recall that $P^p(j_*\Omega^\bullet_U)$ is the following complex

$$\Omega^p_X(Y) \to \Omega^{p+1}_X(2Y) \to \ldots \to \Omega^i_X((i-p+1)Y) \to \ldots$$

with the first non-zero component placed in degree p.

If $\mathrm{H}^i(X,\Omega^p_X(nY)) = 0$ for $i > 0$, $n > 0$, and $p \geqslant 0$, then we simply have

$$\mathbb{H}^\bullet(X, P^p(j_*\Omega^\bullet_U)) = \mathrm{H}^\bullet(\Gamma(X, P^p(j_*\Omega^\bullet_U)))$$

whence the following result.

Proposition 9.2.4. *Suppose that* $\mathrm{H}^i(X, \Omega^p_X(nY)) = 0$ *for* $i > 0$, $n > 0$, *and* $p \geqslant 0$ *(which is the case if* Y *is sufficiently ample). Then a cohomology class* $c \in \mathrm{H}^d(U, \mathbb{C})$ *is of Hodge filtration* $\geqslant p$ *if and only if it can be represented by a closed* d*-form* α *on* U, *with a pole of order* $\leqslant d - p + 1$ *along* Y. *If* α *is a closed form on* U, *with a pole of order* k *along* Y, *and if the class of* α *in* $\mathrm{H}^d(U, \mathbb{C})$ *is zero, then* $\alpha = d\beta$, *with* β *presenting a pole of order* $\leqslant k - 1$ *along* Y*; for* $k \leqslant 1$, *we have that* $\alpha = 0$.

It is well known (by a theorem of Bott) that the hypotheses of Proposition 9.2.4 are satisfied for Y a hypersurface in the projective space $\mathbb{P}^n(\mathbb{C})$ (for $n \geqslant 1$), i.e. that

$$\mathrm{H}^i(\mathbb{P}^n(\mathbb{C}), \Omega^j(m)) = 0 \quad \text{for } i > 0 \text{ and } m > 0$$

(a demonstration appears in [SGA 7, XI], for example). Furthermore, for $n > 1$, the residue map identifies $\mathrm{H}^n(U, \mathbb{C})$ with the primitive part of the cohomology of $\mathrm{H}^{n-1}(Y, \mathbb{C})$. Thus ([G1969, (8.6)]):

Proposition 9.2.6. *Let* Y *be a smooth hypersurface in* $\mathbb{P}^n(\mathbb{C})$ *(with* $n > 1$*) and* $U = \mathbb{P}^n(\mathbb{C}) \setminus Y$.

(i) *For a primitive cohomology class* $c \in \mathrm{H}^{n-1}(Y, \mathbb{C})$ *to be of Hodge filtration* $\leq p$, *it is necessary and sufficient that it be the residue of a regular differential* n*-form on* U, *with a pole of order* $\leqslant n - p$ *along* Y.

(ii) *Let* α *be a regular differential* n*-form on* U, *with a pole of order* $k \geqslant 1$ *along* Y. *For* $\mathrm{Res}(\alpha) \in \mathrm{H}^{n-1}(Y, \mathbb{C})$ *to be zero, it is necessary and sufficient that* $\alpha = d\beta$ *with* β *an* $(n-1)$*-form on* U *with a pole of order* $\leqslant k - 1$ *along* Y.

11.3. Construction of complexes of first-order differential operators

This section depends only on §7 and §8. We will prove the following theorem.

Proposition 9.3.1. *Let* X *be a quasi-projective scheme (over* $\mathbb{C}$*). Then there exists on* X *a complex* K *of sheaves that satisfies the following properties.*

(a) *The* K^n *are coherent algebraic sheaves, and* $K^n = 0$ *for* $n < 0$.

(b) *The differentials* $\mathrm{d}\colon K^n \to K^{n+1}$ *are algebraic differential operators of order 1.*

(c) *The complex* K^{an} *of coherent analytic sheaves on* X^{an} *is a resolution of the constant sheaf* $\underline{\mathbb{C}}$. *More precisely,* $\mathscr{H}^i(K) = 0$ *for* $i > 0$, *and there exists*

$$f\colon \Omega^\bullet_X \to K$$

with f^n *linear algebraic, such that the composite map* $\underline{\mathbb{C}} \to \mathcal{O}_X \xrightarrow{f^0} K^0$ *identifies* $\underline{\mathbb{C}}$ *with* $\mathscr{H}^0(K)$.

(d) *The natural map*

$$\mathbb{H}^i(X, K) \to \mathbb{H}^i(X^{\mathrm{an}}, K^{\mathrm{an}}) \underset{(c)}{\xleftarrow{\sim}} \mathbb{H}^i(X^{\mathrm{an}}, \mathbb{C})$$

is an isomorphism.

As a corollary, we recover a particular case of a theorem of Bloom and Herrera [BH1969].

We will first prove the corollary. The proposition Proposition 9.3.1 will be obtained by redoing the proof at the level of complexes.

Corollary 9.3.2. *The map*

$$\mathrm{H}^\bullet(X^{\mathrm{an}}, \mathbb{C}) \to \mathbb{H}^\bullet(X^{\mathrm{an}}, \Omega^\bullet_{X^{\mathrm{an}}})$$

induced by $\underline{\mathbb{C}} \to \Omega^\bullet_{X^{\mathrm{an}}}$ *identifies* $\mathrm{H}^\bullet(X^{\mathrm{an}}, \mathbb{C})$ *with a direct factor of* $\mathrm{H}^\bullet(X^{\mathrm{an}}, \Omega^\bullet_{X^{\mathrm{an}}})$.

Proof. Let $\varepsilon \colon X_\bullet \to X$ be an augmented simplicial scheme over X. Suppose that ε is a proper hypercover, and that the X_n are smooth. We have the commutative diagram:

The arrow (1) is bijective by (II) and (IV) of §7.3. The arrow (2) is bijective since $\Omega^\bullet_{X_\bullet}$ is a resolution of $\underline{\mathbb{C}}$ (the Poincaré lemma on the X_n). The diagram Equation 9.3.2.1 gives a retract, compatible with the cup product, of $\mathbb{H}^\bullet(X^{\mathrm{an}}, \Omega^\bullet_X)$ onto $\mathrm{H}^\bullet(X^{\mathrm{an}}, \mathbb{C})$, and this proves Corollary 9.3.2. □

The complex K that we wish to construct is, in the derived category, exactly $\mathrm{sR}\varepsilon_{\bullet *}\Omega^\bullet_{X_\bullet}$. We will only deal with the case where the augmented simplicial space $X_\bullet$ is s-split, defined as in §8.2 by proper and surjective maps, with smooth quasi-projective sources

$$f'_k \colon N_k \to (\operatorname{cosq}\operatorname{sq}_{k-1}(X_\bullet))_k$$

for $k \geqslant 0$.

We will construct a "compatible system" of affine open covers of the X_n. It will be relevant, for this construction, to consider an open cover of a space Y as a diagram

$$Y \xleftarrow{u} U \xrightarrow{\varphi} I$$

where I is discrete, u is surjective, and $u_i \colon U_i \coloneqq \varphi^{-1}(i) \hookrightarrow Y$ is an open embedding. Proceeding as in §8.2, we construct a diagram

$$X_\bullet \xleftarrow{u} U_\bullet \xrightarrow{\varphi} I_\bullet \tag{9.3.3.1}$$

of the following type.

(a) $I_\bullet$ is a discrete simplicial set. The I_n are finite.

(b) For all $k \geqslant 0$, Equation 9.3.3.1 induces a diagram

$$\begin{tikzcd}
(\operatorname{cosq}\operatorname{sq}_{k-1}(X_\bullet))_k & (\operatorname{cosq}\operatorname{sq}_{k-1}(U_\bullet))_k \arrow[l] \arrow[r, "\varphi"] & (\operatorname{cosq}\operatorname{sq}_{k-1}(I_\bullet))_k \\
N_k \arrow[u, "f'_k"] & N_k(U_\bullet) \arrow[l] \arrow[u] \arrow[r, "\varphi"'] & N_k(I_\bullet) \arrow[u, "a'_k"']
\end{tikzcd}$$

(c) In the diagram in (b), for $i \in \operatorname{cosq}\operatorname{sq}_{k-1}(I_\bullet)_k$, the $\varphi^{-1}(j)$ for $a'_k(j) = i$ form an affine open cover of

$$N_k \times_{(\operatorname{cosq}\operatorname{sq}_{k-1}(X_\bullet))_k} \varphi^{-1}(i).$$

(d) (Follows by induction on (c)). The first (and thus the second) row of the diagram in (b) defines an open cover. The $X_k \leftarrow U_k \to I_k$ thus also form open covers.

(e) (Follows from a suitable choice of affine covers in (c)). $U_{n_i} = \varphi_n^{-1}(i)$ is the complement of a subscheme $D_{n,i}$ of X_n, given by the union of a divisor and some connected components of X_n. For every coherent sheaf $\mathscr{F}$ on X_n and integers $(a_i)_{i\in I_n}$ (with $(a_i \geqslant 0)$), we set

$$\mathscr{F}(\textstyle\sum_i a_i D_{n,i}) = \begin{cases} 0 & \text{on the components of } X_n \text{ contained inside some } D_{n,i} \text{ with } a_i > 0 \\ \mathscr{F} \otimes \mathcal{O}(\sum_{a_i>0} a_i D_{n,i}) & \text{on the others.} \end{cases}$$

In the inductive construction of Equation 9.3.3.1, the open covers in (c) are chosen arbitrarily. This allows us to suppose the following.

(f) For any $a_i > 0$ (for $i \in I_k$),

$$\mathrm{R}^\ell \varepsilon_{k*}\big(\Omega^m_{X_k}(\textstyle\sum_i a_i D_{k,i})\big) = 0$$

for $\ell > 0$.

(g) For any $\alpha\colon \Delta_m \to \Delta_n$ and $i \in I_n$, we have that

$$X_\bullet(\alpha)^{-1}(D_{m,\alpha(i)}) \subset D_{n,i}$$

(as schemes),

For each k, the complex of sheaves

$$\mathrm{R}\varepsilon_{k*}(\Omega^\bullet_{X_k})$$

on X can be calculated in a Čech manner. If, for $P \subset I_k$, j_P is the inclusion of $U_P = \bigcap_{i\in P} U_{k_i}$ in X_k, then this is the simple complex associated to the double complex of the

$$\check{K}^{k,p,q} = \sum_{\#P=p+1} \varepsilon_{k*}\Omega^q_{X_k}\big((q+1)\textstyle\sum_{i\in P} D_{k,i}\big).$$

For fixed k and fixed q, it follows from (f) that the inclusion of $K^{k,\bullet,q}$ into $\check{K}^{k,\bullet,q}$ is a quasi-isomorphism. For fixed k, the inclusion of $K^{k,\bullet,\bullet}$ into $\check{K}^{k,\bullet,\bullet}$ is also a quasi-isomorphism. Finally, for varying k, the $K^{k,\bullet,\bullet}$ form, by (g), a simplicial double complex. The associated simple complex K agrees with $\mathrm{R}\varepsilon_{\bullet*}\Omega^\bullet_{X_\bullet}$ in the derived category.

We now prove that K satisfies Proposition 9.3.1. Properties (a) and (b) of K are clear; we define $f\colon \Omega^\bullet_X \to K$ by the evident maps from the Ω^q_X to the $K^{0,0,q}$.

In the analytic category, the complex K^{an} of coherent analytic sheaves is given by the same formulas as for K. We have a commutative diagram

$$\begin{tikzcd} \mathbb{H}^i(X,K) \arrow[r, "\sim"] \arrow[d] & \mathbb{H}^i(X_\bullet,\Omega^\bullet_{X_\bullet}) \arrow[d, "(1)"] \\ \mathbb{H}^i(X^{\mathrm{an}},K^{\mathrm{an}}) \arrow[r, "\sim"] & \mathbb{H}^i(X^{\mathrm{an}}_\bullet,\Omega^\bullet_{X^{\mathrm{an}}_\bullet}) \end{tikzcd}$$

and (1) is an abutment of a morphism of spectral sequences

$$\begin{tikzcd} \mathbb{H}^a(X_b,\Omega^\bullet_{X_\bullet}) \arrow[r, Rightarrow] \arrow[d] & \mathbb{H}^{a+b}(X_\bullet,\Omega^\bullet_{X_\bullet}) \arrow[d] \\ \mathbb{H}^a(X^{\mathrm{an}}_b,\Omega^\bullet_{X^{\mathrm{an}}_\bullet}) \arrow[r, Rightarrow] & \mathbb{H}^{a+b}(X^{\mathrm{an}}_\bullet,\Omega^\bullet_{X^{\mathrm{an}}_\bullet}). \end{tikzcd}$$

This morphism is an isomorphism by [G1966], whence (d).

To prove (c), we consider the commutative diagram

$$\begin{tikzcd}
\underline{\mathbb{C}} \arrow[rr, "\sim", "(1)"'] \arrow[d] & & \mathrm{R}\varepsilon_{\bullet *}\underline{\mathbb{C}} \arrow[d, "(2)"', "\sim"] \\
{\Omega^{\bullet}_{X_\bullet}}^{\mathrm{an}} \arrow[r, "f"'] & K^{\mathrm{an}} \arrow[r, "\sim"] & \mathrm{R}\varepsilon_{\bullet *}{\Omega^{\bullet}_{X_\bullet}}^{\mathrm{an}}
\end{tikzcd}$$

The fact that (1) is a quasi-isomorphism follows from §7.3. The fact that (2) is a quasi-isomorphism follows from the Poincaré lemma on the X_n. Claim (c) thus follows.

The same construction gives resolutions that satisfy (a) and (b) for sheaves $\mathscr{F}$ on X^{an} other than the constant sheaf $\underline{\mathbb{C}}$ (we can probably take $\mathscr{F}$ to be any "algebraically constructible" sheaf). Unfortunately, I can not prove that the resolutions thus obtained are essentially unique (in a suitable derived category). In the absence of such uniqueness, Proposition 9.3.1 is not of great interest.

12. Hodge theory in level $\leqslant 1$

12.1. 1-Motives

Definition 10.1.1.

(i) A mixed Hodge structure H is ***torsion free*** if $H_{\mathbb{Z}}$ is torsion free.

(ii) If H is torsion free, then we again denote by W the filtration of $H_{\mathbb{Z}} \subset \mathrm{H}_{\mathbb{Q}}$ induced by W. We denote by $\mathrm{Gr}_n^W(H)$ the torsion free $\mathbb{Z}$-module $\mathrm{Gr}_n^W(H_{\mathbb{Z}})$ endowed with its Hodge structure of weight n.

Definition 10.1.2. A ***1-motive*** over an algebraically closed field k consists of the following:

(a) a free $\mathbb{Z}$-module X of finite type, an abelian variety A, and a torus T;

(b) an extension G of A by T;

(c) a homomorphism $u\colon X \to G(k)$.

It is often useful to consider a 1-motive M as being a complex of group schemes, concentrated in degrees 0 and 1:

$$M = [X \xrightarrow{u} G].$$

Construction 10.1.3. We will construct an equivalence $M \mapsto T(M)$ from the category of 1-motives over $\mathbb{C}$ to the category of torsion free mixed Hodge structures H of type

$$\big\{(0,0),(0,-1),(-1,0),(-1,-1)\big\}$$

such that $\mathrm{Gr}_{-1}^W(H)$ is polarisable. Furthermore, we will construct isomorphisms

$$\begin{aligned}
\mathrm{H}_1(T,\mathbb{Z}) &\simeq \mathrm{Gr}_{-2}^W T(M)_{\mathbb{Z}} \\
\mathrm{H}_1(A,\mathbb{Z}) &\simeq \mathrm{Gr}_{-1}^W T(M) \qquad \text{(isomorphism of Hodge structures)} \\
X &\simeq \mathrm{Gr}_0^W T(M)_{\mathbb{Z}}
\end{aligned}$$

that are natural in $M = (X, A, T, G, u)$.

Let M be a 1-motive. The exponential map $\exp\colon \mathrm{Lie}(G) \to G$ has $\mathrm{H}_1(G)$ as its kernel. We define $T_{\mathbb{Z}}(M)$ as the fibre product of $\mathrm{Lie}(G)$ and X over G, whence

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{H}_1(G) \arrow[r] & \mathrm{Lie}(G) \arrow[r, "\exp"] & G \arrow[r] & 0 \\
0 \arrow[r] & \mathrm{H}_1(G) \arrow[r] \arrow[u, equal] & T_{\mathbb{Z}}(M) \arrow[r, "\beta"'] \arrow[u, "\alpha"] & X \arrow[r] \arrow[u, "u"'] & 0
\end{tikzcd} \tag{10.1.3.1}$$

We set

$$W_{-1}(T_{\mathbb{Z}}(M)) = \mathrm{Ker}(\beta) = \mathrm{H}_1(G)$$
$$W_{-2}(T_{\mathbb{Z}}(M)) = \mathrm{H}_1(T) = \mathrm{Ker}\left(\mathrm{H}_1(G) \to \mathrm{H}_1(A)\right).$$

This defines the *weight filtration*. The morphism α extends to

$$\alpha_{\mathbb{C}}\colon T_{\mathbb{Z}}(M) \otimes \mathbb{C} \to \mathrm{Lie}(G).$$

We set $F^0(T_{\mathbb{Z}}(M) \otimes \mathbb{C}) = \mathrm{Ker}(\alpha_{\mathbb{C}})$, and this defines the *Hodge filtration* on

$$T_{\mathbb{C}}(M) = T_{\mathbb{Z}}(M) \otimes \mathbb{C}.$$

Lemma 10.1.3.2. *The triple $T(M) = (T_{\mathbb{Z}}(M), W, F)$ is a torsion-free mixed Hodge structure of type $\{(0,0), (0,-1), (-1,0), (-1,-1)\}$, and $\mathrm{Gr}^W_{-1}(T(M))$ is polarisable.*

Proof. (a) We have that $W_{-2}(T_{\mathbb{C}}(M)) \cap F^0 = 0$, since (1) is injective; thus $(W_{-2}(T_{\mathbb{C}}(M)), F)$ is a Hodge structure of type $(-1,-1)$.

(b) Since (2) is surjective, F induces on $\mathrm{Gr}^W_{-1}(T_{\mathbb{Z}}(M)) \otimes \mathbb{C} \simeq \mathrm{H}_1(A, \mathbb{Z}) \otimes \mathbb{C}$ the Hodge filtration of $\mathrm{H}_1(A, \mathbb{Z}) \otimes \mathbb{C}$; this is a polarisable Hodge structure of type $(-1,0) + (0,1)$ (cf. Reminder 4.4.3).

(c) Consider the diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & & \\
0 \arrow[r] & W_{-1}(T_{\mathbb{C}}(M)) \cap F^0 \arrow[r] \arrow[d] & F^0 \arrow[r] \arrow[d] & V \arrow[r] \arrow[d, "\sim"] & 0 \\
0 \arrow[r] & \mathrm{H}_1(G, \mathbb{Z}) \otimes \mathbb{C} \arrow[r] \arrow[d] & T_{\mathbb{Z}}(M) \otimes \mathbb{C} \arrow[r] \arrow[d] & X \otimes \mathbb{C} & 0 \\
 & \mathrm{Lie}(G) \arrow[r, equal] \arrow[d] & \mathrm{Lie}(G) \arrow[d] & & \\
 & 0 & 0 & &
\end{tikzcd}$$

Since F^0 is sent to $\mathrm{Gr}^W_0(T_{\mathbb{C}}(M))$, it is thus of type $(0,0)$.

This finishes the construction of $T(M)$. It is clear that $T(M)$ is functorial in M. $\square$

Suppose that $\mathrm{Gr}^W_{-1}(H)$ is polarisable; the complex torus

$$A = H_{\mathbb{Z}} \backslash \mathrm{Gr}^W_{-1}(H_{\mathbb{C}})/F^0\,\mathrm{Gr}^W_{-1}(H_{\mathbb{C}})$$

is then an abelian variety (cf. Reminder 4.4.3). Let T be the torus with character group the dual of $\mathrm{Gr}^W_{-2}(H_{\mathbb{Z}})$, i.e.

$$\mathrm{H}_1(T) = \mathrm{Gr}^W_{-2}(H_{\mathbb{Z}}).$$

The complex analytic group

$$G = W_{-1}(H_{\mathbb{Z}})\backslash W_{-1}(H_{\mathbb{C}})/F^0 \cap W_{-1}(H_{\mathbb{C}})$$

is an extension of A by T.

Lemma 10.1.3.3. *The functor $E \mapsto E^{\mathrm{an}}$ is an equivalence from the category of algebraic group extensions of A by T to the category of analytic group extensions of A by T.*

Proof. We reduce to the case where $T = \mathbb{G}_{\mathrm{m}}$. By GAGA, $E \mapsto E^{\mathrm{an}}$ is more general an equivalence from the category of $\mathbb{G}_{\mathrm{m}}$-torsors over A to the category of $\mathbb{G}_{\mathrm{m}}$-torsors over A^{an}.

We thus have G, an extension of A by T. Let $X = \mathrm{Gr}^W_0(H_{\mathbb{Z}})$. We similarly define $u\colon X \to G$ (cf. §2.2):

$$\begin{tikzcd}
0 \arrow[r] & W_{-1}(H_{\mathbb{Z}}) \arrow[r] \arrow[dd, equal] & W_{-1}(H_{\mathbb{C}})/(W_{-1}(H_{\mathbb{C}}) \cap F^0) \arrow[r] \arrow[d, "\sim"] & G \arrow[r] & 0 \\
& & H_{\mathbb{C}}/F^0 & & \\
0 \arrow[r] & W_{-1}(H_{\mathbb{Z}}) \arrow[r] & H_{\mathbb{Z}} \arrow[r] \arrow[u] & X \arrow[r] \arrow[uu] & 0
\end{tikzcd}$$

It is clear that the above construction, which associates to H a torsion-free 1-motive, is the inverse of T, and that it is functorial. □

If G_i (for $i = 1, 2$) is an extension of an abelian variety A_i by a torus A_i, then, for every morphism $u\colon G_1 \to G_2$, we have that $u(T_1) \subset T_2$ and $\mathrm{Ker}(u\colon A_1 \to A_2)^0 = \mathrm{Im}(\mathrm{Ker}(u\colon G_1 \to G_2) \to A_1)^0$. This corresponds to the fact that a morphism of mixed Hodge $\mathbb{Q}$-structures is strictly compatible with the weight filtration.

Let H be a torsion-free mixed Hodge structure of type

$$\big\{(0,0),(0,-1),(-1,0),(-1,-1)\big\}.$$

The filtration W of $H_{\mathbb{Z}}$ thus defines a filtration on H by mixed Hodge substructures. At the same time, if $M = (X, A, T, G, u)$ is a 1-motive, then we denote by W the following increasing filtration on M:

$$\begin{cases}
W_i(M) = 0 & \text{for } i < -2; \\
W_i(M) = M & \text{for } i \geqslant 0; \\
W_{-1}(M) = G \quad \text{i.e. } = (\{0\}, A, T, G, 0); & \\
W_{-2}(M) = T \quad \text{i.e. } = (\{0\}, 0, T, T, 0). &
\end{cases}$$

In an evident way, the successive $\mathrm{Gr}^W_i(M)$ for $i = 0, -1, -2$ are X, A, and T (respectively).

The construction in Construction 10.1.3 that associates to a 1-motive M over $\mathbb{C}$ the object $T_{\mathbb{Z}}(M)$ is transcendental. We will show that

$$\widehat{T}(M) = T_{\mathbb{Z}}(M) \otimes \widehat{\mathbb{Z}} = \prod_{\ell} T_{\mathbb{Z}}(M) \otimes \mathbb{Z}_\ell$$

can be defined in a purely algebraic way.

Let M be a 1-motive over an algebraically closed field k of characteristic 0. We identify M with a complex $[X \xrightarrow{u} G]$ (concentrated in degrees 0 and 1). For every integer $n > 0$, consider the complex $[\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z}]$ (concentrated in degrees -1 and 0), and let $T_{\mathbb{Z}/n\mathbb{Z}}(M)$ be the H^0 of the complex $[X \xrightarrow{u} G] \otimes [\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z}]$, i.e.

$$\begin{array}{ccc} X & \xrightarrow{u} & G \\ {\scriptstyle n}\uparrow & & \uparrow{\scriptstyle -n} \\ X & \xrightarrow[u]{} & G \end{array}$$

We have that

$$T_{\mathbb{Z}/n\mathbb{Z}}(M) = \big\{(x, g) \mid u(x) = ng\big\}/\big\{(nx, u(x)) \mid x \in X\big\}.$$

In the derived category, we have that

$$T_{\mathbb{Z}/n\mathbb{Z}}(M) = \mathrm{H}^0(M \overset{\mathrm{L}}{\otimes} \mathbb{Z}/n\mathbb{Z}).$$

For $n = md$, we define transition morphisms

$$\begin{aligned} \varphi_{m,n} \colon T_{\mathbb{Z}/n\mathbb{Z}}(M) &\to T_{\mathbb{Z}/m\mathbb{Z}}(M) \\ (x, g) &\mapsto (x, dg). \end{aligned}$$

We set

$$\widehat{T}(M) = \varprojlim T_{\mathbb{Z}/n\mathbb{Z}}(M).$$

The filtration W of M induces a filtration W of the $T_{\mathbb{Z}/n\mathbb{Z}}(M)$ and of $\widehat{T}(M)$. We have that

$$\begin{aligned} &\mathrm{Gr}_0^W(\widehat{T}(M)) = X \otimes \widehat{\mathbb{Z}} \\ &\mathrm{Gr}_{-1}^W(\widehat{T}(M)) = \varprojlim A_n = \widehat{T}(A) \\ &\mathrm{Gr}_{-2}^W(\widehat{T}(M)) = \varprojlim T_n = Y \otimes \widehat{\mathbb{Z}}(1), \end{aligned}$$

where Y is the dual of the character group of $T = W_{-2}(M)$.

Construction 10.1.6. Let $M = [X \xrightarrow{u} G]$ be a 1-motive on $\mathbb{C}$. We have that

$$\widehat{T}(M) \simeq T_{\mathbb{Z}}(M) \otimes \widehat{\mathbb{Z}}. \tag{10.1.6.1}$$

The natural morphism $[T_{\mathbb{Z}}(M) \to \mathrm{Lie}(G)] \to [X \to G]$ is a quasi-isomorphism. The quasi-isomorphisms

$$\begin{bmatrix} X & \xrightarrow{u} & G \\ {\scriptstyle n}\uparrow & & \uparrow{\scriptstyle -n} \\ X & \xrightarrow[u]{} & G \end{bmatrix} \longleftarrow \begin{bmatrix} T_{\mathbb{Z}} & \xrightarrow{u} & \mathrm{Lie}(G) \\ {\scriptstyle n}\uparrow & & \sim\uparrow{\scriptstyle -n} \\ T_{\mathbb{Z}} & \xrightarrow[u]{} & \mathrm{Lie}(G) \end{bmatrix} \longrightarrow \begin{bmatrix} T_{\mathbb{Z}} & \longrightarrow & 0 \\ {\scriptstyle n}\uparrow & & \uparrow \\ T_{\mathbb{Z}} & \xrightarrow[c]{} & 0 \end{bmatrix}$$

give isomorphisms

$$T_{\mathbb{Z}/n\mathbb{Z}}(M) \simeq T_{\mathbb{Z}}/nT_{\mathbb{Z}}. \tag{10.1.6.2}$$

Under this isomorphism, to $t \in W_{-1}(T_{\mathbb{Z}})$ we associate $\exp(t/n) \in G_n$. By passing to the limit, we deduce Equation 10.1.6.1 from Equation 10.1.6.2.

Techniques of Grothendieck allow us to transpose to de Rham cohomology what we have just done in ℓ-adic cohomology. Let M be a 1-motive over an algebraically closed field k.

Construction 10.1.7. We will construct a vector space $T_{\mathrm{DR}}(M)$ over k, endowed with filtrations W and F.

Let $M = (X, A, T, G, u)$, and consider M as a complex concentrated in degrees -1 and 0. An *extension* of M by $\mathbb{G}_{\mathrm{a}}$ is an extension of M by the complex consisting of $\mathbb{G}_{\mathrm{a}}$ concentrated in degree 0. We have that $\mathrm{Hom}(G, \mathbb{G}_{\mathrm{a}}) = \mathrm{Ext}^1(X, \mathbb{G}_{\mathrm{a}}) = 0$. Thus:

(a) extensions of M by $\mathbb{G}_{\mathrm{a}}$ have no automorphisms;

(b) the sequence

$$0 \to \mathrm{Hom}(X, \mathbb{G}_{\mathrm{a}}) \to \mathrm{Ext}^1(M, \mathbb{G}_{\mathrm{a}}) \to \mathrm{Ext}^1(G, \mathbb{G}_{\mathrm{a}}) \to 0$$

is exact.

We have that $\mathrm{Hom}(T, \mathbb{G}_{\mathrm{a}}) = \mathrm{Ext}^1(T, \mathbb{G}_{\mathrm{a}}) = 0$, and so

(c) $\mathrm{Ext}^1(A, \mathbb{G}_{\mathrm{a}}) \xrightarrow{\sim} \mathrm{Ext}^1(G, \mathbb{G}_{\mathrm{a}})$.

The vector space $\mathrm{Ext}^1(M, \mathbb{G}_{\mathrm{a}})$ is thus of finite dimension, and, by (a), there thus exists a universal extension of M by a vector group; we denote it by $M^\natural \colon X \to G^\natural$.

$$\begin{tikzcd}
 & 0 \arrow[r] \arrow[d] & X \arrow[r, equal] \arrow[d] & X \arrow[d] & \\
0 \arrow[r] & \mathrm{Ext}^1(M, \mathbb{G}_{\mathrm{a}})^* \arrow[r] & G^\natural \arrow[r] & G \arrow[r] & 0
\end{tikzcd}$$

We set

$$\begin{aligned}
T_{\mathrm{DR}}(M) &= \mathrm{Lie}(G^\natural) \\
F^0 T_{\mathrm{DR}}(M) &= \mathrm{Ker}(\mathrm{Lie}\, G^\natural \to \mathrm{Lie}\, G) \simeq (\mathrm{Ext}^1(M, \mathbb{G}_{\mathrm{a}}))^*.
\end{aligned}$$

Then T_{DR} is functorial in M (and thus also in $M^\natural$), and we define the filtration W of $T_{\mathrm{DR}}(M)$ as coming from the filtration W of M.

Construction 10.1.8. Let M be a 1-motive over $\mathbb{C}$. We have that

$$(T_{\mathrm{DR}}(M), F, W) \simeq (T_{\mathbb{C}}(M), F, W).$$

Proof. The $\mathrm{Ext}^i(X, \mathbb{G}_{\mathrm{a}})$, $\mathrm{Ext}^i(A, \mathbb{G}_{\mathrm{a}})$, and $\mathrm{Ext}^i(T, \mathbb{G}_{\mathrm{a}})$ (for $i = 0, 1$) are the same in the algebraic and analytic categories. Consider the map

$$T_{\mathbb{C}}(M) = T_{\mathbb{Z}}(M) \otimes \mathbb{C} \to \mathrm{Lie}(G) \to G.$$

The diagram

$$\begin{array}{ccc} X & \longrightarrow & T_{\mathbb{C}}(M)/\operatorname{H}_1(G) \\ \| & & \downarrow \\ X & \longrightarrow & G \end{array}$$

defines an extension of M^{an} by the vector group $F^0 T_{\mathbb{C}} M$, and thus an extension of M by this group. We have to prove that this is the universal extension. For this, it suffices to verify that the category of extensions of $[X \to T_{\mathbb{C}}(M)/\operatorname{H}_1(G)]$ by $\mathbb{G}_{\mathrm{a}}$ is trivial (i.e. that it has a single object, and no automorphisms). This is also the category of extensions of $T_{\mathbb{C}}(M)/T_{\mathbb{Z}}(M)$ by $\mathbb{G}_{\mathrm{a}}$, and, indeed,

$$\operatorname{Ext}^i(T_{\mathbb{C}}(M)/T_{\mathbb{Z}}(M), \mathbb{G}_{\mathrm{a}}) = 0 \qquad \text{for } i = 0, 1.$$

□

Remark 10.1.9. The universal extension $M^\natural$ is characterised by the commutative diagram

$$\begin{array}{ccccc} X & \longrightarrow & G^\natural & \longrightarrow & G \\ \uparrow & & \uparrow & & \uparrow \\ T_{\mathbb{Z}}(M) & \longrightarrow & \operatorname{Lie}(G^\natural) & \longrightarrow & \operatorname{Lie}(G) \end{array}$$

whose exterior square is Equation 10.1.3.1, and that induces an isomorphism $T_{\mathbb{C}}(M) \xrightarrow{\sim} \operatorname{Lie}(G^\natural)$.

Variant 10.1.10. A *smooth 1-motive* M over a scheme S consists of the following:

(a) a group scheme X over S that is locally (for the étale topology) a constant group scheme defined by a free $\mathbb{Z}$-module of finite type; an abelian scheme A over S, and a torus T over S;

(b) an extension G of A by T;

(c) a morphism $u \colon X \to G$.

For every integer n, a construction analogous to §12.1 associates to a smooth 1-motive M over S a group scheme $T_{\mathbb{Z}/n\mathbb{Z}}(M)$ that is finite and flat over S. For ℓ invertible over S, the projective system

$$T_\ell(M) = \varprojlim T_{\mathbb{Z}/\ell^n\mathbb{Z}}(M)$$

is a $\mathbb{Z}_\ell$-sheaf on S. In general, $T_\ell(M)$ corresponds to an ℓ-divisible group (i.e. a Barsotti–Tate group) over S.

The filtration W of M, defined as in §12.1, defines a filtration W on $T_\ell(M)$. We again have that

$$\begin{aligned} \operatorname{Gr}^W_0(T_\ell(M)) &= X \otimes \widehat{\mathbb{Z}} \\ \operatorname{Gr}^W_{-1}(T_\ell(M)) &= T_\ell(A) \\ \operatorname{Gr}^W_{-2}(T_\ell(M)) &= Y \otimes \widehat{\mathbb{Z}}_\ell(1), \end{aligned}$$

where $Y = \underline{\operatorname{Hom}}(T, \mathbb{G}_{\mathrm{m}})^\vee$.

Similarly, a construction analogous to that of Construction 10.1.7 associates to M a vector bundle $T_{\mathrm{DR}}(M)$.

Terminology 10.1.11. Let M be a 1-motive. We call $T(M)$, $T_\ell(M)$, and $T_{\mathrm{DR}}(M)$ the *Hodge*, *ℓ-adic*, and *de Rham realisations* of M (respectively).

12.2. 1-Motives and bi-extensions

The results of this section will not be used elsewhere. We will make use of the notion of biextensions [SGA 7, VII, (2.1)] and of the following generalisation.

Let $K_i\colon [A_i \to B_i]$ (for $i = 1, 2$) be complexes of sheaves, concentrated in degrees 0 and -1 (in an arbitrary topos). A *biextension* of K_1 and K_2 by an abelian sheaf H consists of the following:

(a) a biextension P of B_1 and B_2 by H, i.e. an H-torsor P over $B_1 \times B_2$, with $P_{b_1 b_2}$ depending biadditively on b_1 and b_2;

(b) a trivialisation (i.e. a biadditive section) of the biextension of B_1 and A_2 by H, given by the inverse image of P;

(c) a trivialisation of the biextension of A_1 and B_2 by H, given by the inverse image of P;

where we demand that the trivialisations in (b) and (c) agree on $A_1 \times A_2$.

We denote by $\mathbf{Biext}(K_1, K_2; H)$ the category of biextensions of K_1 and K_2 by H. This is a Picard category (the objects add). The group $\mathrm{Biext}^0(K_1, K_2; H)$ of automorphisms of an arbitrary biextension of K_1 and K_2 by H is

$$\mathrm{Biext}^0(K_1, K_2; H) = \mathrm{Hom}(\mathrm{H}^0(K_1) \otimes \mathrm{H}^0(K_2), H).$$

We denote by $\mathrm{Biext}^1(K_1, K_2; H)$ the group of isomorphism classes of biextensions. We can show, as in [SGA 7, VII, (3.6.5)], that

$$\mathrm{Biext}^i(K_1, K_2; H) = \mathrm{Ext}^i(K_1 \overset{\mathrm{L}}{\otimes} K_2; H) \qquad \text{for } i = 0, 1.$$

From this formula, or by elementary means, we can show that, if $K_i' \to K_i$ is a quasi-isomorphism (for $i = 1, 2$), then giving a biextension of K_1 and K_2 by H is equivalent to giving a biextension of K_1' and K_2' by H.

This also applies to replacing "sheaves" by "algebraic groups" (resp. by "complex analytic groups"): we can interpret these as sheaves on a suitable site.

In this section, we identity a 1-motive $M = (X, T, A, G, u)$ with a complex concentrated in *degrees* 0 *and* -1

$$M\colon [X \to G].$$

Over $\mathbb{C}$, we denote by M^{an} the complex of complex analytic groups $[X \to G^{\mathrm{an}}]$. We have the following rigidity result:

Lemma 10.2.2.1. *Let* $M_i = (X_i, A_i, T_i, G_i, u_i)$ *(for* $i = 1, 2$*) be 1-motives. Then*

$$\mathrm{Biext}^0(M_1, M_2; \mathbb{G}_{\mathrm{m}}) = 0.$$

Over $\mathbb{C}$*, we similarly have that* $\mathrm{Biext}^0(M_1^{\mathrm{an}}, M_2^{\mathrm{an}}; \mathbb{G}_{\mathrm{m}}) = 0$.

Proof. Every biadditive morphism $G_1 \times G_2 \to \mathbb{G}_{\mathrm{m}}$ is trivial. □

Construction 10.2.3. Let $M_i = (X_i, A_i, T_i, G_i, u_i)$ (for $i = 1, 2$) be 1-motives over $\mathbb{C}$. We have that

$$\mathrm{Biext}^1(M_1, M_2; \mathbb{G}_{\mathrm{m}}) = \mathrm{Hom}(T(M_1) \otimes T(M_2), \mathbb{Z}(1)).$$

Proof. We will first calculate $\mathrm{Biext}^1(M_1^{\mathrm{an}}, M_2^{\mathrm{an}}; \mathbb{G}_\mathrm{m})$.
If V and W are vector spaces, then, ***in the analytic category***,

$$\mathrm{Biext}^0(V, W; \mathbb{G}_\mathrm{m}) \xleftarrow[\exp]{\sim} \mathrm{Hom}(V \otimes W, \mathbb{C}) \tag{10.2.3.1(an)}$$

$$\mathrm{Biext}^1(V, W; \mathbb{G}_\mathrm{m}) = 0. \tag{10.2.3.2(an)}$$

From the fact that an extension of W by $\mathbb{G}_\mathrm{m}$, over an arbitrary base, is always locally trivial, it follows, essentially formally, that

$$\begin{aligned}\mathrm{Biext}^i(V, W; \mathbb{G}_\mathrm{m}) &= \mathrm{Ext}^i(V, \underline{\mathrm{Hom}}(W, \mathbb{G}_\mathrm{m})) \\ &= \mathrm{Ext}^i(V, W^*) \qquad \text{for } i = 1, 2.\end{aligned}$$

Let $V_\mathbb{Z}$ and $W_\mathbb{Z}$ be free $\mathbb{Z}$-modules of finite rank. Since

$$[V_\mathbb{Z} \to V_\mathbb{C}] \to [0 \to V_\mathbb{C}/V_\mathbb{Z}]$$

is a quasi-isomorphism, the "inverse image" functor is an equivalence.

$$\mathbf{Biext}(V_\mathbb{C}/V_\mathbb{Z}, W_\mathbb{C}/W_\mathbb{Z}; \mathbb{G}_\mathrm{m}) \simeq \mathbf{Biext}([V_\mathbb{Z} \to V_\mathbb{C}], [W_\mathbb{Z} \to W_\mathbb{C}]; \mathbb{G}_\mathrm{m}). \tag{10.2.3.3(an)}$$

Every biadditive morphism from $V_\mathbb{C}/V_\mathbb{Z} \times W_\mathbb{C}/W_\mathbb{Z}$ to $\mathbb{G}_\mathrm{m}$ is trivial, whence (a) in the following:

Lemma 10.2.3.4.

(a) $\mathrm{Biext}^0(V_\mathbb{C}/V_\mathbb{Z}, W_\mathbb{C}/W_\mathbb{Z}; \mathbb{G}_\mathrm{m}) = 0$

(b) $\mathrm{Biext}^1(V_\mathbb{C}/V_\mathbb{Z}, W_\mathbb{C}/W_\mathbb{Z}; \mathbb{G}_\mathrm{m}) = \mathrm{Hom}(V_\mathbb{Z} \otimes W_\mathbb{Z}, \mathbb{Z}(1))$.

Proof. Let $\psi_i \colon V_\mathbb{C} \otimes W_\mathbb{C} \to \mathbb{C}$ be bilinear maps, with $\psi = \psi_1 - \psi_2$ taking values in $\mathbb{Z}(1) = 2\pi i\mathbb{Z}$ on $V_\mathbb{Z} \otimes W_\mathbb{Z}$. Let $P(\psi_1, \psi_2)$ be the following biextension of $[V_\mathbb{Z} \to V_\mathbb{C}]$ and $[W_\mathbb{Z} \to W_\mathbb{C}]$ by $\mathbb{G}_\mathrm{m}$: the trivial biextension of $V_\mathbb{C}$ and $W_\mathbb{C}$ by $\mathbb{G}_\mathrm{m}$, endowed with the trivialisations $\varphi_i = \exp \psi_i \colon V_\mathbb{Z} \otimes W_\mathbb{C} \to \mathbb{G}_\mathrm{m}$ for $i = 1, 2$. By Equation 10.2.3.2(an), every biextension is of this form; by Equation 10.2.3.1(an), $P(\psi_1, \psi_2) \simeq P(\psi_1', \psi_2')$ if and only if $\psi_1' - \psi_2' = \psi_1 - \psi_2$. The $P(\psi, 0)$ thus represent each biextension exactly once. ☐

Under the general hypotheses of §12.2, let $K_i = [A_i + F_i \to B_i]$ be complexes (for $i = 1, 2$). Let K_i' be the subcomplex $[A_i \to B_i]$. We can show that giving a biextension of K_1 and K_2 by H is equivalent to giving (both of) the following:

(a) a biextension P of K_1' and K_2' by H; and

(b) trivialisations of the biextensions of $[0 \to F_1]$ and K_2' (resp. K_1' and $[0 \to F_2]$) by H, given by the inverse images of P, that agree on $F_1 \times F_2$.

In the particular case where

$$\mathrm{Biext}^i(F_1, K_2'; H) = 0$$
$$\mathrm{Biext}^i(K_1', F_2; H) = 0$$

for $i = 1, 2$, there exists exactly one trivialisation φ_1, a section of P over $F_1 \times B_2$ (resp. φ_2, a section of P over $B_1 \times F_2$) as in (b). The category $\mathbf{Biext}(K_1, K_2; H)$ can then be identified with the subcategory of $\mathbf{Biext}(K_1', K_2'; H)$ consisting of the P for which $\varphi_1 = \varphi_2$ on $F_1 \times F_2$.

Let $V_{\mathbb{Z}}$ and $W_{\mathbb{Z}}$ be free $\mathbb{Z}$-modules of finite type, and F and G vector subspaces of $V_{\mathbb{C}}$ and $W_{\mathbb{C}}$ (respectively). We have that

$$\mathrm{Biext}^i(F, [W_{\mathbb{Z}} \to W_{\mathbb{C}}]; \mathbb{G}_{\mathrm{m}}) = 0.$$

For $i = 0$, this follows from Equation 10.2.3.1(an) and from the fact that a bilinear map

$$B\colon F \times W_{\mathbb{C}} \to \mathbb{C} \quad \text{such that} \quad B(F, W_{\mathbb{Z}}) \subset \mathbb{Z}(1)$$

is zero. For $i = 1$, this implies that every biadditive morphism $F \times W_{\mathbb{Z}} \to \mathbb{G}_{\mathrm{m}}$ comes from a biadditive morphism $F \times W_{\mathbb{C}} \to \mathbb{G}_{\mathrm{m}}$. In other words, we use the long exact sequence

$$\begin{aligned}0 &\to \mathrm{Biext}^0(F, [W_{\mathbb{Z}} \to W_{\mathbb{C}}]; \mathbb{G}_{\mathrm{m}}) \to \mathrm{Biext}^0(F, W_{\mathbb{C}}; \mathbb{G}_{\mathrm{m}}) \xrightarrow{\sim} \mathrm{Biext}^0(F, W_{\mathbb{Z}}; \mathbb{G}_{\mathrm{m}}) \\ &\to \mathrm{Biext}^1(F, [W_{\mathbb{Z}} \to W_{\mathbb{C}}]; \mathbb{G}_{\mathrm{m}}) \to \mathrm{Biext}^1(F, W_{\mathbb{C}}; \mathbb{G}_{\mathrm{m}}).\end{aligned}$$

We can thus apply §12.2 to the complexes $K_1 = [V_{\mathbb{Z}} \oplus F \to V_{\mathbb{C}}]$ and $K_2 = [W_2 \oplus G \to W_{\mathbb{C}}]$. We find that the biextensions of K_1 and K_2 by $\mathbb{G}_{\mathrm{m}}$ can be identified with certain biextensions of $V_{\mathbb{C}}/V_{\mathbb{Z}}$ and $W_{\mathbb{C}}/W_{\mathbb{Z}}$ (respectively) by $\mathbb{G}_{\mathrm{m}}$, i.e. with certain forms $\psi\colon V_{\mathbb{Z}} \otimes W_{\mathbb{Z}} \to \mathbb{Z}(1)$ (Lemma 10.2.3.4). For $P(\psi_1, \psi_2)$ (Lemma 10.2.3.4) to come from a biextension of K_1 and K_2 by $\mathbb{G}_{\mathrm{m}}$, it is necessary and sufficient that $\exp\psi_2\colon F \times W_{\mathbb{C}} \to \mathbb{G}_{\mathrm{m}}$ and $\exp\psi_1\colon V_{\mathbb{C}} \times G \to \mathbb{G}_{\mathrm{m}}$ agree on $F \times G$, i.e. that $\psi(F, G) = 0$.

Lemma 10.2.3.7. *With the above notation, we have that* $\mathrm{Biext}^0(K_1, K_2; \mathbb{G}_{\mathrm{m}}) = 0$, *and* $\mathrm{Biext}^1(K_1, K_2; \mathbb{G}_{\mathrm{m}})$ *can be identified with the set of forms* $\psi\colon V_{\mathbb{Z}} \otimes W_{\mathbb{Z}} \to \mathbb{Z}(1)$ *with complexifications satisfying* $\Psi(F, G) = 0$.

Proof. For ψ as in Lemma 10.2.3.7, with $\psi = \psi_1 - \psi_2$, the biextension corresponding to ψ is given by the following trivialisations of the trivial biextension of $V_{\mathbb{C}}$ and $W_{\mathbb{C}}$ by $\mathbb{G}_{\mathrm{m}}$:

- on $(V_{\mathbb{Z}} \oplus F) \times W_{\mathbb{C}} : \exp\psi_1(v_{\mathbb{Z}}, w_{\mathbb{C}}) \cdot \exp\psi_2(f, w_{\mathbb{C}})$
- on $V_{\mathbb{C}} \times (W_{\mathbb{C}} \oplus G) : \exp\psi_2(v_{\mathbb{C}}, w_{\mathbb{Z}}) \cdot \exp\psi_2(v_{\mathbb{C}}, g)$.

□

For the 1-motives M_i, we have quasi-isomorphisms

$$\begin{tikzcd} T_{\mathbb{Z}}(M_i) \oplus F^0 T_{\mathbb{C}}(M_i) \arrow[r] \arrow[d] & T_{\mathbb{Z}}(M_i) \arrow[r] \arrow[d] & X_i \arrow[d] \\ T_{\mathbb{C}}(M_i) \arrow[r] & \mathrm{Lie}(G_i) \arrow[r] & G_i \end{tikzcd}$$

The group $\mathrm{Biext}^1(M_1^{\mathrm{an}}, M_2^{\mathrm{an}}; \mathbb{G}_{\mathrm{m}})$ can thus be identified with the group of maps

$$\psi\colon T_{\mathbb{Z}}(M_1) \otimes T_{\mathbb{Z}}(M_2) \to \mathbb{Z}(1)$$

whose complexification is compatible with the Hodge filtration.

Lemma 10.2.3.9. *The map* $\mathrm{Biext}^1(M_1, M_2; \mathbb{G}_{\mathrm{m}}) \to \mathrm{Biext}^1(M_1^{\mathrm{an}}, M_2^{\mathrm{an}}; \mathbb{G}_{\mathrm{m}})$ *is injective. Its image consists of those biextensions* P *whose restriction to* $G_1 \times G_2$ *is the inverse image of a biextension of* A_1 *and* A_2 *by* $\mathbb{G}_{\mathrm{m}}$.

Proof. We have that

$$\mathrm{Biext}^1(A_1, A_2; \mathbb{G}_\mathrm{m}) \xrightarrow{\sim} \mathrm{Biext}^1(G_1, G_2; \mathbb{G}_\mathrm{m})$$
$$\mathrm{Biext}^1(A_1, A_2; \mathbb{G}_\mathrm{m}) \xrightarrow{\sim} \mathrm{Biext}^1(A_1^{\mathrm{an}}, A_2^{\mathrm{an}}; \mathbb{G}_\mathrm{m})$$

where the former follows from [SGA 7, VIII, (3.5)] and the latter from GAGA.

Let P be a biextension of M_1 and M_2 by $\mathbb{G}_\mathrm{m}$. If P^{an} is trivial, then:

(a) The restriction of P to $G_1 \times G_2$ is the image of a biextension P_0 of A_1 and A_2 by $\mathbb{G}_\mathrm{m}$; the latter is trivial. If this were not the case, then it would be analytically non-trivial, and would thus give rise to a non-zero bilinear form ψ_0, and the form ψ corresponding to P would be ***a fortiori*** non-zero. The restriction of P to $G_1 \times G_2$ is thus trivial (and uniquely trivialisable, even analytically).

(b) P is thus defined by biadditive maps $X_1 \times G_2 \to \mathbb{G}_\mathrm{m}$ and $G_1 \times X_2 \to \mathbb{G}_\mathrm{m}$. These are zero in the analytic category, and thus thus zero, and so P is trivial.

Let P be a biextension of M_1^{an} and M_2^{an} by $\mathbb{G}_\mathrm{m}$ that satisfies the condition Lemma 10.2.3.9. It remains to prove that P is algebrisable. By hypothesis, and GAGA, its restriction P_1 to $G_1 \times G_2$ is algebrisable. We can conclude by showing that a trivialisation of P_1 on $X_1 \times G_2$ (resp. $G_1 \times X_2$) is automatically algebraic: P_1 can be thought of as an extension of $X_1 \otimes G_2$ by $\mathbb{G}_\mathrm{m}$, which allows us to reduce to showing that

$$\mathrm{Ext}^i(G_1, \mathbb{G}_\mathrm{m}) \xrightarrow{\sim} \mathrm{Ext}^i(G_1^{\mathrm{an}}, \mathbb{G}_\mathrm{m}) \qquad \text{for } i = 1, 2.$$

This follows from the same claim for an abelian variety (GAGA) and for $\mathbb{G}_\mathrm{m}$.

By Lemma 10.2.3.9, $\mathrm{Biext}^1(M_1, M_2; \mathbb{G}_\mathrm{m})$ can be identified with the set of

$$\psi\colon T_\mathbb{Z}(M_1) \otimes T_\mathbb{Z}(M_2) \to \mathbb{Z}(1)$$

such that

(a) ψ is compatible with the Hodge filtration (§12.2); and

(b) the restriction of ψ to $W_{-1}T_\mathbb{Z}(M_1) \otimes W_{-1}T_\mathbb{Z}(M_2)$ comes from a form on

$$\mathrm{Gr}^W_{-1} T_\mathbb{Z}(M_1) \otimes \mathrm{Gr}^W_{-1} T_\mathbb{Z}(M_2).$$

Condition (b) implies that ψ is compatible with W, and this concludes Construction 10.2.3. □

□

Remark 10.2.4. A biextension P of M_1 and M_2 by $\mathbb{G}_\mathrm{m}$ defines, in an evident way, a biextension P^0 of M_2 and M_1 by $\mathbb{G}_\mathrm{m}$.

If ψ (resp. ψ^0) is the bilinear form defined by P (resp. by P^0), then $\psi(x,y) = -\psi^0(y,x)$. This is clear from the equations in Lemma 10.2.3.7.

Construction 10.2.3 has purely algebraic analogues in ℓ-adic or de Rham cohomology.

Construction 10.2.5. Let $M_i = (X_i, A_i, T_i, G_i, u_i)$ (for $i = 1, 2$) be 1-motives over an algebraically closed field k of characteristic 0. Let P be a biextension of M_1 and M_2 by $\mathbb{G}_\mathrm{m}$. We associate to P a morphism

$$T_{\mathbb{Z}/n\mathbb{Z}}(M_1) \otimes T_{\mathbb{Z}/n\mathbb{Z}}(M_2) \to \mathbb{Z}/n\mathbb{Z}(1).$$

Proof. Let $m_i \in T_{\mathbb{Z}/n\mathbb{Z}}(M_1)$ be represented by (x_i, g_i), with $u_i(x_i) = ng_i$. We will construct isomorphisms a_1 and a_2 from $P^{\otimes n}_{g_1,g_2}$ to the trivial torsor $\mathbb{G}_{\mathrm{m}}$:

$$a_1\colon P^{\otimes n}_{g_1,g_2} \xrightarrow{\sim} P_{ng_1,g_2} = P_{u(x_1),g_2} \xrightarrow{\sim} \mathbb{G}_{\mathrm{m}}$$
$$a_2\colon P^{\otimes n}_{g_1,g_2} \xrightarrow{\sim} P_{g_1,ng_2} = P_{g_1,u(x_2)} \xrightarrow{\sim} \mathbb{G}_{\mathrm{m}}.$$

We set $a_2 = \Phi(m_1, m_2)a_1$. We can verify that $\Phi(m_1, m_2)$ does not depend on the choice of the (x_i, g_i), and is an n-th root of unity. This is the desired form. □

Proposition 10.2.6. *On $\mathbb{C}$, the form in Construction 10.2.5 is induced by the form in Construction 10.2.3 by reduction modulo n.*

Proof. This can be proven by using the formulas given in Lemma 10.2.3.7. □

Construction 10.2.7. Let $M_i = (X_i, A_i, T_i, G_i, u_i)$ (for $i = 1, 2$) be 1-motives over an algebraically closed field k. Let P be a biextension of M_1 and M_2 by $\mathbb{G}_{\mathrm{m}}$. We associate to P a morphism

$$T_{\mathrm{DR}}(M_1) \otimes_k T_{\mathrm{DR}}(M_2) \to k.$$

Proof. ♮♮

A ♮-extension of a smooth commutative group G by a smooth commutative group H consists of the following:

(a) an extension E of G by H; if $\mu\colon G \times G \to G$ is the addition, then we consider E as an H-torsor over G endowed with a morphism

$$\nu\colon\ \mathrm{pr}_1^* E + \mathrm{pr}_2^* E \to \mu^* E$$

of H-torsors over $G \times G$ (addition);

(b) a connection on the H-torsor E, such that the map ν is horizontal. This connection is automatically flat.

If G is the extension of an abelian variety by a torus, then every extension of G by $\mathbb{G}_{\mathrm{m}}$ admits a ♮-structure; two ♮-structures differ by an invariant form on G (with values in the Lie algebra of $\mathbb{G}_{\mathrm{m}}$). We define, in an evident way, the Baer sum of two ♮-extensions.

Let P be a biextension of G_1 and G_2 by H. Consider P as an H-torsor P over $G_1 \times G_2$, endowed with morphisms

$$\nu_1\colon\ \mathrm{pr}_{13}^* P + \mathrm{pr}_{23}^* P \to (\mu_1 \times \mathrm{Id})^* P \quad \text{on} \quad G_1 \times G_1 \times G_2$$
$$\nu_2\colon\ \mathrm{pr}_{12}^* P + \mathrm{pr}_{13}^* P \to (\mathrm{Id} \times \mu_2)^* P \quad \text{on} \quad G_1 \times G_2 \times G_2.$$

A *♮-1-structure* on P is a connection on the H-torsor P with respect to $G_1 \times G_2 \to G_2$ (the covariant derivative is defined only for vector fields that are parallel to G_1) such that ν_1 and ν_2 are horizontal. We similarly define *♮-2-structures.* A *♮-structure* is the data of a ♮-1-structure and of a ♮-2-structure. This is also a connection on the H torsor P such that ν_1 and ν_2 are horizontal.

The *curvature* R of a ♮-biextension P is the curvature of the connection of P. This is a 2-form on $G_1 \times G_2$ that is invariant under translation, with values in Lie(H), i.e. it is an alternating form on Lie$(G_1) \times$ Lie(G_2), with values in Lie(H). Its restriction to Lie(G_1) and Lie(G_2) is zero, and so R defines a pairing

$$\Phi\colon\ \mathrm{Lie}(G_1) \otimes \mathrm{Lie}(G_2) \to \mathrm{Lie}(H) \tag{10.2.7.3}$$

(also called the ***curvature*** of P), with

$$R(g_1 + g_2, g_1' + g_2') = \Phi(g_1, g_2') - \Phi(g_1', g_2).$$

For complexes of smooth groups $K_i\colon [A_i \to B_i]$, a ***$\natural$-biextension*** (resp. a ***$\natural$-i-extension***) of K_1 and K_2 by H is a $\natural$-biextension (resp. $\natural$-i-biextension) of B_1 and B_2 by H, endowed with trivialisations (as $\natural$-biextensions, resp. $\natural$-i-biextensions) as in §12.2.

Proposition 10.2.7.4. *Let $P^\natural$ be the biextension of $M_1^\natural$ and $M_2^\natural$ by $\mathbb{G}_{\mathrm{m}}$ given by the inverse image of P. Then there exists exactly one $\natural$-structure on $P^\natural$.*

Proof. We first prove uniqueness: we have to show that, on the trivial biextension of $M_1^\natural$ and $M_2^\natural$ by $\mathbb{G}_{\mathrm{m}}$, every $\natural$-structure is trivial. A $\natural$-structure is, first of all, a connection on the trivial $\mathbb{G}_{\mathrm{m}}$-torsor over $G_1^\natural \times G_2^\natural$, i.e. a field of forms on $G_1^\natural \times G_2^\natural$, say $\Gamma_{g_1,g_2}(t_1 + t_2)$. The axiom of $\natural$-structures implies that

$$\Gamma_{g_1,g_2}(t_1 + t_2) = \Phi_1(g_1, t_2) + \Phi_2(t_1, g_2)$$

where Φ_1 and Φ_2 are biadditive. Now, $G_i^\natural$ is an extension of $X_i \otimes k$ by the universal extension G_i' of G_i by an additive group. Every morphism $G_i' \to \mathbb{G}_{\mathrm{a}}$ is trivial. Furthermore, Φ_1 factors through $(X_1 \otimes k) \otimes \mathrm{Lie}(G_2^\natural)$. Finally, the restriction of Γ to $X_1 \times G_2^\natural$ must be trivial, and thus so too must be the restriction of Φ_1 to $X_1 \times \mathrm{Lie}(G_2^\natural)$. We thus have that $\Phi_1 = 0$. Similarly, $\Phi_2 = 0$, and so $\Gamma = 0$.

To prove existence, it suffices to construct a $\natural$-2-structure on the biextension of $M_1^\natural$ and M_2 by $\mathbb{G}_{\mathrm{m}}$ given by the inverse image of P: by the inverse image, we will obtain a $\natural$-2-structure on $P^\natural$ and, symmetrically, a $\natural$-1-structure.

For $g \in G_1$, P_g is an extension of G_2 by $\mathbb{G}_{\mathrm{m}}$. Let C_g be the set of $\natural$-structures on P_g. This is a torsor under $\mathrm{Lie}(G_2)^*$. For $g \in G_1$, the C_g are the fibres of a torsor C over G_1; the Baer sum of $\natural$-extensions further makes C an extension of G_1 by $\mathrm{Lie}(G_2)^*$

We lift $u_1\colon X \to G_1$ to $u_1'\colon X \to C$ by sending $x \in X$ to the trivial connection on the trivial extension $P_{u_1(x)}$. We thus obtain a $\natural$-2-structure on the biextension of $[X \to C]$ and M_2 by $\mathbb{G}_{\mathrm{m}}$ given by the inverse image of P.

By the universal property of $M_1^\natural$, there exists a unique commutative diagram

$$\begin{tikzcd} & {[X \longrightarrow G]} & \\ M_1^\natural \arrow[ur] \arrow[rr, "v"'] & & {[X \longrightarrow C]} \arrow[ul] \end{tikzcd}$$

by taking an inverse image under v, we find the desired $\natural$-2-structure.

By definition, the pairing in Construction 10.2.7 is the negative of the curvature of the $\natural$-biextension Proposition 10.2.7.4. ☐

☐

Proposition 10.2.8. *Over $\mathbb{C}$, the pairing in Construction 10.2.7 is the complexification of the pairing in Construction 10.2.3.*

Proof. $\natural P^\natural \natural [T_{\mathbb{Z}}(M_1) \to T_{\mathbb{C}}(M_1)][T_{\mathbb{Z}}(M_2) \to T_{\mathbb{C}}(M_2)]\mathbb{G}_{\mathrm{m}}$ ☐

Lemma 10.2.9. *Let $V_{\mathbb{Z}}$ and $W_{\mathbb{Z}}$ be two free $\mathbb{Z}$-modules of finite rank, and P a biextension of $[V_{\mathbb{Z}} \to V_{\mathbb{C}}]$ and $[W_{\mathbb{Z}} \to W_{\mathbb{C}}]$ by $\mathbb{G}_{\mathrm{m}}$ with underlying biextension P. Then the curvature $\psi\colon V_{\mathrm{c}} \otimes W_{\mathbb{C}} \to \mathbb{C}$ of $P^{\natural}$ agrees with the negative of the complexification of the form corresponding to P under Lemma 10.2.3.4.*

Proof. $P = P(\psi_1, \psi_2)\natural P \mathbb{G}_{\mathrm{m}} V_{\mathbb{C}} \times W_{\mathbb{C}} \Gamma_{v,w}(v' + w') V_{\mathbb{C}} \times W_{\mathbb{C}} \natural$

$$\Gamma_{v,w}(v' + w') = \Phi_1(v, w') + \Phi_2(v', w)$$

$\Phi_1 \Phi_2 \exp \psi_1(v_{\mathbb{Z}}, w_{\mathbb{C}}) \exp \psi_2(v_{\mathbb{C}}, w_{\mathbb{Z}}) \Phi_i = -\psi_i \mathrm{d}\Gamma$

$$R_{v,w}(v' + w', v'' + w'') = \Phi_1(v', w'') + \Phi_2(v'', w') - \Phi_1(v'', w') - \Phi_2(v', w'').$$

P

$$\Phi_1(v', w'') - \Phi_2(v', w'') = -(\psi_1(v', w'') - \psi_2(v', w'')) = -\psi(v', w''). \qquad \square$$

Let M be a 1-motive over $\mathbb{C}$. Then M has a corresponding mixed Hodge structure $T(M)$ of type $\{(0,0), (0,-1), (-1,0), (-1,-1)\}$. The mixed Hodge structure $\mathrm{Hom}(T(M), \mathbb{Z}(1))$ is again of this type, and thus defines a 1-motive M^*. The evident map $T(M) \otimes T(M^*) \to \mathbb{Z}(1)$ defines a biextension P of M and M^* by $\mathbb{G}_{\mathrm{m}}$. The 1-motive M^*, endowed with the biextension P, is the *Cartier dual* of M. We call P the *Poincaré biextension.*

The construction in §12.2 can be made purely algebraic. To a 1-motive $M = (X, A, T, G, u)$ over an algebraically closed field k, we will associated a *Cartier dual* $M^* = (X', A', T', G', u')$:

(a) X' is the character group of T, A' is the dual abelian variety of A, and T' is the character group of X.

(b) We first consider the case where $T = 0$. An extension of M by $\mathbb{G}_{\mathrm{m}}$ then consists of an extension E of A by $\mathbb{G}_{\mathrm{m}}$, along with a trivialisation $\widetilde{u}$ of the extension u^*E of X by $\mathbb{G}_{\mathrm{m}}$.

$$\begin{tikzcd} & & & X \arrow[dl, "\widetilde{u}"'] \arrow[d, "u"] & \\ 0 \arrow[r] & \mathbb{G}_{\mathrm{m}} \arrow[r] & E \arrow[r] & A \arrow[r] & 0 \end{tikzcd}$$

We have an exact sequence

$$0 \to \mathrm{Hom}(X, \mathbb{G}_{\mathrm{m}}) \to \mathrm{Ext}^1(M, \mathbb{G}_{\mathrm{m}}) \to \mathrm{Ext}^1(A, \mathbb{G}_{\mathrm{m}}) \to 0.$$

The extensions of M by $\mathbb{G}_{\mathrm{m}}$ have no non-trivial automorphisms. The functor of isomorphism classes of extensions is representable; we define G' as the algebraic group that represents this functor; it is an extension of A' by T'. We have a biextension P of M and G' by $\mathbb{G}_{\mathrm{m}}$.

(c) In the general case, we set $G' = \mathrm{Ext}^1(M/W_{-2}M, \mathbb{G}_{\mathrm{m}})$. The natural biextension P'' of $M/W_{-2}M$ and G' by $\mathbb{G}_{\mathrm{m}}$ induces a biextension P' of M and G' by $\mathbb{G}_{\mathrm{m}}$. For each $\chi \in X'$, the extension M of $M/W_{-2}M$ by T defines an extension of $M/W_{-2}M$ by $\mathbb{G}_{\mathrm{m}}$, whence $u'(\chi) \in G'$.

$$\begin{tikzcd} 0 \arrow[r] & T \arrow[r] \arrow[d] & M \arrow[r] \arrow[d, "v"] & M/W_{-2}M \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & \mathbb{G}_{\mathrm{m}} \arrow[r] & P''_{u(\chi)} \arrow[r] & M/W_{-2}M \arrow[r] & 0 \end{tikzcd}$$

The trivial map v trivialises the extension $P'_{u(\chi)}$ of M by $\mathbb{G}_{\mathrm{m}}$

$$\begin{tikzcd}
0 \arrow[r] & \mathbb{G}_{\mathrm{m}} \arrow[r] \arrow[d, equal] & P_{u(\chi)} \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] \arrow[dl, "v"'] & 0 \\
& \mathbb{G}_{\mathrm{m}} \arrow[r] & P'_{u(\chi)} \arrow[r] & M/W_{-2}M &
\end{tikzcd}$$

These trivialisations make P' into a biextension of M and $M^* = [X' \xrightarrow{u'} G']$ by $\mathbb{G}_{\mathrm{m}}$. The 1-motive M^*, endowed with P', is the desired Cartier dual.

This construction allows us to give a more symmetric definition of 1-motives.

Let $M = (X, A, T, G, u)$ be a 1-motive, with Cartier dual $M^* = (X', A', T', G', u')$, and let P_0 be the Poincaré biextension. By passing to the quotient, the morphisms u and u' define morphisms

$$\overline{u}\colon X \to A \quad \text{and} \quad \overline{u}'\colon X' \to A'. \tag{10.2.12.1}$$

The biextension of G and G' by $\mathbb{G}_{\mathrm{m}}$ induced by P_0 is the inverse image of the Poincaré biextension P of A and A' by $\mathbb{G}_{\mathrm{m}}$. By hypothesis, the trivialisations (of the inverse images) of P on $X \times G'$ and $G \times X'$ agree on $X \times X'$, whence

$$\text{a trivialisation } \psi \text{ of } (\overline{u} \times \overline{u}')^* P. \tag{10.2.12.2}$$

A morphism $f\colon M_1 \to M_2$, with transpose (in an evident sense) $M_2^* \to M_1^*$, gives rise to commutative diagrams

$$\begin{tikzcd}
X_1 \arrow[r] \arrow[d, "f_X"'] & A_1 \arrow[d, "f_A"] & X'_1 \arrow[r] & A'_1 \\
X_2 \arrow[r] & A_2 & X'_2 \arrow[r] \arrow[u, "f'_X"] & A'_2 \arrow[u, "f'_A"']
\end{tikzcd} \tag{10.2.12.3}$$

Furthermore, f_A and f'_A are transpose to one another, and, via the isomorphism $(1, f'_A)^* P_1 \simeq (f_A, 1)^* P_2$, we have

$$\psi_1(x_1, f'_X(x'_2)) = \psi_2(f_X(x_1), x'_2) \tag{10.2.12.4}$$

$$\begin{tikzcd}
X_1 \times X'_1 \arrow[r, "\psi_1"] & (P_1 \text{ on } A_1 \times A'_1) \\
X_1 \times X'_2 \arrow[r] \arrow[u] \arrow[d] & (1, f'_A)^* P_1 = (f_A, 1)^* P_2 \text{ on } A_1 \times A'_2 \arrow[u] \arrow[d] \\
X_2 \times X'_2 \arrow[r, "\psi_2"'] & (P_2 \text{ on } A_2 \times A'_2)
\end{tikzcd}$$

We thus obtain a functor form the category of 1-motives to the following category. Its objects consist of:

(a) two abelian varieties A and A', dual to one another;

(b) two free $\mathbb{Z}$-modules X and X', of finite rank;

(c) two morphisms $v\colon X \to A$ and $v'\colon X' \to A'$;

(d) a trivialisation ψ of the inverse image under (v, v') of the Poincaré biextension of A and A' by $\mathbb{G}_{\mathrm{m}}$.

Its arrows consist of diagrams of the form Equation 10.2.12.3 that satisfy the conditions of §12.2

Proposition 10.2.14. *The functor in §12.2 is an equivalence of categories.*

Proof. □

12.3. Algebraic interpretation of the mixed H^1: the case of curves

Let X_0 be an algebraic curve over an algebraically closed field k. Let $p\colon X' \to X_0$ be the normalisation of X_0, and let $\overline{X}'$ be the non-singular projective curve in which X' is a dense open. We denote by X (resp. by $\overline{X}$) the curve induced from X' (resp. from $\overline{X}'$) by contracting each of the finite sets $p^{-1}(s)$ (for $s \in X_0$) to a point.

$$\begin{tikzcd}
X' \arrow[r, hook, "j'"] \arrow[d, "r"'] & \overline{X'} \arrow[d, "\overline{r}"] \\
X \arrow[r, hook, "j"'] \arrow[d, "p_0"'] & \overline{X} \\
X_0
\end{tikzcd}$$

We can characterise X and $\overline{X}$ by the following properties.

(a) p_0 if finite and radicial, and thus induces isomorphisms

$$\mathrm{H}^\bullet(X_0, \mathbb{Z}_\ell) \to \mathrm{H}^\bullet(X, \mathbb{Z}_\ell)$$

(with ℓ coprime to the characteristic).

(b) The singularities of X are analytically isomorphic to those presented by a union of coordinate axes in affine space.

(c) $\overline{X}$ is projective and $\overline{X} \setminus X$ is a finite set of points where $\overline{X}$ is smooth.

The Picard group $\mathrm{Pic}(\overline{X})$ ***TO-DO: unfolds?*** as follows.

(a) Let I be the set of irreducible components of $\overline{X}$ (which is equal to the set of irreducible components of X_0). For each irreducible component Y of $\overline{X}$, the degree of the restriction to Y of an invertible sheaf is a homomorphism

$$\begin{aligned}\deg_Y\colon\ \mathrm{Pic}(\overline{X}) &\to \mathbb{Z} \\ \mathscr{L} &\mapsto \deg_Y(\mathscr{L}).\end{aligned}$$

We thus obtain

$$\deg\colon\ \mathrm{Pic}(\overline{X}) \to \mathbb{Z}^I.$$

This homomorphism is surjective, with kernel $\mathrm{Pic}^0(\overline{X})$.

(b) The morphism from $\mathrm{Pic}^0(\overline{X})$ to the abelian variety $\mathrm{Pic}^0(\overline{X}')$ is surjective, and its kernel is a torus.

Let S be the finite set $\overline{X} \setminus X$, and $\alpha_0 \colon S \to I$ be the map that sends $s \in S$ to the unique irreducible component of $\overline{X}$ such that $s \in \alpha(s)$; let $\alpha \colon \mathbb{Z}^S \to \mathbb{Z}^I$ be induced by α_0.

Each $s \in S$ defines an invertible sheaf $\mathcal{O}(s)$, which defines $u_0 \colon \mathbb{Z}^S \to \operatorname{Pic}(\overline{X})$. We have that $\deg u_0 = \alpha$.

Definition 10.3.4. The *motivic* H^1 *of* X_0, denoted $\mathrm{H}^1_{\mathrm{m}}(X_0)(1)$, is the 1-motive

$$\operatorname{Ker}(\alpha) \to \operatorname{Pic}^0(\overline{X}).$$

By definition, we thus have that $\mathrm{H}^1_{\mathrm{m}}(X_0)(1) = \mathrm{H}^1_{\mathrm{m}}(X)(1)$.

Remarks 10.3.5.

(i) We have that

$$\begin{aligned}\operatorname{Gr}^W_0(\mathrm{H}^1_{\mathrm{m}}(X)(1)) &= \operatorname{Ker}(\alpha \colon \mathbb{Z}^S \to \mathbb{Z}^I)\\ \operatorname{Gr}^W_{-1}(\mathrm{H}^1_{\mathrm{m}}(X)(1)) &= \operatorname{Pic}^0(\overline{X}').\end{aligned}$$

(ii) The cokernel of α is torsion free: it can be identified with the free $\mathbb{Z}$-module with basis $I \setminus \alpha_0(S)$. We thus have a short exact sequence of complexes and a quasi-isomorphism

$$\begin{tikzcd} 0 \arrow[r] & {[\operatorname{Ker}(\alpha) \to \operatorname{Pic}^0(\overline{X}_0)]} \arrow[r] & {[\mathbb{Z}^S \to \operatorname{Pic}(\overline{X}_0)]} \arrow[r] & {[\operatorname{Im}(\alpha) \to \mathbb{Z}^I]} \arrow[r] \arrow[d, "\simeq"] & 0 \\ & & & {[0 \to \operatorname{Coker}(\alpha)]} & \end{tikzcd}$$

Construction 10.3.6. For $n > 0$ integer and coprime to the characteristic of k, we will construct an isomorphism

$$\mathrm{H}^1(X, \mathbb{Z}/n\mathbb{Z})(1) = \mathrm{H}^1(X, \mu_n) \cong T_{\mathbb{Z}/n\mathbb{Z}}(\mathrm{H}^1_{\mathrm{m}}(X)(1)).$$

We know that $\mathrm{H}^1(X, \mu_n)$ can be identified with the isomorphism classes of invertible sheaves $\mathscr{L}$ on X, endowed with an isomorphism $\mathscr{L}^{\otimes n} \cong \mathcal{O}$. Since $p_0 \colon X_0 \to X$ is finite and radicial, we have that $\mathrm{H}^1(X, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} \mathrm{H}^1(X_0, \mathbb{Z}/n\mathbb{Z})$, and $\mathrm{H}^1(X, \mu_n)$ is again the set of isomorphism classes of pairs $(\mathscr{L}, \alpha)$ with $\mathscr{L}$ an invertible sheaf on X_0 and α an isomorphism $\alpha \colon \mathscr{L}^{\otimes n} \xrightarrow{\sim} \mathcal{O}$.

Now consider such a pair $(\mathscr{L}, \alpha)$. The invertible sheaf $\mathscr{L}$ extends to $\overline{\mathscr{L}}$ on $\overline{X}$, and there exists a divisor D of support S such that α extends to $\overline{\alpha} \colon \overline{\mathscr{L}}^{\otimes n} \xrightarrow{\sim} \mathcal{O}(D)$. If there exists an isomorphism β from $\overline{\mathscr{L}}^{\otimes n}$ to $\mathcal{O}(E)$, then it is uniquely determined up to multiplication by an element of the divisible group $\mathrm{H}^0(\overline{X}, \mathcal{O}^*)$. We thus deduce that $(\overline{\mathscr{L}}, D)$ determines $(\mathscr{L}, \alpha)$ up to isomorphism. For a pair $(\overline{\mathscr{L}}, D)$ to come from a suitable pair $(\mathscr{L}, \alpha)$, it is necessary and sufficient that $n[\mathscr{L}] = [\mathcal{O}(D)]$ in $\operatorname{Pic}(\overline{X})$; it comes from $(\mathcal{O}_{X_0}, 0)$ if and only if it is of the form $(\mathcal{O}_{X_0}(D), nD)$.

This identifies $\mathrm{H}^1(X, \mu_n)$ with the H^0 of the tensor product of $[\mathbb{Z}^S \to \operatorname{Pic}(\overline{X})]$ (concentrated in degrees 0 and 1) with $[\mathbb{Z} \xrightarrow{n} \mathbb{Z}]$ (concentrated in degrees -1 and 0):

$$\begin{tikzcd} \mathbb{Z}^S \arrow[r, "u_0"] & \operatorname{Pic}(\overline{X}) \\ \mathbb{Z}^S \arrow[r, "u_0"'] \arrow[u, "n"] & \operatorname{Pic}(\overline{X}) \arrow[u, "-n"'] \end{tikzcd}$$

$$\mathrm{H}^1(X,\mu_n) \cong \mathrm{H}^0\left([\mathbb{Z}^S \to \mathrm{Pic}(\overline{X})] \overset{\mathrm{L}}{\otimes} \mathbb{Z}/n\mathbb{Z}.\right)$$

Since $\mathrm{Coker}(\alpha)$ is torsion free, we read from (ii) of Remarks 10.3.5 that

$$\begin{aligned}\mathrm{H}^1(X,\mu_n) &\cong \mathrm{H}^0\left([\mathbb{Z}^S \to \mathrm{Pic}(\overline{X})] \overset{\mathrm{L}}{\otimes} \mathbb{Z}/n\mathbb{Z}.\right)\\ &\xleftarrow{\sim} \mathrm{H}^0\left(\mathrm{H}^1_{\mathrm{m}}(X) \overset{\mathrm{L}}{\otimes} \mathbb{Z}/n\mathbb{Z}\right)\\ &\cong T_{\mathbb{Z}/n\mathbb{Z}}(\mathrm{H}^1_{\mathrm{m}}(X)(1)).\end{aligned}$$

Variant 10.3.7. For ℓ coprime to the characteristic of k, we have that

$$\mathrm{H}^1(X_0,\mathbb{Z}_\ell(1)) \simeq T_\ell(\mathrm{H}^1_{\mathrm{m}}(X_0)(1)).$$

Construction 10.3.8. Let X_0 be a curve on $\mathbb{C}$. We will construct an isomorphism of mixed Hodge structures

$$\mathrm{H}^1(X_0)(1) \simeq T(\mathrm{H}^1_{\mathrm{m}}(X_0)(1)).$$

Proof. $X = X_0$ □

Construction 10.3.9.

(i) Let $j_*^{\mathrm{m}}\mathcal{O}_X^*$ be the subsheaf of meromorphic functions in $j_*\mathcal{O}_X^*$. We have that

$$\mathrm{H}^1(X,\mathbb{Z}(1)) = \mathbb{H}^1(\overline{X},[\mathcal{O}_{\overline{X}} \xrightarrow{\exp} j_*^{\mathrm{m}}\mathcal{O}_X^*]).$$

(ii) We have that

$$\mathrm{H}^1(X,\mathbb{C}) = \mathbb{H}^1(\overline{X},[\mathcal{O}_{\overline{X}} \xrightarrow{\mathrm{d}} r_*\Omega^1_{\overline{X}'}(\log S)]).$$

(iii) The inclusion of $\mathrm{H}^1(X,\mathbb{Z})$ into $\mathrm{H}^1(X,\mathbb{C})$ is defined by the morphism of complexes

$$\begin{tikzcd} \mathcal{O}_{\overline{X}} \arrow[r, "\exp"] \arrow[d, "\simeq"'] & j_*^{\mathrm{m}}\mathcal{O}_X^* \arrow[d, "\mathrm{d}f/f"] \\ \mathcal{O}_{\overline{X}} \arrow[r] & r_*\Omega^1_{\overline{X}'}(\log S) \end{tikzcd}$$

Proof. Consider the commutative diagram

$$\begin{tikzcd} \mathrm{H}^1(X,\mathbb{Z}(1)) \arrow[r, "\sim"] \arrow[d] \arrow[dr, phantom, "(1)"] & \mathbb{H}^1(X,[\mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^*]) \arrow[d] \arrow[dr, phantom, "(2)"] & \mathbb{H}^1(\overline{X},[j_*\mathcal{O}_X \xrightarrow{\exp} j_*\mathcal{O}_X^*]) \arrow[l, "\sim"'] \arrow[d] \arrow[dr, phantom, "(3)"] & \mathbb{H}^1(\overline{X},[\mathcal{O}_{\overline{X}} \xrightarrow{\exp} j_*^{\mathrm{m}}\mathcal{O}_X^*]) \arrow[l, "\sim"'] \arrow[d] \\ \mathrm{H}^1(X,\mathbb{C}) \arrow[r] & \mathbb{H}^1(X,[\mathcal{O}_X \xrightarrow{\mathrm{d}} r_*\Omega^1_{\overline{X}'}]) & \mathbb{H}^1(\overline{X},[j_*\mathcal{O}_X \to j_*r_*\Omega^1_{X'}]) \arrow[l] & \mathbb{H}^1(\overline{X},[\mathcal{O}_{\overline{X}} \to r_*\Omega^1_{\overline{X}'}(\log S)]) \arrow[l, "\sim"'] \end{tikzcd}$$

The horizontal arrows in the square labelled (1) are isomorphisms, since the sequences

$$0 \to \mathbb{Z}(1) \to \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^* \to 0$$
$$0 \to \mathbb{C} \to \mathcal{O}_X \xrightarrow{\mathrm{d}} r_*\Omega^1_{X'} \to 0$$

are exact. Those in (2) are isomorphisms since $\mathrm{R}^1 j_*\mathcal{O}_X = 0$. Those in (3) are isomorphisms since the morphisms of the complexes that define them are quasi-isomorphisms. This diagram defines the desired construction. □

Recall that if $d\colon F \to G$ is a morphism of abelian sheaves on a space Z, then $\mathbb{H}^1(Z, [F \to G])$ can be identified with the set of isomorphism classes of pairs (P, α), where P is an F-torsor (i.e. a principal homogeneous space under F) and α is a trivialisation of the G-torsor dP.

In particular:

(a) $\mathbb{H}^1(\overline{X}, [\mathcal{O}^*_{\overline{X}} \to r_*\Omega^1_{\overline{X}'}(\log S)])$ can be identified with the set of isomorphism classes of pairs $(\mathscr{L}, \alpha)$, where $\mathscr{L}$ is an invertible sheaf on $\overline{X}$ and α is a connection on $\overline{r}^*\mathscr{L}$ that is holomorphic on X' and can present a simple pole at the points of S. Let a be the projection from $\overline{X}$ to $\mathrm{Spec}(\mathbb{C})$ (i.e. a point). In the fppf topology, the sheaf $\mathrm{R}^1 a_*([\mathscr{O}^*_{\overline{X}} \to r_*\Omega^1_{\overline{X}'}(\log S)])$ is representable by a group scheme $\mathrm{Pic}^\natural(X)$, of which $\mathbb{H}^1(\overline{X}, [\mathscr{O}^*_{\overline{X}} \to r_*\Omega^1_{\overline{X}'}(\log S)])$ is the set of points. $\mathrm{Pic}^\natural(X)$ is an extension of a subgroup of $\mathrm{Pic}(\overline{X})$, containing $\mathrm{Pic}^0(\overline{X})$, by $\mathrm{H}^0(\overline{X}', \Omega^1_{\overline{X}'}(\log S))$.

We define a map

$$\mathbb{Z}^S \to \mathrm{Pic}^\natural(X) \tag{10.3.10.1}$$

by sending each divisor D of $\overline{X}$, concentrated in S, to the invertible sheaf $\mathcal{O}(D)$.

(b) $\mathrm{H}^1(X, \mathbb{C}) = \mathbb{H}^1(\overline{X}, [\mathcal{O}_{\overline{X}} \to r_*\Omega^1_{\overline{X}'}(\log S)])$ is the Lie algebra of $\mathrm{Pic}^\natural(X)$. It is the set of isomorphism classes of pairs (P, α), where P is an $\mathcal{O}_{\overline{X}}$-torsor and α is a connection on P as in (a).

(c) $\mathrm{H}^1(X, \mathbb{Z}(1)) = \mathbb{H}^1(\overline{X}, [\mathcal{O}_{\overline{X}} \to j^{\mathrm{m}}_*\mathcal{O}^*_{\overline{X}}])$ is the set of isomorphism classes of pairs (P, α), where P is an $\mathcal{O}_{\overline{X}}$-torsor and α is an isomorphism from the invertible sheaf $\exp(P)$ to an invertible sheaf $\mathcal{O}_X(D)$ for some D concentrated in S. The map

$$\mathrm{Aut}(P) = \mathbb{C} \to \mathrm{Aut}(\exp(P)) = \mathbb{C}^*$$

is surjective. This allows us to also identify $\mathrm{H}^1(X, \mathbb{Z}(1))$ with the set of pairs (p, d), where p is an isomorphism class of $\mathcal{O}_{\overline{X}}$-torsors (i.e. an element of $\mathrm{Lie}(\mathrm{Pic}\,\overline{X})$) and $d \in \mathrm{Ker}(\alpha)$, defining a divisor concentrated in S with whose image in $\mathrm{Pic}^0(\overline{X})$ is $\exp(p)$. In other words, we have defined an isomorphism

$$\mathrm{H}^1(X, \mathbb{Z}(1)) \xrightarrow{\sim} T_{\mathbb{Z}}(\mathrm{H}^1_{\mathrm{m}}(X)(1)). \tag{10.3.10.2}$$

From part (i) of Lemma 10.3.11, we deduce that this isomorphism is compatible with the weight filtration.

Lemma 10.3.11.

(i) We have that

$$W^1(\mathrm{H}^1(X,\mathbb{Z})) = \mathrm{Im}\left(\mathrm{H}^1(\overline{X},\mathbb{Z}) \to \mathrm{H}^1(X,\mathbb{Z})\right)$$
$$W^0(\mathrm{H}^1(X,\mathbb{Z})) = \mathrm{Ker}\left(\mathrm{H}^1(\overline{X},\mathbb{Z}) \to \mathrm{H}^1(\overline{X}',\mathbb{Z})\right).$$

(ii) The spectral sequence defined by the stupid filtration of $[\mathcal{O}_{\overline{X}} \to \overline{r}_*\Omega^1_{\overline{X}'}(\log S)]$ *degenerates and abuts to the Hodge filtration of* $\mathrm{H}^*(X,\mathbb{C})$.

Proof. We first prove (i). It suffices to prove the first claim: the second follows from Proposition 8.2.5. We have a comparison of exact sequences

$$\begin{tikzcd}
\mathrm{H}^1(\overline{X},\,\mathbb{Z}) \arrow[r] \arrow[d] & \mathrm{H}^1(X,\,\mathbb{Z}) \arrow[r, "\partial"] \arrow[d] & \mathrm{H}^2(\overline{X} \bmod X,\,\mathbb{Z}) \arrow[d, "\simeq"] & \\
\mathrm{H}^1(\overline{X}',\,\mathbb{Z}) \arrow[r] & \mathrm{H}^1(X',\,\mathbb{Z}) \arrow[r, "\partial"] & \mathrm{H}^2(\overline{X}' \bmod X',\,\mathbb{Z}) \arrow[r] & \mathrm{H}^2(\overline{X}',\,\mathbb{Z})
\end{tikzcd}$$

It follows from Example 8.3.7 and Corollary 3.2.17 that $\mathrm{H}^2(\overline{X}' \bmod X,\mathbb{Z})$, and thus $\mathrm{H}^2(\overline{X} \bmod X,\mathbb{Z})$, is pure of weight 2. Taking Theorem 8.2.4, this proves (i).

To prove (ii), we have to return to §10.2 and use the fact that the simplicial scheme $((X'/X)^{\Delta_n})_{n\geqslant 0}$ is smooth (the $(n+1)$-tuple $(X'/X)^{\Delta_n}$ given by the iterated fibre product is the disjoint sum of X', the diagonal, and a finite number of points) and that it admits $((\overline{X}'/\overline{X})^{\Delta_n})_{n\geqslant 0}$ as a smooth compactification. Let $\varepsilon\colon ((\overline{X}'/\overline{X})^{\Delta_n})_{n\geqslant 0} \to \overline{X}$ the augmentation morphism. Let (K,F) be the complex $\mathcal{O} \to \Omega^1(\log S)$ on $((\overline{X}'/\overline{X})^{\Delta_n})_{n\geqslant 0}$ endowed with the stupid filtration. We have an augmentation morphism

$$\alpha\colon [\mathcal{O}_{\overline{X}} \to \overline{r}_*\Omega^1_{\overline{X}'}(\log S)] \to \mathrm{s}\varepsilon_{\bullet *}K.$$

The diagram

$$\begin{tikzcd}
\mathrm{H}^1(X,\mathbb{C}) \arrow[r, equal] \arrow[d, equal] & \mathbb{H}^1\left(((\overline{X}'/\overline{X})^{\Delta_n})_\bullet, K\right) & \mathbb{H}^1(\overline{X}, \mathrm{s}\varepsilon_{\bullet *}K) \arrow[l, "(1)"'] \\
\mathrm{H}^1(X,\mathbb{C}) \arrow[rr, equal] & & \mathbb{H}^1\left(\overline{X}, [\mathcal{O}_{\overline{X}} \to \overline{r}_*\Omega^1_{\overline{X}'} * \log S]\right) \arrow[u]
\end{tikzcd}$$

commutes, and the arrow labelled (1) is an isomorphism since the ε_n are finite, so $\mathrm{R}^i\varepsilon_{n*} = 0$ for $i > 0$. The claim then follows from the fact that α is a filtered quasi-isomorphism. □

We now realise Construction 10.3.8. The diagram

$$\begin{tikzcd}
\mathrm{H}^1(X,\mathbb{Z}(1)) \arrow[r, "\simeq"] \arrow[d, equal] & T_{\mathbb{Z}}(\mathrm{H}^1_{\mathrm{m}}(X)(1)) \arrow[r] & \mathrm{Lie}\,\mathrm{Pic}(\overline{X}) = \mathrm{H}^1(\overline{X},\mathcal{O}_{\overline{X}}) \arrow[d, equal] \\
\mathrm{H}^1(X,\mathbb{C}) \arrow[r, "\simeq"] & \mathrm{Lie}\,\mathrm{Pic}^\natural(X) \arrow[r] & \mathrm{Lie}\,\mathrm{Pic}(\overline{X}) = \mathrm{H}^1(\overline{X},\mathcal{O}_{\overline{X}})
\end{tikzcd}$$

commutes. By (ii) of Lemma 10.3.11 and the definition of Hodge filtrations, Equation 10.3.10.2 is compatible with the Hodge filtration.

Corollary 10.3.13. $\mathrm{H}^1_{\mathrm{m}}(X)(1)^\natural$ *is the extension*

$$\mathrm{Ker}(\alpha) \to \mathrm{Pic}^\natural(X)^0$$

of $\mathrm{H}^1_{\mathrm{m}}(X)(1)$ *induced by Equation 10.3.10.1.*

Proof.

$$\begin{tikzcd}
\operatorname{Ker}(\alpha) \arrow[r] & \operatorname{Pic}^{\natural}(X)^0 \arrow[r] & \operatorname{Pic}^0(\overline{X}) \\
\mathrm{H}^1(X,\mathbb{Z}) \arrow[r] \arrow[u] & \operatorname{Lie}\operatorname{Pic}^{\natural}(X) \arrow[r] \arrow[u] & \operatorname{Lie}\operatorname{Pic}(\overline{X}) \arrow[u]
\end{tikzcd}$$

□

Exercise 10.3.14. Show that the diagram

$$\begin{tikzcd}
\mathrm{H}^1(X,\mathbb{Z}(1)) \arrow[r, equal] \arrow[d] & T_{\mathbb{Z}}(\mathrm{H}^1_{\mathrm{m}}(X)(1)) \arrow[d] \\
\mathrm{H}^1(X,\mu_n) \arrow[r, equal] & T_{\mathbb{Z}/n\mathbb{Z}}(\mathrm{H}^1_{\mathrm{m}}(X)(1))
\end{tikzcd}$$

commutes.

Remark 10.3.15. The isomorphism induced by Construction 10.3.8 and (ii) of Construction 10.3.9, between $\mathbb{H}(\overline{X},\mathscr{O}_{\overline{X}} \to r_*\Omega^1_{\overline{X}'}(\log S))$ (which we may call $\mathrm{H}^1_{\mathrm{DR}}(X)$) and $T_{\mathrm{DR}}(\mathrm{H}^1_{\mathrm{m}}(X)(1))$, equal to $\operatorname{Lie}\operatorname{Pic}^{\natural}(X)$ by Corollary 10.3.13, has been defined in a purely algebraic manner.

12.4. Translation of a theorem of Picard

Let $\mathscr{H}$ be a mixed Hodge structure such that $h^{pq} = 0$ for $p < 0$ or $q < 0$ (resp. for $p > n$ or $q > n$). I provisionally denote by $\mathrm{I}(\mathscr{H})$ (resp. by $\mathrm{II}_n(\mathscr{H})$) the 1-motive defined by the largest mixed Hodge sub-structure of $(\mathscr{H}_{\mathbb{Z}}/\text{torsion})(1)$ (resp. by the largest quotient of $(\mathscr{H}_{\mathbb{Z}}/\text{torsion})(n)$) that is of pure of type $\{(-1,-1),(-1,0),(0,-1),(0,0)\}$. If X in a complex algebraic variety of dimension $\leqslant N$, I conjecture that the 1-motives $\mathrm{I}(\mathrm{H}^n(X,\mathbb{Z}))$, $\mathrm{II}_n(\mathrm{H}^n(X,\mathbb{Z}))$ (for $n \leqslant N$), and $\mathrm{II}_n(\mathrm{H}^n(X,\mathbb{Z}))$ (for $n \geqslant N$) admit a purely algebraic definition. The morphisms

$$\begin{aligned}
T_\ell(\mathrm{I}(\mathrm{H}^n(X,\mathbb{Z}))) &\to (\mathrm{H}^n(X,\mathbb{Z}_\ell)/\text{torsion})(1) \\
T_\ell(\mathrm{II}_n(\mathrm{H}^n(X,\mathbb{Z}))) &\leftarrow \mathrm{H}^n(X,\mathbb{Z}_\ell)(n) \quad \text{for } n \leqslant N \\
T_\ell(\mathrm{II}_N(\mathrm{H}^n(X,\mathbb{Z}))) &\leftarrow \mathrm{H}^n(X,\mathbb{Z}_\ell)(N) \quad \text{for } n \geqslant N
\end{aligned}$$

(and their analogues in de Rham cohomology) should also admit a purely algebraic definition. This is what we have shown for $\dim(X) = 1$. Here is another example.

Let S be a smooth projective surface, U a dense Zariski open of S, and $C = S \setminus U$. Set

$$\mathrm{H}^{\log}_2(U,\mathbb{Z}) = \operatorname{Ker}\big(\mathrm{H}_2(U,\mathbb{Z}) \to \mathrm{H}_2(S,\mathbb{Z})\big).$$

By [G1968, §9], this group does not depend on the chosen smooth compactification S of U. Modulo finite groups, this is nothing more (Corollary 3.2.17) than $W_{-3}\,\mathrm{H}_2(U,\mathbb{Z})$ ($\mathrm{H}_2(U,\mathbb{Q})$, the dual of $\mathrm{H}^2(U,\mathbb{Q})$, is endowed with the dual mixed Hodge structure). We can show that the 1-motive defined by $(\mathrm{H}^{\log}_2(U,\mathbb{Z})/\text{torsion})(-1)$ admits the following description:

(a) By Poincaré duality, we have that $\mathrm{H}_2(U,\mathbb{Z}) = \mathrm{H}^2_{\mathrm{c}}(U,\mathbb{Z})$. The long exact sequence of cohomology of (S,C) gives an exact sequence

$$\mathrm{H}^1(S,\mathbb{Z}) \to \mathrm{H}^1(C,\mathbb{Z}) \to \mathrm{H}^{\log}_2(U,\mathbb{Z}) \to 0.$$

(b) Let $\mathrm{H}^1_{\mathrm{m}}(C)$ be the quotient of $\mathrm{Pic}^0(C)$ by its unipotent radical. We know that $\mathrm{H}^1(C,\mathbb{Z}) = T_{\mathbb{Z}}(\mathrm{H}^1_{\mathrm{m}}(C))$ and $\mathrm{H}^1(S,\mathbb{Z}) = T_{\mathbb{Z}}(\mathrm{Pic}^0(S))$. We thus have a map

$$\mathrm{H}_2^{\log}(U,\mathbb{Z}) \to T_{\mathbb{Z}}(H^1_{\mathrm{m}}(C)/\operatorname{Pic}^0(S))$$

which is an isomorphism modulo finite groups.

(c) The map

$$(\mathrm{H}_2^{\log}(U,\mathbb{Z})/\mathrm{torsion})(-1) \to T_{\mathbb{Z}}(\mathrm{H}^1_{\mathrm{m}}(C)/\operatorname{Pic}^0(S)) \tag{10.4.2.1}$$

induced from (b) is a morphism of mixed Hodge structures (and an isomorphism modulo finite groups).

We can define the isomorphisms (modulo finite groups) induced by Equation 10.4.2.1 between the analogues of $\mathrm{H}_2^{\log}(U,\mathbb{Z})(-1)$ in ℓ-adic or de Rham cohomology, as well as the ℓ-adic or de Rham realisations of $\mathrm{H}^1_{\mathrm{m}}(C)/\operatorname{Pic}^0(S)$, in a purely algebraic way.

We will not prove the above constructions and compatibilities. The seed of the idea is in the following theorem of E. Picard ([PS1895, p. 62]). Here P and A denote polynomials.

Theorem 10.4.3.1. *Every residue of the double integral*

$$\iint \frac{P(x,y)\mathrm{d}x\mathrm{d}y}{A(x,y)}$$

can be thought of as a logarithmic or cyclic period of an abelian integral.

Recall that:

(a) A *k-fold integral* on a non-singular projective variety V of dimension d is a closed rational k-form on V, i.e. it is an element $\alpha \in \mathrm{H}^0(U,\Omega^k)$, where U is a dense open of V, such that $\mathrm{d}\alpha = 0$. Here, $k = 2$ and $V = \mathbb{P}^2$, and so $k = \dim V$ and the condition $\mathrm{d}\alpha = 0$ is trivial.

(b) A k-fold integral α defines an element, again denoted by α, of $\mathrm{H}^k(U,\mathbb{C})$. The *periods* of α are the numbers $\langle \alpha, c\rangle$ for $c \in \mathrm{H}_k(U,\mathbb{Z})$. If c is defined by a singular chain C, then

$$\langle \alpha, c\rangle = \int_C \alpha.$$

(c) The *residues* of a k-fold integral $\alpha \in \mathrm{H}^0(U,\Omega^k)$ are the numbers $\langle \alpha, c\rangle$ for

$$c \in \operatorname{Ker}\big(\mathrm{H}_k(U,\mathbb{Z}) \to \mathrm{H}_k(V,\mathbb{Z})\big).$$

In the case considered by Picard, the map $\mathrm{H}_2(U,\mathbb{Z}) \to \mathrm{H}_2(\mathbb{P}^3,\mathbb{Z})$ is zero, and there is no need to distinguish between periods and residues. For the general case, see [L1924, note I].

(d) To Picard, "logarithmic or cyclic period of an abelian integral" simply means "period of a simple integral β on a curve Y". It is implicit that β and Y are rationally determined by P and A.

Let us translate this theorem. Let S and U be as in §12.4 (for example, $S = \mathbb{P}^2(\mathbb{C})$), and let $\beta \in \mathrm{H}^0(U, \Omega_U^2)$ be a double integral and $c \in \mathrm{H}_2^{\log}(U, \mathbb{Z})$ a cycle. By definition,

$$\langle \beta, c \rangle = \int_c \beta$$

is a residue of β. Let $A = \mathrm{H}_{\mathrm{m}}^1(C)(1) / \operatorname{Pic}^0(S)$ (an extension of an abelian variety by a torus). The 2-form β defines a cohomology class $\mathrm{cl}(\beta) \in \mathrm{H}_{\mathrm{DR}}^2(U)$ and induces a linear form β' on $\mathrm{H}_{\mathrm{DR},2}^{\log}(U) = T_{\mathrm{DR}}(A)$.

The map $\beta \mapsto \beta'$ is rational. The class c defines $c' \in T_{\mathbb{Z}}(A) \otimes \mathbb{Q}$, and we have that

$$\langle \beta, c \rangle = 2\pi i \langle \beta', c' \rangle. \tag{10.4.4.1}$$

We can represent c' by a cycle on C, and β' as a simple integral on C.

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